

Adjoint entropy and measure regularity for Hörmander operators

Results organized by geometric setting

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Abstract

We collect estimates and structural results for nonnegative adjoint solutions of second-order operators with bounded measurable horizontal diffusion matrices. The full parabolic measure-to-density and local $L \log L$ estimate is proved on Carnot groups of step at most two, including all Heisenberg groups, uniformly in an added coordinate viscosity. For arbitrary smooth Hörmander fields we prove scalar occupation and hypersurface nullity estimates, localization identities, and a boundedness theorem when the matrix is a measurable scalar multiple of a fixed smooth matrix. For Carnot groups of arbitrary step we prove qualitative stationary absolute continuity and a stationary entropy theorem for commuting horizontal quadratic forms. Determinant-weighted stationary estimates cover Grushin fields and finite determinant degeneracy. Further sections provide central smoothing estimates, quantitative coefficient-oscillation bounds, control identities, and precisely scoped obstructions. Conditional criteria state their additional premises explicitly. No general entropy theorem for arbitrary smooth Hörmander fields or unrestricted matrices on higher-step groups is asserted.

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1 Common problem, conventions, and guide to the results

1.1 The parabolic measure problem

Let $\Omega \subset \mathbb{R}^n$ be open and let $X_1, \dots, X_m$ be fixed smooth real vector fields such that

$$\text{span}\{Y(x) : Y \in \text{Lie}(X_1, \dots, X_m)\} = \mathbb{R}^n \quad (x \in \Omega).$$

Fix $T > 0$ and $0 < \lambda \leq \Lambda < \infty$. For a symmetric matrix field $A = (a_{ij})$, set

$$\mathcal{L}_A \varphi = \frac{1}{2} \sum_{i,j=1}^m a_{ij} (X_i X_j + X_j X_i) \varphi.$$

There is no additional first-order term. In Euclidean coordinates the vector-field products themselves may contain first-order coefficients. Symmetry allows the equivalent notation $\sum_{i,j} a_{ij} X_i X_j$. The formal adjoint is always defined by duality, with respect to the stated volume measure; derivatives of the Borel coefficients are never assumed to exist as functions.

The *parabolic measure hypotheses* mean the following. The family $(\nu_t)_{0 < t < T}$ consists of nonnegative finite Borel measures on Ω , with $\nu_t(\Omega) \leq 1$ for almost every t , and is weakly measurable: $t \mapsto \int \psi d\nu_t$ is measurable for every $\psi \in C_c(\Omega)$. Set $d\nu(t, x) = dt d\nu_t(x)$. The symmetric matrix A is Borel measurable and $\lambda I_m \leq A \leq \Lambda I_m$ for ν -almost every (t, x) . For $\varepsilon \in [0, 1]$ and a finite nonnegative Borel measure μ on $(0, T) \times \Omega$, the equation is

$$\int (\partial_t \varphi + \mathcal{L}_A \varphi + \varepsilon \Delta_x \varphi) d\nu = \int \varphi d\mu \quad (\varphi \in C_c^\infty((0, T) \times \Omega)). \quad (1.1)$$

Equivalently $\partial_t \nu = \mathcal{L}_A^* \nu + \varepsilon \Delta_x \nu - \mu$. The desired conclusion is a density $p \geq 0$ with

$$\nu = p dt dx, \quad \int_0^T \int_\Omega p \leq T, \quad \int_0^T \int_K p \log(1 + p) \leq C_K \quad \text{for every } K \Subset \Omega, \quad (1.2)$$

where C_K depends only on $T, K, \Omega, X_1, \dots, X_m, \lambda, \Lambda$. It is independent of A, ε, ν, μ . In particular there is no initial density or initial entropy assumption, no continuity assumption on A , and no smallness assumption on Λ/λ .

1.2 The stationary problem and geometric conventions

The stationary measure problem replaces (1.1) by

$$\int_\Omega (\mathcal{L}_A \varphi + \varepsilon \Delta_x \varphi) d\sigma = \int_\Omega \varphi d\mu_0 \quad (\varphi \in C_c^\infty(\Omega)), \quad (1.3)$$

where $\sigma \geq 0$ has mass at most one, $A = A(x)$ has the same σ -almost-everywhere cone bounds, and μ_0 is a finite nonnegative Borel measure. Thus the stationary sign is $\mathcal{L}_A^* \sigma + \varepsilon \Delta_x \sigma = \mu_0 \geq 0$. A theorem stated for a *stationary density* assumes in addition $\sigma = p dx$. Whenever only a local equation is needed, the theorem explicitly permits a locally finite source and bounds the mass on a specified larger set. Neither an a priori density assumption nor a local normalization is silently promoted to the full measure assertion.

A Carnot group is a connected simply connected nilpotent Lie group with a stratification $\mathfrak{g} = V_1 \oplus \dots \oplus V_s$, $[V_1, V_j] = V_{j+1}$, $V_{s+1} = 0$. Exponential coordinates identify it with Euclidean space, Haar measure with Lebesgue measure, and the homogeneous dilation with $\delta_r(v_1 + \dots + v_s) = r v_1 + \dots + r^s v_s$. Its homogeneous dimension is $\sum_j j \dim V_j$. The letters D or Q for this dimension are local to a section and are distinguished there from a stress tensor. Horizontal fields on a Carnot group are left invariant unless explicitly designated as right invariant. They are divergence free for Haar measure. Our convolution convention is $(f * g)(x) = \int f(y) g(y^{-1} x) dy$.

A componentwise convolution on one side and the heat semigroup on the other side have different derivative-commutation properties; the proofs retain this distinction.

The notation $K \Subset \Omega$ means that K is compact and contained in the open set Ω . Matrix norms without a subscript can be replaced by the Frobenius norm in finite dimensions. Constants may change from line to line, but their admissible dependencies are fixed by the statement in which they occur. The fixed fields are part of the geometry. In the step-two viscosity theorem the phrase *coordinate Laplacian* means the literal Euclidean Laplacian in standard exponential coordinates.

1.3 Which settings give which conclusions?

The following guide separates complete PDE estimates from auxiliary and conditional statements. A conditional criterion is itself a proved implication, but its additional premise is not asserted for every admissible coefficient field.

Setting	Conclusion and additional hypotheses	Reference
Arbitrary smooth Hörmander fields	Parabolic measure traces, scalar occupation bounds, and nullity of smooth spacetime hypersurfaces; $0 \leq \varepsilon \leq 1$.	Propositions 4.1, 4.3, 4.5.
Arbitrary smooth Hörmander fields	Stationary local boundedness for density input when $A = a(x)B(x)$, with B fixed smooth positive definite and a Borel between fixed positive constants; $\varepsilon = 0$.	Theorem 6.1.
Arbitrary smooth fields, density input	Stationary determinant-weighted power and entropy estimates; no bracket hypothesis needed for the weighted inequality itself.	Theorem 12.4.
Grushin and finite determinant type	Stationary local unweighted entropy and specified powers, for density input and arbitrary cross coefficients; uniform coordinate viscosity.	Theorems 12.7, 12.9.
Carnot groups of step at most two	Full parabolic and stationary measure-to-density $L \log L$ estimate, arbitrary Borel matrix cone, $0 \leq \varepsilon \leq 1$.	Theorem 2.1.
Smooth frames in a step-two Carnot distribution	The full conclusions at zero viscosity, also after a specified distribution-preserving smooth chart.	Theorem 3.2, Corollary 3.3.
Every Heisenberg group $\mathbb{H}^d$	The preceding full estimate with the explicit standard fields and coordinate Laplacian.	Corollary 2.2.
Carnot groups of arbitrary step	Qualitative stationary absolute continuity for a positive definite stress, even with a locally finite signed measure on the right; $\varepsilon = 0$. No uniform entropy conclusion is inferred.	Theorem 8.2.

Setting	Conclusion and additional hypotheses	Reference
Carnot groups of arbitrary step	Stationary entropy for coefficients in a fixed space of horizontal quadratics commuting with one fixed positive sub-Laplacian; $\varepsilon = 0$.	Theorem 14.3, Corollary 14.4.
Carnot groups of arbitrary step	Quantitative small-set estimate with an explicit coefficient-shape oscillation error; $\varepsilon = 0$ and stationary density input.	Section 19.
The first Heisenberg group $\mathbb{H}^1$	Stronger power estimates under axisymmetry; sharp relative ball exponent at zero viscosity; exact stationary power examples.	Theorems 16.2, 16.4, 16.5.
The first Heisenberg group $\mathbb{H}^1$	Exact complex-variable stationary stresses with uniform powers below 2; failure of a uniform strong L^2 endpoint.	Theorems 17.2, 17.3.
Heisenberg heat smoothing	Central coefficient square function on $\mathbb{H}^d$ and enhanced Fisher estimates on $\mathbb{H}^1$ for arbitrary integrable inputs.	Theorems 13.2, 13.3.
Step-three Carnot groups	Exact triangular intertwining algebra and a homogeneity obstruction. An unrestricted entropy theorem is not claimed.	Propositions 14.11, 14.12.

Other sections prove localization, reference-semigroup, control, Poisson-kernel, directional-variance, and structural duality results. Their individual statements specify whether a PDE is assumed. The geometric counterexamples in Section 16 show limitations of scalar or convex testing budgets; they are explicitly distinguished from counterexamples to the adjoint PDE estimate.

1.4 Proof conventions and analytic inputs

The proofs of the collected results are given in this manuscript. Classical analytic inputs are stated where used: the fixed smooth Hörmander subelliptic estimate, the fixed smooth Carnot heat-kernel estimates, convex transport and Alexandrov differentiation, and the elementary singular-integral and Sobolev consequences of those inputs. Their hypotheses and normalizations are given explicitly; proofs of these foundational background theorems are not reproduced in their entirety. All regularization, localization, positivity, and entropy arguments specific to the results here are supplied. No appeal to a measurable-coefficient subelliptic heat-kernel theorem is made. References identify background results, and do not serve as a claim of priority or novelty for this collection.

2 The full estimate on Carnot groups of step at most two

The result in this section includes absolute continuity of the unknown measure. The viscosity is the Euclidean coordinate Laplacian in the coordinates specified below. All dependence on the geometry is on the fixed group and the fixed generating family.

Theorem 2.1. *Let $G = \mathbb{R}^n$ be a fixed Carnot group of step at most two in standard exponential coordinates, and let $X_1, \dots, X_m$ be fixed left-invariant degree-one generators. Redundant generators are allowed. Let $\Omega \subset G$ be open, $T > 0$, and $0 < \lambda \leq \Lambda < \infty$. Define*

$$L_A \varphi = \frac{1}{2} \sum_{i,j=1}^m a_{ij} (X_i X_j + X_j X_i) \varphi.$$

Suppose $(\nu_t)_{0 < t < T}$ is a weakly measurable family of nonnegative finite Borel measures on Ω , with $\nu_t(\Omega) \leq 1$ for almost every t . Here weak measurability means that $t \mapsto \int \psi d\nu_t$ is measurable for every $\psi \in C_c(\Omega)$. Put $d\nu = dt d\nu_t$. Let A be a Borel symmetric matrix field on $(0, T) \times \Omega$, with $\lambda I \leq A \leq \Lambda I$ ν -almost everywhere. If $0 \leq \varepsilon \leq 1$, μ is a finite nonnegative Borel measure on $(0, T) \times \Omega$, and

$$\int (\partial_t \varphi + L_A \varphi + \varepsilon \Delta_{\text{coord}} \varphi) d\nu = \int \varphi d\mu \quad (\varphi \in C_c^\infty((0, T) \times \Omega)),$$

then $\nu = p dt dx$ with $p \geq 0$, $p \in L^1$, and

$$\int_0^T \int_\Omega p \leq T, \quad \int_0^T \int_K p \log(1 + p) \leq C \quad (K \Subset \Omega).$$

Here C depends only on $T, K, \Omega, G, X_1, \dots, X_m, \lambda, \Lambda$, and is independent of ε, A, ν, μ . In particular no initial density or initial entropy is assumed.

The stationary version also holds: if σ is a nonnegative finite Borel measure with $\sigma(\Omega) \leq 1$, $A = A(x)$ is Borel and symmetric with $\lambda I \leq A \leq \Lambda I$ σ -almost everywhere, and $L_A^ \sigma + \varepsilon \Delta_{\text{coord}} \sigma = \mu_0 \geq 0$ for a finite Borel measure μ_0 , then $\sigma = p dx$ and $\int_K p \log(1 + p) \leq C$ with the same allowed spatial and ellipticity dependencies.*

2.1 A reference family that retains the coordinate viscosity

Write exponential coordinates as $(h, z) \in V_1 \oplus V_2$, where $d_1 = \dim V_1$ and $d_2 = \dim V_2$. Put $Z_\alpha = \partial_{z_\alpha}$. The X_i are affine, divergence-free fields and

$$[X_i, X_j] = \sum_\alpha c_{ij}^\alpha Z_\alpha, \quad [X_i, Z_\alpha] = 0.$$

For fixed ε enlarge the family to

$$(W_1, \dots, W_N) = (X_1, \dots, X_m, \sqrt{\varepsilon} \partial_{h_1}, \dots, \sqrt{\varepsilon} \partial_{h_{d_1}}, \sqrt{\varepsilon} \partial_{z_1}, \dots, \sqrt{\varepsilon} \partial_{z_{d_2}}).$$

Thus $N = m + n$. These fields are divergence-free and have constant central brackets

$$[W_a, W_b] = \sum_\alpha c_{ab}^{\alpha, \varepsilon} Z_\alpha, \quad [W_a, Z_\alpha] = 0, \quad \sup_{\varepsilon \in [0, 1]} |c_{ab}^{\alpha, \varepsilon}| < \infty.$$

Zero fields at $\varepsilon = 0$ cause no difficulty. Define

$$\begin{aligned} L &= \sum_a W_a^2 = L_0 + \varepsilon \Delta_{\text{coord}}, & L_0 &= \sum_i X_i^2, & H_s &= e^{sL}, \\ Dv &= \sum_a W_a v_a, & \mathcal{T}Q &= \sum_{a,b} W_a W_b Q_{ab}. \end{aligned}$$

The original stress becomes $\text{diag}(A, I_n)\nu$, whose cone bounds are independent of ε .

The affine diffusion is nonexplosive: its horizontal coordinates are Brownian linear combinations and its central coordinates are linear stochastic integrals and Brownian terms. Hence H_s is positive, conservative and symmetric on Lebesgue measure. Central translations commute with it. For each fixed ε it smooths at positive times: for $\varepsilon = 0$ this is the fixed Carnot heat kernel, and for $\varepsilon > 0$ it follows from ordinary local parabolic regularity of the smooth elliptic reference operator. No derivative estimate uniform in ε is used.

Let $Q_h = d_1 + 2d_2$ be the homogeneous dimension. The fixed Carnot heat bound gives the Nash inequality for L_0 . One can see this directly from

$$\|f\|_2^2 \leq C t^{-Q_h/2} \|f\|_1^2 + t \sum_i \|X_i f\|_2^2$$

and optimization in t . The form of L dominates that of L_0 , so the same Nash constant applies to every ε . The differential inequality for $\|H_s f\|_2^2$, then self-adjointness and the semigroup property, yield

$$\|H_s\|_{1 \rightarrow \infty} \leq C s^{-Q_h/2} \quad (2.1)$$

uniformly in ε . The only external heat input here is the fixed smooth Carnot heat estimate; Gaussian and derivative versions are given in Biagi–Bramanti, Theorem 2.4 [2]. The one-dimensional abelian case is the ordinary Gaussian.

2.2 Exact scalar, vector and tensor regularization

Let κ be a large fixed parameter. In the central variables use

$$c_s(z) = \prod_{\alpha=1}^{d_2} \frac{1}{2\kappa s} \operatorname{sech} \frac{\pi z_\alpha}{2\kappa s}, \quad P_s = H_s C_s,$$

where C_s is central convolution. Empty central products are the identity. Set

$$(J_\alpha)_{ab} = -c_{ab}^{\alpha, \varepsilon}, \quad K_\alpha B = J_\alpha B - B J_\alpha.$$

The J_α are skew-symmetric, and K_α preserves real symmetric matrices and annihilates I . Constant-bracket expansion gives

$$LD = D \left(L - 2 \sum_\alpha Z_\alpha J_\alpha \right), \quad L\mathcal{T} = \mathcal{T} \left(L - 2 \sum_\alpha Z_\alpha K_\alpha \right). \quad (2.2)$$

For example $[W_a, L] = 2 \sum_{b, \alpha} c_{ab}^{\alpha, \varepsilon} Z_\alpha W_b$; expanding $[W_a W_b, L]$ proves the second formula after symmetric contraction.

Define

$$U_s = H_s C_s \exp \left(-2s \sum_\alpha Z_\alpha J_\alpha \right), \quad S_s = H_s C_s \exp \left(-2s \sum_\alpha Z_\alpha K_\alpha \right).$$

The products have ordinary integrable matrix kernels, despite the fact that the isolated correction exponentials are not positive operators. Here are the required bounds and their justification. For real z , the product sech kernel obeys

$$|c_s(z+w)| \leq C_* c_s(z) \quad (|w_\alpha| \leq c_* \kappa s), \quad |\partial^\beta c_s(z)| \leq C_* \beta! (c_* \kappa s)^{-|\beta|} c_s(z).$$

The first inequality follows coordinatewise from the elementary complex formula for $\cosh$ in a strip; the second is Cauchy's estimate. For bounded real matrices M_α , expand the single ordered exponential kernel as

$$\sum_{j=0}^{\infty} \frac{(-2s)^j}{j!} \left(\sum_\alpha M_\alpha \partial_{z_\alpha} \right)^j c_s(z).$$

The derivative factorial is at most $j!$, while the sum of the norms of ordered matrix words is at most $(\sum_\alpha \|M_\alpha\|)^j$. Thus the series converges absolutely pointwise and in L^1 for large κ , and differs from $c_s I$ by at most $C\kappa^{-1}c_s$. Termwise Fourier transformation identifies the claimed operator. Its total integral is I . No commutation among the matrices is assumed. All constants are uniform in ε because their matrix norms are uniformly bounded.

It follows that

$$|U_s F| \leq C P_s |F|, \quad |S_s Q - P_s Q|_F \leq C \kappa^{-1} P_s |Q|_F. \quad (2.3)$$

The central difference kernel, before applying H_s , commutes with every W_a . The full central matrix operator commutes with L . Therefore (2.2) gives, exactly,

$$D U_s = P_s D, \quad \mathcal{T} S_s = P_s \mathcal{T}. \quad (2.4)$$

These identities can first be verified with compact central Fourier support and smooth data; the integrable kernel bounds give the distributional passage. The regularizers are approximate identities, since their kernels have bounded total variation, mass I , and shrink to the origin, while H_s tends to the identity.

2.3 Where the bracket controls entropy

For $q \geq 0$ with $\int q \leq 1$, put $r_s = 1 + P_s q$. Write

$$F(r) = r \log r - r + 1, \quad E(s) = \int F(r_s), \quad I(s) = \int \frac{\sum_a |W_a r_s|^2}{r_s}.$$

Differentiation of the central multiplier $\text{sech}(\kappa s \xi)$ gives the positive symmetric jump generator with symbol $-\kappa \xi \tanh(\kappa s \xi)$. Its Levy density in each central coordinate is

$$n_s(z) = \frac{\pi}{4\kappa s^2} \frac{\cosh t}{\sinh^2 t}, \quad t = \frac{\pi|z|}{2\kappa s}. \quad (2.5)$$

For completeness, integrate the elementary identity for $\log(1 + \xi^2/b^2)$ and sum the product expansion of $\cosh$ to get

$$\log \cosh(a\xi) = \int_{\mathbb{R}} \frac{1 - \cos(\xi z)}{2|z| \sinh(\pi|z|/(2a))} dz.$$

Differentiating at $a = \kappa s$ proves (2.5); the integrands are positive and the second difference removes the singularity at zero.

Central convolution gives $|Z_\alpha \log r_s| \leq \pi/(2\kappa s)$. The scalar entropy identity is

$$-E'(s) = I(s) + J_s(r_s),$$

where J_s is the sum of the central symmetric jump entropy forms. For positive u, v ,

$$(u - v)(\log u - \log v) \leq [4 + |\log(u/v)|](\sqrt{u} - \sqrt{v})^2.$$

Also $z^2 n_s(z)(4 + t) \leq C\kappa$. Consequently, with $h_s = \sqrt{r_s} - 1$,

$$J_s(r_s) \leq C\kappa \sum_{\alpha} \langle h_s, |Z_\alpha| h_s \rangle.$$

The crucial bracket inequality is

$$\sum_{\alpha} \langle h, |Z_\alpha| h \rangle \leq C_G \sum_i \|X_i h\|_2^2 \leq C_G \sum_a \|W_a h\|_2^2. \quad (2.6)$$

Indeed, Fourier transform in all central coordinates. Put $\omega_{ij}(\xi) = \sum_{\alpha} c_{ij}^{\alpha} \xi_{\alpha}$. Bracket spanning implies $|\xi|_1 \leq C \sum_{i < j} |\omega_{ij}(\xi)|$. Expanding the nonnegative square of the magnetic derivatives $X_{i,\xi} - i \operatorname{sign}(\omega_{ij}) X_{j,\xi}$ gives

$$|\omega_{ij}(\xi)| \|h_{\xi}\|_2^2 \leq \|X_{i,\xi} h_{\xi}\|_2^2 + \|X_{j,\xi} h_{\xi}\|_2^2.$$

Summation, Plancherel and form closure prove (2.6), including redundant generators. Since $h_s^2 \leq P_s q$, it belongs to L^2 despite the constant baseline. We have proved

$$J_s(r_s) \leq C \kappa I(s), \quad E(0) \leq C + (1 + C \kappa) \int_0^1 I(s) ds. \quad (2.7)$$

The terminal bound $E(1) \leq C$ uses (2.1) and $F(1+q) \leq q \log(1+q)$. All baseline entropies here are finite whenever the initial perturbation is bounded and integrable; no entropy is assigned to the constant itself.

2.4 A global estimate with a finite localization flux

First consider smooth data $q(t, x) \geq 0$, symmetric $Q(t, x)$, and a generalized source equation

$$\mathcal{T}Q = \partial_t q + \beta + DF_0 + G_0, \quad (2.8)$$

where $\beta \geq 0$, and, uniformly in t ,

$$\int q \leq 1, \quad \lambda_* q I \leq Q \leq \Lambda_* q I, \quad \int \frac{|F_0|^2}{1+q} \leq B, \quad \|G_0\|_1 \leq B.$$

Assume initially finite entropy and the regularity obtained by the presmoothing in the next section. Constants below may depend on B, λ_*, Λ_* but not on the entropy value or ε . Set

$$r_0 = 1 + q, \quad R_0 = I + Q, \quad r_s = P_s r_0, \quad R_s = S_s R_0, \quad a_s = R_s / r_s.$$

Choose κ so large that (2.3) gives $c r_s I \leq R_s \leq C r_s I$, where c, C stay fixed as κ increases further. The finite coefficient perspectives for the two common scalar evolutions

$$b_s = \frac{H_s R_0}{H_s r_0}, \quad \bar{a}_s = \frac{P_s R_0}{r_s}$$

give

$$\int_0^1 \int (H_s r_0) |\nabla_W b_s|_F^2 ds \leq C, \quad \int_0^1 \int r_s |\nabla_W \bar{a}_s|_F^2 ds \leq C. \quad (2.9)$$

The perspectives used are $\int (H_s r_0) |b_s - I|^2$ and $\int r_s |\bar{a}_s - I|^2$, whose initial values are bounded by $C \int q$. For heat, differentiation gives minus twice the displayed energy; the positive central jump generator adds a nonpositive perspective term. Thus the infinite baseline never appears as an infinite variance.

Write $E_s^{\text{mat}} = R_s - P_s R_0$ and $e_s = E_s^{\text{mat}} / r_s$. The relative central kernel bound and weighted Cauchy–Schwarz imply

$$|e_s| \leq C / \kappa, \quad \int \frac{|W_a E_s^{\text{mat}}|^2}{r_s} \leq \frac{C}{\kappa^2} \int \frac{|W_a H_s R_0|^2}{H_s r_0}.$$

Commutation with central convolution is essential in this derivative estimate. Expand $W_a H_s R_0 = (H_s r_0) W_a b_s + b_s W_a H_s r_0$ and use the quotient rule. Since $a_s = \bar{a}_s + e_s$, (2.9) and ordinary heat entropy dissipation give

$$\int_0^1 \int r_s |\nabla_W a_s|_F^2 ds \leq C + \frac{C}{\kappa^2} E_0(t) + \frac{C}{\kappa^2} \int_0^1 I(s, t) ds, \quad E_0(t) = \int F(1 + q(t)). \quad (2.10)$$

Here the ordinary heat Fisher integral is at most $E_0(t)$. The constants in (2.10) do not grow with κ .

By (2.4), equation (2.8) becomes

$$\mathcal{T}R_s = \partial_t r_s + P_s \beta + DU_s F_0 + P_s G_0.$$

Weighted kernel Jensen gives $\int |U_s F_0|^2 / r_s \leq CB$. Test with the nonnegative $\log r_s$ and integrate once in the horizontal fields. The principal term is $-\int r_s a_{s,ab} (W_a \log r_s) (W_b \log r_s)$; the coefficient cross term costs a small fraction of I and $C \int r_s |\nabla_W a_s|^2$. The positive source has the favorable sign. The signed scalar source costs at most $CB(1 + |\log s|)$, since (2.1) gives

$$\|\log r_s(t)\|_\infty \leq C(1 + |\log s|), \quad E(s, t) \leq C(1 + |\log s|). \quad (2.11)$$

Thus, for $0 \leq \theta \leq 1$, $\theta \in C_c^\infty(0, T)$, temporal integration by parts yields

$$\int \theta(t) I(s, t) dt \leq C \int \theta(t) \int r_s |\nabla_W a_s|^2 dt + C(T + \|\theta'\|_1)(1 + |\log s|). \quad (2.12)$$

The temporal contribution is $\int \theta' E(s, t)$, controlled by (2.11); no value of the entropy at $t = 0$ or T is used.

All global tests can be made first with Carnot cutoffs tending to one. For radius $R \geq 1$ their first and second W derivatives are bounded by C/R and C/R^2 uniformly in ε . At positive smoothing times, $(\log r_s)^2 r_s \leq C_s(r_s - 1)$, weighted coefficient gradients and fluxes are integrable, and the scalar source is integrable. Cauchy–Schwarz removes the first-order cutoff terms and the finite perturbation removes the second-order terms. The constant tensor itself has zero double divergence. The symmetric jump identities follow by truncating their kernels, using convexity, and passing by monotone convergence or finite perturbation domination. These arguments need no uniform bounds on heat-kernel derivatives.

Put $\mathbf{E} = \int \theta E_0$ and $\mathbf{I} = \int \theta(t) \int_0^1 I(s, t) ds dt$. Before absorption these are finite for the presmoothed data; in particular the positive scalar entropy identity gives $\mathbf{I} \leq \mathbf{E}$. Integrating (2.10) and (2.12), then absorbing $C\kappa^{-2}\mathbf{I}$, gives

$$\mathbf{I} \leq C(T + \|\theta'\|_1) + C\kappa^{-2}\mathbf{E}.$$

On the other hand (2.7) gives $\mathbf{E} \leq CT + (1 + C\kappa)\mathbf{I}$. A second choice of sufficiently large fixed κ absorbs the coefficient $C(1 + C\kappa)/\kappa^2$ of $\mathbf{E}$. Therefore

$$\int \theta(t) \int F(1 + q(t, x)) dx dt \leq C(T + \|\theta'\|_1). \quad (2.13)$$

This is the complete entropy closure. Quadratic smallness of the tensor error and linear growth of the central entropy cost are both used.

2.5 Localization, rough measures and endpoints

Return to the measures in Theorem 2.1. Fix $\eta \in C_c^\infty(\Omega)$, $0 \leq \eta \leq 1$, equal to one near K , and put $\chi = \eta^2$. On all space define

$$q = \chi\nu, \quad Q = \chi \operatorname{diag}(A, I_n)\nu,$$

extending the finite measures by zero. The product rule, with $\widehat{Q} = \operatorname{diag}(A, I_n)\nu$, gives exactly

$$\begin{aligned} \mathcal{T}Q &= \partial_t q + \beta + DF_0 + G_0, \quad \beta = \chi\mu \geq 0, \\ (F_0)_a &= 2 \sum_b (W_b \chi) \widehat{Q}_{ab}, \quad G_0 = - \sum_{a,b} (W_a W_b \chi) \widehat{Q}_{ab}. \end{aligned}$$

Symmetry justifies the derivative ordering. On each time slice let $r_0 = dx + q_t$. These flux measures satisfy

$$\int \left| \frac{dF_{0,t}}{dr_{0,t}} \right|^2 dr_{0,t} \leq C, \quad |G_{0,t}|(G) \leq C. \quad (2.14)$$

Indeed $|\nabla_W \chi|^2 / \chi \leq 4|\nabla_W \eta|^2$ on $\chi > 0$, while $W\chi = 0$ where $\chi = 0$. All cutoff constants are uniform in ε . No absolute continuity of ν is needed for this Radon–Nikodym calculation.

An ordinary group mollifier need not commute with the coordinate viscosity. Instead choose a fixed sufficiently large presmoothing parameter κ_{pre} and apply the exact regularizers to these already localized measures:

$$q_\delta = P_\delta q, \quad Q_\delta = S_\delta Q, \quad \beta_\delta = P_\delta \beta, \quad F_{0,\delta} = U_\delta F_0, \quad G_{0,\delta} = P_\delta G_0.$$

Their equation is unchanged in form. The scalar mass is at most one, the matrix cone is enlarged by fixed factors only, $\beta_\delta \geq 0$, and (2.14) survives by positive-kernel weighted Jensen and (2.3). The scalar source keeps its L^1 bound. The tensor cone comparison is with q_δ itself; no artificial finite density is inserted into the target measure. The baseline is introduced only in the entropy proof.

Convolve also in time with a positive compact smooth kernel. Extend the localized slice measures by zero outside $(0, T)$ for this convolution, but use the equation only on interior time slabs whose distance from the endpoints exceeds the convolution radius. There are consequently no endpoint source terms. For each fixed compactly supported θ , the resulting triples are smooth, have finite entropy by (2.1), retain all the preceding uniform bounds, and satisfy the generalized equation on its support. Apply (2.13), with a second large parameter chosen for the already enlarged cone. The constants are independent of both smoothing radii and of ε .

The bound is superlinear in the scalar density. On compact space-time sets, uniform integrability and weak L^1 compactness identify the measure limit $\chi\nu$ with a density. Both scalar regularizers are approximate identities, so the weak measure limit is the original $\chi\nu$. Convex lower semicontinuity gives the same entropy bound. Exhausting the spatial domain and interior time interval identifies a single density p of ν . Taking cutoffs $\theta_j \rightarrow 1$ on $(0, T)$ with $0 \leq \theta_j \leq 1$ and $\|\theta'_j\|_1 \leq 2$, and using Fatou, yields the estimate over the entire time interval. Finally,

$$p \log(1 + p) \leq F(1 + p) + p,$$

and the total mass equals $\nu((0, T) \times \Omega) \leq T$. This proves the parabolic assertion. Apply it with a time-independent stationary measure and $\mu = dt \mu_0$ on a time interval of length one to obtain the stationary assertion. $\square$

2.6 Explicit Heisenberg specialization

Corollary 2.2 (Heisenberg adjoint entropy). *Let $d \geq 1$ and equip $\mathbb{H}^d = \mathbb{R}^{2d+1}$ with*

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + \tfrac{1}{2}(x \cdot y' - y \cdot x')),$$

and the horizontal fields

$$X_j = \partial_{x_j} - \tfrac{1}{2}y_j \partial_z, \quad Y_j = \partial_{y_j} + \tfrac{1}{2}x_j \partial_z \quad (1 \leq j \leq d).$$

For this family of $2d$ fields, every assertion of Theorem 2.1 holds, with $dx dy dz$ as Haar measure and $\Delta_{\text{coord}} = \sum_j (\partial_{x_j}^2 + \partial_{y_j}^2) + \partial_z^2$. Thus arbitrary Borel symmetric $2d \times 2d$ diffusion matrices with fixed positive ellipticity bounds satisfy the parabolic and stationary local $L \log L$ estimates, uniformly for $0 \leq \varepsilon \leq 1$.

Proof. The displayed fields are left invariant for the displayed group law, and $[X_j, Y_k] = \delta_{jk}Z$, where $Z = \partial_z$, while the other horizontal brackets vanish. They generate a step-two Carnot group with dilations $\delta_r(x, y, z) = (rx, ry, r^2z)$ and homogeneous dimension $2d+2$. Consequently Theorem 2.1 applies with exactly these coordinates, measure, and viscosity. No change of the second-order normalization is needed: symmetry of A makes the factor $1/2$ in the symmetrized definition cancel the two equal summed terms. $\square$

Proposition 2.3 (Central energy on $\mathbb{H}^d$). *For every f in the closure of $C_c^\infty(\mathbb{H}^d)$ under the norm $(\|f\|_2^2 + \sum_j (\|X_j f\|_2^2 + \|Y_j f\|_2^2))^{1/2}$,*

$$d\langle f, |Z|f \rangle \leq \sum_{j=1}^d (\|X_j f\|_2^2 + \|Y_j f\|_2^2).$$

Proof. Use the unitary Fourier transform in z . At frequency ξ , the horizontal operators are $P_j = \partial_{x_j} - i\xi y_j/2$ and $Q_j = \partial_{y_j} + i\xi x_j/2$. Skew-adjointness and $[P_j, Q_j] = i\xi$ give, for $\xi \neq 0$,

$$\|(P_j - i \operatorname{sgn}(\xi) Q_j)u\|_2^2 = \|P_j u\|_2^2 + \|Q_j u\|_2^2 - |\xi| \|u\|_2^2.$$

Nonnegativity, summation in j , and Plancherel prove the assertion on test functions. Form closure proves the stated extension. $\square$

Remark 2.4 (Geometric limits of this proof). The tensor intertwiners above use centrality of every horizontal bracket. They do not establish an entropy estimate for unrestricted matrices on step-three groups or for arbitrary smooth Hörmander families. The viscosity argument uses the affine form of step-two generators in standard exponential coordinates. It does not replace a coordinate Laplacian by a different operator after a nonlinear coordinate change.

3 Variable frames in a fixed step-two distribution

The zero-viscosity theorem also transfers to certain non-left-invariant families. The extra geometric assumption in this section is a fixed step-two Carnot horizontal distribution, or a specified smooth local identification with one. No such identification is asserted for an arbitrary bracket-generating distribution.

Proposition 3.1 (A locally bounded horizontal drift). *Under the hypotheses of Theorem 2.1, set $\varepsilon = 0$ and allow a Borel horizontal drift $b = (b_1, \dots, b_m)$ such that, for every $K' \Subset \Omega$, $|b(t, x)| \leq B_{K'}$ for ν -almost every $(t, x) \in (0, T) \times K'$. Replace the equation by*

$$\int \left(\partial_t \varphi + \mathcal{L}_A \varphi + \sum_i b_i X_i \varphi \right) d\nu = \int \varphi d\mu.$$

Then the same absolute continuity, mass, and local entropy conclusions hold, with constants also depending on the local drift bounds. The stationary version holds with the same modification.

Proof. Use the notation and square cutoff of the proof of Theorem 2.1. At zero viscosity $W_i = X_i$, $D\nu = \sum_i X_i \nu_i$, and the drift adjoint is $-D(b\nu)$. The localized identity becomes

$$\mathcal{T}Q = \partial_t q + \chi \mu + D(F_0 + \chi b\nu) + G_0 - \sum_i (X_i \chi) b_i \nu.$$

Indeed $\chi D(b\nu) = D(\chi b\nu) - \sum_i (X_i \chi) b_i \nu$. On a time slice let $r_0 = dx + \chi \nu_t$. The Radon–Nikodym derivative of the additional vector measure obeys

$$\int \left| \frac{d(\chi b\nu_t)}{dr_0} \right|^2 dr_0 \leq B_{\operatorname{supp} \chi}^2 \int \chi d\nu_t \leq B_{\operatorname{supp} \chi}^2.$$

The additional scalar source has uniformly bounded total variation, since the cutoff derivatives and the local drift are bounded. Thus these terms have exactly the finite flux energy and scalar mass required in (2.8) and (2.14). The vector intertwiner, positive scalar smoothing, logarithmic test, and absorption therefore give (2.13) with the new fixed constants. The same approximation and lower semicontinuity prove the measure assertion. Time-independent data give the stationary case. No derivative of b is used. $\square$

Theorem 3.2 (A smooth full-rank horizontal frame). *Let G be a fixed Carnot group of step at most two, let $H_1, \dots, H_d$ be a fixed basis of its first layer, and let $\Omega \subset G$ be open. Suppose*

$$\tilde{X}_i = \sum_{a=1}^d s_{ia}(x) H_a, \quad 1 \leq i \leq m,$$

where $S = (s_{ia})$ is smooth and has column rank d at every point of Ω . Form the no-additional-drift operator

$$\tilde{\mathcal{L}}_A = \frac{1}{2} \sum_{i,j} a_{ij} (\tilde{X}_i \tilde{X}_j + \tilde{X}_j \tilde{X}_i).$$

For this fixed family, all parabolic and stationary conclusions of Theorem 2.1 hold at $\varepsilon = 0$, with the original measure, measurability, ellipticity, mass, and positive-source hypotheses. The constant may depend on S and its derivatives on a compact neighborhood of the set being estimated.

Proof. Symmetry of A gives the exact expansion on test functions

$$\tilde{\mathcal{L}}_A = \sum_{a,b} \hat{a}_{ab} H_a H_b + \sum_b \hat{b}_b H_b, \quad \hat{A} = S^t A S, \quad \hat{b}_b = \sum_{i,j,a} a_{ij} s_{ia} H_a s_{jb}.$$

For a fixed relatively compact open neighborhood U of K , the smallest singular value of S has a positive lower bound on $\bar{U}$, and its largest singular value has a finite upper bound. Consequently $\hat{A}$ has fixed positive ellipticity bounds there. The derivatives of the fixed smooth S and the bound on A also give a fixed bound for $\hat{b}$ on U . All these assertions hold with respect to the original measure; no derivative of A is taken. Apply Proposition 3.1 on U . Its hypotheses only require the restricted subprobability measures and the restricted positive source, so the same local conclusion follows. Exhausting Ω identifies a single density, whose total mass remains at most T in the parabolic case and at most one in the stationary case. The drift in this proof is produced by expanding the stipulated operator; it is not an additional assumption on that operator. $\square$

Corollary 3.3 (A specified distribution-preserving chart). *Let the fixed smooth fields of the original problem, on a neighborhood U of a compact set K , be pushed by a fixed smooth diffeomorphism $\Phi : U \rightarrow V \subset G$ to a smooth spanning family for the horizontal distribution of a fixed step-two Carnot group. Then the zero-viscosity parabolic and stationary absolute continuity and local entropy conclusions hold on K , with constants also depending on the chart. The conclusion holds on any domain covered locally by such charts.*

Proof. The pushed-forward fields have a smooth full-column-rank coefficient matrix in a fixed Carnot horizontal basis, so Theorem 3.2 applies. Push the measures forward and transform the operator on test functions. Distributional duality then gives precisely the transformed nondivergence operator; no Jacobian derivative is added to its action on tests. The pushforward preserves the mass bounds and positivity of the source. If the pushed density is q , the original

density is $p(x) = q(\Phi(x))|\det D\Phi(x)|$. On a compact neighborhood both the Jacobian and its reciprocal are bounded. If $J = |\det D\Phi| \leq C_0$, then

$$\begin{aligned} \int_K p \log(1+p) dx &= \int_{\Phi(K)} q(y) \log(1 + J(\Phi^{-1}y)q(y)) dy \\ &\leq \int_{\Phi(K)} q \log(1+q) + \log(\max\{1, C_0\}) \int_{\Phi(K)} q. \end{aligned}$$

This proves the transfer. Finite chart covers handle any fixed compact set. This corollary makes no statement about how a coordinate viscosity would transform, since it assumes $\varepsilon = 0$. $\square$

4 Consequences for arbitrary smooth Hörmander fields

Throughout this section we impose exactly the parabolic measure hypotheses of the main problem, with arbitrary fixed smooth bracket-generating fields on an open subset of $\mathbb{R}^n$. In particular, no constant-rank assumption is imposed. All bounds are uniform over the admitted A, ε, ν, μ . Write $D = \mathcal{L}_A + \varepsilon \Delta$ and

$$\Gamma(g, h) = \sum_{i,j} a_{ij}(X_i g)(X_j h) + \varepsilon \nabla g \cdot \nabla h, \quad \Gamma(g) = \Gamma(g, g).$$

All identities involving A below are pointwise ν -almost everywhere. Symmetry of A gives the exact product and chain rules

$$D(gh) = gDh + hDg + 2\Gamma(g, h), \quad Dh(f) = h'(f)Df + h''(f)\Gamma(f), \quad \Gamma(f) \geq \lambda \sum_i |X_i f|^2.$$

No differentiation of A occurs.

Proposition 4.1 (Mass, traces, and local sink bound). *The spacetime measure has mass at most T . There are finite nonnegative Radon measures ν_0, ν_T on Ω , each of mass at most one, such that for every smooth φ up to the temporal endpoints with a common compact spatial support,*

$$\int (\partial_t \varphi + D\varphi) d\nu = \int \varphi d\mu + \int \varphi(T, x) d\nu_T(x) - \int \varphi(0, x) d\nu_0(x). \quad (4.1)$$

For $0 \leq \chi \leq 1$, $\chi \in C_c^\infty(\Omega)$, $\chi = 1$ near K , put

$$M_\chi = \sup_{x \in \text{supp } \chi, \lambda I \leq B \leq \lambda I, 0 \leq e \leq 1} |\mathcal{L}_B \chi(x) + e \Delta \chi(x)|.$$

Then $M_\chi < \infty$ and $\mu((0, T) \times K) \leq 1 + TM_\chi$.

Proof. Weak measurability extends from continuous compactly supported spatial tests to bounded Borel tests by the monotone-class theorem. Tonelli and the slice mass bound give $\nu((0, T) \times \Omega) \leq T$.

For $\psi \in C_c^\infty(\Omega)$ set $F_\psi(t) = \int \psi d\nu_t$ and $G_\psi(t) = \int D\psi d\nu_t$. The latter is measurable and bounded on $(0, T)$ by a constant depending on the fixed compact support and derivatives of ψ . In temporal distributions, the equation says

$$F'_\psi = G_\psi dt - m_\psi, \quad m_\psi(E) = \int_{E \times \Omega} \psi(x) d\mu(t, x).$$

The right side is a finite signed measure, so F_ψ has a BV representative and finite one-sided limits at both endpoints. The endpoint limits define linear functionals, positive for $\psi \geq 0$, and bounded in absolute value by $\|\psi\|_\infty$. Density in $C_0(\Omega)$ and the Riesz representation theorem

produce the asserted subprobability measures. A countable dense family of spatial tests supplies one common exceptional-time set outside which the endpoint convergence is vague convergence against every C_0 test.

To verify (4.1), apply the original identity to $\eta_\delta(t)\varphi(t, x)$, where η_δ rises monotonically from zero to one in a layer of width $O(\delta)$ near zero and falls back to zero similarly near T . The spatial and ordinary temporal terms converge by dominated convergence. In the η'_δ term replace $\varphi(t, x)$ by its endpoint value in the two boundary layers. The replacement error is $O(\delta)$, using bounded $\partial_t\varphi$, the slice mass bound, and $\int |\eta'_\delta| = 2$. The limits of that term are $\int \varphi(0) d\nu_0 - \int \varphi(T) d\nu_T$, proving the signs in (4.1).

Finally use $\varphi(t, x) = \chi(x)$ in (4.1). Nonnegativity of ν_T gives

$$\int \chi d\mu = \int D\chi d\nu + \int \chi d\nu_0 - \int \chi d\nu_T \leq TM_\chi + 1.$$

The bound uses only the permitted fixed data and the chosen cutoff. $\square$

4.1 Quantitative scalar occupation

Lemma 4.2 (Weighted scalar estimate). *Let $K \Subset \Omega$ and let f be smooth on a neighborhood of a compact neighborhood of K . There is $C_f < \infty$, independent of A, ε, ν, μ , such that every Borel $E \subset \mathbb{R}$ satisfies*

$$\int_{(0,T) \times K} \mathbf{1}_E(f(x)) \sum_i |X_i f(x)|^2 d\nu \leq C_f |E|.$$

Proof. Choose $0 \leq \chi \leq 1$ smooth, compactly supported in the region where f is defined, with $\chi = 1$ near K . Choose a further compactly supported cutoff $\theta = 1$ near $\text{supp } \chi$. For nonnegative $g \in C_c^\infty(\mathbb{R})$ choose h with $h'' = g$, $\|h'\|_\infty \leq \|g\|_1$, and $|h| \leq C_f \|g\|_1$ on the bounded range of f on $\text{supp } \chi$. This is obtained by integrating g from $-\infty$ once and normalizing the second primitive at one point of that range.

The product rule gives

$$D(\chi h(f)) = h(f)D\chi + \chi h'(f)Df + 2h'(f)\Gamma(\chi, f) + \chi g(f)\Gamma(f).$$

Use this function in (4.1). The two endpoint terms are bounded by $C_f \|g\|_1$ because the traces are subprobabilities. The μ term is bounded by the local sink bound on $\text{supp } \chi$. The other three spatial terms are bounded by $C_f \|g\|_1$ using the bounded coefficients, smooth fields and cutoffs on a compact neighborhood, and ν mass at most T . Isolating the last term gives

$$\int \chi g(f)\Gamma(f) d\nu \leq C_f \|g\|_1.$$

Ellipticity and $\chi = 1$ on K prove the smooth-test estimate. The finite pushforward measure $f_\#(\mathbf{1}_K \sum_i |X_i f|^2 \nu)$ is therefore dominated by C_f times Lebesgue measure, by regularity and approximation of continuous tests with nonnegative smooth tests. This proves the assertion for Borel sets. $\square$

Proposition 4.3 (Scalar submersion estimate). *If in addition $df \neq 0$ on K , there exist $C_f < \infty$ and an integer $s \geq 1$ such that, with $\alpha = (2^s - 1)^{-1}$,*

$$\nu((0, T) \times (K \cap f^{-1}(E))) \leq C_f |E|^\alpha \quad (E \text{ Borel, } |E| \leq 1). \quad (4.2)$$

The constants depend on the fixed structural data, finitely many derivatives of f , and its finite-word nondegeneracy on K . In particular, this scalar pushforward has a density r_f and $\int_{\mathbb{R}} r_f \log(1 + r_f) < \infty$, with the corresponding uniform bound.

Proof. The bracket assumption and $df \neq 0$ imply that at every point of K some iterated bracket Y satisfies $Yf \neq 0$. A finite subcover and expansion of each bracket as a signed sum of words give an integer s and $c > 0$ with

$$\max_{1 \leq \ell \leq s, |w|=\ell} |X_w f(x)| \geq c \quad (x \in K).$$

For $d = |E| \in (0, 1]$ set

$$q_\ell = \frac{2^{s-\ell} - 1}{2^s - 1}, \quad h_\ell = \frac{c}{2} d^{q_\ell}, \quad 1 \leq \ell \leq s.$$

At each point of K there is a first length ℓ for which some word $X_w f$ of that length has magnitude greater than h_ℓ . Such a length exists because all thresholds are at most $c/2$.

On the first-length-one part of $\{f \in E\}$, Lemma 4.2, summed over first derivatives, bounds the mass by Cd/h_1^2 . For first length $\ell \geq 2$, write the selected word as $X_i(X_v f)$ with $|v| = \ell - 1$. Minimality gives $|X_v f| \leq h_{\ell-1}$. Apply the weighted lemma to the smooth function $X_v f$ and the interval $[-h_{\ell-1}, h_{\ell-1}]$, then sum over the finitely many words. This part has mass at most $Ch_{\ell-1}/h_\ell^2$. Since

$$1 - 2q_1 = \alpha, \quad q_{\ell-1} - 2q_\ell = \alpha,$$

summing gives (4.2). Null sets are treated by open supersets of arbitrarily small Lebesgue measure. The case $s = 1$ is included.

Absolute continuity follows by taking $|E| = 0$. If $\alpha = 1$, the density is bounded. Otherwise set $q = (1 - \alpha)^{-1} > 1$. Since its total mass is at most T , for sufficiently large v the set $E_v = \{r_f > v\}$ has measure at most one. Applying (4.2) gives $v|E_v| \leq C_f |E_v|^\alpha$, hence $|E_v| \leq (C_f/v)^q$. Layer-cake integration of $r_f \log(1 + r_f)$ now has an integrable tail bounded by a constant times $\int_1^\infty (1 + \log v) v^{-q} dv$; the remaining part uses the finite interval containing $f(K)$ and finite mass. $\square$

Corollary 4.4 (Uniform linear projections). *For fixed $K \subseteq \Omega$, the scalar set and entropy bounds in Proposition 4.3 hold for $f_\omega(x) = \omega \cdot x$, uniformly over $|\omega| = 1$, with one exponent $\alpha > 0$.*

Proof. The bracket condition and compactness on $K \times S^{n-1}$ give a common finite word length and positive lower bound for the maximum of $|X_w f_\omega|$. All necessary derivatives of f_ω and these words are uniformly bounded on a fixed compact neighborhood. Every constant in the threshold proof can therefore be chosen uniformly. $\square$

Proposition 4.5 (Smooth hypersurfaces receive no mass). *Every smooth embedded hypersurface of $(0, T) \times \Omega$ has zero ν mass.*

Proof. For a smooth time-dependent $f(t, x)$, the proof of the weighted lemma on a compact time-space neighborhood gains just the bounded term $h'(f)\partial_t f$. Thus the word argument also excludes $\{f = 0\}$ on compact regions where $d_x f \neq 0$. Locally describe a hypersurface as $F(t, x) = 0$ with $d_{t,x} F \neq 0$. Its part with $d_x F \neq 0$ is null by this observation and a compact exhaustion. On the remaining part, $\partial_t F \neq 0$, so the implicit-function theorem writes the surface as $t = h(x)$. There, $d_x F = 0$ means $dh = 0$. Sard's theorem says that the corresponding time values $h(\{dh = 0\})$ form a Lebesgue-null subset of $\mathbb{R}$. The time projection of ν is bounded by dt , so this part is null too. Finish with a countable chart cover. $\square$

4.2 An exponential-duality criterion

Proposition 4.6 (Conditional exponential-duality criterion). *Let $D_0 = (0, T) \times U$, where $U \Subset \Omega$ is bounded and open. Suppose that, in addition to the original hypotheses, there is a uniform constant C_0 such that every bounded nonnegative Borel g supported in D_0 satisfies*

$$\int_{D_0} g \, d\nu \leq C_0 + \int_{D_0} (e^g - 1) \, dt \, dx. \quad (4.3)$$

Then ν has a density on D_0 , and for every compact $K \subset U$, $\int_0^T \int_K p \log(1 + p) \leq C_0 + T$.

Proof. For every Lebesgue-null Borel $E \subset D_0$, apply the premise to $g = N \mathbf{1}_E$ and let $N \rightarrow \infty$. Thus $\nu(E) = 0$. For the resulting density, set $g = \mathbf{1}_{(0,T) \times K} \min\{N, \log(1 + p)\}$. Then $e^g - 1 \leq p$ on its support. Monotone convergence and the mass bound prove the entropy assertion. $\square$

Remark 4.7. The premise (4.3) is a sufficient condition; it is not established here for arbitrary smooth fields and arbitrary Borel matrix coefficients. Scalar projection regularity and hypersurface nullity do not imply joint absolute continuity.

5 Stationary identities in arbitrary smooth geometry

In this section $X_1, \dots, X_m$ are arbitrary fixed smooth real bracket-generating fields on $\Omega \subset \mathbb{R}^n$. Unless a statement explicitly says otherwise, the stationary assumptions are

$$\begin{aligned} p \geq 0, \quad p \in L^1(\Omega), \quad \int_{\Omega} p \leq 1, \quad \lambda I \leq A \leq \Lambda I \quad p \, dx\text{-a.e.}, \\ (\mathcal{L}_A + \varepsilon \Delta)^* p = \mu \geq 0, \quad 0 \leq \varepsilon \leq 1. \end{aligned} \quad (5.1)$$

where A is Borel and symmetric and μ is a finite measure. Its values on the exceptional $p \, dx$ -null set can be replaced by λI ; this leaves the equation unchanged and permits uniform pointwise bounds on cutoff expressions. No derivative of this representative is used. Constants may depend on fixed compact neighborhoods, the fields, and λ, Λ , and are independent of p, A, μ, ε .

5.1 Two exact localizations

Proposition 5.1 (Positive-stress localization). *Set $Q_{ij} = a_{ij}p$, $d_i = \operatorname{div} X_i$, $P = \sum_i X_i^* X_i$, $c = (\Lambda + \lambda)/2$ and $\rho = (\Lambda - \lambda)/(\Lambda + \lambda) < 1$. For a fixed $0 \leq \chi \leq 1$ in $C_c^\infty(\Omega)$, put $r = \chi p$ and $E = \chi(Q - cpI)$. Then*

$$-cprI \leq E \leq cprI \quad (5.2)$$

and

$$(I + cP - \varepsilon \Delta)r = \sum_{i,j} X_j^* X_i^* E_{ij} + \sum_j X_j^* f_j + \sum_k \partial_k^* g_k + f_0 - \chi \mu, \quad (5.3)$$

where

$$f_j = 2 \sum_i (X_i \chi) Q_{ij} - cd_j r, \quad g_k = 2\varepsilon p \partial_k \chi, \quad f_0 = r + \sum_{i,j} (X_i X_j \chi) Q_{ij} + \varepsilon p \Delta \chi.$$

These functions have uniformly bounded L^1 norms, $\|g\|_1 \leq C\varepsilon$, and $\int \chi \, d\mu \leq C$.

Proof. For a distribution F and smooth b , $X_i^*(bF) = bX_i^*F - (X_i b)F$. Applying this twice and using symmetry gives

$$\sum_{i,j} X_j^* X_i^* (\chi Q_{ij}) = \chi \sum_{i,j} X_j^* X_i^* Q_{ij} - 2 \sum_j X_j^* \left(\sum_i (X_i \chi) Q_{ij} \right) - \sum_{i,j} (X_i X_j \chi) Q_{ij}.$$

The analogous identity is $\Delta(\chi p) = \chi \Delta p - 2 \sum_k \partial_k^* (p \partial_k \chi) - p \Delta \chi$. Finally $\sum_i (X_i^*)^2 r = -Pr - \sum_i X_i^* (d_i r)$. Substitute $\chi Q = crI + E$ and the stationary equation to obtain (5.3). The cone follows from ellipticity. All remaining coefficients are fixed smooth bounded functions on a compact set. If $\eta \geq 0$ equals one near $\text{supp } \chi$, then

$$\int \chi d\mu \leq \int \eta d\mu = \int p(\mathcal{L}_A + \varepsilon \Delta) \eta \leq C.$$

This proves every asserted norm bound. $\square$

Proposition 5.2 (Square cutoff with finite covariance energy). *Write $X_i = \sigma_i \cdot \nabla$ and let σ have these columns. The coordinate form of the prescribed operator is*

$$D = \mathcal{L}_A + \varepsilon \Delta = B : D^2 + b \cdot \nabla, \quad B = \sigma A \sigma^\top + \varepsilon I, \quad b = \sum_{i,j} a_{ij} (D\sigma_j) \sigma_i.$$

Fix $0 \leq \chi \leq 1$ with $\chi \in C_c^\infty(\Omega)$. For $h = \chi^2$, $q = hp$, define $c_h = 2B\nabla h/h$ on $\{h > 0\}$ and zero elsewhere. Then

$$(D + c_h \cdot \nabla)^* q = \kappa - \beta, \quad \kappa = h\mu + (Dh)^- p dx, \quad \beta = (Dh)^+ p dx, \quad \kappa(\Omega) = \beta(\Omega) \leq C,$$

and, with the Moore–Penrose inverse of B ,

$$\int q c_h^\top B^\dagger c_h = 16 \int p \nabla \chi^\top B \nabla \chi \leq C. \quad (5.4)$$

The vector c_h need not be bounded.

Proof. The coordinate drift displayed above is obtained by expanding $X_i X_j$; it is already part of the original operator. The exact product rule is $D(h\phi) = hD\phi + 2B\nabla h \cdot \nabla \phi + \phi Dh$. Since $qc_h = 2pB\nabla h$, testing proves the balance identity. Testing the original equation with h gives $\int h d\mu = \int p Dh$, hence equality of the two masses. On $\{\chi > 0\}$, use $BB^\dagger B = B$ in $c_h = 4B\nabla \chi/\chi$ to obtain (5.4). On $\{\chi = 0\}$, smooth nonnegativity gives $\nabla \chi = 0$, so that identity remains valid. Compact coefficient bounds and $\int p \leq 1$ finish the proof. $\square$

5.2 A second-derivative identity specific to step two

Proposition 5.3 (Horizontal Hessian and central brackets). *Suppose, in addition, that the fields are divergence-free and their commutators $T_{ij} = [X_i, X_j]$ commute with every X_k . For every real $u \in C_c^\infty$, with $L_0 = \sum_i X_i^2$ and $H_{ij}u = (X_i X_j + X_j X_i)u/2$,*

$$\sum_{i,j} \|H_{ij}u\|_2^2 = \|L_0 u\|_2^2 + \frac{3}{4} \sum_{i,j} \|T_{ij}u\|_2^2. \quad (5.5)$$

In particular this holds for left-invariant horizontal fields on every step-two Carnot group. For the standard $\mathbb{H}^1$, it reads $\|H_H u\|_{L^2(\mathbb{F})}^2 = \|\Delta_H u\|_2^2 + \frac{3}{2} \|Zu\|_2^2$.

Proof. Integration by parts and centrality give

$$\begin{aligned} \sum_{i,j} \|X_i X_j u\|_2^2 - \|L_0 u\|_2^2 &= - \sum_{i,j} \langle X_j u, [X_i^2, X_j] u \rangle \\ &= 2 \sum_{i,j} \langle X_i X_j u, T_{ij} u \rangle = \sum_{i,j} \|T_{ij} u\|_2^2. \end{aligned}$$

For the last equality, exchange i, j in half of the sum. The symmetric array $H_{ij}u$ and anti-symmetric array $T_{ij}u$ are orthogonal after summing indices. Since $X_i X_j u = H_{ij}u + T_{ij}u/2$, subtract one quarter of the commutator norm to obtain (5.5). In $\mathbb{H}^1$ the two nonzero ordered commutators are Z and $-Z$. $\square$

6 A measurable scalar amplitude in arbitrary Hörmander geometry

Theorem 6.1 (Local boundedness for a fixed smooth matrix times a Borel scalar). *Let $X_1, \dots, X_m$ be arbitrary fixed smooth bracket-generating fields on Ω , and let $B(x)$ be a fixed smooth symmetric positive definite $m \times m$ matrix, with positivity locally uniform. Suppose that $p \geq 0$, $\int_{\Omega} p \leq 1$, a is Borel, $0 < \lambda \leq a \leq \Lambda$ for $p dx$ -almost every x , and*

$$\mathcal{L}_{aB}^* p = \mu \geq 0 \quad \text{in } \mathcal{D}'(\Omega),$$

where μ is finite. Then, for every $K \Subset \Omega$,

$$\|p\|_{L^\infty(K)} \leq C(K, \Omega, X, B, \lambda, \Lambda), \quad \int_K p \log(1+p) \leq C.$$

This assertion has $\varepsilon = 0$ and permits changing horizontal rank. The constant may depend on derivatives of the fixed smooth matrix B .

We first give the precise fixed smooth kernel input and the construction needed in the proof. The underlying local Gaussian estimates for a smooth sum of squares are those of Capogna–Citti, Theorem 1.3 and Section 5.1; their horizontal matrix is constant, and we use it only with the identity matrix after changing the fixed smooth fields.

Lemma 6.2 (Separated source-kernel bounds). *A fixed smooth operator $L = \sum b_{ij} X_i X_j$ as in Theorem 6.1 can be extended from a prescribed relatively compact neighborhood to a smooth conservative diffusion generator $\tilde{L}$ on a compact connected torus M . For its nonnegative kernel, with convention $P_t \phi(y) = \int_M k_t(y, x) \phi(x) dx$, one has*

$$\sup_{y,x} k_1(y, x) < \infty, \quad \int_0^1 \sup_{y \in E, x \in K_0} (k_t(y, x) + |X_j^y k_t(y, x)|) dt < \infty$$

whenever the compact sets E, K_0 are separated and lie in the retained coordinate region. The adjoint semigroup is strongly continuous on $L^1(M)$.

Proof. Retain the fields near the prescribed neighborhood, cut them off farther away, and add fields $\eta \partial_k$ which vanish near that neighborhood and span wherever the original fields have changed. Interpolate B with the identity outside the retained region. The resulting operator is $\tilde{L} = \sum C_{ij} W_i W_j$ for a fixed smooth positive matrix C and a globally bracket-generating family W . Write $C = S^\top S$ and $Z_\alpha = \sum_i S_{\alpha i} W_i$. Direct expansion gives

$$\tilde{L} = H + V, \quad H = \sum_{\alpha} Z_{\alpha}^2, \quad V = \sum_{\alpha} v_{\alpha} Z_{\alpha},$$

with fixed smooth bounded coefficients v_{α} . Invertibility of S preserves the smooth module and bracket generation. This horizontal drift is only the algebraic correction in rewriting the fixed operator.

For completeness we describe why local Gaussian estimates give the stated compact, drifted bounds. Glue local kernels q_t^ℓ for H using source partitions $\psi_\ell(x)$ and target cutoffs $\theta_\ell(y)$ equal to one near their supports: $K_t(y, x) = \sum_{\ell} \theta_\ell(y) q_t^\ell(y, x) \psi_\ell(x)$. Its error $R_t = (\partial_t - H_y) K_t$ contains derivatives of kernels at separated points. The local Gaussian estimates imply $R_t = O(t^M)$ uniformly, with any fixed derivatives, for every M . The Volterra series $\Phi = -R + R * R - \dots$ converges by factorial time-simplex bounds, and $q = K + K * \Phi$ is the compact kernel. The integral norms in either variable of K_t and $Z_{\alpha}^y K_t$ are bounded by C and $Ct^{-1/2}$ respectively; this follows by integrating the volume-normalized Gaussian estimates over doubling annuli.

Consequently $q - K$ and $Z_\alpha^y(q - K)$ are $O(t^M)$ for every M . A compact finite-step volume lower bound now gives, for some finite N , every $R > 0$, and $\sigma = 0, 1$,

$$|D_y^{(\sigma)} q_t(y, x)| \leq C_R t^{-N-\sigma/2} (1 + d(y, x)/\sqrt{t})^{-R}, \quad \|D_y^{(\sigma)} q_t\|_{L^1 \text{ in either variable}} \leq C t^{-\sigma/2}. \quad (6.1)$$

Here $D^{(1)}$ is any one of the Z_α and d is their control distance. No constant-rank assumption is needed for a nonsharp finite N .

Add V by the convergent Duhamel series

$$P_t = \sum_{j \geq 0} \int_{\tau_0 + \dots + \tau_j = t} Q_{\tau_0} V Q_{\tau_1} \dots V Q_{\tau_j} d\tau.$$

Its time singularities have integrals

$$\int_{\sum \tau_i = t} \prod_{i=1}^j \tau_i^{-1/2} d\tau = \frac{\Gamma(1/2)^j}{\Gamma(1+j/2)} t^{j/2}, \quad \int_{\sum \tau_i = t} \prod_{i=0}^j \tau_i^{-1/2} d\tau = \frac{\Gamma(1/2)^{j+1}}{\Gamma((j+1)/2)} t^{(j-1)/2}.$$

The second is used for one source derivative. If $d(y, x) \geq \delta > 0$, at least one of the $j+1$ edges of a broken path has length at least $\delta/(j+1)$. Apply (6.1) with arbitrarily large $R > 2N$ on that edge, and integrate all other edges using the integral bounds there. Summing over the choice of long edge bounds its spatial integral by

$$C_R \delta^{-R} C^j (j+1)^{R+1} t^{R/2-N} \prod_{i=0}^j \tau_i^{-\sigma_i/2},$$

where $\sigma_0 = 0$, $\sigma_i = 1$ for $i \geq 1$, or all $\sigma_i = 1$ for a source derivative. The Gamma denominators sum the series and give separated bounds $C t^{R/2-N}$ and $C t^{R/2-N-1/2}$. Take $R > 2N + 2$ to integrate them at zero. Original fields are bounded smooth combinations of the Z_α on the retained region. Smoothness at positive time gives the time-one bound. The positive maximum principle identifies the constructed kernel with the conservative nonnegative diffusion semigroup. Its adjoint is an L^1 contraction; convergence on smooth densities and density of smooth functions prove strong L^1 continuity. $\square$

Proof of Theorem 6.1. Set $r = ap$. Then $r \geq 0$, $\|r\|_1 \leq \Lambda$, and $L^* r = \mu$ for the fixed smooth $L = \mathcal{L}_B$. This is direct multiplication of the density, requiring no product rule for a . Work in one of finitely many coordinate neighborhoods covering K . Take $\chi = 1$ near the compact set K_0 under consideration and put $F = \chi r$, extended by zero to the reference torus. Exact localization gives

$$\tilde{L}^* F = \chi \mu + \sum_j X_j^* J_j - h dx, \quad J_j = -2r \sum_i b_{ij} X_i \chi, \quad h = r L \chi.$$

All J_j, h are supported in a fixed compact set E separated from K_0 , and their L^1 norms sum to at most $C \|r\|_1$. For a nonnegative smooth ϕ supported in the interior neighborhood being estimated and $0 < \delta < 1$, differentiation of the reference semigroup gives

$$\begin{aligned} \langle P_\delta^* F, \phi \rangle &= \langle P_1^* F, \phi \rangle - \int_\delta^1 \int \chi P_t \phi d\mu dt \\ &\quad - \sum_j \int_\delta^1 \int J_j X_j P_t \phi dx dt + \int_\delta^1 \int h P_t \phi dx dt. \end{aligned}$$

The sink term is nonpositive and can be dropped before taking any limit. Lemma 6.2 bounds the positive-time term and the separated cutoff terms by $C \|r\|_1 \|\phi\|_1$, independently of δ . Strong L^1 continuity and $F = r$ on the test support yield $\int r \phi \leq C \|r\|_1 \int \phi$. Thus $r \leq C \Lambda$ there almost everywhere. A finite cover and $p \leq r/\lambda$ prove the claimed bound. Finally $\int_K p \log(1+p) \leq \log(1 + \|p\|_{L^\infty(K)})$. $\square$

7 Reference-semigroup identities and conditional entropy criteria

These are exact reductions under (5.1), not proofs of their additional commutator premises. The parameter s is a smoothing scale, not physical time. Extend the original fields to a torus M , adding fixed smooth fields that vanish near a chosen cutoff support and preserve global bracket generation. Add $\sqrt{\varepsilon} \partial_k$ to the list and call the full fixed-length list $Z_1, \dots, Z_d$. The subfamily excluding viscosity is fixed and bracket generating. Set

$$L_\varepsilon = \sum_i Z_i^2, \quad P_s = e^{sL_\varepsilon}, \quad \mathcal{T}Q = \sum_{i,j} Z_j^* Z_i^* Q_{ij}, \quad \mathcal{D}U = \sum_i Z_i^* U_i.$$

Augment A by the identity on additional directions, obtaining $\lambda_0 I \leq \mathcal{A} \leq \Lambda_0 I$, where $\lambda_0 = \min\{\lambda, 1\}$ and $\Lambda_0 = \max\{\Lambda, 1\}$. This represents the original operator on the cutoff support.

Lemma 7.1 (Uniform reference estimates). *For $h_s = P_s^* 1$ there are constants $c, C > 0$ and finite $N > 2$, independent of ε , such that*

$$c \leq h_s \leq C, \quad \|\nabla h_s\|_\infty \leq C \quad (0 \leq s \leq 1), \quad \|P_s^*\|_{1 \rightarrow \infty} \leq C s^{-N/2} \quad (0 < s \leq 1).$$

Also $h_s \rightarrow 1$ uniformly as $s \downarrow 0$.

Proof. Put $d_i = \operatorname{div} Z_i$. Direct expansion gives

$$L_\varepsilon^* = \sum_i Z_i^2 + 2 \sum_i d_i Z_i + V_\varepsilon, \quad V_\varepsilon = \sum_i (Z_i d_i + d_i^2).$$

All smooth coefficients have uniform bounds; constant viscosity fields have zero divergence. For the smooth diffusion W_s with generator $\sum_i Z_i^2 + 2 \sum_i d_i Z_i$, Feynman–Kac gives $h_s(x) = \mathbb{E}_x \exp(\int_0^s V_\varepsilon(W_u) du)$. This proves the two-sided bounds and convergence to one. The variational stochastic equation has bounded coefficients, so Itô’s formula and Gronwall give $\sup_{s \leq 1, x, \varepsilon} \mathbb{E} |D_x W_s|^2 \leq C$. Differentiating this expectation proves the gradient estimate.

The fixed-field Sobolev theorem of Chen–Chen–Li, Theorems 1.1 and 1.2(2), a finite partition of unity, and interpolation give a nonsharp Nash inequality

$$\|v\|_2^{2+4/N} \leq C \left(\sum_{\text{fixed } i} \|Z_i v\|_2^2 + \|v\|_2^2 \right) \|v\|_1^{4/N}.$$

Indeed choose a common local Sobolev exponent $r > 2$ and set $N = 2r/(r-2)$; in a critical chart use any finite permitted exponent. In dimension one bracket generation makes the fixed energy elliptic. No viscosity-dependent fields enter this inequality. Integration by parts gives

$$\langle v, L_\varepsilon v \rangle = - \sum_i \|Z_i v\|_2^2 + \frac{1}{2} \int V_\varepsilon v^2,$$

and the adjoint has the same quadratic form. The adjoint semigroup is an L^1 contraction and $\|P_s f\|_1 \leq \int |f| h_s \leq C \|f\|_1$. Thus for either evolution $y = \|u_s\|_2^2$ satisfies $y' \leq -c \|u_0\|_1^{-4/N} y^{1+2/N} + C y$. An exponential change of unknown and integration give the $1 \rightarrow 2$ bound $C s^{-N/4}$. Duality and composition at $s/2$ give the asserted $1 \rightarrow \infty$ bound. No uniform kernel derivative estimate is used. $\square$

Proposition 7.2 (Coefficient variance and the full commutator). *For $q = \chi^2 p$, put $r_0 = 1 + q$, $Q_0 = I + q\mathcal{A}$, $U_{0,i} = 4\chi p \sum_j \mathcal{A}_{ij} Z_j \chi$, and define*

$$r_s = P_s^* r_0, \quad Q_s = P_s^* Q_0, \quad U_s = P_s^* U_0, \quad a_s = Q_s / r_s, \quad f_s = \log(r_s / h_s),$$

$$E(s) = \int [r_s \log(r_s/h_s) - r_s + h_s], \quad I(s) = \int r_s \sum_i |Z_i f_s|^2.$$

Then $f_s \geq 0$, $\lambda_0 I \leq a_s \leq \Lambda_0 I$, and

$$-E'(s) = I(s), \quad \int_0^1 \int r_s \sum_{i,j,k} |Z_k a_{s,ij}|^2 \leq C, \quad \int |U_s|^2 / r_s \leq C. \quad (7.1)$$

For the full commutator

$$\mathcal{C}_s = \mathcal{T}(P_s^* Q_0) - P_s^*(\mathcal{T}Q_0) + \mathcal{D}(P_s^* U_0) - P_s^*(\mathcal{D}U_0), \quad (7.2)$$

one has, uniformly for $0 < \delta < 1$,

$$\frac{\lambda_0}{2} \int_\delta^1 I(s) ds \leq C - \int_\delta^1 \langle \mathcal{C}_s, f_s \rangle ds. \quad (7.3)$$

All constants are independent of the stationary data and viscosity.

Proof. The square-cutoff product rule gives

$$\mathcal{T}(qA) + \mathcal{D}U_0 = \kappa - \beta, \quad \kappa = \chi^2 \mu + (D\chi^2)^- p \, dx, \quad \beta = (D\chi^2)^+ p \, dx,$$

with $\kappa(M) = \beta(M) \leq C$ and $\int |U_0|^2 / (1+q) \leq 16\Lambda_0^2 \int p |Z\chi|^2 \leq C$. The baseline contributes $\mathcal{T}I = L_\varepsilon^* 1 = V_\varepsilon$:

$$\mathcal{T}Q_0 + \mathcal{D}U_0 = \kappa - \beta + V_\varepsilon \, dx =: S. \quad (7.4)$$

Positive componentwise evolution preserves the matrix cone and $r_s \geq h_s$. Lemma 7.1 yields $\|f_s\|_\infty \leq C(1 + |\log s|)$ and $\int r_s \leq |M| + 1$.

Differentiate relative entropy and move adjoints onto f_s and e^{f_s} . The identity $L_\varepsilon e^f = e^f L_\varepsilon f + e^f |Zf|^2$ gives $E' = -I$. For a possibly signed scalar entry,

$$\frac{d}{ds} \int Q_{s,ij}^2 / r_s = 2 \int Q_{s,ij} L_\varepsilon a_{s,ij} - \int r_s L_\varepsilon (a_{s,ij}^2) = -2 \int r_s |Z a_{s,ij}|^2.$$

Kernel Cauchy-Schwarz gives $|P_s^* Q_0|^2 / (P_s^* r_0) \leq P_s^* (|Q_0|^2 / r_0)$; its integral is bounded by $d\Lambda_0^2 \int r_0$. Integrate from δ to one and let $\delta \downarrow 0$ to obtain the variance budget. The same kernel inequality for U_0 proves the flux budget.

The smoothed equation is $\mathcal{T}Q_s + \mathcal{D}U_s = P_s^* S + \mathcal{C}_s$. For $b_i = \operatorname{div} Z_i + Z_i \log h_s$, which is uniformly bounded, integration by parts gives

$$\begin{aligned} \langle \mathcal{T}Q_s + \mathcal{D}U_s, f_s \rangle &= - \int r_s \sum_{i,j} a_{s,ij} (Z_i f_s) (Z_j f_s) \\ &\quad - \int r_s \sum_j \left[\sum_i Z_i a_{s,ij} + \sum_i a_{s,ij} b_i - U_{s,j} / r_s \right] Z_j f_s. \end{aligned}$$

Ellipticity, Young's inequality and the budgets imply

$$\frac{\lambda_0}{2} \int_\delta^1 I(s) ds \leq C - \int_\delta^1 \langle P_s^* S + \mathcal{C}_s, f_s \rangle ds.$$

The κ pairing is nonnegative. The β pairing is bounded by $C(1 + |\log s|)$ because its mass is bounded. Finally $|P_s^* V_\varepsilon| \leq C h_s$ and $\int h_s f_s \leq \int (r_s - h_s) = \int q \leq 1$. These source bounds are integrable and prove (7.3). $\square$

Corollary 7.3 (Conditional commutator criterion). *If the full commutator additionally satisfies*

$$-\int_{\delta}^1 \langle C_s, f_s \rangle ds \leq \eta \int_{\delta}^1 I(s) ds + C, \quad \eta < \lambda_0/2,$$

with uniform constants, then $\int_K p \log(1+p) \leq C$. A sufficient additional premise is

$$C_s = \sum_i Z_i^*(r_s v_{s,i}), \quad \int_0^1 \int r_s |v_s|^2 \leq C.$$

Neither premise is asserted here for unrestricted matrix coefficients.

Proof. Absorb the premise in (7.3), integrate $-E' = I$, and use $E(1) \leq C$. For each viscosity, $r_{\delta} \rightarrow 1+q$ in L^1 and $h_{\delta} \rightarrow 1$ uniformly. Fatou gives $\int [(1+q) \log(1+q) - q] \leq C$. On K , $q = p$, and $p \log(1+p) \leq (1+p) \log(1+p) - p + p$ proves the conclusion. The flux premise implies the first by $-\langle C_s, f_s \rangle \leq \eta I(s) + (4\eta)^{-1} \int r_s |v_s|^2$. $\square$

Proposition 7.4 (Signed entropy defect at zero viscosity). *Assume (5.1) with $\varepsilon = 0$ and fix $c > 0$. Extend $c \sum_i X_i^2$ to a compact smooth bracket-generating generator $L_0 = c \sum_i \tilde{X}_i^2 + \sum_k Y_k^2$ as above. Let $P_s = e^{sL_0}$ and let π be its smooth positive invariant probability density. Set*

$$F = \chi p, \quad z_s = 1 + P_s^* F / \pi, \quad \mathcal{E}(s) = \int \pi(z_s \log z_s - z_s + 1), \quad g_s = P_s \log z_s.$$

Then $\mathcal{E}(1) \leq C$ and

$$\left| \mathcal{E}(\delta) - \mathcal{E}(1) + \int_{\delta}^1 \int (cI - A) \chi p : H_X g_s dx ds \right| \leq C \quad (0 < \delta < 1). \quad (7.5)$$

A uniform upper bound for the negative of this signed tensor integral is therefore sufficient for local entropy, but remains an additional premise.

Proof. The fixed smooth kernel and horizontal derivative estimates follow from the sum-of-squares construction in Lemma 6.2. Near-diagonal lower bounds concatenated along horizontal paths give positive transition densities. Cesàro averages on the compact torus yield an invariant measure; positive-time smoothing gives a smooth positive density π , bounded above and away from zero, with $L_0^* \pi = 0$. Localization gives

$$L_0^* F = \sum_{i,j} X_j^* X_i^* R_{ij} + \sum_j X_j^* B_j + G, \\ R = (cI - A)F, \quad B_j = -2p \sum_i a_{ij} X_i \chi, \quad G = \chi \mu - p \mathcal{L}_A \chi dx.$$

Their L^1 or total-variation norms are bounded by the local sink bound. The nonreversible entropy identity is

$$\mathcal{E}'(s) = \int \pi z_s L_0 \log z_s = - \int \pi \Gamma_0(z_s) / z_s =: -\mathcal{I}(s),$$

since $\int \pi L_0 z_s = 0$. Commuting the generator with its own semigroup,

$$-\mathcal{I}(s) = \sum_{i,j} \int R_{ij} X_i X_j g_s + \sum_j \int B_j X_j g_s + \int g_s dG.$$

The fixed kernel estimates give $\|g_s\|_{\infty} \leq C(1 + |\log s|)$ and $\|X_j g_s\|_{\infty} \leq C s^{-1/2}(1 + |\log s|)$. Their integrability proves (7.5) by integration. Strong L^1 continuity, Fatou, and Jensen for relative densities give $\lim_{s \downarrow 0} \mathcal{E}(s) = \int \pi \Phi(1 + F/\pi)$, also for the value $+\infty$. This controls the desired local entropy because $\chi = 1$ on K and π has fixed positive upper and lower bounds. $\square$

8 Qualitative absolute continuity on Carnot groups of arbitrary step

Here G is a Carnot group, dx is Haar measure, and $X_1, \dots, X_m$ are fixed left-invariant generators of the first layer. A redundant generating list is allowed. Let N be the homogeneous dimension and d a fixed homogeneous control distance. Thus $|B(x, r)| = vr^N$. Left-invariant fields are divergence-free. These exact translation and dilation properties are additional hypotheses; the following theorem is not asserted for arbitrary smooth fields.

Lemma 8.1 (Fixed-matrix Carnot resolvent estimates). *Let A_0 be a fixed symmetric positive definite matrix and $P = -\sum_{i,j}(A_0)_{ij}X_iX_j$, $\mathcal{R} = (I + P)^{-1}$. Then:*

1. *Each $\mathcal{R}X_iX_j$ is of weak type $(1, 1)$ on Haar measure.*
2. *Images under $\mathcal{R}$ and $\mathcal{R}X_i$ of signed measures of uniformly bounded total variation supported in a fixed compact set are relatively compact in L^1 on each fixed compact target set.*
3. *For $\phi \in C_c^\infty(G)$, $(\mathcal{R}X_iX_j)^*\phi$ is bounded on every fixed compact set.*

Constants may depend on the fixed matrix and group. More precisely, the weak-type bound depends only on the group geometry and finitely many size, smoothness, cancellation, and integral remainder bounds for the kernels appearing in the proof. Uniform bounds for these quantities therefore give a uniform weak-type constant.

Proof. Choose a basis $Y_1, \dots, Y_h$ of the first layer. Let the columns of a surjective matrix $S : \mathbb{R}^m \rightarrow \mathbb{R}^h$ express the X_i in this basis, so $X_i = \sum_\alpha S_{\alpha i} Y_\alpha$. The matrix $SA_0S^\top$ is positive definite, so an invertible fixed recombination makes P a sum of squares of horizontal generators. The fixed homogeneous heat kernel and arbitrary source and target horizontal derivatives satisfy the Gaussian estimates of Biagi–Bramanti, Theorem 2.4. We give the singular-integral and compactness deductions required here.

First suppose $N > 2$. Integrating the heat kernel in time gives a fundamental solution Γ homogeneous of degree $2 - N$, smooth away from the identity. Its two source derivatives are homogeneous distributions of degree $-N$. Their off-identity kernels have zero mean on every homogeneous annulus: pairing a degree- $-N$ homogeneous distribution against the difference of the same radial cutoff at two scales gives zero, whereas polar integration would yield its spherical mean times the logarithm of the scale ratio. Their principal-value extensions consequently exist; the difference from the differentiated distribution is supported at the identity and has homogeneous degree $-N$, hence is a multiple of δ_e .

Decompose a principal kernel into smooth mean-zero dyadic annular pieces K_ℓ . Their L^1 norms are bounded uniformly. Cancellation at the finer scale and smoothness at the coarser scale give, in both orders,

$$\|K_\ell^* * K_k\|_1 + \|K_\ell * K_k^*\|_1 \leq C2^{-\eta|k-\ell|}$$

for some $\eta > 0$. Indeed rescale the coarser annulus to unit size, subtract its untranslated value, and use the mean zero of the smaller kernel; left and right translations are smooth on this fixed compact annulus. Young's inequality and the Cotlar–Stein almost-orthogonality lemma give uniform L^2 bounds for partial annular sums. The same rescaled smoothness gives the integral translated-difference estimate outside a ball comparable to the translation size. Apply the Haar-dyadic Calderón–Zygmund decomposition of an L^1 function at height a : the L^2 estimate controls the good part; mean zero and this integral estimate control each bad part outside its enlarged cube; these cubes have total measure at most $C\|f\|_1/a$. Chebyshev therefore proves weak type $(1, 1)$.

The resolvent kernel is the heat integral with factor e^{-t} . Its second source derivative differs locally from the preceding principal kernel by an integrable kernel: at small times

$$\int_0^1 (1 - e^{-t}) \|D^2 h_t\|_1 dt \leq C \int_0^1 t t^{-1} dt < \infty.$$

The large-time difference is smooth locally, and outside a fixed identity neighborhood the resolvent derivative has integrable decay by the Gaussian bound and e^{-t} . A local truncation of the principal kernel has the same annular proof. Delta and integrable remainders are L^1 bounded, proving (1).

Near the identity the resolvent and its first source derivative have bounds Cd^{2-N} and Cd^{1-N} , whose integrals over a radius- δ ball are $O(\delta^2)$ and $O(\delta)$. Remove a smooth diagonal neighborhood. On fixed compact source and target sets the remaining kernels are jointly continuous, so images of bounded measures are bounded and equicontinuous. Arzelà–Ascoli and uniform smallness of the removed part prove (2). Subtracting the value of a smooth test at the singular point in the principal-value integral proves (3).

The derivative side matters. With $(f * g)(x) = \int f(y)g(y^{-1}x)dy$ and $\mathcal{R}f = f * r$, integration by parts gives

$$\mathcal{R}X_i X_j f = f * (X_j^R X_i^R r), \quad (\mathcal{R}X_i X_j)^* = X_j X_i \mathcal{R}.$$

Here X^R denotes the associated right-invariant field. These are source kernel derivatives, which the cited Gaussian theorem covers. No horizontal derivative is commuted through the resolvent.

If $N \leq 2$, the group is abelian. For $N = 2$, the elliptic constant-coefficient fundamental solution is a fixed multiple of $-\log |B^{-1/2}x|$, where $B = SA_0S^\top$. Its second derivatives are homogeneous distributions of degree -2 , so the same annular argument applies; the lower-order diagonal bounds are $C(1 + |\log d|)$ and C/d , which have vanishing local integrals. The resolvent correction is handled by the same heat calculation. For $N = 1$, after writing $P = -b\partial_x^2$, the resolvent kernel is $e^{-|x|/\sqrt{b}}/(2\sqrt{b})$; its second derivative is an integrable function plus a delta, and its first derivative is locally integrable. The three claims follow directly. This covers all dimensions and redundant generating lists. $\square$

Theorem 8.2 (Positive stress implies absolute continuity on every Carnot group). *Let $U \subset G$ be open. Let ν be a positive locally finite measure and $Q = (Q_{ij})$ a symmetric matrix of locally finite measures such that*

$$\lambda\nu I \leq Q \leq \Lambda\nu I, \quad \sum_{i,j} X_j^* X_i^* Q_{ij} = \mu \quad \text{in } \mathcal{D}'(U),$$

where $0 < \lambda \leq \Lambda < \infty$ and μ is a locally finite signed measure. Then $\nu \ll dx$ locally in U . This theorem assumes $\varepsilon = 0$ and gives qualitative absolute continuity; it asserts no entropy bound or uniform-integrability modulus.

Proof. Write $Q = A\nu$ with $\lambda I \leq A \leq \Lambda I$. We first choose scales at almost every point of the hypothetical singular part ν^s . Use finitely many adjacent nested dyadic systems on the geometrically doubling metric space (G, d) , with the properties that cubes have diameter comparable to their scale, contain comparable balls, and every ball satisfies

$$B(x, r) \subset C \subset B(x, C_d r)$$

for a cube in one system. These geometric systems are supplied by Hytönen–Kairema, Theorems 2.2 and 4.1; they do not require the measure ν to be doubling. For each system the shrinking partitions generate the Borel sigma algebra, since cube diameters tend to zero. On bounded

neighborhoods apply martingale convergence to the finite dominating measure $\nu + |\mu| + dx$. Taking ratios of its differentiated densities gives at ν^s -almost every x , along the cubes containing x in every system,

$$\frac{\int_C |A - A(x)| d\nu}{\nu(C)} \rightarrow 0, \quad \frac{\nu^a(C)}{\nu(C)} \rightarrow 0, \quad \frac{|\mu|(C)}{\nu(C)} \rightarrow f(x) < \infty, \quad \frac{|C|}{\nu(C)} \rightarrow 0. \quad (8.1)$$

For the first limit differentiate each entry A_{ij} and $|A_{ij}|^2$ and use Cauchy–Schwarz; thus absolute oscillation, not just mean convergence, is obtained. The part of $|\mu|$ singular with respect to ν contributes zero at ν -almost every point.

Fix $L > 4C_d$. At ν -almost every point there are arbitrarily small radii $r = L^{-k}$ with

$$\nu(B(x, Lr)) \leq L^{N+1} \nu(B(x, r)). \quad (8.2)$$

Otherwise iteration would give $\nu(B(x, r)) \leq C_x r^{N+1}$ at all sufficiently small radii. For points in a bounded region with common bound $j r^{N+1}$ for $r < 1/j$, a maximal r -separated set has at most $C r^{-N}$ centers by Haar-volume packing. The radius- $2r$ balls cover the set and have total ν -mass at most $C j r$, which tends to zero. A countable union over the bounds and bounded regions proves (8.2) almost everywhere.

For good radii, an adjacent cube containing $B(x, 4r)$ lies in $B(x, Lr)$ and has ν -mass at most $L^{N+1} \nu(B(x, r))$. Applying (8.1) gives

$$\frac{\int_{B(x, 4r)} |A - A(x)| d\nu + \nu^a(B(x, 4r))}{\nu(B(x, r))} \rightarrow 0, \quad \sup_r \frac{|\mu|(B(x, 4r))}{\nu(B(x, r))} < \infty. \quad (8.3)$$

Switching among finitely many systems causes no loss.

If $\nu^s \neq 0$, choose x_0 and $r_j \downarrow 0$ with these properties. Put $F_j(y) = \delta_{1/r_j}(x_0^{-1}y)$, $c_j = 1/\nu(B(x_0, r_j))$, and

$$\nu_j = c_j(F_j)_* \nu^s, \quad Q_j = c_j(F_j)_* Q, \quad \mu_j = r_j^2 c_j(F_j)_* \mu, \quad A_0 = A(x_0), \quad E_j = A_0 \nu_j - Q_j.$$

Exact degree-one homogeneity of the fields gives $\mathcal{T}Q_j = \mu_j$, where $\mathcal{T}Q = \sum X_j^* X_i^* Q_{ij}$. Equation (8.3) yields on $B_4 = B(e, 4)$

$$\|E_j\|_{\text{TV}} \rightarrow 0, \quad \|\mu_j\|_{\text{TV}} \rightarrow 0, \quad \nu_j(B_1) \rightarrow 1, \quad \sup_j \nu_j(B_4) < \infty.$$

The source limit uses its factor r_j^2 . Each ν_j is singular because group translations and dilations preserve Haar-null sets.

Restrict the measures to B_4 and smooth by a positive compactly supported left approximate identity: $u_j = \eta_{\delta_j} * \nu_j$, $V_j = \eta_{\delta_j} * E_j$, $s_j = \eta_{\delta_j} * \mu_j$. With the convolution convention above, $X_i(\eta * f) = \eta * (X_i f)$. Thus, for sufficiently small δ_j , on B_3

$$-P u_j = \mathcal{T} V_j + s_j, \quad P = - \sum (A_0)_{ij} X_i X_j, \quad (8.4)$$

where $u_j \geq 0$, $\|u_j\|_1$ is bounded and $\|V_j\|_1 + \|s_j\|_1 \rightarrow 0$. Uniform continuity of multiplication on compact sets keeps the boundary of the restricted measures outside B_3 after smoothing.

Choose a compact Haar-null $N_j \subset B_1$ with $\nu_j(N_j) > 3/4$, and an open $O_j \subset B_{3/2}$ containing it with $|O_j| < 1/j$. Approximate-identity convergence permits δ_j so small that also

$$\int_{O_j} u_j > 2/3. \quad (8.5)$$

This retains concentration before taking any tangent limit.

Take $0 \leq \chi \leq 1$ supported in B_3 and equal to one on B_2 . Symmetry and the product rule in (8.4) give

$$(I + P)(\chi u_j) = -\mathcal{T}(\chi V_j) + \sum_b X_b F_{b,j} + G_j,$$

where explicitly

$$F_{b,j} = 2 \sum_a (X_a \chi)(V_{ab,j} - (A_0)_{ab} u_j),$$

$$G_j = \chi u_j - \chi s_j + \sum_{a,b} (X_a X_b \chi)((A_0)_{ab} u_j - V_{ab,j}).$$

These lower-order sources have bounded L^1 norms and fixed compact support. Applying Lemma 8.1, write

$$\chi u_j = f_j + h_j, \quad f_j = -\mathcal{R}\mathcal{T}(\chi V_j), \quad h_j = \sum_b \mathcal{R}X_b F_{b,j} + \mathcal{R}G_j.$$

Then $\|f_j\|_{L^{1,\infty}} \rightarrow 0$, $f_j \rightarrow 0$ distributionally, and h_j is locally L^1 precompact. Distributional convergence follows by testing V_j against the bounded adjoint image of a smooth compact test.

Positivity gives $f_j^- \leq |h_j|$. Consequently f_j^- is locally uniformly integrable and tends to zero in measure; hence it tends to zero in local L^1 . For a nonnegative smooth test ψ , $\int \psi f_j^+ = \langle f_j, \psi \rangle + \int \psi f_j^- \rightarrow 0$. Choosing $\psi \geq 1$ on a prescribed compact set shows $f_j \rightarrow 0$ in local L^1 . Thus χu_j is locally L^1 precompact and uniformly integrable on $B_{3/2}$. This contradicts (8.5) since $|O_j| \rightarrow 0$, and proves $\nu^s = 0$. $\square$

9 Fixed directions, control packets, and conditional occupation transfer

Lemma 9.1 (Finite strictly positive direction decomposition). *There are finitely many unit vectors $v_1, \dots, v_M \in \mathbb{R}^m$, including the coordinate vectors, constants $0 < c < C < \infty$, and Borel functions a_ℓ on $\{\lambda I \leq A \leq \Lambda I\}$ such that*

$$A = \sum_{\ell=1}^M a_\ell(A) v_\ell v_\ell^\top, \quad c \leq a_\ell(A) \leq C.$$

Consequently $\mathcal{L}_A = \sum_\ell a_\ell(A) V_\ell^2$, where $V_\ell = \sum_i (v_\ell)_i X_i$ are fixed fields generating the same Lie algebra.

Proof. Choose a finite δ -net of unit vectors, containing the coordinate vectors, and let $\mathcal{C}$ be their finitely generated rank-one cone. For H in its dual cone, approximation gives $v^\top H v \geq -2\delta \|H\|_{\text{op}}$ for every unit v . If $\delta < 1/2$ and $H \neq 0$, its largest absolute eigenvalue is positive; hence $\text{tr } H_- \leq 2m\delta \text{tr } H_+$. For $\lambda I/2 \leq R \leq \Lambda I$ and $\delta \leq \lambda/(8m\Lambda)$, it follows that $\text{tr}(RH) \geq (\lambda/4) \text{tr } H_+ \geq 0$. Finite-cone separation gives $R \in \mathcal{C}$. Now let $S_0 = \sum v_\ell v_\ell^\top$ and $\eta = \lambda/(2M)$. Then $A - \eta S_0$ has these bounds, so it has a nonnegative conic representation with coefficients β_ℓ . Taking traces gives $\beta_\ell \leq m\Lambda$. Set $a_\ell = \eta + \beta_\ell$. For a Borel choice enumerate all linearly independent subsets of generators of cardinality at most $m(m+1)/2$ and select the first with nonnegative coordinates representing $A - \eta S_0$. Conic Carathéodory reduction gives one; each eligibility set is Borel and its coordinate map is linear on its span. The vector coefficients are constant, so the operator formula introduces no derivatives. Inclusion of the coordinate directions proves Lie-algebra equality. $\square$

Proposition 9.2 (A finite full-rank horizontal control packet). *For arbitrary smooth bracket-generating fields, every point x_0 has neighborhoods $W \Subset U \Subset \Omega$ such that any $x, y \in W$ are joined in time one by a piecewise constant horizontal control whose length and energy are uniformly bounded and whose trajectory stays a fixed distance from ∂U . Moreover, a randomization of finitely many bounded control parameters gives an endpoint law of uniformly bounded Lebesgue density for every $x \in W$.*

Proof. The span of $(\Phi_* X_i)(x_0)$, where Φ ranges over sufficiently short finite flow words, is a closed finite-dimensional space. Differentiating successive pushforwards at zero shows that it contains every iterated bracket value. Choose n such pushed fields $Z_j = (\Phi_j)_* X_{i_j}$ that form a basis at x_0 . Define conjugate flows $F_j(t) = \Phi_j \circ \exp(tX_{i_j}) \circ \Phi_j^{-1}$ and endpoint map $E(x, t) = F_n(t_n) \circ \dots \circ F_1(t_1)(x)$. Its parameter derivative at $(x_0, 0)$ is invertible. The parametric inverse function theorem, applied to $(x, t) \mapsto (x, E(x, t))$, supplies the connecting parameters uniformly for x, y in a smaller W . Shrink all neighborhoods so intermediate flow points stay inside a compact subset of U . Assign fixed positive time slots to the finitely many flow pieces; bounded parameters then give bounded length and energy. On a smaller parameter box J the map $t \mapsto E(x, t)$ is injective and $c_0 \leq |\det D_t E(x, t)| \leq C_0$. For uniform $t \in J$, change of variables gives density $\mathbf{1}_{E(x, J)}(y)/(|J| |\det D_t E(x, E_x^{-1}y)|)$, proving the last claim. $\square$

Proposition 9.3 (Accessibility with arbitrary adapted covariance). *Let $W \Subset U$ be as in Proposition 9.2. Suppose an existing continuous semimartingale, up to its exit time τ_U , satisfies*

$$dZ_t = b_{A_t}(Z_t)dt + \sqrt{2}\sigma(Z_t)A_t^{1/2}dW_t + \sqrt{2\varepsilon}dB_t,$$

where W, B form an independent Brownian pair, A_t is progressively measurable with $\lambda I \leq A_t \leq \Lambda I$, and $b_A = \sum a_{ij}(D\sigma_j)\sigma_i$. There exist fixed $c, C, r_0 > 0$ such that, for $x, y \in W$, $0 < r < r_0$, and $\tau = cr^2$,

$$\mathbb{P}_x\{Z_\tau \in B(y, r), \tau_U > \tau\} \geq \frac{1}{4}e^{-C/r^2}.$$

The bound is uniform over the adapted covariance and $\varepsilon \in [0, 1]$. Existence or uniqueness of such a semimartingale is not a conclusion.

Proof. Choose a deterministic control u joining x to y in unit time, with length at most L and energy at most E , and rescale it as $u_t^\tau = u(t/\tau)/\tau$. Change the first Brownian coordinate by the adapted shift $\eta_t = 2^{-1/2}A_t^{-1/2}u_t^\tau$, stopped at exit. Its squared integral is bounded deterministically by $E/(2\lambda\tau)$, so Novikov holds. Under the changed law Q , the process has the added drift $\sigma(Z_t)u_t^\tau$, and $D(Q\|P) \leq E/(4\lambda\tau)$. Compare it to the accelerated deterministic curve. Lipschitz bounds for σ , bounded geometric drift and covariance, pathwise Gronwall using $\int |u_t^\tau|dt \leq L$, and Doob's inequality yield

$$\mathbb{E}_Q \sup_{t \leq \tau \wedge \tau_U} |Z_t - \gamma(t/\tau)|^2 \leq C\tau.$$

Choose r_0 below the fixed boundary margin and then c small; the desired event F has $Q(F) \geq 1/2$. Binary relative entropy gives $D(Q\|P) \geq Q(F)\log(1/P(F)) - \log 2$. Combining the inequalities proves the displayed lower bound. No derivative of A_t has been used. $\square$

Proposition 9.4 (Conditional drift-energy transfer). *Let U have finite Lebesgue measure. Suppose a nonnegative occupation density q has a representation*

$$q(dx) = M \mathbb{E}_Q \int_0^\tau e^{-t} \mathbf{1}_{\{Z_t \in dx\}} dt, \quad 0 < M \leq M_0,$$

and let S be an independent rate-one exponential time. Suppose additionally that comparison with a law P without a specified extra drift is justified on the marked stopped path space up to $\tau \wedge S$, and

$$D(Q_{\tau \wedge S} \| P_{\tau \wedge S}) \leq E/(4M), \quad 0 \leq E \leq E_0.$$

The mark retains whether S or τ occurs first. Let $Y = Z_S$ if $S < \tau$ and a cemetery point otherwise. If the non-cemetery part of the reference law has a density k with $\|k\|_{L^s(U)} \leq C_s$ for some $s > 1$, then, with $s' = s/(s-1)$, every bounded nonnegative f supported in U satisfies

$$\frac{1}{M} \int f q \leq \frac{E}{4M} + \log \left(1 + C_s \left(\int_U e^{s'f} \right)^{1/s'} \right).$$

For $q = \chi^2 p$, $\chi = 1$ on K , and $\int p \leq 1$, these premises imply a uniform bound on $\int_K p \log(1+p)$. Alternatively, a reference upper modulus $P(Y \in F) \leq \omega(|F|) < 1$ gives

$$\int_F q \leq \frac{E/4 + M \log 2}{\log(1/\omega(|F|))}.$$

The occupation representation, relative-entropy comparison and upper reference bounds are explicit additional premises.

Proof. The non-cemetery Q -law of Y is q/M . Data processing transfers the assumed path entropy bound to Y . The entropy variational inequality gives $M^{-1} \int f q \leq E/(4M) + \log \mathbb{E}_P e^{f(Y)}$. The reference exponential moment is at most $1 + \int_U e^f k$; Hölder gives the asserted bound. Take $f_N = s'^{-1} \mathbf{1}_K \min\{N, \log(1+p)\}$. Then $\int_U e^{s'f_N} \leq |U| + 1$, so monotone convergence proves entropy. Finally binary relative entropy gives $Q(Y \in F) \log(1/P(Y \in F)) \leq E/(4M) + \log 2$; multiply by M . A zero reference probability is treated by absolute continuity. $\square$

Remark 9.5. The accessibility estimate is a lower bound for probabilities of balls. It supplies neither of the upper reference bounds assumed in Proposition 9.4. A deterministic full-rank packet also need not preserve an arbitrary stationary measure. These distinctions prevent the conditional transfer statement from being read as a general entropy theorem.

10 Transport of the full stress and a covariant tensor energy

The next identities are valid in arbitrary smooth geometry. The stochastic statements are made on a compact flat torus with fixed smooth reference fields, so their flow derivatives have finite moments. They concern an auxiliary smooth flow, not a stochastic flow with Borel coefficients A . For a local original equation one must first retain the cutoff sources in Proposition 5.2 or 5.1.

Proposition 10.1 (Exact deterministic transport). *For a symmetric matrix density R and vector density b , put $\mathcal{S}(R, b) = \text{Div Div } R - \text{div } b$. For a smooth diffeomorphism F , define*

$$R^F = F_*[DF R D F^\top], \quad b^F = F_*[DF b + D^2 F : R].$$

Pushforward includes the density Jacobian. Then

$$\mathcal{S}(R^F, b^F) = F_* \mathcal{S}(R, b), \quad \text{Div } R^F - b^F = F_*[DF(\text{Div } R - b)].$$

The latter formula is an identity of distributions even if its current has no function representative. For the original horizontal stress,

$$R = \sigma Q \sigma^\top, \quad Q = A p, \quad b = \sum_{i,j} Q_{ij} (D \sigma_j) \sigma_i,$$

its current has the exact form

$$\text{Div } R - b = - \sum_i \sigma_i \sum_j X_j^* Q_{ij}. \quad (10.1)$$

If $\varepsilon p I$ is added to R , add $\varepsilon \nabla p$ to this current.

Proof. Pair against a smooth scalar test and use

$$D^2(\phi \circ F) = DF^\top(D^2\phi)(F)DF + \sum_a (\partial_a \phi)(F) D^2 F^a.$$

This proves the first identity. Testing against a smooth vector field gives $\text{Div } R^F = F_*[DF \text{Div } R + D^2 F : R]$; subtract the transported b . For (10.1), expand

$$\text{Div}(\sigma Q \sigma^\top) = \sum_i \sigma_i \sum_j (X_j Q_{ij} + (\text{div } X_j) Q_{ij}) + \sum_{i,j} Q_{ij} (D\sigma_i) \sigma_j.$$

Symmetry makes the last term equal to b , and $X_j^* = -X_j - \text{div } X_j$. All calculations extend to integrable densities or distributions by duality. $\square$

Proposition 10.2 (Averaged cones and transported drift energy). *Let $p \geq 0$, $\lambda I \leq A \leq \Lambda I$ on its support, and let random smooth diffeomorphisms F have sufficient local first- and second-derivative moments for all following measures to be finite. With $B = \sigma A \sigma^\top + \varepsilon I$, define*

$$q = \mathbb{E}F_*(p \, dx), \quad R = \mathbb{E}F_*[p DF B DF^\top dx],$$

$$G = \mathbb{E}F_*[p DF \sigma \sigma^\top DF^\top dx], \quad V = \mathbb{E}F_*[p DF DF^\top dx].$$

Then $\lambda G + \varepsilon V \leq R \leq \Lambda G + \varepsilon V$. The componentwise average $\bar{Q} = \mathbb{E}F_(p A \, dx)$ instead satisfies $\lambda q I \leq \bar{Q} \leq \Lambda q I$ and defines a Borel ratio $\bar{A} = d\bar{Q}/dq$ in that cone. In general this does not identify R with $q(\sigma \bar{A} \sigma^\top + \varepsilon I)$.*

If an extra measurable drift c lies in $\text{range } B$ and $\int p c^\top B^\dagger c \leq E_0$, its averaged current $C = \mathbb{E}F_[p DF c \, dx]$ obeys*

$$C \in \text{range } R \quad \text{a.e.}, \quad \int C^\top R^\dagger C \leq E_0.$$

All covariance and current statements use their Lebesgue densities, which exist because each diffeomorphism pushes forward an integrable density.

Proof. Congruence, pushforward, and expectation preserve positive matrix order. This proves the cone assertions and the Radon–Nikodym ratio statement. For a single diffeomorphism, write $B = LL^\top$. The value $c^\top B^\dagger c$ is the minimum of $|u|^2$ over $Lu = c$; invertibility of DF makes this equivalent to $DFLu = DFc$. Change of variables therefore preserves the integral of the covariance energy. For mixtures, use the convex extended perspective

$$\sup_v \{2v \cdot z - v^\top M v\} = \begin{cases} z^\top M^\dagger z, & z \in \text{range } M, \\ +\infty, & z \notin \text{range } M. \end{cases}$$

A supremum of the average is bounded by the average of suprema. Applying this pointwise and then integrating proves both the energy and range assertions. This controls the transported covariance metric only. $\square$

Proposition 10.3 (Necessary range preservation and a geometric limitation). *At $\varepsilon = 0$, in the preceding notation let $H_y = \text{range } \sigma(y)$. If $R(y) \leq Cq(y)\sigma(y)\sigma(y)^\top$ almost everywhere, then $DF(x)H_x \subset H_{F(x)}$ for almost every density-weighted contributing pair (F, x) . This assertion permits changing rank. Separately, on an open proper constant-rank bundle H spanned by the X_i , preservation by every sufficiently short local flow of every X_i would force H to be involutive. Such universal preservation is incompatible with bracket generation when $\text{rank } H < n$.*

Proof. Use the Borel orthogonal projection $\Pi(y) = I - \sigma(y)\sigma(y)^\dagger$ onto $H_y^\perp$. The assumed inequality gives $\text{tr}(\Pi R \Pi) = 0$. For $w_F(y) = p(F^{-1}y)|\det DF^{-1}(y)|$, positivity implies

$$0 \geq \lambda \mathbb{E}[w_F(y) \|\Pi(y) DF(F^{-1}y) \sigma(F^{-1}y)\|_{\mathbb{F}}^2] \geq 0.$$

The nonnegative integrand vanishes almost everywhere, proving the inclusion. For the separate bundle assertion, differentiate $(\exp(tX_i))_* X_j$ at zero to obtain, up to sign, $[X_i, X_j] \in H$. A local subset of the original fields is a smooth frame. The bracket product rule then gives closure for all horizontal sections; all iterated brackets remain in the proper bundle, a contradiction. A condition holding for a single weighted random law is not identified with this universal flow-preservation premise. $\square$

Proposition 10.4 (Finite-energy correction criterion). *Suppose integrable densities $q \geq 0$, R, J satisfy $\text{Div Div } R - \text{div } J = \mu$. Fix a locally bounded positive semidefinite Borel covariance B^0 and a drift b^0 for which qB^0, qb^0 are locally integrable, and set*

$$D_0 = B^0 : D^2 + b^0 \cdot \nabla, \quad E = R - qB^0, \quad H = J - qb^0, \quad \mathcal{F}(\phi) = \int E : D^2 \phi + \int H \cdot \nabla \phi.$$

For a chosen compactly supported smooth test space, there exists a measurable $c \in \text{range } B^0$, q -almost everywhere, satisfying

$$(D_0 + c \cdot \nabla)^* q = \mu, \quad \int q c^\top (B^0)^\dagger c \leq C_0$$

if and only if

$$|\mathcal{F}(\phi)|^2 \leq C_0 \int q \nabla \phi^\top B^0 \nabla \phi \tag{10.2}$$

for every test. Independently of this condition, the canonical distributional current $j_0 = H - \text{Div } E$ satisfies $D_0^ q - \text{div } j_0 = \mu$; it need not be a measurable drift times q .*

Proof. Expanding the balance gives the canonical current identity. A correction must satisfy $\int qc \cdot \nabla \phi = \mathcal{F}(\phi)$, so necessity is weighted Cauchy–Schwarz with the stated range condition. Conversely, map tests into $L^2(q, dx; \mathbb{R}^n)$ by $T\phi = (B^0)^{1/2} \nabla \phi$. The bound makes $\mathcal{F}$ a functional of norm at most $\sqrt{C_0}$ on the closure of the image, vanishing on its null space. Riesz gives v in that closure with $\mathcal{F}(\phi) = \int qv \cdot (B^0)^{1/2} \nabla \phi$ and $\|v\|_2^2 \leq C_0$. Set $c = (B^0)^{1/2} v$. Then c lies in the covariance range and $\int qc^\top (B^0)^\dagger c \leq \|v\|_2^2$. Local boundedness of B^0 and Cauchy–Schwarz ensure local integrability of qc . Testing proves the equation. This construction permits all currents with the same divergence, including divergence-free modifications of j_0 . $\square$

Theorem 10.5 (Covariant quadratic energy for smooth reference transport). *Let $Y_1, \dots, Y_k$ be fixed smooth bracket-generating fields on a compact connected flat torus M . Let F_s be their smooth Stratonovich stochastic flow driven by $\sqrt{2}Y_i$, and average the tensor transport of Proposition 10.1. Write $L = \sum_i Y_i^2$, $p_s = P_s^* p_0$ and $R_s = \mathbb{E} R_0^{F_s}$, where $p_0 \geq 0$ is integrable, $R_0 = p_0 C_0$ is symmetric, and $\int p_0 |C_0|_{\mathbb{F}}^2 < \infty$. For $J_i = DY_i$ define*

$$K_i C = J_i C + C J_i^\top, \quad K_i^\dagger C = J_i^\top C + C J_i, \quad \nabla_i^{\text{tr}} C = Y_i C - K_i C.$$

Then $C_s = R_s/p_s$, with the zero-mass case interpreted as zero, satisfies

$$\sup_{0 \leq s \leq 1} \int p_s |C_s|_{\mathbb{F}}^2 + \int_0^1 \int p_s \sum_i |\nabla_i^{\text{tr}} C_s|_{\mathbb{F}}^2 \leq C \int p_0 |C_0|_{\mathbb{F}}^2. \tag{10.3}$$

The derivatives for rough initial tensors are weak derivatives. The constant depends only on the smooth reference fields and the time interval. For smooth positive data the exact identity is

$$\begin{aligned} \frac{d}{ds} \int p_s |C_s|^2 = & -2 \sum_i \int p_s |\nabla_i^{\text{tr}} C_s|^2 \\ & + 2 \sum_i \int p_s \left[|K_i C_s|^2 + \langle C_s, (Y_i K_i^\dagger + (K_i^\dagger)^2) C_s \rangle \right], \end{aligned} \quad (10.4)$$

where $Y_i K_i^\dagger$ differentiates the coefficient endomorphism only. The estimates remain uniform on adjoining the constant fields $\sqrt{\varepsilon} \partial_j$, $0 \leq \varepsilon \leq 1$, to a fixed bracket-generating family.

Proof. For one field Y , differentiation of deterministic transport gives

$$\mathcal{R}_Y R = Y^* R + (DY)R + R(DY)^\top, \quad \mathcal{B}_Y(R, b) = Y^* b + (DY)b + D^2 Y : R.$$

Let $\mathcal{T}_Y(R, b)$ denote this pair. The chain rule in Proposition 10.1 gives $\mathcal{S}\mathcal{T}_Y = Y^* \mathcal{S}$. The Stratonovich normalization $\sqrt{2}$ gives the averaged generators $\sum_i \mathcal{T}_{Y_i}^2$ for the pair and L^* for the scalar. Thus

$$\mathcal{S}(R_s, b_s) = P_s^* \mathcal{S}(R_0, b_0). \quad (10.5)$$

The matrix generator is $\sum_i (Y_i^* + K_i)^2$, whose adjoint on matrix functions is $\sum_i (Y_i + K_i^\dagger)^2$. Differentiate $\int |R_s|^2 / p_s$. A single field contributes

$$\int p_s \{ 2 \langle C_s, (Y_i + K_i^\dagger)^2 C_s \rangle - Y_i^2 |C_s|^2 \}.$$

Expanding this expression gives $-2|Y_i C_s|^2 + 4\langle K_i C_s, Y_i C_s \rangle + 2\langle C_s, (Y_i K_i^\dagger + (K_i^\dagger)^2) C_s \rangle$. Completing the square proves (10.4). Boundedness of its coefficient endomorphisms gives $\mathcal{E}' + 2\mathcal{V} \leq C\mathcal{E}$; Gronwall and integration prove (10.3) for smooth data.

We justify the smooth transport and rough-data limit. The derivative flow satisfies $d(DF_s) = \sqrt{2} \sum_i DY_i(F_s) DF_s \circ dW_s^i$. Its Itô form is linear with bounded smooth coefficients. Itô's formula for every fixed even matrix moment, localization and Gronwall give uniform moments of DF_s . The inverse flow has the corresponding linear equation; its derivatives have the same bounds. Second and higher spatial derivatives satisfy linear equations with lower-derivative forcing, so induction supplies every fixed moment needed to differentiate transported smooth data. In particular conditional Cauchy–Schwarz gives

$$\int |R_s|^2 / p_s \leq \int p_0(x) \mathbb{E} |DF_s(x) C_0(x) DF_s(x)^\top|^2 dx \leq C \int p_0 |C_0|^2.$$

Simultaneously convolve p_0, R_0 on the torus and add a vanishing positive scalar baseline. Kernel Cauchy–Schwarz does not increase the initial quadratic perspective. Apply the preceding estimate to these smooth data. For a fixed $\tau > 0$, scalar smooth-reference heat smoothing makes the scalar approximations converge in $C^\infty([\tau, 1] \times M)$ to p_s . For nonzero mass, positivity of the bracket-generating heat kernel gives a positive minimum there. The energy estimate therefore gives ordinary weak L^2 compactness of C_s and its covariant derivatives on this cylinder. Testing stochastic transport against a smooth tensor and using the derivative moments identifies the distributional limit of the tensors with R_s . Thus the weak quotient limit is R_s/p_s , and testing derivatives identifies its weak covariant derivatives. Weak lower semicontinuity gives the budget on $[\tau, 1]$ and at each fixed positive time. Let $\tau \downarrow 0$. This uses only scalar smoothing, not an unproved smooth matrix kernel.

The added constant viscosity fields have zero spatial derivatives and bounded amplitudes. All coefficient and moment constants above are therefore uniform in ε , and the fixed subfamily supplies scalar Hörmander smoothing for every parameter. $\square$

Corollary 10.6 (Moving reference tensor and its entropy identity). *Let $Y_i = X_i = \sigma_i \cdot \nabla$ in Theorem 10.5, $S = \sum_i \sigma_i \sigma_i^\top$, and transport the initial pair*

$$M_0 = p_0 S, \quad b_0^0 = p_0 \sum_i (D\sigma_i) \sigma_i.$$

Then

$$\text{Div Div } M_s - \text{div } b_s^0 = L^* p_s = \partial_s p_s.$$

If initially $\lambda M_0 \leq R_0 \leq \Lambda M_0$, then $\lambda M_s \leq R_s \leq \Lambda M_s$. Both R_s/p_s and M_s/p_s obey (10.3). For positive smooth data, writing $G_s = M_s/p_s$, $h_s = b_s^0/p_s$ gives the exact identity

$$-\frac{d}{ds} \int p_s \log p_s = \int p_s (\nabla \log p_s)^\top G_s (\nabla \log p_s) + \int p_s (\text{Div } G_s - h_s) \cdot \nabla \log p_s. \quad (10.6)$$

Finally $[\nabla_X^{\text{tr}}, \nabla_Y^{\text{tr}}] = \nabla_{[X, Y]}^{\text{tr}}$ on contravariant symmetric tensors.

Proof. Initially $\mathcal{S}(M_0, b_0^0) = L^* p_0$. Apply (10.5) and commutation of L^* with its semigroup. Congruence and expectation preserve the matrix order. The energy conclusion is Theorem 10.5 for each tensor. Conservation of scalar mass and integration by parts in the displayed Fokker–Planck equation give

$$\frac{d}{ds} \int p_s \log p_s = \int p_s G_s : D^2 \log p_s + \int p_s h_s \cdot \nabla \log p_s.$$

One further integration by parts is exactly (10.6). The last formula is the representation identity for Lie derivatives; it also follows by expanding $XC - (DX)C - C(DX)^\top$ twice and collecting $[X, Y]$ and its Jacobian. $\square$

Remark 10.7. The tensor budget controls Frobenius derivatives along the fixed driving fields. It is not a bound for the relative flux $\text{Div } G_s - h_s$ in the inverse moving tensor metric. The second integral in (10.6) remains present. Consequently these exact transport identities and finite energies do not by themselves prove the general stationary entropy estimate. They also do not assert that $M_s = p_s S$ or that its range remains the original horizontal space. In the viscous version the initial reference tensor is $p_0(S + \varepsilon I)$ and its cone constants are $\min\{\lambda, 1\}$ and $\max\{\Lambda, 1\}$.

Proposition 10.8 (Heat commutator identity and homogeneous order). *For a fixed smooth reference generator L with semigroup P_t and a smooth differential operator Z , on smooth tests one has*

$$[Z, P_t] = \int_0^t P_{t-s} [Z, L] P_s ds.$$

For degree-one left-invariant generators on a Carnot group and the homogeneous $L = \sum_i X_i^2$, the second-order commutator satisfies

$$[X_i X_j, P_t](f \circ \delta_{t^{-1/2}}) = t^{-1}([X_i X_j, P_1]f) \circ \delta_{t^{-1/2}}.$$

If this commutator is nonzero on a bounded smooth test, no uniform $L^\infty \rightarrow L^\infty$ bound with improved order $Ct^{-1+\eta}$, $\eta > 0$, can hold for all small t .

Proof. Differentiate $P_{t-s} Z P_s$ in s and integrate. The second assertion follows from $X_i(f \circ \delta_r) = r(X_i f) \circ \delta_r$ and $P_t(f \circ \delta_r) = (P_{r^2 t} f) \circ \delta_r$. The sup norm is invariant under dilation. Apply a hypothetical improved bound to a test with nonzero time-one commutator and let $t \downarrow 0$. $\square$

11 From a quantitative small-set modulus to a power gain

Proposition 11.1 (Qualitative compactness does not imply uniform integrability). *There exists a convex weak-* compact class of absolutely continuous probability measures on $[0, 1]$ which is not uniformly integrable with respect to Lebesgue measure. No differential equation is asserted for this class.*

Proof. Let $E_n = \bigcup_{k=0}^{n-1} [k/n, k/n + 1/n^2)$ and $p_n = n\mathbf{1}_{E_n}$. Then $|E_n| = 1/n$ and $\int_{E_n} p_n = 1$. Uniform continuity of continuous tests and Riemann sums show $p_n dx \rightarrow dx$. On the one-point compactification $S = \mathbb{N} \cup \{\infty\}$, set $\nu_n = p_n dx$ and $\nu_\infty = dx$. This is a weak-* continuous family. The barycentric image of the compact convex set of probability measures on S is compact and convex. Every element has density $\alpha_\infty + \sum_{n \geq 1} \alpha_n p_n$ with nonnegative coefficients summing to one, by Tonelli. The class contains all p_n and is therefore not uniformly integrable. $\square$

Theorem 11.2 (A local modulus self-improves without weight doubling). *Let $w \geq 0$ be integrable on an open $V \subset \mathbb{R}^n$, with $\int_V w \leq M$. Suppose a nondecreasing ω with $\omega(t) \rightarrow 0$ as $t \downarrow 0$ satisfies*

$$\int_E w \leq \omega(|E|/|Q|) \int_{2Q} w \quad (11.1)$$

for every axis-parallel cube Q with $2Q \Subset V$ and every Borel $E \subset Q$. There is $\theta > 0$, depending only on n, ω , such that for every bounded $U \Subset V$ and $0 < s < \theta$, $\int_U w^{1+s} \leq C_{U,V,n,\omega,s,M}$. Homogeneously, on each cube Q_0 with $2Q_0 \subset V$, the premise need only hold on strictly interior testing doubles in $2Q_0$, and

$$\left(\frac{1}{|Q_0|} \int_{Q_0} w^{1+s} \right)^{1/(1+s)} \leq \frac{C}{|2Q_0|} \int_{2Q_0} w.$$

No doubling assumption on $w dx$ is required.

Proof. The zero-mass case is trivial. Choose a finite buffer radius $d > 0$ between $\bar{U}$ and V^c , and extend $f = w\mathbf{1}_U$ by zero. Let $O_a = \{\mathcal{M}f > a\}$ for the uncentered cube maximal function. For $a \geq a_0 = C_n M/d^n$, with a sufficiently large dimensional constant, $\bar{O}_a \Subset V$: every cube meeting U and a point at distance at least $d/2$ from it has volume at least a fixed multiple of d^n and average at most $C_n M/d^n$.

Decompose O_a into maximal dyadic Whitney cubes Q with $8Q \subset O_a$. Their interiors are disjoint, their doubles lie compactly in O_a , and the doubles overlap at most $N_0(n)$ times. Indeed distance to O_a^c on $2Q$ is comparable to $\ell(Q)$, so intersecting doubles have comparable sizes, and geometric packing gives bounded overlap. Maximality supplies a point $z_Q \notin O_a$ at distance at most $C_n \ell(Q)$ from Q . Enlarging $3Q$ to contain z_Q gives $\int_{3Q} f \leq C_n a|Q|$. Any averaging cube containing $x \in Q$ with side at least $\ell(Q)$ can likewise be enlarged to contain z_Q , so its average is at most $A_0 a$ for a fixed $A_0 > 1$. For $A > A_0$, the level $\mathcal{M}f > Aa$ on Q must therefore be witnessed inside $3Q$. The maximal weak $(1, 1)$ inequality gives $|O_{Aa} \cap Q| \leq C_0 |Q|/A$. Apply (11.1) and sum to obtain

$$\int_{O_{Aa}} w \leq N_0 \omega(C_0/A) \int_{O_a} w.$$

Choose $A > A_0$ so this coefficient is at most $1/2$, and put $\theta = \log 2 / \log A$. Iteration and $w \leq \mathcal{M}f$ almost everywhere on U give

$$\int_{\{x \in U : w(x) > R\}} w(x) dx \leq 2M(R/a_0)^{-\theta} \quad (R \geq a_0).$$

Weighted layer cake yields, for $0 < s < \theta$,

$$\int_U w^{1+s} \leq \left(1 + \frac{2s}{\theta - s}\right) M a_0^s.$$

For $U = \text{int } Q_0$, $V = \text{int}(2Q_0)$, take $a_0 = C_n w(2Q_0)/|Q_0|$, without replacing it by one. This is exactly the homogeneous assertion. Only strictly interior Whitney doubles are tested. $\square$

Corollary 11.3 (Fixed Hörmander control-metric version). *On a fixed compact neighborhood of arbitrary smooth bracket-generating fields, replace cubes by sufficiently small control balls and $2Q$ by τB for a fixed $\tau > 1$. If the corresponding local modulus holds whenever that testing dilate is admissible and compactly contained in the region, then the same local positive power gain holds. For sufficiently small B_0 with a fixed outer dilate κB_0 , $\kappa > 1$, it takes the homogeneous weak reverse Hölder form with outer average on κB_0 . Constants depend on the fixed local control geometry, τ, κ, ω .*

Proof. Extend the fields to a compact bracket-generating ambient manifold, retaining them near the chosen region. The fixed control metric is proper, and Lebesgue control-ball volumes and the metric are locally doubling. Small balls in the retained neighborhood use the original geometry. For $f = w\mathbf{1}_U$, a positive control-distance buffer and the positive minimum volume of a fixed small ball give a finite high-level threshold placing $O_a = \{\mathcal{M}f > a\}$ compactly inside V . The maximal weak-type bound makes $|O_a|$ smaller than the minimum volume of a ball at the largest admissible Whitney scale; hence no Whitney radius exceeds that scale.

For $x \in O_a$ put $\delta(x) = d(x, O_a^c)$ and take balls $b_x = B(x, \delta(x)/L)$ with $L > 100\tau$. Select disjoint b_j whose fivefold dilates B_j cover O_a . Then $\tau B_j \subseteq O_a$ and these dilates have bounded overlap: the one-Lipschitz function δ makes intersecting dilates have comparable radii, and geometric doubling plus disjointness bounds their multiplicity. A point of O_a^c is within a fixed multiple of each radius. Enlarging balls to include that point and using volume doubling gives the same large-witness bound as in the cube proof. The weak-type maximal estimate then gives $|O_{Aa} \cap B_j|/|B_j| \leq C/A$. Sum the assumed modulus and use bounded overlap to obtain weighted level decay, and then the same layer-cake bound.

For $U = B_0$, $V = \kappa B_0$, every ball meeting B_0 and the complement of $\kappa' B_0$, $\kappa' = (1 + \kappa)/2$, has radius at least a fixed multiple of that of B_0 . A fixed dilation contains B_0 , so volume doubling bounds its volume below by $c|B_0|$. Its $w\mathbf{1}_{B_0}$ average is at most $Cw(\kappa B_0)/|B_0|$. This is the homogeneous threshold in the preceding proof. Sufficient smallness of B_0 ensures every Whitney testing dilate is within the allowed local scale. No weight-doubling estimate has entered. $\square$

Remark 11.4. The stationary equation has not been shown here to imply (11.1) with uniform constants in arbitrary smooth geometry. The theorem proves the self-improvement after that quantitative premise; qualitative absolute continuity from Theorem 8.2 alone does not supply it.

12 Stationary estimates for general fields and determinant degeneracy

This section uses Euclidean convexity to obtain estimates that apply to arbitrary fixed smooth fields. No Lie group structure is required. The determinant appearing below is the determinant of the original diffusion matrix, not a determinant formed from commutators. The resulting statements include uniformly elliptic systems, every one-dimensional bracket-generating system, every Grushin order, and a finite-type determinant class. All the stationary estimates in this section retain the original constant coordinate viscosity uniformly for $0 \leq \varepsilon \leq 1$.

12.1 Setting and scalar testing

Let $\Omega \subset \mathbb{R}^n$ be open, and let $X_i = \sigma_i \cdot \nabla$, $1 \leq i \leq m$, have fixed smooth real coefficients. Write $\sigma = (\sigma_1, \dots, \sigma_m)$ and

$$S = \sigma \sigma^\top, \quad d = \det S \geq 0. \quad (12.1)$$

For a symmetric Borel matrix $A = (a_{ij})$, put

$$D\varphi = \frac{1}{2} \sum_{i,j=1}^m a_{ij} (X_i X_j + X_j X_i) \varphi + \varepsilon \Delta \varphi.$$

An admissible stationary measure in this section is a nonnegative finite Borel measure ρ such that $\rho(\Omega) \leq 1$, $\lambda I_m \leq A \leq \Lambda I_m$ ρ -almost everywhere, and

$$\int_{\Omega} D\varphi d\rho = \int_{\Omega} \varphi d\mu \quad \text{for every } \varphi \in C_c^\infty(\Omega), \quad \mu \geq 0, \quad \mu(\Omega) < \infty. \quad (12.2)$$

Here $0 < \lambda \leq \Lambda < \infty$ are fixed and $\varepsilon \in [0, 1]$. When a density is assumed, we explicitly write $d\rho = p dx$, with $p \geq 0$ and $\int_{\Omega} p dx \leq 1$. No bracket hypothesis is imposed until a statement specifically asks for it. Constants denoted by C_K depend only on $K \Subset \Omega$, Ω , the fixed fields, λ , and Λ ; additional dependence is stated when needed. They never depend on A, ρ, μ , or ε .

After changing A on a ρ -null Borel set, we may arrange the displayed matrix inequalities everywhere. Symmetry gives the exact expansion

$$D\varphi = B : D^2\varphi + b \cdot \nabla \varphi, \quad B = \sigma A \sigma^\top + \varepsilon I_n, \quad b = \sum_{i,j} a_{ij} (D\sigma_j) \sigma_i. \quad (12.3)$$

Thus $B \geq \lambda S$, and B and b are bounded on each fixed compact subset by the allowed data. The vector b is already part of the specified symmetrized operator. No independent drift has been added. For scalar functions set

$$\Gamma_B(f, g) = \nabla f^\top B \nabla g, \quad \Gamma_B(f) = \Gamma_B(f, f).$$

The ordinary product and chain rules give

$$D(fg) = fDg + gDf + 2\Gamma_B(f, g), \quad D(h \circ f) = h'(f)Df + h''(f)\Gamma_B(f). \quad (12.4)$$

Lemma 12.1 (Local sink and scalar occupation estimates). *Let ρ satisfy (12.2). For every compact $K \Subset \Omega$,*

$$\mu(K) \leq C_K. \quad (12.5)$$

If f is smooth on a neighborhood of K , then for every Borel set $E \subset \mathbb{R}$,

$$\int_K 1_E(f(x)) \sum_{i=1}^m |X_i f(x)|^2 d\rho(x) \leq C_{K,f} |E|. \quad (12.6)$$

The constant also depends on f and its derivatives on one fixed compact neighborhood of K . In particular, if $\sum_i |X_i f|^2 \geq c_0 > 0$ on K , then

$$\rho(K \cap \{|f| < s\}) \leq C_{K,f,c_0} s \quad (s > 0). \quad (12.7)$$

These assertions do not assume absolute continuity.

Proof. Choose $0 \leq \chi \leq 1$, $\chi \in C_c^\infty(\Omega)$, equal to one near K . Equation (12.2) and (12.3) imply

$$\mu(K) \leq \int \chi d\mu = \int D\chi d\rho \leq \|D\chi\|_{L^\infty} \rho(\Omega) \leq C_K.$$

For the scalar estimate take the support of χ inside the neighborhood where f is defined. Let $g \in C_c^\infty(\mathbb{R})$ be nonnegative, and set $M = \int_{\mathbb{R}} g$. Choose $h'' = g$ by first taking $h'(s) = \int_{-\infty}^s g(t) dt$ and then fixing the value of h at one point of the bounded range of f on $\text{supp } \chi$. Thus $|h'| \leq M$ and $|h| \leq C_f M$ on that range. With $R = C_f M$ sufficiently large, $\varphi = \chi(R - h(f))$ is a nonnegative smooth compactly supported test. Its equation reads

$$0 \leq \int \varphi d\mu = \int [(R - h(f))D\chi - \chi h'(f)Df - 2h'(f)\Gamma_B(\chi, f) - \chi g(f)\Gamma_B(f)] d\rho.$$

All terms except the last have absolute integral at most CM , using the mass bound and fixed derivatives on the cutoff support. Since $\Gamma_B(f) \geq \lambda \sum_i |X_i f|^2$, this proves

$$\int_K g(f) \sum_i |X_i f|^2 d\rho \leq C_{K,f} \int_{\mathbb{R}} g.$$

The measure on the left, pushed forward by f , is a finite Borel measure on $\mathbb{R}$. The inequality against nonnegative smooth tests implies domination by $C_{K,f}$ times Lebesgue measure: first approximate nonnegative compactly supported continuous functions, then use regularity of finite Borel measures. This is (12.6) for every Borel E . Taking $E = (-s, s)$ and using the assumed lower bound proves (12.7). $\square$

12.2 Convex tests and a determinant lemma

For a finite convex function u on $\mathbb{R}^n$, denote its Alexandrov Hessian by H_u . We use the classical Alexandrov differentiability theorem in the following precise form: for almost every x there are a vector $\nabla u(x)$ and a positive semidefinite matrix $H_u(x)$ such that

$$u(x+h) = u(x) + \nabla u(x) \cdot h + \frac{1}{2} h^\top H_u(x) h + o(|h|^2).$$

The distributional Hessian is a positive semidefinite matrix-valued Radon measure and its absolutely continuous density is H_u . The convex case of the theorem and the relevant differentiation-of-measures facts are recalled in [1]. In particular, if u_δ is ordinary convolution with a nonnegative smooth approximate identity, then $D^2 u_\delta \rightarrow H_u$ almost everywhere. The latter follows by differentiating the absolutely continuous and singular parts of the Hessian measure; the singular part has zero Lebesgue density almost everywhere. We use this standard differentiability input, but do not use a regularity theorem for optimal transport.

Lemma 12.2 (Stationary convex-test estimate). *Assume (12.2) with $d\rho = p dx$. For every globally convex function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ with Lipschitz constant at most one,*

$$\int_K p \operatorname{tr}(S H_u) dx \leq C_K. \quad (12.8)$$

Proof. First suppose that u is smooth. Subtract its value at one fixed point and choose R so large that $R - u \geq 0$ on $\text{supp } \chi$, where χ is the preceding cutoff. The Lipschitz bound makes R uniform in u . Testing with $\chi(R - u)$ gives

$$0 \leq \int [(R - u)D\chi - \chi Du - 2\Gamma_B(\chi, u)] p dx.$$

The value of u and its first derivatives are uniformly bounded on the cutoff support, so (12.3) yields $\int \chi p(B : D^2 u) \leq C_K$. Convexity and $B \geq \lambda S$ imply (12.8). For general convex one-Lipschitz u , its Euclidean mollifications remain convex and one-Lipschitz. Normalize their values at the same fixed point and apply the smooth result. The preceding almost-everywhere Hessian convergence also holds $p dx$ -almost everywhere. All the integrands are nonnegative, so Fatou's lemma proves the assertion. This last passage is where the density hypothesis is used. $\square$

Lemma 12.3 (Convex transport determinant identity). *Let $q \geq 0$ be a compactly supported probability density on $\mathbb{R}^n$, and let $v_n = |B_1(0)|$. There is a finite globally convex one-Lipschitz function u such that*

$$\det H_u(x) = v_n q(x) \quad \text{for almost every } x \in \mathbb{R}^n. \quad (12.9)$$

No smoothness, strict positivity, or entropy hypothesis on q is required.

Proof. Let F be a compact set supporting q . Among probability couplings of $q(x) dx$ and $v_n^{-1} 1_{B_1}(y) dy$, choose one maximizing $\int x \cdot y$. Such a coupling exists by weak compactness on $F \times \overline{B_1}$ and continuity of the cost. Write its support as a relation $\mathcal{C}$ of pairs (y, x) .

For any finite cycle $(y_j, x_j) \in \mathcal{C}$, with $y_{k+1} = y_0$, optimality implies

$$\sum_{j=0}^k x_j \cdot (y_{j+1} - y_j) \leq 0. \quad (12.10)$$

Indeed, if a strict reverse inequality held, it would persist in small product neighborhoods of these pairs. Remove equal sufficiently small positive masses of the coupling from those neighborhoods, taking their sum no larger than the original coupling. Replace each removed piece by the product of its x marginal and the next piece's y marginal, divided by their common mass. Both marginals are preserved and the cost strictly increases, a contradiction.

Fix $(y_0, x_0) \in \mathcal{C}$. Define

$$v(y) = \sup \left\{ \sum_{j=0}^{k-1} x_j \cdot (y_{j+1} - y_j) + x_k \cdot (y - y_k) : k \geq 0, (y_j, x_j) \in \mathcal{C} \right\},$$

where every chain starts at the fixed pair. Closing the chain at y_0 and using (12.10) shows that its value there is at most zero. The length-zero chain attains zero. All its affine slopes lie in F , so, with $R_F = \sup_{x \in F} |x|$, this supremum is finite, convex, and globally R_F -Lipschitz. Appending any $(\bar{y}, \bar{x}) \in \mathcal{C}$ to chains evaluated at $\bar{y}$ gives

$$v(y) \geq v(\bar{y}) + \bar{x} \cdot (y - \bar{y}), \quad \text{hence} \quad \bar{x} \in \partial v(\bar{y}).$$

Convex functions are differentiable almost everywhere, in particular by the Alexandrov theorem recalled above. The y marginal is absolutely continuous, so the coupling is concentrated on $x = \nabla v(y)$. Consequently

$$(\nabla v)_\#(v_n^{-1} 1_{B_1} dy) = q dx. \quad (12.11)$$

Put

$$u(x) = \sup_{y \in \overline{B_1}} \{x \cdot y - v(y)\}.$$

This is finite, convex, and one-Lipschitz; all its subgradients belong to $\overline{B_1}$. We spell out the elementary duality needed here. For $y \in B_1$, existence of a supporting hyperplane to the finite convex function v at y implies

$$u^*(y) := \sup_x \{x \cdot y - u(x)\} = v(y).$$

The inequality $u^*(y) \leq v(y)$ follows directly from the definition of u , and a vector in $\partial v(y)$ gives the reverse inequality. The definition of a subgradient now gives

$$y \in \partial u(x) \iff u(x) + u^*(y) = x \cdot y \iff x \in \partial v(y), \quad y \in B_1.$$

For the last implication, maximality at an interior point of the ball first gives the supporting inequality locally; convexity extends it to all of $\mathbb{R}^n$.

Except on the null set of nondifferentiability of v and the sphere ∂B_1 , this equivalence identifies the subgradient image of any Borel set E with $\{y \in B_1 : \nabla v(y) \in E\}$. Thus

$$|\partial u(E)| = v_n \int_E q(x) dx. \quad (12.12)$$

This also proves measurability of each subgradient image in the completed Lebesgue sigma algebra: it differs from the measurable set $\{y \in B_1 : \nabla v(y) \in E\}$ by a subset of the fixed null set consisting of the nondifferentiability points of v and the sphere.

It remains to identify the density in (12.12), without assuming that the transport map is smooth. At a point x of Alexandrov twice differentiability and of Lebesgue differentiation for q , write $H = H_u(x)$ and set

$$u_r(z) = r^{-2} [u(x + rz) - u(x) - r \nabla u(x) \cdot z].$$

Then $u_r \rightarrow z^\top H z / 2$ uniformly on every fixed ball. We claim

$$|\partial u_r(B_1)| \longrightarrow v_n \det H. \quad (12.13)$$

To prove the upper bound, let e be a unit vector and compare the subgradient inequality at $z \in B_1$ with values at $z \pm te \in B_2$. If the uniform error on B_2 is $\eta_r \rightarrow 0$, each subgradient $\xi \in \partial u_r(z)$ satisfies $|\xi - Hz| \leq C(t + \eta_r/t)$, with a constant depending on H, n . Choosing $t \downarrow 0$ with $\eta_r/t \rightarrow 0$ gives $\partial u_r(B_1) \subset H\bar{B}_1 + o(1)\bar{B}_1$. If H is singular, the enclosing sets have volume tending to zero. If H is positive definite, fix $0 < s < 1$ and $y \in HB_s$. The unique minimizer of $z^\top H z / 2 - y \cdot z$ lies in B_s ; its value is separated from all boundary values on ∂B_1 by at least $\lambda_{\min}(H)(1-s)^2/2$. Uniform convergence therefore places a minimizer of $u_r(z) - y \cdot z$ over $\bar{B}_1$ in B_1 , uniformly for $y \in HB_s$, for small r . Hence $HB_s \subset \partial u_r(B_1)$. Letting $s \uparrow 1$ proves (12.13) in the positive definite case as well.

Subgradient scaling and (12.12) identify the left side as

$$|\partial u_r(B_1)| = r^{-n} |\partial u(B_r(x))| = r^{-n} v_n \int_{B_r(x)} q \longrightarrow v_n^2 q(x).$$

Comparing with (12.13) proves $\det H_u = v_n q$ almost everywhere. $\square$

12.3 Weighted integrability and elliptic consequences

Theorem 12.4 (Weighted stationary estimates for arbitrary smooth fields). *Assume (12.2) with $d\rho = p dx$ and define d by (12.1). No bracket-generating assumption is needed. If $n \geq 2$, then*

$$\int_K p^{n/(n-1)} d^{1/(n-1)} dx \leq C_K. \quad (12.14)$$

For every $n \geq 1$, if $d > 0$ for $p dx$ -almost every point of K , then

$$\int_K p \log(1+p) dx \leq C_K + \int_K p \log_+(1/d) dx, \quad \log_+ s = \max(0, \log s). \quad (12.15)$$

The right side may be infinite; whenever its determinant penalty is uniformly bounded, this is the desired uniform entropy estimate.

Proof. For (12.14) set $h = p^{n/(n-1)}d^{1/(n-1)}1_K$, $h_N = \min(h, N)1_K$, and $M_N = \int h_N$. When $M_N > 0$, apply Lemma 12.3 to $q_N = h_N/M_N$. For positive semidefinite matrices S, H ,

$$\mathrm{tr}(SH) = \mathrm{tr}(S^{1/2}HS^{1/2}) \geq n(\det S \det H)^{1/n}.$$

The inequality remains valid if a determinant vanishes. Together with Lemma 12.2, this yields

$$C_K \geq nv_n^{1/n} M_N^{-1/n} \int_K p d^{1/n} h_N^{1/n} dx.$$

On K , $p d^{1/n} = h^{(n-1)/n}$, so the integral on the right is at least $\int h_N = M_N$. Hence $M_N^{(n-1)/n} \leq C_K$. The zero-mass cases require no estimate, and monotone convergence proves (12.14). This normalization used only bounded truncations, not the sought integrability.

For (12.15), let $M = \int_K p$. If $M = 0$ the claim is immediate. Otherwise $0 < M \leq 1$; apply Lemma 12.3 to $q = p1_K/M$. At $p dx$ -almost every point of K with $p > 0$, the assumed positivity of d and the determinant identity make both S and H_u positive definite. For $C = S^{1/2}H_u S^{1/2}$ use $\log \det C \leq \mathrm{tr} C - n$ to obtain

$$\begin{aligned} \log p &= \log M - \log v_n + \log \det H_u \\ &\leq \mathrm{tr}(SH_u) + \log_+(1/d) - n - \log v_n. \end{aligned}$$

Enlarge the fixed additive constant to be nonnegative; this bounds $\log_+ p$ by the resulting nonnegative right side. Multiply by p , integrate, and apply Lemma 12.2 and $\log(1+p) \leq \log 2 + \log_+ p$. All inequalities are pointwise or use nonnegative integrals, so finite entropy has not been assumed. $\square$

Corollary 12.5 (Every locally uniformly elliptic density setting). *In the density setting of Theorem 12.4, suppose $n \geq 2$ and $d \geq d_0 > 0$ on K . Then*

$$\int_K p^{n/(n-1)} dx \leq C_{K,d_0}, \quad \int_K p \log(1+p) dx \leq C_{K,d_0}.$$

In particular these conclusions hold on every compact set when the original fields themselves span $\mathbb{R}^n$ at every point. These statements remain uniform as $\varepsilon \downarrow 0$.

Proof. Remove the determinant weight in (12.14) using its positive lower bound; alternatively use (12.15) for the entropy conclusion. If the original fields span everywhere, S is positive definite everywhere and continuity supplies the required compact lower bound. $\square$

Theorem 12.6 (The complete one-dimensional stationary measure case). *Let $\Omega \subset \mathbb{R}$ be open, let $X_i = b_i(x)\partial_x$ be smooth real fields whose Lie brackets span $\mathbb{R}$ at every point, and let ρ satisfy (12.2). Then $d\rho = p dx$ with $p \in L^1(\Omega)$, $p \geq 0$, $\int_\Omega p \leq 1$, and*

$$\|p\|_{L^\infty(K)} \leq C_K, \quad \int_K p \log(1+p) dx \leq C_K \quad (K \Subset \Omega).$$

All coefficients, sinks, and viscosities allowed in (12.2) are included.

Proof. If all b_i vanished at a point, every iterated bracket would vanish there as well. Indeed $[f\partial_x, g\partial_x] = (fg' - gf')\partial_x$, and induction applies whenever both input coefficients vanish. The bracket hypothesis therefore implies $S = \sum_i b_i^2 > 0$ everywhere. Let $c_K = \min_K S > 0$. Applying (12.6) to $f(x) = x$ gives, for every Borel $E \subset \mathbb{R}$,

$$c_K \rho(K \cap E) \leq C_K |E|.$$

Thus $\rho|_K$ is dominated by a fixed multiple of Lebesgue measure. A countable exhaustion of Ω gives $d\rho = p dx$ globally, and the displayed domination gives the local essential bound. Finally $\int_K p \log(1+p) \leq \log(1 + \|p\|_{L^\infty(K)}) \int_K p$. $\square$

12.4 Grushin fields and finite determinant type

Theorem 12.7 (Every Grushin order, with arbitrary cross coefficients). *Let $\Omega \subset \mathbb{R}^2$ be open and fix an integer $k \geq 1$. Set*

$$X_1 = \partial_x, \quad X_2 = x^k \partial_y.$$

Assume (12.2) with $dp = p dx$, an arbitrary symmetric Borel 2×2 matrix A , and all the mass, ellipticity, sink, and viscosity hypotheses stated there. For every compact $K \Subset \Omega$,

$$\int_K p^2 |x|^{2k} dx dy \leq C_K, \quad (12.16)$$

$$\int_{K \cap \{p > R\}} p dx dy \leq C_K R^{-1/(2k+1)} \quad (R \geq 1), \quad (12.17)$$

$$\int_K p^{1+\theta} dx dy \leq C_{K,\theta} \quad \left(0 < \theta < \frac{1}{2k+1}\right), \quad (12.18)$$

$$\int_K p \log(1+p) dx dy \leq C_K. \quad (12.19)$$

No diagonal assumption on A , and no smallness of Λ/λ , is made. In particular all these constants are uniform in $\varepsilon \in [0, 1]$. The tail estimate does not assert strong integrability at the excluded endpoint in (12.18).

Proof. These fields are bracket generating: k commutators with ∂_x turn $x^k \partial_y$ into $k! \partial_y$. Their original covariance and determinant are $S = \text{diag}(1, x^{2k})$ and $d = x^{2k}$. Since $X_1 x = 1$, Lemma 12.1 gives

$$\int_{K \cap \{|x| < s\}} p dx dy \leq C_K s. \quad (12.20)$$

Theorem 12.4 gives (12.16). On $\{p > R, |x| \geq s\}$, $p \leq R^{-1} s^{-2k} p^2 |x|^{2k}$, and hence

$$\int_{K \cap \{p > R\}} p \leq C_K s + C_K R^{-1} s^{-2k}.$$

Choose $s = R^{-1/(2k+1)}$ to prove (12.17). The pointwise inequality

$$p^\theta \leq 1 + \theta \int_1^\infty R^{\theta-1} 1_{\{p > R\}} dR$$

and Tonelli's theorem then yield (12.18) precisely in the displayed strict range. Any one fixed exponent in that range implies (12.19), using $\log(1+s) \leq s^\theta/\theta$ for $s \geq 0$, $0 < \theta \leq 1$. One can also see the logarithmic bound directly from (12.20):

$$\int_K p \log_+(1/|x|) = \int_0^\infty \int_{K \cap \{|x| < e^{-t}\}} p dt \leq C_K.$$

The determinant penalty in (12.15) is exactly $2k$ times this integral. This proof has used no derivative of A . $\square$

Proposition 12.8 (A transverse determinant lower bound). *Let $n \geq 2$ and assume the density setting of Theorem 12.4. Suppose that a smooth function f defined near K and fixed constants $c_0, c_1 > 0$, $\gamma > 0$ satisfy*

$$\sum_i |X_i f|^2 \geq c_0, \quad d \geq c_1 |f|^\gamma \quad \text{on } K.$$

Then

$$\int_{K \cap \{p > R\}} p dx \leq C_K R^{-1/(n-1+\gamma)} \quad (R \geq 1),$$

and $\int_K p^{1+\theta} dx \leq C_{K,\theta}$ for every $0 < \theta < 1/(n-1+\gamma)$. In particular entropy is bounded. The constants may also depend on f, c_0, c_1, γ .

Proof. The scalar occupation estimate gives $\int_{K \cap \{|f| < s\}} p \leq Cs$. On the complementary portion of $\{p > R\}$, write

$$p \leq R^{-1/(n-1)} c_1^{-1/(n-1)} s^{-\gamma/(n-1)} p^{n/(n-1)} d^{1/(n-1)}.$$

Combining with (12.14) and choosing $s = R^{-1/(n-1+\gamma)}$ gives the asserted tail bound. The same layer-cake calculation used in Theorem 12.7 proves the strong powers and entropy. $\square$

For a word $w = (i_1, \dots, i_\ell)$ set $X_w = X_{i_1} \cdots X_{i_\ell}$; the empty word is the identity. The next condition concerns horizontal derivatives of the original determinant d .

Theorem 12.9 (Finite horizontal type of the determinant). *Assume (12.2) with $d\rho = p dx$. Let $K \Subset \Omega$ and suppose that, for a fixed integer $r \geq 1$ and $c > 0$,*

$$\max_{0 \leq |w| \leq r} |X_w d(x)| \geq c \quad \text{for every } x \in K. \quad (12.21)$$

Then, with $\alpha = (2^r - 1)^{-1}$,

$$\int_{K \cap \{d < \delta\}} p dx \leq C\delta^\alpha \quad (0 < \delta \leq 1). \quad (12.22)$$

Consequently $d > 0$ $p dx$ -almost everywhere on K and

$$\int_K p \log_+(1/d) dx + \int_K p \log(1+p) dx \leq C.$$

If $n \geq 2$, there are in addition the bounds

$$\begin{aligned} \int_{K \cap \{p > R\}} p dx &\leq CR^{-1/(n+2^r-2)} \quad (R \geq 1), \\ \int_K p^{1+\theta} dx &\leq C_\theta \quad \left(0 < \theta < \frac{1}{n+2^r-2}\right). \end{aligned} \quad (12.23)$$

Constants may depend additionally on r, c ; all previous uniformity in A, p, μ, ε is retained.

Proof. It suffices initially to consider $0 < \delta < \min(1, c/2)$. Define

$$q_\ell = \frac{2^{r-\ell} - 1}{2^r - 1}, \quad h_\ell = \frac{c}{2} \delta^{q_\ell}, \quad 1 \leq \ell \leq r, \quad h_0 = \delta.$$

For $x \in K \cap \{d < \delta\}$, condition (12.21) guarantees a positive-length word of absolute value at least c . Because $h_\ell \leq c/2$, there is a first length ℓ at which some word exceeds its threshold. Let E_ℓ be the set where this first length is ℓ . If the exceeding word is $X_i X_v d$ with $|v| = \ell - 1$, then $|X_v d| \leq h_{\ell-1}$ on E_ℓ ; for $\ell = 1$ this means $d < \delta = h_0$. Applying (12.6) to the fixed smooth function $f = X_v d$ and the scalar interval $[-h_{\ell-1}, h_{\ell-1}]$ yields

$$\int_{E_\ell \cap \{|X_i X_v d| > h_\ell\}} p dx \leq C h_{\ell-1} h_\ell^{-2}.$$

Only finitely many words of length at most r occur; all their scalar-occupation constants are controlled by the stated data. For $\ell = 1$ the exponent of δ is $1 - 2q_1 = \alpha$, and for $\ell \geq 2$ it is $q_{\ell-1} - 2q_\ell = \alpha$. Summing over the finitely many first crossings proves (12.22) for small δ . Enlarging the constant and using $\int p \leq 1$ covers the remaining $\delta \leq 1$.

Letting $\delta \downarrow 0$ shows $\int_{K \cap \{d=0\}} p = 0$. The layer-cake identity and (12.22) give

$$\int_K p \log_+(1/d) dx = \int_0^\infty \int_{K \cap \{d < e^{-t}\}} p dx dt \leq C \int_0^\infty e^{-\alpha t} dt < \infty.$$

The entropy assertion now follows from (12.15).

For the additional powers suppose $n \geq 2$. Split the mass of $K \cap \{p > R\}$ at the determinant level δ . On its part where $d \geq \delta$, (12.14) gives

$$\int_{K \cap \{p > R\}} p \leq C\delta^\alpha + CR^{-1/(n-1)}\delta^{-1/(n-1)}.$$

Choose $\delta = R^{-1/(1+\alpha(n-1))}$. Both terms are bounded by $CR^{-\alpha/(1+\alpha(n-1))}$; the exponent equals $1/(n+2^r-2)$. Layer cake gives exactly the strict strong-power range in (12.23). $\square$

Remark 12.10 (Scope of the determinant method). Condition (12.21) is an additional geometric assumption, not a consequence of bracket generation. For the horizontal fields of a nonabelian Carnot group whose first layer has dimension less than the manifold, S is everywhere rank deficient and $d \equiv 0$. In particular this happens on every standard Heisenberg group and on every genuinely subelliptic two-step Carnot group. Their determinants do not satisfy (12.21); the weighted theorem supplies no entropy conclusion there by itself. Those groups require the separate hypoelliptic arguments developed elsewhere in this manuscript.

13 Heat smoothing and central coefficient energy on Heisenberg groups

This section concerns the fixed sub-Laplacian on $\mathbb{H}^d$ in the normalization of Corollary 2.2. No stationary equation is assumed in the smoothing theorems. In particular these results apply to an arbitrary nonnegative integrable input density; by themselves they do not supply an entropy bound for that input.

Write $w = (x, y)$, $Z = \partial_z$, $L = \sum_{j=1}^d (X_j^2 + Y_j^2)$, and $P_s f = f * h_s$. Convolution is $(f * g)(x) = \int f(y)g(y^{-1}x) dy$. Thus the entropy dissipation of P_s uses the left horizontal fields.

13.1 A central logarithmic derivative of the heat kernel

Lemma 13.1. *For every $s > 0$ and every $x \in \mathbb{H}^d$,*

$$|Zh_s(x)| \leq \pi s^{-1} h_s(x).$$

For each integer $k \geq 1$ there is $C_{d,k}$ such that $|Z^k h_s| \leq C_{d,k} s^{-k} h_s$. Moreover

$$\int h_s |\nabla_H \log h_s|^2 = \frac{d+1}{s}.$$

Proof. The partial Fourier transform in z , with transform $\widehat{f}(\xi) = \int e^{-i\xi z} f(z) dz$, is

$$\widehat{h}_s(w, \xi) = (4\pi s)^{-d} \left(\frac{s\xi}{\sinh(s\xi)} \right)^d \exp \left[-\frac{|w|^2}{4s} s\xi \coth(s\xi) \right]. \quad (13.1)$$

Here the quotients are continued at zero. To verify the formula, the angular first-order part of the magnetic operator annihilates a radial function of w . The transformed equation is consequently $\partial_s H = \Delta_w H - \xi^2 |w|^2 H/4$. Substitute the Gaussian ansatz $a(s)e^{-b(s)|w|^2}$: the equations $b' = \xi^2/4 - 4b^2$ and $a'/a = -4db$, with the Euclidean short-time normalization, give (13.1). Its horizontal marginal is $g_s(w) = (4\pi s)^{-d} e^{-|w|^2/(4s)}$.

Let $a = |w|^2/(4s)$. The conditional characteristic function of z/s given w is

$$\phi_a(u) = \left(\frac{u}{\sinh u} \right)^d e^{-a(u \coth u - 1)}.$$

The density $\ell(v) = \frac{\pi}{4} \operatorname{sech}^2(\pi v/2)$ has characteristic function $u/\sinh u$, variance $1/3$, and $|\ell'| \leq \pi\ell$. These one-dimensional identities follow by integrating the hyperbolic secant squared, or by a rectangular contour with imaginary height $2i$. The partial fraction expansion

$$u \coth u - 1 = 2 \sum_{n=1}^{\infty} \frac{u^2}{u^2 + \pi^2 n^2}$$

follows by differentiating the product for $\sinh u/u$. For $b > 0$, $b^2/(b^2 + u^2)$ is the characteristic function of the centered Laplace density $(b/2)e^{-b|v|}$. Thus $\exp[-2au^2/(u^2 + b^2)]$ is a centered compound Poisson characteristic function, of variance $4a/b^2$. Taking independent variables with $b = \pi n$, their partial sums converge in L^2 because $\sum_n 4a/(\pi n)^2 = 2a/3$. It follows that ϕ_a is the characteristic function of

$$q_a = \ell * \mu_a, \quad \int v^2 d\mu_a(v) = \frac{d-1+2a}{3},$$

where μ_a is a probability measure. In particular $h_s(w, z) = g_s(w)s^{-1}q_a(z/s)$ and $|q'_a| \leq \pi q_a$. This proves the first assertion without estimating an oscillatory integral by a positive kernel. Each ratio $\ell^{(k)}/\ell$ is a bounded polynomial in $\tanh(\pi v/2)$, proving the higher derivative assertion in the same way.

We also record the integrability required for the Fisher identity. Since $\ell(v) \geq (\pi/4)e^{-\pi|v|}$, Jensen's inequality gives

$$\frac{\pi}{4} \exp \left[-\pi|v| - \pi\sqrt{(d-1+2a)/3} \right] \leq q_a(v) \leq \frac{\pi}{4}.$$

Consequently $|\log h_s(w, z)| \leq C(1 + |w|^2 + |z|)$ on every compact positive-time interval. Formula (13.1), differentiated and integrated by parts in ξ , shows that the heat kernel and its fixed space and time derivatives are Schwartz functions on such intervals. Indeed $u \coth u \geq 1$, the quotients are smooth at zero, and the hyperbolic factors decay exponentially at large $|u|$; every fixed number of derivatives is bounded by an integrable exponential times a polynomial. These bounds also justify weighted differentiation of the Fourier inversion formula. It follows that the entropy of h_s is finite and differentiable. Testing Lh_s with a compactly supported cutoff times $\log h_s$, and then letting the cutoff tend to one, gives

$$\frac{d}{ds} \int h_s \log h_s = - \int \frac{|\nabla_H h_s|^2}{h_s}.$$

There is no prior Fisher integrability assumption in this step: the left side and the boundary terms are integrable by the preceding polynomial logarithmic bound and Schwartz estimates, and the nonnegative interior term then has a finite limit. Finally $h_s(x) = s^{-(d+1)}h_1(\delta_{s^{-1/2}x})$, so that $\int h_s \log h_s = \int h_1 \log h_1 - (d+1) \log s$. Differentiating proves the identity. $\square$

Theorem 13.2 (A central coefficient square function). *Let $p \geq 0$ belong to $L^1(\mathbb{H}^d)$, with mass M , and let A be any bounded measurable matrix field on $\{p > 0\}$. Put*

$$p_s = P_s p, \quad Q_s = P_s(pA), \quad a_s = Q_s/p_s.$$

If $M = 0$ interpret all the integrals below as zero. For every $S > 0$,

$$\int_0^S s \int p_s |Za_s|_F^2 dx ds \leq 4\pi^2 \log 2 \|A\|_{L^\infty(p, dx); F}^2 M.$$

In particular for a symmetric $2d \times 2d$ field with $\lambda I \leq A \leq \Lambda I$ p -dx-almost everywhere, the right side is at most $4\pi^2 \log 2 (2d\Lambda^2)M$. No initial entropy or derivative of A is required.

Proof. Assume $M > 0$; the positive heat kernel gives $p_s > 0$. The perspective $V(s) = \int p_s |a_s|_F^2$ is nonincreasing by the semigroup property and conditional Jensen, and $0 \leq V(s) \leq \|A\|_\infty^2 M$. Set

$$d_s(x) = \int h_{s/2}(y^{-1}x) p_{s/2}(y) |a_{s/2}(y) - a_s(x)|_F^2 dy.$$

Expanding the square and using preservation of mass gives $\int d_s = V(s/2) - V(s)$. Differentiating the quotient in the central direction and subtracting its conditional mean gives

$$Za_s(x) = \frac{1}{p_s(x)} \int (Zh_{s/2})(y^{-1}x) p_{s/2}(y) [a_{s/2}(y) - a_s(x)] dy.$$

Lemma 13.1 and Cauchy–Schwarz imply $p_s |Za_s|_F^2 \leq 4\pi^2 s^{-2} d_s$. For $0 < \delta < S$,

$$\int_\delta^S \frac{V(s/2) - V(s)}{s} ds = \int_{\delta/2}^\delta \frac{V(u)}{u} du - \int_{S/2}^S \frac{V(u)}{u} du \leq (\log 2) \sup V.$$

Monotone convergence finishes the proof. $\square$

13.2 An enhanced Fisher estimate on $\mathbb{H}^1$

Theorem 13.3. *Let $p \geq 0$ be integrable on $\mathbb{H}^1$, of mass $M > 0$. For $s > 0$ put $p_s = P_s p$, $u_s = \log p_s$, and*

$$\begin{aligned} I(s) &= \int p_s [(Xu_s)^2 + (Yu_s)^2], & J(s) &= \int p_s (Zu_s)^2, \\ H(s) &= \int p_s \sum_{U,V \in \{X,Y\}} (UVu_s)^2, & R(s) &= \int p_s [(XZu_s)^2 + (YZu_s)^2]. \end{aligned}$$

Let $F(s) = sI(s) + s^2J(s)/4$. Then

$$I(s) \leq \frac{2M}{s}, \quad J(s) \leq \frac{\pi^2 M}{s^2}, \quad F(s) \leq (2 + \pi^2/4)M.$$

For all $0 < \delta < S < \infty$,

$$\int_\delta^S \left(sH(s) + \frac{s^2}{4} R(s) \right) ds \leq 17 \int_\delta^S I(s) ds + F(\delta) - F(S). \quad (13.2)$$

The right side is finite. Where the differentiated identities below are justified, they read

$$J' = -2R, \quad I' = -2H - 4 \int p_s (Yu_s XZu_s - Xu_s YZu_s), \quad F' \leq 17I - sH - s^2R/4.$$

Proof. Differentiating the heat convolution and applying conditional Jensen gives $I(s) \leq MI(h_s) = 2M/s$. The central kernel bound gives $J(s) \leq \pi^2 M/s^2$. For smooth positive data for which integrations by parts are justified, $u'_s = Lu_s + |\nabla_H u_s|^2$. The identities $[L, X] = -2YZ$, $[L, Y] = 2XZ$, and $[L, Z] = 0$ give

$$J' = -2R, \quad I' = -2H - 4K, \quad K = \int p_s (Yu_s XZu_s - Xu_s YZu_s).$$

For clarity, the general differentiated horizontal Fisher identity is

$$I' = -2 \int p_s \left\{ \sum_{U,V} (UVu_s)^2 + \sum_V (Vu_s)[L, V]u_s \right\};$$

substitution yields the displayed formula. Also $(XYu_s - YXu_s)^2 = (Zu_s)^2$ implies $J \leq 2H$. The cross term satisfies

$$4s|K| \leq 4s\sqrt{IR} \leq 16I + \frac{1}{4}s^2R.$$

Thus

$$F' = I - 2sH - 4sK + \frac{1}{2}sJ - \frac{1}{2}s^2R \leq 17I - sH - \frac{1}{4}s^2R.$$

Integration proves (13.2) in this regularity class.

Here is a justification for an arbitrary L^1 input, which may have infinite entropy. On a fixed interval $[\delta, S]$, replace p_s by $r_s = p_s + \epsilon$, $\epsilon > 0$, in every differentiated expression. It still solves $\partial_s r_s = Lr_s$. All fixed left-invariant spatial derivatives of positive order, and the required heat-time derivatives, lie in $L^2 \cap L^\infty$, by convolution with the corresponding Schwartz heat-kernel derivatives. Consequently the derivative expressions for r_s are integrable, and compactly supported cutoffs justify the identities above. Only derivative expressions occur: no integral of the constant ϵ is used. The Fisher integrals with r_s increase to $I(s), J(s)$ as $\epsilon \downarrow 0$, and are bounded by the already proved kernel estimates. Their endpoint terms therefore converge. Fatou's lemma for the nonnegative Hessian terms and monotone convergence for the Fisher terms prove (13.2). This argument asserts the integrated inequality for arbitrary L^1 input without presupposing pointwise differentiability of its energy. $\square$

Proposition 13.4 (Quartic horizontal score). *Let $p > 0$ be smooth and integrable on $\mathbb{H}^1$. If*

$$I = \int p |\nabla_H \log p|^2 < \infty, \quad H = \int p \sum_{U,V \in \{X,Y\}} (UV \log p)^2 < \infty,$$

then

$$\int p |\nabla_H \log p|^4 \leq (\sqrt{2} + 2)^2 H.$$

Consequently the quantities in Theorems 13.2 and 13.3 satisfy, for any $\eta > 0$ and $0 < \delta < S$,

$$\int_\delta^S s \int p_s \left| \sum_{i,j} (Za_{s,ij})(U_i u_s)(U_j u_s) \right| \leq \eta \int_\delta^S s H(s) ds + \frac{(\sqrt{2} + 2)^2}{4\eta} \int_\delta^S s \int p_s |Za_s|_F^2,$$

where $(U_1, U_2) = (X, Y)$.

Proof. Put $q = \nabla_H \log p$, $t = |q|^2$, and $\psi_N(t) = Nt/(N + t)$. Then $0 \leq t\psi'_N \leq \psi_N$ and $\psi_N^2 \leq t\psi_N$. The finite quantity $E_N = \int p t \psi_N(t)$ is at most NI . Integrate the horizontal divergence of $p\psi_N(t)q$ using cutoffs. The boundary term vanishes since $\psi_N \leq N$ and $\int p|q| \leq \sqrt{MI}$. With $H_u = (U_i U_j \log p)_{ij}$ this yields

$$E_N = - \int p \{ \psi_N(t) \operatorname{tr} H_u + 2\psi'_N(t) q^t H_u q \} \leq (\sqrt{2} + 2) E_N^{1/2} H^{1/2}.$$

The estimates also justify the integrability of the interior terms. Divide when $E_N > 0$ and let $N \rightarrow \infty$. This proves the quartic bound without assuming the desired fourth moment in advance. Finally apply Cauchy–Schwarz to the matrix contraction and then Young's inequality, using the quartic bound for almost every s . $\square$

13.3 Exact identities for adaptive kernels

The following identities are algebraic statements for smooth data with integrable displayed terms. They do not assert bounds for the additional parameter derivatives that appear in them.

Proposition 13.5 (Frozen output parameter and heat chain rule). *Let h_s^B be the heat kernel of the constant positive definite horizontal matrix B on a Carnot group. Let $B_s(x)$ be a smooth positive definite matrix field and set*

$$\rho_s(x) = \int h_s^{B_s(x)}(y^{-1}x)p(y) dy.$$

For a smooth compactly supported weight ℓ_s , define $\varphi_s(y) = \int h_s^{B_s(x)}(y^{-1}x)\ell_s(x) dx$. When the integrals and integrations by parts are justified,

$$\begin{aligned} \int p(y)A(y) : D_{H,y}^2 \varphi_s(y) dy &= \int \ell_s(x) \int p(y) \{ \partial_s^{\text{frozen}} h_s^{B_s(x)}(y^{-1}x) \\ &\quad + [A(y) - B_s(x)] : D_{H,R}^2 h_s^{B_s(x)}(y^{-1}x) \} dy dx. \end{aligned}$$

Here D_H^2 is the symmetric horizontal Hessian and R denotes right invariant derivatives. Moreover

$$\partial_s \rho_s = \int p(y) \partial_s^{\text{frozen}} h_s^{B_s(x)}(y^{-1}x) dy + (\partial_s B_s(x)) : \partial_B \rho_{s,B}(x) \big|_{B=B_s(x)}.$$

Independently, if p_s, Q_s evolve by the same fixed symmetric sum-of-squares heat operator L and $a_s = Q_s/p_s$, then every smooth $F(s, x, B)$ with justified integrations satisfies

$$\frac{d}{ds} \int p_s F(s, x, a_s) = \int p_s \left[(\partial_s + L_x) F(s, x, a_s) - \sum_i D_B^2 F(s, x, a_s) [X_i a_s, X_i a_s] \right]. \quad (13.3)$$

In $L_x F$ the parameter B is held fixed.

Proof. A derivative in y of $h(y^{-1}x)$ is the negative of the corresponding right derivative of h . Two derivatives give the right symmetric Hessian. The left and right constant- B sub-Laplacians have the same action on their heat kernel, both equal to its heat-time derivative. Adding and subtracting $B_s(x)$ proves the first formula. The second is the ordinary chain rule in s . For the last identity the quotient equation is $a'_s = La_s + 2 \sum_i (X_i \log p_s) X_i a_s$. Differentiate the integral, use $p'_s = Lp_s$, and move L from p_s to the composite $F(s, x, a_s)$. The resulting two copies of $F_B La_s$, together with $2F_B \nabla_H \log p_s \cdot \nabla_H a_s$, integrate to minus twice the explicit spatial/parameter mixed term and minus twice the parameter Hessian term. The composite chain rule had supplied one Hessian term and twice the mixed term. The mixed terms cancel, and the remaining Hessian has the negative sign in (13.3). $\square$

Proposition 13.6 (A central-parameter divergence on $\mathbb{H}^1$). *Let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. For a symmetric matrix C set $D_C = (CJw) \cdot \partial_w$. For the fixed heat kernel h_s^B , or the fixed Poisson kernel k_r^B , write k^B for either family. Then*

$$L_C^L - L_C^R = 2ZD_C, \quad D_C k^B = -D_B k^B [CJB - BJC].$$

*If Q is a smooth symmetric horizontal tensor and $\mathcal{T}Q = \mu$, define $Q_r^B = Q * k_r^B$ and $\mathcal{R}_B C = CJB - BJC$. With divergence in the three independent coordinates of a symmetric 2×2 matrix,*

$$\mathcal{T}Q_r^B - \mu * k_r^B = -2Z \operatorname{div}_B(\mathcal{R}_B Q_r^B).$$

The same assertion holds for heat kernels with r replaced by s . The equalities extend distributionally whenever the convolutions exist.

Proof. Expanding $X = \partial_x - yZ/2$ and $Y = \partial_y + xZ/2$, and the right fields with the opposite central signs, gives the first formula. Since $\operatorname{tr}(CJ) = 0$, the map $(w, z) \mapsto (e^{tCJ}w, z)$ is a volume-preserving Heisenberg automorphism. Covariance transport of the fixed kernel under this automorphism changes B to $e^{tCJ}B e^{-tJC}$. Differentiating its kernel identity at $t = 0$

gives the second formula. If $B = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ and $C = \begin{pmatrix} u & v \\ v & w \end{pmatrix}$, the three coordinates of $\mathcal{R}_B C$ are $(2av - 2bu, aw - cu, 2bw - 2cv)$. Their divergence with respect to (a, b, c) is zero. Apply the first two identities componentwise to the convolution defining Q_r^B , and use this zero divergence to put the parameter derivative in divergence form. This proves the last formula. Distributional approximation gives the stated extension. $\square$

Remark 13.7. The adaptive identities retain both the change of the kernel parameter and any failure of an output-dependent kernel to preserve mass. No entropy conclusion is drawn by discarding those terms. In particular, ordinary componentwise heat convolution does not commute with every horizontal derivative, even on $\mathbb{H}^1$.

14 Further results on Carnot groups of arbitrary step

This section separates two kinds of conclusions. The commuting-quadratic result below is a stationary entropy theorem for a restricted coefficient class in every Carnot step. The subsequent Poisson estimates are estimates for fixed, constant-coefficient reference operators; they do not assert that substituting a measurable matrix parameter preserves an adjoint stationary equation. Every assertion in this section is intrinsic: the coordinate Euclidean viscosity is not included.

14.1 Conventions and fixed analytic input

A Carnot group is a connected simply connected nilpotent Lie group whose Lie algebra has a stratification $\mathfrak{g} = V_1 \oplus \cdots \oplus V_s$, with $[V_1, V_j] = V_{j+1}$ and $V_{s+1} = 0$. Its graded dilations are denoted by δ_r , and its homogeneous dimension by $D = \sum_{j=1}^s j \dim V_j$. We use Haar measure dx , which is Lebesgue measure in exponential coordinates. Fix left-invariant fields $X_1, \dots, X_m$ spanning V_1 ; the list may be redundant. These fields are divergence free. For a constant symmetric matrix B , set

$$L_B = \sum_{i,j} B_{ij} X_i X_j, \quad L = L_I = \sum_i X_i^2, \quad \Gamma_B(f, g) = \sum_{i,j} B_{ij} (X_i f)(X_j g).$$

Write $\Gamma_B(f) = \Gamma_B(f, f)$, $\nabla_H f = (X_i f)_i$, and, for a symmetric tensor distribution Q , write $\mathcal{T}Q = \sum_{i,j} X_i X_j Q_{ij}$. Symmetry and divergence freeness make this the adjoint of $B \mapsto L_B$. Our convolution convention is

$$(f * g)(x) = \int_G f(y) g(y^{-1}x) dy.$$

Thus $H_t f = f * h_t = e^{tL} f$ for the fixed positive probability heat kernel h_t . We use the following standard fixed-operator fact: for a homogeneous control distance d , h_t has two-sided Gaussian bounds $ct^{-D/2} e^{-Cd(x)^2/t} \leq h_t(x) \leq Ct^{-D/2} e^{-cd(x)^2/t}$; a fixed word of homogeneous derivatives of total degree q has upper bound $C_q t^{-(D+q)/2} e^{-c_q d(x)^2/t}$. The corresponding source and target derivative bounds hold for $h_t(y^{-1}x)$. Positivity, inverse symmetry, unit mass, and homogeneous scaling are included in this fact. These are facts about one smooth reference operator, as in [2, Theorems 2.4 and 3.4]. Redundancy causes no problem: writing the fields in a horizontal basis makes L a fixed positive-definite quadratic form, which a fixed linear change of basis reduces to a sum of squares. In horizontal dimension one the ordinary Gaussian proves the same statements. All remaining estimates below are proved from this fixed input.

14.2 Exact convolution regularization and its limitation

Proposition 14.1 (Cone-preserving regularization and scaling in every step). *Let σ be a nonnegative finite measure and Q a symmetric matrix measure on an open set $\Omega \subset G$, with*

$\lambda\sigma I \leq Q \leq \Lambda\sigma I$ and $\mathcal{T}Q = \mu \geq 0$ in Ω . Let $\eta \in C_c^\infty(G)$ be nonnegative with integral one and $\eta_r(x) = r^{-D}\eta(\delta_{1/r}x)$. On every fixed $U \Subset \Omega$, for all sufficiently small r ,

$$\mathcal{T}(\eta_r * Q) = \eta_r * \mu, \quad \lambda(\eta_r * \sigma)I \leq \eta_r * Q \leq \Lambda(\eta_r * \sigma)I. \quad (14.1)$$

Here the equation is local, so only values inside Ω are sampled. For $g \in G$, $r, M > 0$, put $F(x) = \delta_{1/r}(g^{-1}x)$. Then

$$\sigma' = M^{-1}F_*\sigma, \quad Q' = M^{-1}F_*Q, \quad \mu' = r^2M^{-1}F_*\mu$$

satisfy the same cone inequalities and $\mathcal{T}Q' = \mu'$ on $F(\Omega)$.

Proof. A left field differentiates the second factor in convolution, hence $X_i(\eta_r * f) = \eta_r * (X_i f)$. Repeating this identity and using duality proves the equation for distributions. Integrating the matrix cone against the positive kernel proves the inequalities. In the local assertion, if x ranges over a compact set, $y^{-1}x \in \Omega$ for all $y \in \text{supp } \eta_r$ and small r . Translation invariance and homogeneity give $X_i(\varphi \circ F) = r^{-1}(X_i \varphi) \circ F$; applying this twice in the weak equation proves the scaling assertion. The convolution is smooth, and on a bounded interior domain one can replace its density and stress by $p_r + \delta$ and $Q_r + c\delta I$, with $c \in [\lambda, \Lambda]$, to obtain a smooth everywhere positive ratio without changing the source. Dividing all three quantities by $1 + \delta|U|$ preserves a unit mass bound if desired. Consequently a uniform local entropy theorem for these smooth triples would pass to density data by Fatou, and to measure data by uniform integrability. This last sentence is a conditional reduction, not an entropy bound. $\square$

Proposition 14.2 (The left and right horizontal forms are different). *On a nonabelian Carnot group there is no universal pointwise inequality $\sum_i |X_i^R f|^2 \leq C \sum_i |X_i^L f|^2$ for smooth f . Moreover the extra terms relating the two frames have horizontal scaling degree one, so dilations do not make this discrepancy lower order.*

Proof. Left trivialization gives $X_i^R(x) = (L_x)_* \text{Ad}_{x^{-1}} e_i$. In a homogeneous left frame $Y_{k,\alpha}$ of degree k this reads

$$X_i^R = X_i^L + \sum_{k \geq 2, \alpha} c_{i,k,\alpha}(x) Y_{k,\alpha}, \quad c_{i,k,\alpha}(\delta_r x) = r^{k-1} c_{i,k,\alpha}(x).$$

At scale r the product of such a coefficient with a degree- k derivative has size $r^{k-1}r^{-k} = r^{-1}$. Since the group is nonabelian, choose $v, w \in V_1$ with $[v, w] \neq 0$. At $x = \exp(tv)$, $t \neq 0$ sufficiently small, $\text{Ad}_{x^{-1}} w$ has a nonzero component outside V_1 . A covector annihilating V_1 but not this vector is the differential of a smooth function at x and contradicts the proposed pointwise inequality. $\square$

For clarity, $f * h_t$ dissipates left-horizontal entropy, whereas $h_t * f$ dissipates right-horizontal entropy and commutes with all left fields. If r_t and a tensor Q_t evolve by the latter heat flow, the coefficient perspective identity involves $\int r_t \sum_i |X_i^R(Q_t/r_t)|^2$. A logarithmic test of the convolved stationary equation involves X_i^L . Proposition 14.2 explains why exact scalar convolution alone is not a proof of entropy.

14.3 A stationary theorem for commuting horizontal quadratics

Theorem 14.3 (Commuting coefficients, arbitrary Carnot step). *Let G and X_i be as above, and let $\mathcal{V}$ be a fixed linear subspace of real symmetric $m \times m$ matrices such that*

$$I \in \mathcal{V}, \quad [L_B, L] = 0 \quad \text{for every } B \in \mathcal{V}. \quad (14.2)$$

Let $\Omega \subset G$ be open, $p \in L^1(\Omega)$ be nonnegative with $\int_{\Omega} p \leq 1$, and let A be Borel with $A(x) \in \mathcal{V}$ and $\lambda I \leq A(x) \leq \Lambda I$ for $p dx$ -almost every x . Suppose $\mathcal{T}(Ap) = \mu$ in Ω , where μ is a finite nonnegative measure. Then for every $K \Subset \Omega$,

$$\int_K p \log(1+p) dx \leq C(G, X_1, \dots, X_m, K, \Omega, \lambda, \Lambda).$$

No derivative bound on A or initial entropy is assumed. The statement has $\varepsilon = 0$. A fixed positive-definite reference matrix can replace I , after a fixed change of the horizontal generators.

Proof. Formal commutation implies $L_B H_t = H_t L_B$. Indeed, for smooth compact data both sides solve the same L^2 heat equation with the same initial value; the fixed Gaussian derivative bounds give the required integrability, and the energy identity gives uniqueness. Duality then extends the identity to compact distributions. Expanding a $\mathcal{V}$ -valued tensor in a constant basis therefore gives

$$\mathcal{T}(H_t Q) = H_t \mathcal{T}Q. \quad (14.3)$$

Individual X_i need not commute with H_t .

Choose $0 \leq \chi, \zeta \leq 1$ in $C_c^\infty(\Omega)$ such that $\zeta = 1$ near K and $\chi = 1$ on a neighborhood of $\text{supp } \zeta$. Set $q_0 = \chi p$, $Q_0 = \chi A p$ and extend them by zero. The product rule gives

$$\mathcal{T}Q_0 = \chi \mu + \sum_i X_i F_i + G_0, \quad F_i = 2 \sum_j (X_j \chi) A_{ij} p, \quad G_0 = - \sum_{i,j} (X_i X_j \chi) A_{ij} p. \quad (14.4)$$

The L^1 norms of F, G_0 are bounded and their supports are separated from $\text{supp } \zeta$. If needed, a fixed nonnegative cutoff θ equal to one near $\text{supp } \chi$ gives $\mu(\text{supp } \chi) \leq \langle Ap, D_H^2 \theta \rangle \leq C$. Here and below D_H^2 denotes the symmetrized horizontal Hessian.

For $0 < t \leq 1$ define

$$q_t = H_t q_0, \quad R_t = I + H_t Q_0, \quad r_t = 1 + q_t, \quad a_t = R_t / r_t, \quad f_t = \log r_t.$$

Then $\lambda_0 I \leq a_t \leq \Lambda_0 I$, with $\lambda_0 = \min(1, \lambda)$ and $\Lambda_0 = \max(1, \Lambda)$. By (14.3),

$$\mathcal{T}R_t = H_t(\chi \mu) + g_t, \quad g_t = H_t \left(\sum_i X_i F_i + G_0 \right).$$

The source-derivative Gaussian bounds and separation of supports give

$$\sup_{0 < t \leq 1} \|g_t\|_{L^\infty(\text{supp } \zeta)} \leq C.$$

Also $\|q_t\|_\infty \leq C t^{-D/2}$ and $\int q_t \leq 1$.

Put $S_t = R_t - r_t I = H_t(Q_0 - q_0 I)$. The scalar denominator and every component of S_t solve the same heat equation. The quotient chain rule and integration by parts give

$$\frac{d}{dt} \int_G \frac{|S_t|_F^2}{r_t} = -2 \int_G r_t \sum_i |X_i a_t|_F^2. \quad (14.5)$$

Positivity of the heat kernel and the perspective form of Jensen's inequality bound the integral on the left by $\int |Q_0 - q_0 I|_F^2 / (1 + q_0) \leq C$. Integrating from a positive lower time and decreasing it to zero yields

$$\int_0^1 \int_G r_t \sum_i |X_i a_t|_F^2 \leq C. \quad (14.6)$$

The perspective is finite because its numerator subtracts the constant baseline. All positive-time integrations by parts follow by spatial exhaustion from Gaussian decay of the smoothed compact data.

Test the equation for R_t with $\zeta^2 f_t \geq 0$. Its right side is at least $-C \int_{\text{supp } \zeta} f_t \geq -C$. One integration by parts on its left side gives the negative principal term $-\int \zeta^2 r_t \langle a_t \nabla_H f_t, \nabla_H f_t \rangle$, and the three error terms

$$\begin{aligned} & - \int \zeta^2 r_t \sum_{i,j} (X_i a_{t,ij}) X_j f_t, \\ & - 2 \int \zeta f_t r_t \langle a_t \nabla_H f_t, \nabla_H \zeta \rangle, \\ & - 2 \int \zeta f_t r_t \sum_{i,j} (X_i a_{t,ij}) X_j \zeta. \end{aligned}$$

Young's inequality absorbs fixed fractions of the principal term and therefore gives

$$\int \zeta^2 r_t |\nabla_H f_t|^2 \leq C + C \int_G r_t \sum_i |X_i a_t|_F^2 + C \int_{\text{supp } \zeta} r_t f_t^2.$$

The last integral is at most $C(1 + |\log t|)^2$, by the heat supremum bound and the finite mass on the fixed cutoff support. Thus

$$\int_0^1 \int \zeta^2 r_t |\nabla_H \log r_t|^2 \leq C. \quad (14.7)$$

Finally set $F(r) = r \log r - r + 1$, $r \geq 1$. Its scalar heat identity is

$$\frac{d}{dt} \int \zeta^2 F(r_t) = \int L(\zeta^2) F(r_t) - \int \zeta^2 r_t |\nabla_H \log r_t|^2.$$

Since $F(1+q) \leq q \log(1+q)$, its global integral is bounded by $C(1 + |\log t|)$. Integrating backwards from 1 to δ and using (14.7) proves $\sup_{0 < \delta < 1} \int \zeta^2 F(1+q_\delta) \leq C$. Approximate-identity convergence and Fatou give $\int_K F(1+p) \leq C$. The inequality $p \log(1+p) \leq F(1+p) + p$ finishes the proof. $\square$

Corollary 14.4 (Measure input for commuting quadratics). *Under the group, generators, coefficient subspace and ellipticity hypotheses of Theorem 14.3, let σ be a finite nonnegative measure on Ω with $\sigma(\Omega) \leq 1$ and let A be Borel, with $A \in \mathcal{V}$ and $\lambda I \leq A \leq \Lambda I$ for σ -almost every point. If $\mathcal{T}(A\sigma) = \mu$ is a finite nonnegative measure, then $\sigma = p dx$ and the same local entropy estimate holds. This corollary has $\varepsilon = 0$ and permits any Carnot step.*

Proof. Apply Theorem 8.2 to the positive stress $Q = A\sigma$ to obtain local absolute continuity. An exhaustion of Ω gives a global nonnegative density p with $\int_\Omega p = \sigma(\Omega) \leq 1$. The coefficient restrictions hold $p dx$ -almost everywhere, and the same distributional equation becomes $\mathcal{T}(Ap) = \mu$. Theorem 14.3 applies directly. $\square$

Proposition 14.5 (Explicit non-scalar commuting classes). *In a step-two group with horizontal basis X_i and central basis Z_α , write $[X_i, X_j] = \sum_\alpha c_{ij}^\alpha Z_\alpha$. For a constant symmetric B ,*

$$[L_B, L] = 2 \sum_\alpha Z_\alpha L_{[B, c^\alpha]}.$$

Consequently $[L_B, L] = 0$ if and only if $[B, c^\alpha] = 0$ for every α . On $\mathbb{H}^d$ this is the real representation of the complex Hermitian $d \times d$ matrices, a real vector space of dimension d^2 ; it contains non-scalar positive matrices when $d \geq 2$, and is scalar when $d = 1$. On a direct product of Carnot groups of arbitrary steps, matrices that are scalar on each factor's horizontal block also satisfy Theorem 14.3. Each allowed coefficient may depend measurably on all coordinates.

Proof. Centrality gives $[X_i, X_k^2] = 2 \sum_{\alpha} c_{ik}^{\alpha} Z_{\alpha} X_k$. Expanding $[X_i X_j, X_k^2]$ and using symmetry of B gives the displayed formula, because $\text{sym}(B c^{\alpha}) = [B, c^{\alpha}]/2$. For necessity, at the identity the principal third-order symbol is a nonzero fixed scalar multiple of $\sum_{\alpha} \tau_{\alpha} \xi^T [B, c^{\alpha}] \xi$. Independence of the central frequencies and symmetry of each commutator force every one to vanish. On $\mathbb{H}^d$, the single structure matrix is the standard complex structure J ; commuting real symmetric matrices are exactly $\begin{pmatrix} S & -T \\ T & S \end{pmatrix}$ with $S^T = S$, $T^T = -T$, the real forms of Hermitian matrices $S + iT$. Finally sub-Laplacians on distinct product factors commute, so scalar linear combinations of them commute with their sum. $\square$

14.4 Frozen Poisson kernels and matrix parameters in every step

Let $\mathcal{K}$ be a compact subset of the positive-definite cone in Sym_m . For $B \in \mathcal{K}$ let k_r^B be the probability convolution kernel of $e^{-r\sqrt{-L_B}}$. Parameter multiindices always refer to fixed linear coordinates on Sym_m ; a spatial derivative is taken with B frozen.

Theorem 14.6 (Uniform frozen-parameter kernel bounds). *For every fixed parameter multiindex α , integer $j \geq 0$, and fixed word E of left-invariant homogeneous derivatives of total degree q , uniformly for $B, C \in \mathcal{K}$, $r > 0$, and $x \in G$,*

$$C^{-1} k_r^C(x) \leq k_r^B(x) \leq C k_r^C(x), \quad (14.8)$$

$$|(r\partial_r)^j \partial_B^{\alpha} E k_r^B(x)| \leq C_{\alpha,j,E} r^{-q} k_r^B(x). \quad (14.9)$$

The kernels are strictly positive. Every finite collection of constants is uniform on a slightly enlarged compact positive-definite family.

Proof. For one fixed operator on a group of homogeneous dimension D , the subordination formula is

$$k_r(x) = \frac{r}{2\sqrt{\pi}} \int_0^{\infty} t^{-3/2} e^{-r^2/(4t)} h_t(x) dt.$$

The Gaussian bounds, after the change of variable $u = (r^2 + d(x)^2)/t$, give

$$c \frac{r}{(r + d(x))^{D+1}} \leq k_r(x) \leq C \frac{r}{(r + d(x))^{D+1}}, \quad |E k_r(x)| \leq C_E \frac{r}{(r + d(x))^{D+1+q}}. \quad (14.10)$$

Every $r\partial_r$ inserts a fixed polynomial in r^2/t in the integrand, leaving the same upper estimate. These majorants justify all finite differentiations.

To obtain parameter bounds without assuming horizontal changes preserve the relations of G , take the free step- s Carnot group $\tilde{G}$ on m generators and its fixed graded quotient map $\pi : \tilde{G} \rightarrow G$, carrying the generators to the given X_i . For $S = B^{1/2}$, freeness extends the horizontal map $e_i \mapsto \sum_j S_{ji} e_j$ to a graded automorphism Φ_B of $\tilde{G}$. If J_B is its Haar Jacobian, the lifted kernel is exactly

$$\tilde{k}_r^B(z) = J_B^{-1} \tilde{k}_r^I(\Phi_B^{-1} z).$$

All finite parameter derivatives of $\Phi_B^{\pm 1}$ and $J_B^{\pm 1}$ are bounded on the compact matrix family. A parameter derivative of the composition acts as a homogeneous degree-zero vector field: in a fixed homogeneous left frame E_a of degrees w_a , it is $\sum_a p_a(B, z) E_a$, where p_a is polynomial homogeneous of degree w_a . Iteration gives derivative words of degree ℓ multiplied by polynomials of degree ℓ ; insertion of the fixed spatial word E changes the differential degree to $\ell + q$ without changing the polynomial degree. The estimate (14.10) absorbs the polynomial by $d(z)^{\ell}/(r + d(z))^{\ell} \leq 1$. Uniform distortion of the homogeneous distance by Φ_B proves both desired estimates upstairs, including radial derivatives.

Let $N = \ker \pi$. With quotient-normalized Haar measures,

$$k_r^B(x) = \int_N \tilde{k}_r^B(zh) dh, \quad \pi(z) = x.$$

This follows by pushing down the heat semigroup and then subordinating. If $\tilde{D}$ and $D_N = \tilde{D} - D$ are the homogeneous dimensions, a coset's intersection with a radius- R ball has N -volume at most CR^{D_N} : translate a point of a nonempty intersection and use homogeneous volume on N . The decay exponent $\tilde{D} + 1$ in (14.10) is thus fiber integrable. Uniform comparison with the fixed upstairs kernel supplies a common integrable majorant for every finite parameter derivative. Integrating the pointwise estimates proves (14.8) and (14.9) downstairs. Spatial differentiation commutes with pushdown: normality of N permits conjugating the fiber, and that conjugation preserves Haar measure because its infinitesimal Jacobian is the trace of a nilpotent adjoint map, hence zero. This also justifies the mixed spatial estimate. Redundant original generators merely enlarge the fixed kernel N . $\square$

For zero input all the energies and weighted score expressions below are defined to be zero, and every corresponding inequality is immediate. For a finite nonnegative measure σ of mass $M > 0$, put $\rho_r^B = \sigma * k_r^B$. Integration of the positive kernel bounds and the quotient rule give, to every fixed finite order,

$$|\partial_B^\alpha \rho_r^B| \leq C_\alpha \rho_r^B, \quad |\partial_B^\alpha \log \rho_r^B| \leq C_\alpha (|\alpha| > 0), \quad |w_\alpha^B| + |r \partial_r w_\alpha^B| \leq C_\alpha, \quad w_\alpha^B = \frac{\partial_B^\alpha \rho_r^B}{\rho_r^B}. \quad (14.11)$$

In particular for every measurable $a(r, x) \in \mathcal{K}$, after including I in the compact parameter family,

$$\left| \int_G \rho_r^I(x) \log \frac{\rho_r^{a(r,x)}(x)}{\rho_r^I(x)} dx \right| \leq CM.$$

This finite relative expression is meaningful without separately assuming finite entropies. It is not a chain rule for a measurable a .

Theorem 14.7 (Radial Fisher comparison and frozen-energy comparison). *For $u = \rho_r^B$ define*

$$I_r^B(r) = \int_G \frac{u_r^2}{u}, \quad I_H^B(r) = \int_G \frac{\Gamma_B(u)}{u}, \quad J_B = I_r^B + I_H^B.$$

Uniformly for $0 < \delta < R < \infty$, finite nonnegative input measures, and B, C in a fixed compact positive-definite family,

$$\int_\delta^R r I_r^B(r) dr \leq 3 \int_\delta^R r I_H^B(r) dr + CM, \quad (14.12)$$

$$\left| \int_\delta^R r (J_B - J_C) dr \right| \leq CM. \quad (14.13)$$

In particular, if the family contains I ,

$$\int_\delta^R r J_B(r) dr \leq 4 \int_\delta^R r I_H^I(r) dr + CM. \quad (14.14)$$

No entropy or moment assumption on σ is required.

Proof. The subordination weight is nonnegative and its logarithmic radial derivative is $1 - r^2/(2t)$. Therefore $ru_r \leq u$. The extension solves $u_{rr} = -L_B u$. Set $K = I_r^B - I_H^B$. Differentiating both energies and integrating by parts yields

$$K' = - \int_G \frac{u_r}{u} \left(\frac{u_r^2}{u} + \frac{\Gamma_B(u)}{u} \right).$$

Indeed the mixed terms $2\Gamma_B(u, u_r)/u$ cancel, while the remaining horizontal term is $-u_r\Gamma_B(u)/u^2$. Consequently $(r^2K)' \geq r(I_r^B - 3I_H^B)$ and

$$\int_{\delta}^R rI_r^B \leq 3 \int_{\delta}^R rI_H^B + [r^2(I_r^B - I_H^B)]_{\delta}^R.$$

The kernel score bounds imply $r^2(I_r^B + I_H^B) \leq CM$, proving (14.12). All differentiation is legitimate on compact positive- r intervals: first and second scores are bounded there, and dilated spatial cutoffs have first derivatives $O(N^{-1})$, so their errors vanish against the finite mass.

We prove (14.13) first for compactly supported σ of mass one. Write $\text{Ent}(f) = \int f \log f$ and $h(f) = -\text{Ent}(f)$. Let X have law σ , and let Y, Z be independent of X and one another with densities k_r^B, k_r^C . All output and joint entropies in the following calculation are finite: the kernels are bounded, have the polynomial tails in (14.10), and compact translation preserves an integrable logarithmic moment. Conditioned on XZ , multiplication of XY by $(XZ)^{-1}$ preserves Haar measure and gives $Z^{-1}Y$. Thus

$$h(XY \mid XZ) \leq h(Z^{-1}Y), \quad h(XZ \mid XY) \geq h(XZ \mid XY, X) = h(Z).$$

The conditional entropy chain rule gives $h(XY) - h(XZ) \leq h(Z^{-1}Y) - h(Z)$. Reversing Y, Z gives the opposite bound. By inversion symmetry and kernel comparison the density of $Z^{-1}Y$ is comparable to $k_r^I * k_r^I = k_{2r}^I$. Dilation shows all noise entropies equal $D \log r$ plus a uniformly bounded unit-scale term. It follows that

$$|\text{Ent}(\rho_r^B) - \text{Ent}(\rho_r^C)| \leq CM. \quad (14.15)$$

The mass- M form follows because the $M \log M$ terms cancel. The same argument with radii $r/2$ and r for one operator, and convolution Jensen, gives $0 \leq E_B(r/2) - E_B(r) \leq CM$, where $E_B(r) = \text{Ent}(\rho_r^B)$.

The Poisson equation and integration by parts give $E_B'' = J_B \geq 0$; E_B decreases by the semigroup property and Jensen. Convexity and the preceding decrement imply $0 \leq -rE_B'(r) \leq 2(E_B(r/2) - E_B(r)) \leq CM$. Consequently

$$\int_{\delta}^R rJ_B(r) dr = [rE_B'(r) - E_B(r)]_{\delta}^R.$$

Subtracting the formula for C and using (14.15) proves (14.13) for compact input. For arbitrary σ , restrict it to increasing compact sets. Kernel derivatives converge pointwise, and the Fisher integrands on $[\delta, R]$ are bounded by $Cr^{-2}\rho_r^B$, an integrable majorant. Dominated convergence proves the claimed inequality without subtracting undefined entropies of the original input. Finally combine (14.13) with (14.12) at $B = I$ to obtain (14.14). $\square$

Theorem 14.8 (Higher frozen-parameter gradient energies). *Let $\mathcal{K} \Subset U \Subset \text{Sym}_m^+$, where U is bounded, and include I in a larger fixed compact family. For every positive-order parameter multiindex α , uniformly for $0 < \delta < R < \infty$,*

$$\int_{\delta}^R r \int_G \rho_r^I \sup_{B \in \mathcal{K}} \left(|\partial_r w_{\alpha}^B|^2 + \sum_i |X_i w_{\alpha}^B|^2 \right) dx dr \leq C_{\alpha} \left(M + \int_{\delta}^R r I_H^I(r) dr \right).$$

The same holds for w_{α}^B replaced by $\partial_B^{\alpha} \log \rho_r^B$. In particular frozen derivatives may be evaluated at any measurable matrix choice $a(r, x) \in \mathcal{K}$.

Proof. Choose a fixed orthonormal parameter basis H_ℓ of Sym_m , set $w_0 = 1$, and write $\rho = \rho_r^B$. Differentiating the Poisson equation gives the exact relation

$$(\partial_r^2 + L_B)(\rho w_\alpha) = - \sum_{\ell} \alpha_\ell L_{H_\ell}(\rho w_{\alpha-e_\ell}).$$

For $W_\alpha(r) = \int \rho w_\alpha^2$, twice differentiating and integrating by parts gives

$$\begin{aligned} W_\alpha'' &= 2 \int \rho \{(\partial_r w_\alpha)^2 + \Gamma_B(w_\alpha)\} \\ &\quad + 2 \sum_{\ell} \alpha_\ell \int \rho \{ \Gamma_{H_\ell}(w_\alpha, w_{\alpha-e_\ell}) + w_{\alpha-e_\ell} \Gamma_{H_\ell}(w_\alpha, \log \rho) \}. \end{aligned}$$

There is no radial source term, since B occurs only in L_B . Uniform bounds (14.11) and Young's inequality imply, with $\mathcal{E}_\alpha = \int \rho \{(\partial_r w_\alpha)^2 + \Gamma_B(w_\alpha)\}$,

$$\mathcal{E}_\alpha \leq W_\alpha'' + C_\alpha \left(I_H^B + \sum_{|\beta| < |\alpha|} \mathcal{E}_\beta \right).$$

Moreover $|W_\alpha| + |rW_\alpha'| \leq C_\alpha M$ directly from (14.11). Multiplication by r , integration, and $\int rW_\alpha'' = [rW_\alpha' - W_\alpha]$ prove by induction

$$\int_\delta^R r \mathcal{E}_\alpha \leq C_\alpha \left(M + \int_\delta^R r I_H^B \right).$$

All operations for finite input are justified by the bounded positive-time scores and spatial cutoffs as in Theorem 14.7.

For the supremum, take an integer $k > m(m+1)/4$, the half parameter dimension. The quotient formula $\partial_{B_\ell} w_\alpha = w_{\alpha+e_\ell} - w_\alpha w_{e_\ell}$ shows that a parameter derivative through order k of a spatial or radial derivative of w_α is a bounded linear combination of spatial or radial derivatives of w_β , $|\beta| \leq |\alpha| + k$. Sobolev embedding on finitely many interior parameter cubes, kernel comparison, and the preceding energy estimate therefore bound the supremum by $C(M + \int_U \int_\delta^R r I_H^B dr dB)$. Theorem 14.7 reduces this to the displayed single reference energy. Positive-order derivatives of $\log \rho$ are polynomials in the w_β , and the identical argument applies. $\square$

Proposition 14.9 (Adaptive radial chain rule). *Let $p > 0$ and a finite-dimensional vector or symmetric-matrix field Q be smooth solutions of $(\partial_r^2 + L)p = 0$ and $(\partial_r^2 + L)Q = 0$ on an interval times G . Put $a = Q/p$ and $h = p_r/p$. If $F(r, x, B)$ is C^2 and all displayed differentiations and integrations by parts are justified, then $G'(r) = \int_G p(r, x) F(r, x, a(r, x)) dx$ satisfies*

$$\begin{aligned} G''(r) &= \int_G p [F_{rr} - L_x F + 2hF_r + 2F_{rB} \cdot a_r \\ &\quad + D_B^2 F[a_r, a_r] + \sum_i D_B^2 F[X_i a, X_i a]]_{B=a}. \end{aligned}$$

Every derivative of F on the right is taken with B frozen. For example it suffices that F and its relevant derivatives have a fixed compact spatial support and are bounded on the attained parameter range on every compact radial interval.

Proof. The quotient equation is $a_{rr} + La + 2ha_r + 2 \sum_i (X_i \log p) X_i a = 0$. Expanding G'' and substituting $p_{rr} = -Lp$ gives the displayed terms except that the spatial parameter-Hessian term initially has a minus sign, together with

$$-2 \sum_i (X_{i,x} F_B) \cdot X_i a + F_B \cdot (a_{rr} - La + 2ha_r).$$

The quotient equation replaces the last coefficient by $-2(La + \sum_i (X_i \log p) X_i a)$. Because $pLa + \sum_i (X_i p) X_i a = \sum_i X_i (p X_i a)$, integration by parts converts its contribution to $2 \int p \sum_i [(X_{i,x} F_B) \cdot X_i a + D_B^2 F[X_i a, X_i a]]$. This cancels the explicit spatial mixed term and reverses the spatial Hessian sign, proving the identity. $\square$

For the Poisson smoothing of finite positive data with a bounded matrix ratio, let $p = \sigma * k_r^I$, $Q = (A\sigma) * k_r^I$, and $a = Q/p$. If $F(r, x, B) = \log \rho_r^B(x)$, kernel comparison and scores imply that the finite correction

$$\Delta G(r) = \int p(r, x) [F(r, x, a(r, x)) - \log p(r, x)] dx$$

satisfies $|\Delta G(r)| + |r \Delta G'(r)| \leq CM$. Indeed the bracketed difference is bounded, $|rp_r| \leq Cp$, $|ra_r| \leq C$, and $|rF_r| + |F_B| \leq C$. These bounds justify differentiating this correction even when the separate entropies are undefined. The adaptive radial identity itself is used only under its stated integration hypotheses; it does not close a stationary equation.

14.5 A zero-order frozen-parameter score square function

Theorem 14.10 (Cancellation controls parameter scores). *In the setting of Theorem 14.8, for every positive integer ℓ and $0 < \delta < 1$,*

$$\int_{\delta}^1 \frac{dr}{r} \int_G \rho_r^I(x) \sup_{B \in \mathcal{K}} |D_B^{\ell} \log \rho_r^B(x)|^2 dx \leq C_{\ell} \left(M + \int_{\delta}^1 r I_H^I(r) dr \right).$$

Any fixed finite collection of parameter directions may be used in the norm. The derivatives are frozen derivatives, so measurable evaluation $B = a(r, x)$ is allowed but differentiating that composition is not asserted. The input may be any finite nonnegative measure.

Proof. Fix a positive parameter multiindex β and put $\eta_r = \partial_B^{\beta} k_r^B$. By Theorem 14.6,

$$\int \eta_r = 0, \quad |\eta_r| \leq C k_r^I, \quad |E \eta_r| \leq C r^{-q} k_r^I$$

for every fixed homogeneous derivative word of degree q needed below. Right derivatives have the same bounds by inversion symmetry of the kernels. Fix $1/2 < \alpha < 1$. We first prove

$$|X_i^R(h_t * \eta_r)(x)| \leq \begin{cases} C r^{-1} k_r^I(x), & t \leq r^2, \\ C r^{\alpha} t^{-(1+\alpha)/2} k_{\sqrt{t}}^I(x), & t \geq r^2. \end{cases} \quad (14.16)$$

For the first case, differentiating $\eta_r(y^{-1}x)$ in the right horizontal direction produces direction $\text{Ad}_{y^{-1}} e_i$. Its degree- q component has coefficient at most $C d(y)^{q-1}$. The comparison $k_r^I(y^{-1}x) \leq C(1 + d(y)/r)^{D+1} k_r^I(x)$ and Gaussian moments of h_t bound each contribution by $C r^{-q} t^{(q-1)/2} k_r^I(x) \leq C r^{-1} k_r^I(x)$.

For $t \geq r^2$, dilation reduces the claim to $t = 1$, $r \leq 1$. Set $F = X_i^R h_1$, a smooth rapidly decreasing function. Cancellation gives

$$X_i^R(h_1 * \eta_r)(x) = \int_G [F(xz^{-1}) - F(x)] \eta_r(z) dz.$$

For $d(z) \leq 1$, a horizontal path to z and rapid decay of the mixed left and right derivatives of F bound the difference by $C d(z)^{\alpha} k_1^I(x)$. The truncated α moment of $|\eta_r|$ is at most $C r^{\alpha}$, since $\alpha < 1$. For $d(z) > 1$, its bound $C r d(z)^{-D-1}$ convolved with $|F|$ is at most $C r k_1^I(x)$. To verify this last estimate, split into $d(xz^{-1}) \leq c d(x)$, where $d(z)$ is comparable to $d(x)$, and its complement, where arbitrary polynomial decay of F applies. The subtracted $F(x)$ term has the same bound. Since $r \leq r^{\alpha}$, rescaling proves (14.16).

The exact Calderón identity is

$$\sigma * \eta_r = - \int_0^\infty \partial_t [(H_t \sigma) * \eta_r], dt.$$

The endpoints follow respectively from the approximate identity and the heat supremum bound. With $f = H_{t/2} \sigma$ and $K = h_{t/2} * \eta_r$, integration by parts gives the correctly ordered identity

$$\partial_t [(H_t \sigma) * \eta_r] = (Lf) * K = \sum_i (X_i^L f) * (X_i^R K).$$

Put $S = \max(r, \sqrt{t})$. Positive kernel bounds imply $h_{t/2} * k_S^I \leq C k_S^I$ and $k_S^I \leq C(S/r) k_r^I$. Weighted Cauchy–Schwarz in convolution, followed by integration in x , consequently gives

$$\left\| \frac{\partial_t [(H_t \sigma) * \eta_r]}{\sqrt{\rho_r^I}} \right\|_2 \leq C a(r, t) \sqrt{I_H(H_{t/2} \sigma)},$$

where

$$a(r, t) = \begin{cases} r^{-1}, & t \leq r^2, \\ r^{\alpha-1/2} t^{-1/4-\alpha/2}, & t \geq r^2. \end{cases}$$

Indeed the extra factor beyond (14.16) is $(S/r)^{1/2}$, coming from $f * k_S^I \leq C(S/r) \rho_r^I$; the remaining positive convolution preserves the integral of $|\nabla_H f|^2/f$.

Set $s = \sqrt{t}$ and $g(s) = s \sqrt{I_H(H_{s^2/2} \sigma)}$. Minkowski yields

$$\left\| \frac{\sigma * \eta_r}{\sqrt{\rho_r^I}} \right\|_2 \leq C \int_0^\infty K_0(s/r) g(s) \frac{ds}{s}, \quad K_0(z) = \begin{cases} z, & z \leq 1, \\ z^{-(\alpha-1/2)}, & z \geq 1. \end{cases}$$

The kernel K_0 is integrable for dz/z . To use only the relevant scales, split the t integral into $t < \delta^2$, $\delta^2 \leq t \leq 1$, and $t > 1$. The universal heat Fisher bound

$$I_H(H_t \sigma) \leq M I_H(h_t) = \frac{DM}{2t}$$

follows from convolution Cauchy–Schwarz and the heat kernel entropy scaling; Gaussian bounds justify differentiating that finite kernel entropy. The first and third ranges therefore contribute at most $C\sqrt{M}\delta/r$ and $C\sqrt{M}r^{\alpha-1/2}$, whose squared $L^2((\delta, 1), dr/r)$ norms are bounded by CM . Young’s convolution inequality on logarithmic scale for the middle range gives

$$\int_\delta^1 \frac{dr}{r} \int \frac{|\sigma * \partial_B^\beta k_r^B|^2}{\rho_r^I} \leq C_\beta \left(M + \int_{\delta^2}^1 I_H(H_{t/2} \sigma) dt \right). \quad (14.17)$$

For compact input, the heat integral equals twice the heat entropy difference at times $\delta^2/2$ and $1/2$. The independent-noise argument in the proof of Theorem 14.7 also compares heat noise $h_{r^2/2}$ and Poisson noise k_r^I : their individual and product entropies are finite, and dilation cancels the term $D \log r$. Thus their output entropies differ by at most CM , uniformly in r . For $E(r) = \text{Ent}(\rho_r^I)$, the endpoint identity in that proof and (14.12) give

$$E(\delta) - E(1) \leq \int_\delta^1 r J_I(r) dr + CM \leq 4 \int_\delta^1 r I_H^I(r) dr + CM.$$

Combining this with (14.17) and kernel comparison proves the claimed bound for each normalized derivative w_β^B . For arbitrary finite input, compact restriction and bounded positive-time scores permit dominated convergence on the fixed interval $[\delta, 1]$, on both sides.

Finally parameter derivatives of w_β are polynomials in positive-order normalized derivatives, with every monomial containing at least one such factor. All remaining factors are uniformly bounded by (14.11). Parameter Sobolev embedding on interior cubes bounds the supremum by parameter integrals of finitely many $|w_\gamma|^2$ with $|\gamma| > 0$. Applying the bound just proved uniformly on a larger compact family gives the supremum. The positive-order derivatives of $\log \rho$ have the same polynomial form, completing the proof. $\square$

14.6 Exact algebra in step three

The result in this subsection is a differential identity. It does not assert the existence of a positive smoothing evolution for its block operator.

Proposition 14.11 (Triangular intertwining in step at most three). *Let G have step at most three. Choose a basis Y_α of V_2 and write $[X_i, X_j] = \sum_\alpha c_{ij}^\alpha Y_\alpha$ and $Z_{i\alpha} = [X_i, Y_\alpha] \in V_3$. Set*

$$\begin{aligned} J_\alpha &= -c^\alpha, \quad K_\alpha Q = J_\alpha Q - Q J_\alpha, \quad Dv = \sum_i X_i v_i, \\ A_Q &= L - 2 \sum_\alpha Y_\alpha K_\alpha, \quad A_v = L - 2 \sum_\alpha Y_\alpha J_\alpha, \\ (BQ)_i &= 2 \sum_{j,\alpha} Z_{j\alpha} (K_\alpha Q)_{ij}, \quad Cv = \sum_{i,\alpha} Z_{i\alpha} (J_\alpha v)_i. \end{aligned}$$

For smooth symmetric tensors Q , vectors v , and scalars w ,

$$\mathcal{D}(Q, v, w) = \mathcal{T}Q + Dv + w, \quad \mathcal{A}(Q, v, w) = (A_Q Q, BQ + A_v v, Cv + Lw)$$

satisfy exactly

$$\mathcal{D}\mathcal{A} = L\mathcal{D}.$$

Redundant horizontal generators and vanishing or dependent third-layer brackets are allowed.

Proof. The $Z_{i\alpha}$ are central and $[Y_\alpha, Y_\beta] = 0$. Direct expansion gives

$$[X_i, L] = 2 \sum_{k,\alpha} c_{ik}^\alpha Y_\alpha X_k + \sum_{k,\alpha} c_{ik}^\alpha Z_{k\alpha}.$$

Expanding $[X_i X_j, L]$ gives

$$\begin{aligned} & 2 \sum_{k,\alpha} c_{jk}^\alpha Y_\alpha X_i X_k + 2 \sum_{k,\alpha} c_{ik}^\alpha Y_\alpha X_k X_j \\ & + 2 \sum_{k,\alpha} c_{jk}^\alpha Z_{i\alpha} X_k + \sum_{k,\alpha} c_{jk}^\alpha Z_{k\alpha} X_i + \sum_{k,\alpha} c_{ik}^\alpha Z_{k\alpha} X_j. \end{aligned}$$

Contracting with symmetric Q , the first two terms equal $2 \sum_\alpha Y_\alpha \mathcal{T} K_\alpha Q$; the other terms equal

$$\mathcal{R}Q = 2 \sum_{i,j,k,\alpha} c_{jk}^\alpha (Z_{i\alpha} X_k + Z_{k\alpha} X_i) Q_{ij}.$$

On the other hand $[\mathcal{T}, Y_\alpha]Q = 2 \sum_{i,j} Z_{j\alpha} X_i Q_{ij}$. Substituting the entries of $K_\alpha Q$ proves $\mathcal{R} = \sum_\alpha [\mathcal{T}, Y_\alpha] K_\alpha = DB$ and

$$[\mathcal{T}, L] = \sum_\alpha (Y_\alpha \mathcal{T} K_\alpha + \mathcal{T} Y_\alpha K_\alpha), \quad \mathcal{T} A_Q = L\mathcal{T} - DB.$$

The vector calculation from the same basic commutator gives

$$[D, L] = \sum_\alpha (Y_\alpha D J_\alpha + D Y_\alpha J_\alpha), \quad D A_v = LD - C,$$

because $C = \sum_\alpha [D, Y_\alpha] J_\alpha$. Therefore $\mathcal{D}\mathcal{A} = (L\mathcal{T} - DB)Q + D(BQ) + (LD - C)v + Cv + Lw = L\mathcal{D}$. Notice also $B(qI) = 0$, even for nonconstant q , since $K_\alpha(qI) = 0$ before differentiation. $\square$

Proposition 14.12 (Homogeneity obstruction for degree-three centers). *If Z is a nonzero central field of degree three, there is no constant C such that $\langle f, |Z|f \rangle \leq C \sum_i \|X_i f\|_2^2$ for every $f \in C_c^\infty(G)$, where $|Z| = (-Z^2)^{1/2}$.*

Proof. The unitary dilation $U_r f(x) = r^{D/2} f(\delta_r x)$ satisfies $X_i U_r = r U_r X_i$ and $|Z| U_r = r^3 U_r |Z|$. Choose f with $Zf \neq 0$. Its positive $|Z|$ form scales as r^3 and its horizontal energy as r^2 , which contradicts the asserted inequality as $r \rightarrow \infty$. The matching homogeneous power is $|Z|^{2/3}$; this scaling observation alone proves no inequality for that power. $\square$

The step-three intertwining still requires analytic construction of a positive reference smoothing, cone preservation in its tensor slot, and quantitative bounds for the generated flux and scalar slots. Unlike in step two, the second-layer fields are not central. Neither Proposition 14.11 nor its triangular form establishes an entropy estimate for unrestricted matrices in step three.

15 Structural reductions and obstructions

The results here identify exact reductions or limitations of particular proof mechanisms. None is a counterexample to the adjoint entropy estimate. Their settings are stated separately because several are valid for arbitrary smooth vector fields, whereas others use Carnot geometry or specifically $\mathbb{H}^1$.

15.1 Exact stationary duality and qualitative uniformity

For smooth real fields on a neighborhood of $\bar{U} \subset \mathbb{R}^n$, write $H_X \varphi = ((X_i X_j + X_j X_i) \varphi / 2)_{ij}$ and define $\langle \mathcal{T}_X Q, \varphi \rangle = \langle Q, H_X \varphi \rangle$. This definition does not assume the fields are divergence free. Set $\mathcal{M}^+(H) = \max_{\Lambda I \leq B \leq \Lambda I} \text{tr}(BH)$.

Proposition 15.1 (Compact stress duality for smooth fields). *Let U be bounded with compact closure, and let $\mathcal{C}_U$ consist of pairs of measures (σ, Q) on $\bar{U}$ such that $\sigma \geq 0$, $\sigma(\bar{U}) \leq 1$, $\lambda \sigma I \leq Q \leq \Lambda \sigma I$, and $\langle Q, H_X \varphi \rangle \geq 0$ for every nonnegative $\varphi \in C_c^\infty(U)$. Boundary atoms are allowed. For every continuous f supported in U ,*

$$\sup_{(\sigma, Q) \in \mathcal{C}_U} \int f d\sigma = \inf_{\substack{\varphi \in C_c^\infty(U) \\ \varphi \geq 0}} \max \left\{ 0, \sup_{\bar{U}} [f + \mathcal{M}^+(H_X \varphi)] \right\}.$$

No bracket assumption is needed for this convex duality identity.

Proof. The cone and mass constraints alone define a weak-star compact convex set $\mathcal{C}$: the cone gives uniform variation bounds on each entry of Q , and all inequalities are closed under continuous tests. For continuous symmetric H its support function is exactly

$$\sup_{(\sigma, Q) \in \mathcal{C}} \left(\int f d\sigma + \langle Q, H \rangle \right) = \max \{ 0, \sup_{\bar{U}} (f + \mathcal{M}^+(H)) \}.$$

The upper bound is pointwise maximization. For the converse use zero mass, or a unit atom at a maximizing point with a maximizing constant matrix there; no measurable selection is required.

Impose first finitely many positive-test constraints, and map $\mathcal{C}$ to the compact convex set $C \subset \mathbb{R}^{N+1}$ of points $(g_1, \dots, g_N, z)$, where $g_i = \langle Q, H_X \varphi_i \rangle$ and $z = \int f d\sigma$. Let P be its maximum z subject to $g_i \geq 0$. For $a > P$, $(0, a)$ lies outside the closed convex set $C - \mathbb{R}_+^{N+1}$. Strict separation gives a vector b with $b \cdot (0, a) > \sup_{d \in C - \mathbb{R}_+^{N+1}} b \cdot d$. Finiteness of the supremum forces $b \geq 0$, and $b_{N+1} > 0$ because the set contains zero. Dividing by b_{N+1} gives $c_i \geq 0$ with $a >$

$\sup_C(z + \sum_i c_i g_i)$. Weak duality gives the reverse bound; letting $a \downarrow P$ proves finite-constraint duality with the plus sign shown. Finite positive combinations of φ_i remain nonnegative smooth tests. Finally, if a exceeds the supremum with all constraints, compactness of $\mathcal{C} \cap \{\int f d\sigma \geq a\}$ supplies finitely many constraint half-spaces whose intersection is already empty. Thus the fully constrained value is the infimum of the finite values, proving the assertion. $\square$

Proposition 15.2 (Conditional entropy certificates). *Fix $K \Subset \text{int } S \Subset U$. Suppose a constant B bounds the right side of Proposition 15.1 for every nonnegative smooth f supported in S with $\int_U(e^f - 1) \leq 1$. Then every density member $p dx$ of $\mathcal{C}_U$ obeys $\int_K p \log(1 + p) \leq B$. Conversely, if every member of $\mathcal{C}_U$ has an interior density on a neighborhood of S , and those densities have a uniform entropy bound C on S , the stated dual values are bounded by $C + |S| + 1$. These are conditional implications; existence of the certificates is not asserted for arbitrary coefficients.*

Proof. Choose $0 \leq \zeta \leq 1$ with $\zeta = 1$ near K and support compactly inside S . Put $f_N = \zeta \min\{N, \log(1 + p)\}$. Then $e^{f_N} - 1 \leq p$. Smooth mollification with the support margin gives $0 \leq g_j \leq N$, supported in S , converging almost everywhere to f_N . Dominated convergence holds against both dx and $p dx$. If $I_j = \int(e^{g_j} - 1) > 1$, replace g_j by g_j/I_j ; convexity of the exponential restores the constraint, and the factors tend to one. The dual bound passes to f_N . Monotone convergence in N proves the first conclusion. Conversely use the elementary Young bound $pf \leq p \log(1 + p) + e^f$ on S , then exact duality. Boundary atoms make no contribution because f is supported in the interior. $\square$

Proposition 15.3 (Qualitative local entropy implies a uniform bound). *Fix arbitrary smooth fields, ellipticity constants, and a common $\varepsilon \geq 0$. Assume that on every relatively compact open subdomain every stationary density triple of mass at most one satisfying*

$$\mathcal{L}_A^*(p dx) + \varepsilon \Delta(p dx) = \mu \geq 0$$

with a finite source has finite local $L \log L$ entropy. Then, for every fixed $K \Subset \Omega$, the local entropy is bounded uniformly over A, p, μ in this class. This does not give uniformity over varying ε .

Proof. If the bound fails, choose $K \Subset U \Subset \Omega$ and triples with $\int_K p_n \log(1 + p_n) \geq n2^n$. A fixed nonnegative cutoff η equal to one on $\overline{U}$ gives

$$\mu_n(U) \leq \int \eta d\mu_n = \int (\mathcal{L}_{A_n} \eta + \varepsilon \Delta \eta) p_n \leq C.$$

On U , set $p = \sum_n 2^{-n} p_n$, $Q = \sum_n 2^{-n} A_n p_n$ and $\mu = \sum_n 2^{-n} \mu_n|_U$. These define a unit-mass density, a stress with $\lambda p I \leq Q \leq \Lambda p I$, and a finite positive source. Set $A = Q/p$ on $\{p > 0\}$ and any admissible matrix elsewhere. After changing representatives on a null set this is Borel; all weak tests pass through the sums by absolute convergence. Thus it is an admissible triple on U . Since $p \geq w_n p_n$, $w_n = 2^{-n}$, and $1 + w_n z \geq w_n(1 + z)$,

$$\int_K p \log(1 + p) \geq w_n \int_K p_n \log(1 + p_n) + w_n \log w_n \geq n - n2^{-n} \log 2.$$

This contradicts the qualitative premise. For varying viscosities the mixture instead contains $\Delta(\sum_n 2^{-n} \varepsilon_n p_n)$, which need not be a constant multiple of Δp ; this explains the scope restriction. $\square$

Proposition 15.4 (No constant-mass universal scalar supersolution). *Let G be a nontrivial Carnot group and $0 < \lambda < \Lambda$. There is no nonzero nonnegative locally integrable k on $I \times G$, where I is an open time interval, with a common finite spatial mass M for almost every time and $\partial_t k - L_B k \geq 0$ distributionally for every constant symmetric B with $\lambda I \leq B \leq \Lambda I$.*

Proof. The positive distribution $R_B = \partial_t k - L_B k$ is a Radon measure. Let $\chi_R(x) = \chi(\delta_{1/R}x)$, where $0 \leq \chi \leq 1$ is compactly supported and equals one near the identity. Uniformly in B , $\|L_B \chi_R\|_\infty \leq CR^{-2}$. For nonnegative $\theta \in C_c^\infty(I)$,

$$\langle R_B, \theta \chi_R \rangle = - \int k \theta' \chi_R - \int k \theta L_B \chi_R \longrightarrow -M \int \theta' = 0.$$

Positivity forces $R_B = 0$ on every compact set. Differences between constant matrices in the ellipticity interval, which has nonempty interior, imply $H_X k = 0$, and then $\partial_t k = 0$. Hence $k = k_0(x)$ almost everywhere. Left convolution smooths k_0 while preserving mass, nonnegativity and $X_i^2 k_0 = 0$. On each complete horizontal one-parameter flow the smoothed function is affine and nonnegative on all of $\mathbb{R}$, so its slope is zero. Bracket generation and connectedness make it constant. A constant of finite mass on G is zero. Passing to the approximate-identity limit finishes the proof. When $\lambda = \Lambda$ this argument does not apply, and the single reference heat kernel has constant mass as expected. $\square$

15.2 Horizontal Hessians, contact maps, and source exponents

Proposition 15.5 (A contact-Jacobian rank obstruction). *Let $X_1, \dots, X_m$ have coefficient matrix $\sigma(x)$ of rank $m < n$. Let $F(x, y)$ be a smooth cost with invertible F_{xy} at a point. There are smooth local minimizing envelopes $u(x) = \inf_y \{F(x, y) + b(y)\}$ whose symmetric X -Hessian at that point stays fixed, while the full Jacobian $|\det D_x y(x)|$ of the minimizing contact map tends to infinity. Consequently that Jacobian cannot be bounded pointwise by the symmetric horizontal Hessian alone.*

Proof. At a strict local minimizer let $M = F_{yy} + D^2 b > 0$. The implicit-function theorem and the stationarity equation give

$$D_x y = -M^{-1} F_{yx}, \quad D_x^2 u = F_{xx} - F_{xy} M^{-1} F_{yx}, \quad H_X u = H_{X,x} F - K M^{-1} K^T, \quad K = \sigma^T F_{xy}.$$

The first-order geometric terms in $H_X u$ are those of F because $D_x u = F_x$. Choose a unit vector $v \in \ker K$ and $M_h = I + (h-1)vv^T$, $0 < h < 1$. Prescribe $Db = -F_y$ and $D^2 b = M_h - F_{yy}$ at the minimizing point. These jets are realizable by a smooth b , and $M_h > 0$ supplies a local minimizing branch. Since $K M_h^{-1} K^T = K K^T$ whereas $|\det D_x y| = h^{-1} |\det F_{yx}|$, the assertion follows. The branch neighborhood may shrink with h , which suffices for a pointwise obstruction. No statement about a globally constrained contact set is being made. $\square$

In the next two propositions use precisely $X = \partial_x - (y/2)\partial_z$, $Y = \partial_y + (x/2)\partial_z$ on $\mathbb{H}^1$, and $Z = [X, Y] = \partial_z$. A smooth function is horizontally convex when its symmetric Hessian $H_H u$ is positive semidefinite. For a continuous function, horizontal convexity and the determinant equation below are understood in the viscosity sense: smooth tests from above must have nonnegative horizontal Hessian and determinant at least the source; tests from below with positive-definite horizontal Hessian have determinant at most the source.

Proposition 15.6 (Two different Heisenberg determinant identities). *For smooth u ,*

$$\det \begin{pmatrix} XXu & XYu \\ YXu & YYu \end{pmatrix} = \det H_H u + \frac{1}{4}(Zu)^2.$$

For $u \in C_c^\infty(\mathbb{H}^1)$,

$$\int \det H_H u = -\frac{3}{4} \int (Zu)^2.$$

In particular neither determinant is determined by the symmetric Hessian in the way an ordinary Euclidean gradient Jacobian is.

Proof. Put $P = XYu$ and $R = YXu$. The symmetric off-diagonal entry is $(P+R)/2$ and $Zu = P - R$, proving the first identity. For the integral identity, $\int(P^2 - R^2) = \int Zu(XY + YX)u = 0$: Z is skew-adjoint, $XY + YX$ is self-adjoint, and they commute. Integration by parts and $[X, Y] = Z$ give

$$\int (XXu)(YYu) = \int (2PR - R^2).$$

More explicitly, $\int PR = \int (XXu)(YYu) - \int YuXZu$, and $\int YuXZu = -\int P(P - R)$. These equalities, together with $\int P^2 = \int R^2$, give the displayed formula. Subtracting $\int(P + R)^2/4$ and using $\int P^2 = \int R^2$ yields $-\frac{3}{4}\int(P - R)^2$. $\square$

Proposition 15.7 (Failure of determinant-source bounds through exponent four). *On the fixed uniformly horizontally convex domain $U = \{x^2 + y^2 + z^2 < 4\} \subset \mathbb{H}^1$, there is no constant C_q such that every continuous horizontally convex viscosity solution of*

$$\det H_H v = f^2 \quad \text{in } U, \quad v = 0 \quad \text{on } \partial U, \quad f \geq 0 \text{ smooth},$$

satisfies $\|v\|_\infty \leq C_q \|f\|_{L^q(U)}$, for any $1 \leq q \leq 4$. At $q = 4$ the counterexample sources may all be supported in one fixed compact subset of U . For the radial convex class on a Korányi unit ball, the analogous estimate does hold when $q > 4$ under the radial regularity conditions in the proof.

Proof. Write $s = (x^2 + y^2)^2 + 16z^2$, $\rho = s^{1/4}$, $B = \{\rho < 1\}$, and $r_H^2 = x^2 + y^2$. Direct differentiation gives $H_H s = 12r_H^2 I$ and $|\nabla_H s|^2 = 16r_H^2 s$. For $h > 0$ the smooth function $u_h = (s + h^4)^{1/4} - (1 + h^4)^{1/4}$ vanishes on ∂B , is horizontally convex, and has minimum tending to -1 . Its horizontal Hessian eigenvalues are

$$3r_H^2(s + h^4)^{-3/4}, \quad 3r_H^2 h^4(s + h^4)^{-7/4},$$

so $f_h = (\det H_H u_h)^{1/2} = 3h^2 r_H^2(s + h^4)^{-5/4}$. It satisfies $f_h(\delta_h \xi) = h^{-1} f_1(\xi)$ and $f_1 = O(\rho^{-3})$ at infinity. Polar integration in homogeneous dimension four gives $\|f_h\|_{L^q(U)} \rightarrow 0$ for $q < 4$: its q th power is $O(h^{4-q})$ if $q > 4/3$, $O(h^{8/3} |\log h|)$ if $q = 4/3$, and $O(h^{2q})$ if $q < 4/3$.

For the critical exponent let $L > 1$, $h = e^{-L}$, and choose a smooth nondecreasing function θ , zero on $(-\infty, 0]$, one on $[1, \infty)$, and with positive derivative on $(0, 1)$, flat at both endpoints. Put

$$a_h(r) = \theta(\log(r/h)/L), \quad F_h(r) = -\int_r^1 a_h(t) dt, \quad u_h = F_h(\rho).$$

This is smooth, constant near the identity, horizontally convex, and vanishes on ∂B ; its minimum tends to -1 . For any increasing convex radial F with $a = F'$, direct differentiation gives

$$(\det H_H F(\rho))^{1/2} = \sqrt{3} \left(\frac{a(\rho)a'(\rho)}{\rho} \right)^{1/2} \frac{r_H^2}{\rho^2}. \quad (15.1)$$

Homogeneous polar integration therefore gives

$$\int_B (\det H_H u_h)^2 = \frac{C_*}{L} \int_0^1 \theta(t)^2 \theta'(t)^2 dt \rightarrow 0.$$

For definiteness take $\theta(t) = \eta(t)/(\eta(t) + \eta(1 - t))$, where $\eta(t) = e^{-1/t}$ for $t > 0$ and zero otherwise. The square root in (15.1) is smooth and flat at $\rho = h, 1$. Extending f_h by zero outside B gives a smooth source compactly supported in U .

The defining function of U has horizontal Hessian $2I + 2w \otimes w \geq 2I$, $w = (-y/2, x/2)$. The Dirichlet existence and comparison theorem for continuous nonnegative horizontal determinant sources on such uniformly horizontally convex domains supplies a continuous horizontally convex solution v_h on U ; the precise fixed theorem used is [3, Corollary 3.1, Theorem 4.2 and

Example 4.5]. Its hypotheses apply because the source is continuous, independent of the value and gradient, and bounded for each fixed h . Comparison with zero gives $v_h \leq 0$. On B , it is a subsolution with boundary values at most zero, while u_h is a smooth solution with the same source and zero boundary value. Comparison gives $v_h \leq u_h$, hence $-v_h(e) \geq -u_h(e) \rightarrow 1$, whereas the corresponding source norms tend to zero. This proves failure on U , including the critical compact-support assertion.

For the stated surviving radial class assume $F(1) = 0$, $a = F' \geq 0$, $a' \geq 0$ and $a(0) = 0$. With $b = a^2$, (15.1) gives

$$\|f\|_q^q = C_q \int_0^1 |b'(r)|^{q/2} r^{3-q/2} dr.$$

For $q > 4$, weighted Hölder yields $b(r) \leq C_q \|f\|_q^2 r^{2-8/q}$: the dual weight integral is finite exactly when $q > 4$. Integrating $\sqrt{b}$ gives $-F(0) \leq C_q \|f\|_q$. This is only a theorem for that radial class, not a general supercritical Dirichlet estimate. $\square$

The source-exponent obstruction leaves open the supercritical horizontal Monge–Ampère bound that could imply an entropy theorem. Indeed, conditionally on a construction of smooth horizontally convex u with $(\det H_H u)^{1/m} \geq f$ and bounded oscillation and cutoff horizontal gradient, the arithmetic–geometric mean inequality gives $L_A u \geq m\lambda f$. Testing the stationary equation with $\chi(M - u) \geq 0$, $M \geq \sup u$, bounds $\int_K pf$ by those fixed oscillation and gradient bounds. This is another conditional entropy certificate; the required construction for general exponential-source data is not established here.

15.3 A tensor-cone obstruction on the first Heisenberg group

Proposition 15.8 (The unmodified tensor correction does not preserve positivity). *On $\mathbb{H}^1$ let $L = X^2 + Y^2$ and*

$$KQ = \begin{pmatrix} -2Q_{12} & Q_{11} - Q_{22} \\ Q_{11} - Q_{22} & 2Q_{12} \end{pmatrix}.$$

Then $\mathcal{T}L - L\mathcal{T} = 2Z\mathcal{T}K$, but the formal corrected generator $L - 2ZK$ cannot generate a classical evolution preserving every pointwise positive-semidefinite tensor.

Proof. Centrality gives $[X^2, Y^2] = 2Z(XY + YX)$, $[XY + YX, X^2] = -4ZX^2$, and $[XY + YX, Y^2] = 4ZY^2$. Contraction with Q yields the displayed identity. For the positivity claim set $Q = ww^T$, $w = (f, 1)$, and choose a point where $f = Xf = Yf = 0$, $Zf < 0$. Such a jet exists since Z is outside the horizontal span. At that point

$$((L - 2ZK)Q)_{11} = 2[(Xf)^2 + (Yf)^2] + 4Zf < 0, \quad Q_{11} = 0.$$

A cutoff squared preserves positivity and the jet. A differentiable classical evolution with this initial derivative immediately has a negative diagonal entry. The obstruction concerns this generator alone; additional central smoothing is not excluded. $\square$

15.4 Adapted control: a geometric theorem and a distinct obstruction

The stochastic assertions below use standard Itô calculus, Burkholder–Davis–Gundy inequalities, the exponential martingale inequality, and elementary moment interpolation. They concern a fixed stochastic development driven by a martingale. They do not identify a Malliavin covariance for a Borel state-feedback diffusion.

Theorem 15.9 (Inverse moments of the geometric control Gramian). *Let G be a Carnot group of step s , with horizontal generators $e_1, \dots, e_m$, and fix $0 < \lambda \leq \Lambda$. Let M be any continuous $\mathbb{R}^m$ -valued martingale starting at zero with $d\langle M \rangle_t = 2A_t dt$, where A_t is predictable and $\lambda I \leq$*

$A_t \leq \Lambda I$. Let Z_t be the left-invariant Stratonovich development $dZ_t = \sum_i X_i(Z_t) \circ dM_t^i$, $Z_0 = e$, and define

$$J_t = \text{Ad}_{Z_t}, \quad B_t = (J_t e_1, \dots, J_t e_m), \quad C_T = \int_0^T B_t A_t B_t^T dt.$$

For every $0 < p < \infty$ and $T > 0$, C_T is almost surely invertible and $\sup_A \mathbb{E} \|C_T^{-1}\|^p < \infty$. For $0 < T \leq 1$ one may bound this by $C_p T^{-sp}$. Constants depend only on the fixed group, generators, ellipticity constants and the displayed p, T .

Proof. First let $T = 1$. In the adjoint representation, conversion to Itô form using $d\langle M \rangle = 2A dt$ gives

$$dJ_t = \sum_i J_t \text{ad}_{e_i} dM_t^i + \sum_{i,j} a_{ij}(t) J_t \text{ad}_{e_i} \text{ad}_{e_j} dt.$$

This finite-dimensional linear equation, Itô's formula, BDG and Gronwall imply moments of every order for $\sup_t \|J_t\|$ and increment moments $\mathbb{E} \|J_t - J_u\|^q \leq C_q |t - u|^{q/2}$. The dyadic Hölder argument, using higher moment exponents, therefore gives moments of every order for

$$H = 1 + \sup_{0 \leq t \leq 1} \|J_t\| + [J]_{C^{1/4}([0,1])}.$$

Choose a finite spanning list of bracket words in the e_i , including the shorter words needed below. For a unit covector ξ set $f_Y(t) = \langle \xi, J_t Y \rangle$. It satisfies

$$df_Y = \sum_i f_{[e_i, Y]} dM_t^i + \sum_{i,j} a_{ij}(t) f_{[e_i, [e_j, Y]]} dt. \quad (15.2)$$

On $\{H \leq R\}$ the drifts and the one-quarter Hölder seminorms of this finite family are at most CR .

We record the needed elementary estimate. Suppose $dy = \gamma \cdot dM + b dt$, with drift bounded by CR until the flow norm reaches $2R$, and put $q_t = \int_0^t \gamma^T A \gamma du$. If $\rho \geq C'(\delta^2 + \delta R)$, then

$$\mathbb{P}\{\sup |y| \leq \delta, q_1 \geq \rho, H \leq R\} \leq \exp(-c\rho/\delta^2). \quad (15.3)$$

To prove this, stop when $|y|$ reaches 2δ , the flow norm reaches $2R$, or q reaches ρ . On the event displayed, the last stopping occurs first. For $N = \int y \gamma \cdot dM$, Itô's formula gives $N \leq -\rho/2$ at that time, while $\langle N \rangle \leq 8\delta^2\rho$. The exponential martingale inequality proves (15.3). Only adapted norm stopping has been used; the global Hölder event is used afterwards.

Choose $a_j = 48^{-j}$, $\kappa = 48^{-(s+2)}$ and $R = \epsilon^{-\kappa}$. If $\xi^T C_1 \xi \leq 2\epsilon$, ellipticity gives $\int \sum_i |f_{e_i}|^2 \leq C\epsilon$. A one-quarter Hölder function with seminorm CR obeys

$$\|f\|_\infty \leq C(R^{2/3} \|f\|_2^{1/3} + \|f\|_2).$$

Indeed a maximum of size a persists with size at least $a/2$ on an interval of length at least $c \min(1, (a/R)^4)$. This interpolation gives $\max_i \|f_{e_i}\|_\infty \leq \epsilon^{a_1}$ for small ϵ on $\{H \leq R\}$. If all words of length j satisfy $\|f_Y\|_\infty \leq \epsilon^{a_j}$, apply (15.3) to (15.2) with $\delta = \epsilon^{a_j}$ and $\rho = \epsilon^{a_j/2}$. Its drift requirement follows from $\kappa < a_j/2$. Except with probability at most $C \exp(-c\epsilon^{-3a_j/2})$, the next bracket energies are at most $C\epsilon^{a_j/2}$. Interpolation gives

$$\|f_{[e_i, Y]}\|_\infty \leq C\epsilon^{a_j/12 - 2\kappa/3} \leq \epsilon^{a_{j+1}}.$$

The strict exponent inequality absorbs all fixed constants for small ϵ . A finite union suffices over the word list. After s layers, all spanning Y have $|\langle \xi, Y \rangle| = |f_Y(0)|$ tending to zero, contradicting $|\xi| = 1$. Thus a fixed covector's small quadratic form on $\{H \leq R\}$ has stretched-exponentially small probability.

On that event $\|C_1\| \leq CR^2$. A deterministic sphere net of mesh $c\epsilon/R^2$ detects $\lambda_{\min}(C_1) \leq \epsilon$ at a net point with quadratic form at most 2ϵ ; its cardinality is polynomial in ϵ^{-1} . Union bounds and $\mathbb{P}(H > R) \leq C_q \epsilon^{\kappa_q}$ give

$$\mathbb{P}\{\lambda_{\min}(C_1) \leq \epsilon\} \leq C_N \epsilon^N \quad \text{for every finite } N.$$

Integration proves all inverse moments. Finally time rescaling and graded dilation give exactly $C_T = D_T \tilde{C}_1 D_T$, where $D_T = \text{diag}(T^{j/2} I_{V_j})_{j=1}^s$ and the rescaled martingale has the same covariance bounds. For $T \leq 1$ its smallest squared dilation is T^s , yielding the claimed T^{-sp} estimate. $\square$

Proposition 15.10 (No adapted first-order pure second-layer endpoint control). *On any non-abelian Carnot group, no predictable control u with $\mathbb{E} \int_0^T |u_t|^2 dt < \infty$ can make the frozen-input perturbation $M + \theta \int_0^T u_t dt$ of the preceding development have deterministic first-order endpoint effect $(0, \zeta)$ in the step-two quotient, for nonzero $\zeta \in V_2$.*

Proof. Write $h_t = \int_0^t u_s ds$. In the step-two quotient the endpoint is $(M_T, \frac{1}{2} \int_0^T [M_t, dM_t])$. Differentiating its exact polynomial expansion under $M \mapsto M + \theta h$ and integrating by parts gives

$$\dot{Z}_T^1 = h_T, \quad \dot{Z}_T^2 = \int_0^T [h_t, dM_t] + \frac{1}{2} [M_T, h_T].$$

Zero horizontal endpoint variation forces $h_T = 0$. The remaining bracket integral is a square-integrable martingale terminal value, since its second moment is bounded by $C\Lambda T^2 \mathbb{E} \int_0^T |u_t|^2 dt$. Its expectation is zero, so it cannot equal deterministic nonzero ζ almost surely. The same proof works at a bounded stopping time with vanishing horizontal variation, by stopping the square-integrable bracket martingale. This does not exclude other integration-by-parts weights or density arguments. In particular it does not contradict the geometric Gramian invertibility theorem: a terminal inverse-Gramian control is generally anticipative. $\square$

16 Auxiliary results on the first Heisenberg group

This section records finer power estimates, sharp mass examples, and limitations of several auxiliary proof mechanisms. Unless another gauge is displayed, the group is $\mathbb{H}^1 = \mathbb{R}^3$ with

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + (xy' - yx')/2), \quad X = \partial_x - \frac{y}{2} \partial_z, \quad Y = \partial_y + \frac{x}{2} \partial_z. \quad (16.1)$$

The Haar measure is $dg = dx dy dz$, the dilations are $\delta_r(x, y, z) = (rx, ry, r^2 z)$, and the homogeneous dimension is four. We use the stationary density hypotheses

$$p \geq 0, \quad p \in L^1(\Omega), \quad \int_{\Omega} p \leq 1, \quad \lambda I_2 \leq A \leq \Lambda I_2 \quad p \, dg\text{-a.e.}, \quad (16.2)$$

$$\int_{\Omega} p(A : H_H \varphi + \varepsilon \Delta \varphi) = \int_{\Omega} \varphi \, d\mu.$$

Here $\Omega \subset \mathbb{R}^3$ is arbitrary open, A is Borel symmetric, $\varepsilon \in [0, 1]$ is constant, μ is finite and nonnegative, $\varphi \in C_c^\infty(\Omega)$, and $(H_H \varphi)_{ij} = (X_i X_j + X_j X_i) \varphi / 2$, with $(X_1, X_2) = (X, Y)$. Write $D = \mathcal{L}_A + \varepsilon \Delta$. The Laplacian is the literal coordinate Euclidean Laplacian. All constants in estimates for this class are independent of p, A, μ, ε , except when dependence is expressly displayed. No geometric example below is asserted to be a counterexample to the stationary PDE entropy estimate.

16.1 A translation-invariant Heisenberg quotient

Theorem 16.1 (Translation-invariant quotient). *Use instead $X = \partial_x$, $Y = \partial_y + x\partial_z$ on $\Omega = \{(x, y, z) : y \in I, (x, z) \in D\}$, with I a nonempty open interval and $D \subset \mathbb{R}^2$ open. Suppose the stationary hypotheses (16.2) hold in these coordinates and*

$$p(x, y, z) = q(x, z), \quad A(x, y, z) = \begin{pmatrix} a(x, z) & b(x, z) \\ b(x, z) & c(x, z) \end{pmatrix}$$

almost everywhere for the respective density and stress measures. Then, for $K \Subset \Omega$ and $0 < \theta < 1/3$,

$$\int_K p^{1+\theta} dg \leq C_{K,\theta}, \quad \int_K p \log(1+p) dg \leq C_K.$$

No product assumption on μ is required.

Proof. Direct calculation gives

$$X^2 = \partial_{xx}, \quad Y^2 = \partial_{yy} + 2x\partial_{yz} + x^2\partial_{zz}, \quad XY + YX = 2\partial_{xy} + 2x\partial_{xz} + \partial_z.$$

For $\eta \in C_c^\infty(I)$ and $\psi \in C_c^\infty(D)$, Fubini and y -invariance remove the η', η'' terms. The resulting operator is

$$\mathcal{G}\psi = (a + \varepsilon)\psi_{xx} + 2bx\psi_{xz} + (cx^2 + \varepsilon)\psi_{zz} + b\psi_z,$$

and $\int p(\mathcal{L}_A + \varepsilon\Delta)(\eta\psi) = (\int_I \eta) \int_D q\mathcal{G}\psi$. Choose nonnegative $\eta_0 \in C_c^\infty(I)$ with integral one and define $\bar{\mu}(E) = \int \eta_0(y)1_E(x, z) d\mu$. This is a finite nonnegative measure. The weak equation implies

$$\int \eta(y)\psi(x, z) d\mu = \left(\int_I \eta\right) \int_D \psi d\bar{\mu}.$$

Finite sums of such tensor-product tests are uniformly dense on compact product neighborhoods, so uniqueness of Radon measures proves $\mu = dy \otimes \bar{\mu}$. If $\ell = |I| = \infty$, finite mass forces $q = 0$ and then $\mu = 0$. Otherwise $Q = \ell q$ has mass at most one and satisfies $\mathcal{G}^*Q = \ell\bar{\mu}$; ellipticity descends by Fubini. The operator $\mathcal{G}$ is precisely the symmetrized Grushin operator for $\partial_x, x\partial_z$, including its necessary drift $b\partial_z$. Theorem 12.7 gives $\int_L Q^{1+\theta} \leq C_{L,\theta}$, $L = \pi_{xz}(K) \Subset D$. Therefore $\int_K p^{1+\theta} \leq \ell^{-\theta} \int_L Q^{1+\theta}$. The elementary inequality $s \log(1+s) \leq C_\theta(s + s^{1+\theta})$ proves entropy. This computation preserves the original coordinate viscosity; it does not use a non-linear change of coordinates. $\square$

16.2 Inverse radial moments and axisymmetric power estimates

Theorem 16.2 (Radial-central projection and axisymmetry). *Under (16.2) in the standard gauge (16.1), put $u = (x^2 + y^2)/4$ and $\beta_* = (\Lambda + \lambda)/(2\Lambda)$. For every $K \Subset \Omega$ and every $0 < \beta < \beta_*$,*

$$\int_K p u^{-\beta} dg \leq C_{K,\beta}. \tag{16.3}$$

For $0 \leq \chi \leq 1$, $\chi \in C_c^\infty(\Omega)$, extend χp by zero and define

$$q_\chi(u, z) = 2 \int_0^{2\pi} (\chi p)(2\sqrt{u} \cos t, 2\sqrt{u} \sin t, z) dt \quad (u > 0), \tag{16.4}$$

with value zero for $u < 0$. For every $0 < \theta < \beta_/2 = (\Lambda + \lambda)/(4\Lambda)$,*

$$\int_{\mathbb{R}^2} q_\chi^{1+\theta} du dz \leq C_{\chi,\theta}. \tag{16.5}$$

If, in addition, $p(x, y, z) = P(u, z)$ almost everywhere on Ω for a Borel function $P \geq 0$, then

$$\int_K p^{1+\theta} dg \leq C_{K,\theta}, \quad \int_K p \log(1+p) dg \leq C_K \quad (16.6)$$

for the same strict range of θ . No symmetry of A, μ , or Ω is assumed. No endpoint or optimality assertion is made.

Proof. Inverse radial moments.

First let $\chi = \chi(u, z) \geq 0$ be a smooth rotationally invariant compactly supported cutoff in Ω . Fix $\beta \in (0, \beta_*)$, put $\gamma = 1 - \beta$, and set

$$h_\delta(u) = (u + \delta)^\gamma, \quad 0 < \delta \leq 1. \quad (16.7)$$

Thus $0 < \gamma < 1$ and $\gamma > (1 - \lambda/\Lambda)/2$. Direct calculation gives

$$\nabla_H u = (x/2, y/2), \quad H_H u = \frac{1}{2}I, \quad |\nabla u|^2 = u, \quad \Delta u = 1. \quad (16.8)$$

At $u > 0$, write $a_r = e_r^t A e_r$, $a_t = e_t^t A e_t$, where $e_r = (x, y)/(2\sqrt{u})$ and $e_t = (-y, x)/(2\sqrt{u})$. Both numbers belong to $[\lambda, \Lambda]$. Since $\text{tr } A = a_r + a_t$,

$$\mathcal{L}_A h_\delta = \gamma(u + \delta)^{\gamma-1} \left[\frac{a_r + a_t}{2} + (\gamma - 1) \frac{u}{u + \delta} a_r \right]. \quad (16.9)$$

The bracket is at least $(\gamma - 1/2)a_r + a_t/2$. It is therefore at least

$$c_\gamma = \begin{cases} \gamma\lambda, & \gamma \geq 1/2, \\ (\gamma - 1/2)\Lambda + \lambda/2, & 0 < \gamma < 1/2, \end{cases} \quad c_\gamma > 0. \quad (16.10)$$

The same bound holds on the axis by the direct formula there. Also

$$\Delta h_\delta = \gamma(u + \delta)^{\gamma-1} \left[1 + (\gamma - 1) \frac{u}{u + \delta} \right] \geq \gamma^2(u + \delta)^{\gamma-1} \geq 0. \quad (16.11)$$

Consequently $Dh_\delta \geq \gamma c_\gamma(u + \delta)^{-\beta}$.

Choose a constant C independent of δ with $C - h_\delta \geq 0$ on $\text{supp } \chi$, and test (16.2) with $\phi = \chi(C - h_\delta) \geq 0$. For

$$\Gamma_D(f, g) = (Xf, Yf)^t A(Xg, Yg) + \varepsilon \nabla f \cdot \nabla g, \quad (16.12)$$

the product rule is

$$D\phi = (C - h_\delta)D\chi - \chi Dh_\delta - 2h'_\delta(u)\Gamma_D(\chi, u). \quad (16.13)$$

All the coefficients of $D\chi$ are uniformly bounded on its fixed support. Because $\chi = \chi(u, z)$,

$$X\chi = \frac{x}{2}\chi_u - \frac{y}{2}\chi_z, \quad Y\chi = \frac{y}{2}\chi_u + \frac{x}{2}\chi_z, \quad \nabla\chi \cdot \nabla u = u\chi_u. \quad (16.14)$$

It follows that $|\Gamma_D(\chi, u)| \leq C_\chi u$, uniformly in A, ε . Moreover

$$uh'_\delta(u) = \gamma u(u + \delta)^{\gamma-1} \leq \gamma(u + \delta)^\gamma \quad (16.15)$$

is uniformly bounded on the cutoff support. Since the sink integral is nonnegative, (16.13), the mass bound, and (16.9)–(16.11) imply

$$\int \chi p(u + \delta)^{-\beta} \leq C_{\chi, \beta}. \quad (16.16)$$

Monotone convergence as $\delta \downarrow 0$ proves the moment bound on every region where $\chi = 1$.

This suffices on arbitrary Ω and arbitrary compact K . Cover the compact set $K \cap \{x = y = 0\}$ by finitely many neighborhoods on which rotational cylinder cutoffs, supported in Ω , equal one. The part of K outside these neighborhoods is compact and has $u \geq c > 0$; there the mass bound suffices. If the axis does not meet K , only this latter argument is needed. This proves (16.3). In particular, rotational symmetry of the domain was not used.

Translated-axis variant. The same bound holds for $u_a = ((x - a)^2 + (y - b)^2)/4$, with constants uniform for (a, b) in a fixed compact set. Near this axis use the contact coordinate $z_a = z + (bx - ay)/2$ and a cutoff $\chi = \zeta(u_a, z_a)^2$. Horizontal cutoff cross terms are again $O(h'_\delta u_a)$. The Euclidean cross term satisfies

$$2\varepsilon h'_\delta |\nabla \chi \cdot \nabla u_a| \leq C\varepsilon h'_\delta |\zeta| \sqrt{u_a} \leq \frac{1}{2}\varepsilon \gamma \zeta^2 h'_\delta + C_\gamma \varepsilon h'_\delta u_a. \quad (16.17)$$

The first term is absorbed by the favorable principal term $\varepsilon \chi \Delta h_\delta \geq \varepsilon \gamma \chi h'_\delta$, and the second is uniformly bounded. Fixed compactness and the distance of K from Ω^c permit uniformly sized cylinder cutoffs and a bounded number of them. No translated-coordinate interpretation of the original Laplacian is assumed; (16.17) is computed in the original coordinates.

The change of variables $(x, y) = (2\sqrt{u} \cos t, 2\sqrt{u} \sin t)$ has Jacobian $dx dy = 2 du dt$. Consequently $q = q_\chi$ is compactly supported, has mass at most one, and satisfies $\int q u^{-\beta} \leq C_{\chi, \beta}$.

A changed convex coordinate.

Choose

$$1 - \beta_* < a < 1, \quad v = u^a, \quad g_\delta(u) = (u + \delta)^a - \delta^a, \quad 0 < \delta \leq 1. \quad (16.18)$$

In particular $a > 0$. Let $f(v, z)$ be smooth, globally convex and one-Lipschitz, normalized at one point, and write $F_\delta = f(g_\delta(u), z)$. All these are smooth in the original coordinates. Set $e_r = (\cos t, \sin t)$, $e_t = (-\sin t, \cos t)$, and let $R_t = (e_r, e_t)$. Then

$$H_H F_\delta = M_\delta(D^2 f)(g_\delta(u), z) M_\delta^t + f_v(g_\delta(u), z) H_H g_\delta, \quad (16.19)$$

$$M_\delta = \sqrt{u} R_t \operatorname{diag}(g'_\delta, 1), \quad H_H g_\delta = \frac{1}{2} g'_\delta I + u g''_\delta e_r \otimes e_r. \quad (16.20)$$

Therefore, with $S_\delta = \operatorname{diag}(u(g'_\delta)^2, u)$,

$$\mathcal{L}_A F_\delta \geq \lambda S_\delta : D^2 f(g_\delta(u), z) - C(u + \delta)^{a-1}. \quad (16.21)$$

Here $\|H_H g_\delta\| \leq C_a(u + \delta)^{a-1}$, because $u|g''_\delta| \leq C_a(u + \delta)^{a-1}$. The Hessian contraction against $D^2 f$ is nonnegative. Also ∇g_δ is horizontal in Euclidean coordinates, so

$$\Delta F_\delta = u(g'_\delta)^2 f_{vv} + f_{zz} + (g'_\delta + u g''_\delta) f_v. \quad (16.22)$$

Again the first two terms are nonnegative and the last has absolute value at most $C_a(u + \delta)^{a-1}$. Thus (16.21) also holds for DF_δ , after changing the fixed constant, uniformly in $\varepsilon \in [0, 1]$.

On a fixed compact set,

$$|F_\delta| \leq C, \quad |\nabla F_\delta| + |\nabla_H F_\delta| \leq C_a \{1 + \sqrt{u}(u + \delta)^{a-1}\} \leq C_a(1 + u^{a-1}) \quad (16.23)$$

for $u > 0$, uniformly in δ . Since $1 - a < \beta_*$, (16.3), with $\beta = 1 - a > 0$, controls the integral of the right side against p . In particular, all cutoff product terms in the nonnegative test $\chi(C - F_\delta)$ have uniformly bounded integrals, for any fixed smooth cutoff $\chi \geq 0$. The drift error in (16.21) is controlled by the same moment, since $(u + \delta)^{a-1} \leq u^{a-1}$.

The positive sink sign therefore yields

$$\int \chi p S_\delta : D^2 f(g_\delta(u), z) \leq C_{\chi, a}. \quad (16.24)$$

At every $u > 0$, $g_\delta \rightarrow u^a$, $g'_\delta \rightarrow au^{a-1}$, and the integrands converge to a nonnegative limit. Fatou gives

$$\int \chi p[a^2 u^{2a-1} f_{vv}(u^a, z) + u f_{zz}(u^a, z)] \leq C_{\chi, a}. \quad (16.25)$$

The axis is a Lebesgue null set and causes no exceptional integral.

Weighted transport in the new variable.

Write $q = q_\chi$ as a density in (u, z) . Under $v = u^a$, its pushforward is the compactly supported density

$$\hat{q}(v, z) = \frac{q(u, z)}{au^{a-1}}, \quad u = v^{1/a}, \quad v > 0, \quad (16.26)$$

extended by zero to $v < 0$. It has mass at most one. Equation (16.25) is the convex-test budget

$$\int \hat{q} S_a : H_f \leq C_{\chi, a}, \quad S_a(v, z) = \text{diag}(a^2 u^{2a-1}, u), \quad \det S_a = a^2 u^{2a}. \quad (16.27)$$

For smooth convex f , this is just (16.25). For general globally convex one-Lipschitz f , convolution and Fatou pass to the Alexandrov Hessian: $S_a \geq 0$, is finite at each $v > 0$, and $\hat{q} dv dz$ is absolutely continuous. Neither boundedness nor smoothness of S_a at $v = 0$ is needed for this passage.

The convex transport determinant lemma (Lemma 12.3), applied using $h = \hat{q}^2 \det S_a$, gives

$$\int \hat{q}^2 \det S_a dv dz \leq C_{\chi, a}. \quad (16.28)$$

Explicitly, for $h_N = \min(h, N)$ on a compact support and $M_N = \int h_N > 0$, choose $\det H_f = \pi h_N / M_N$. The matrix arithmetic-geometric mean gives

$$\int \hat{q} S_a : H_f \geq 2\sqrt{\pi} M_N^{-1/2} \int \hat{q} \sqrt{\det S_a} \sqrt{h_N} \geq 2\sqrt{\pi} M_N^{1/2}. \quad (16.29)$$

Monotone convergence proves (16.28). Thus this step does not presume the desired weighted integrability or any boundedness of the coefficient weight.

Since $dv = au^{a-1} du$, (16.26)–(16.28) are exactly

$$\boxed{\int q(u, z)^2 u^{a+1} du dz \leq C_{\chi, a}.} \quad (16.30)$$

Indeed the transformed integrand is

$$\frac{q^2}{a^2 u^{2a-2}} \cdot a^2 u^{2a} \cdot au^{a-1} = a q^2 u^{a+1}. \quad (16.31)$$

Improved tail and reconstruction.

Choose any fixed $0 < \beta < \beta_*$. The inverse moment of q and (16.30) give, for $L \geq 1$, $0 < s \leq 1$,

$$\int_{\{q > L\}} q \leq C s^\beta + C L^{-1} s^{-(a+1)}. \quad (16.32)$$

Choosing $s = L^{-1/(\beta+a+1)}$ yields

$$\int_{\{q > L\}} q \leq C L^{-\beta/(\beta+a+1)}. \quad (16.33)$$

Consequently $q \in L^{1+\theta}$ uniformly for every $0 < \theta < \beta/(\beta + a + 1)$. Letting the available fixed choices approach $\beta \uparrow \beta_*$ and $a \downarrow 1 - \beta_*$, the supremum of these tail exponents is

$$\frac{\beta_*}{\beta_* + (1 - \beta_*) + 1} = \frac{\beta_*}{2}. \quad (16.34)$$

For every strictly smaller θ , choices strictly within both parameter ranges exist, so all constants remain finite. This proves the claimed range (16.5), without an endpoint assertion.

Reconstruction of an axisymmetric density on an arbitrary domain.

Assume the density symmetry in the theorem, and choose χ equal to one in a neighborhood of K . Write $\pi(x, y, z) = (u, z)$, and define

$$W_\chi(u, z) = 2 \int_0^{2\pi} \chi(2\sqrt{u} \cos t, 2\sqrt{u} \sin t, z) dt. \quad (16.35)$$

Extend χ by zero outside Ω . The function W_χ is continuous on the closed half-plane $u \geq 0$, by dominated convergence, including at $u = 0$. It is strictly positive on $\pi(K)$. Indeed, for a point of K off the axis, a nonempty angular arc remains in the neighborhood where $\chi = 1$; at a point on the axis, every angle represents that same point and $W_\chi = 4\pi$. Compactness therefore gives

$$\inf_{\pi(K)} W_\chi = c_K > 0. \quad (16.36)$$

Fubini and the assumed density symmetry give $q_\chi = W_\chi P$ almost everywhere on $u > 0$. For $s = 1 + \theta$, the preceding polar Jacobian identity now gives

$$\begin{aligned} \int_K p^s dx dy dz &\leq 4\pi \int_{\pi(K)} P(u, z)^s du dz \\ &\leq 4\pi c_K^{-s} \int_{\mathbb{R}^2} q_\chi^s du dz \leq C_{K, \theta}. \end{aligned} \quad (16.37)$$

Values or representatives on the axis affect none of these integrals. The elementary inequality $r \log(1 + r) \leq C_\theta(r + r^{1+\theta})$, for $r \geq 0$, proves the entropy assertion in (16.6).

This reconstruction is the sole use of axisymmetry. The radial-central projection estimate alone does not control concentration in the angular variable. □

Corollary 16.3 (Uniform viscosity ball bound). *Fix $K \subset U \Subset \Omega$, with K compact and U open. Under (16.2), for every $1 < d < 1 + \lambda/\Lambda$, there are $C, r_0 > 0$, depending only on the fixed sets, coefficient bounds, and d , such that*

$$\begin{aligned} \int_{B_r(a)} p dg &\leq Cr^d \int_U p dg \quad (a \in K, 0 < r \leq r_0), \\ B_r(a) &= \{g : ((x'^2 + y'^2)^2 + 16z'^2)^{1/4} < r\}, \quad (x', y', z') = a^{-1}g. \end{aligned}$$

The estimate is uniform over every constant $\varepsilon \in [0, 1]$.

Proof. The translated-axis moment proved above is uniform for centers in K , with cutoffs supported in U . Choose $\beta = d/2 < \beta_*$. On $B_r(a)$, $u_a = ((x - a_x)^2 + (y - a_y)^2)/4 < r^2/4$. The moment estimate gives the assertion after dividing p and μ by $\int_U p$ in the open set U . If this mass is zero the assertion is immediate. This normalization also explains the relative factor $\int_U p$; the cutoff argument only uses that local mass. □

16.3 A sharp relative ball estimate and exact stationary powers

Theorem 16.4 (Sharp zero-viscosity relative ball exponent). *Assume (16.2) with $\varepsilon = 0$, and set $d = 1 + 3\lambda/\Lambda$. Whenever $\overline{B_{2R}(a)} \subset \Omega$,*

$$\int_{B_r(a)} p \, dg \leq C(r/R)^d \int_{B_{2R}(a)} p \, dg \quad (0 < r \leq R),$$

where C depends only on λ/Λ . One may take $C = 12^d + (25/4)(\Lambda/\lambda)5^d/(1-2^{-d})$. The exponent is sharp among the stated stationary densities.

Proof. Put $b = 3\lambda/\Lambda$, so $d = 1 + b$.

Exact horizontal Hessian of the gauge.

Write $r_h^2 = x^2 + y^2$ and

$$v = (r_h^2 x - 4yz, r_h^2 y + 4xz). \quad (16.38)$$

Away from the origin direct differentiation gives

$$\nabla_H \rho = \frac{v}{\rho^3}, \quad \Gamma := |\nabla_H \rho|^2 = \frac{r_h^2}{\rho^2} \leq 1, \quad (16.39)$$

$$H\rho = \frac{3r_h^2}{\rho^3} I_2 - \frac{3}{\rho^7} v \otimes v. \quad (16.40)$$

For verification, $H(\rho^4) = 12r_h^2 I_2$; applying the scalar Hessian chain rule to the fourth root gives (16.40). Also $|v|^2 = r_h^2 \rho^4$.

Where $\Gamma > 0$, let $e = \nabla_H \rho / \sqrt{\Gamma}$ and let $e^\perp$ be its quarter-turn. Then

$$H\rho = \frac{3\Gamma}{\rho} e^\perp \otimes e^\perp. \quad (16.41)$$

On the vertical axis away from the origin, both Γ and $H\rho$ vanish, so the formulas below are unaffected by any arbitrary definition of e there. For a smooth one-variable function f ,

$$A : H[f(\rho)] = \Gamma \left(a f''(\rho) + \frac{3c}{\rho} f'(\rho) \right), \quad (16.42)$$

where $a = e^t A e$, $c = (e^\perp)^t A e^\perp$, and $\lambda \leq a, c \leq \Lambda$ almost everywhere for the weighted density. Formula (16.42) holds almost everywhere and extends across that axis in its nonsingular form using (16.40).

All these identities hold for left-translated gauges because the fields are left invariant.

The radial distribution and its flux at the center.

Fix a center and a radius R with $\overline{B_R} \subset \Omega$, and suppress the center in notation. Apply Euclidean coarea to the smooth function ρ away from its origin. Its Euclidean gradient is nonzero there. Define for almost every $t \in (0, R)$

$$m(t) = \int_{\{\rho=t\}} \frac{p \Gamma a}{|\nabla_E \rho|} d\mathcal{H}^2, \quad n(t) = \frac{3}{t} \int_{\{\rho=t\}} \frac{p \Gamma c}{|\nabla_E \rho|} d\mathcal{H}^2. \quad (16.43)$$

Then $m \in L^1(0, R)$, $n \in L^1_{\text{loc}}(0, R)$, both are nonnegative, and

$$n(t) \geq \frac{b}{t} m(t) \quad (16.44)$$

almost everywhere. Moreover

$$\lambda \int_{B_s} p \Gamma \leq \int_0^s m(t) dt \leq \Lambda \int_{B_s} p \Gamma \leq \Lambda \int_{B_s} p. \quad (16.45)$$

Let ν be the pushforward under ρ of $\mu|_{B_R}$, including its possible atom at zero. For every smooth f compactly supported in $[0, R)$ and constant near zero, the function $f(\rho)$ is a legitimate smooth compactly supported test in (16.2). Equations (16.42)–(16.43) give

$$\int_0^R (mf'' + nf') dt = \int_{[0,R)} f d\nu. \quad (16.46)$$

In particular, on the open interval $(0, R)$,

$$(m' - n)' = \nu. \quad (16.47)$$

Thus $F = m' - n$, initially a distribution, has a locally bounded variation representative with $F' = \nu$. Because ν is finite, $F(0+)$ exists and is finite: subtracting the cumulative finite measure from F leaves a constant distribution.

The sign of this boundary value is determined by the original equation, not imposed as an extra boundary condition. Indeed for f as in (16.46), its derivative is supported away from zero, so distributional integration by parts gives

$$\int_0^R (mf'' + nf') = - \int_0^R F f' = F(0+)f(0) + \int_{(0,R)} f d\nu. \quad (16.48)$$

Comparison with (16.46) yields the exact identity

$$F(0+) = \nu(\{0\}) = \mu(\{\text{center}\}) \geq 0. \quad (16.49)$$

Since $F' = \nu \geq 0$, it follows that $F \geq 0$ on $(0, R)$. Also $m' = n + F$ shows that m has a locally absolutely continuous representative. From (16.44),

$$m' \geq \frac{b}{t} m \quad \text{almost everywhere,} \quad \frac{m(t)}{t^b} \text{ is nondecreasing.} \quad (16.50)$$

The representative is nonnegative everywhere since it is continuous and nonnegative almost everywhere. No division by m is needed in (16.50), so zero values cause no difficulty.

Weighted relative mass bound.

For $0 < t < R/2$ and $R/2 < s < R$, (16.50) gives $m(t) \leq (t/s)^b m(s)$. Integrating first over s and then over $0 < t < r \leq R/2$ proves

$$\int_0^r m(t) dt \leq \frac{(r/R)^{b+1}}{1 - 2^{-(b+1)}} \int_{R/2}^R m(s) ds. \quad (16.51)$$

Together with (16.45), this yields

$$\int_{B_r(q)} p \Gamma_q \leq C_w \left(\frac{r}{R}\right)^d \int_{B_R(q)} p, \quad 0 < r \leq R/2, \quad (16.52)$$

where

$$C_w = \frac{\Lambda/\lambda}{1 - 2^{-d}}, \quad \Gamma_q(g) = \frac{|(q^{-1}g)_h|^2}{\rho(q^{-1}g)^2}. \quad (16.53)$$

The value of Γ_q at the center is immaterial for a density.

Removing the angular weight by moving the center.

Suppose $0 < r \leq R/12$, and set

$$q' = q(3r, 0, 0). \quad (16.54)$$

If $g = q(x, y, z) \in B_r(q)$, then

$$|(x, y)| < r, \quad |z| < r^2/4, \quad (16.55)$$

and

$$q'^{-1}g = (x - 3r, y, z - 3ry/2). \quad (16.56)$$

Consequently its horizontal length lies between $2r$ and $4r$, and its central coordinate has magnitude at most $7r^2/4$. Hence

$$\rho(q'^{-1}g)^4 < (4r)^4 + 16(7r^2/4)^2 = 305r^4 < (5r)^4. \quad (16.57)$$

It follows that

$$B_r(q) \subset B_{5r}(q'), \quad \Gamma_{q'} \geq \frac{4}{25} \quad \text{on } B_r(q). \quad (16.58)$$

We also have $B_R(q') \subset B_{2R}(q)$, with strict room at the boundary. Indeed $3r \leq R/4$. For $g = q'(x', y', z') \in \overline{B_R(q')}$, the horizontal coordinate of $q^{-1}g$ has length at most $5R/4$, and its central coordinate has magnitude at most $3R^2/8$. Thus

$$\rho(q^{-1}g)^4 \leq (5R/4)^4 + 16(3R^2/8)^2 = \frac{1201}{256}R^4 < 16R^4. \quad (16.59)$$

In particular the equation is available on a neighborhood of $\overline{B_R(q')}$. Since $5r < R/2$, applying (16.52) at center q' , followed by (16.58)–(16.59), gives

$$\int_{B_r(q)} p \leq \frac{25}{4} C_w 5^d \left(\frac{r}{R}\right)^d \int_{B_{2R}(q)} p. \quad (16.60)$$

For $R/12 < r \leq R$, the trivial inclusion bound gives the claimed relative estimate with constant 12^d . This completes the proof of the claimed relative estimate, including the stated explicit constant. No metric triangle inequality for the gauge was used.

Sharpness follows from the next theorem: its density has homogeneous degree $-\alpha$, with $\alpha = 3 - 3\lambda/\Lambda$, and hence ball mass proportional to $r^{4-\alpha} = r^d$. If $\lambda = \Lambda$, a constant density on a bounded domain gives sharpness of exponent four. Conversely, a nonzero homogeneous stationary density of degree $-\alpha$ must have $\alpha \leq 3 - 3\lambda/\Lambda$, by applying the proved estimate as $r \downarrow 0$. $\square$

Theorem 16.5 (Exact stationary power family). *For each ratio $L = \Lambda/\lambda > 1$, on $\Omega = \{x^2 + y^2 < 1, |z| < 1\}$ there are an exact stationary probability density p and smooth exact stationary probability densities p_δ , with Borel or smooth uniformly elliptic coefficient matrices having the prescribed bounds λ, Λ , $\varepsilon = 0$, and $\mu = 0$, such that, for $K = \{x^2 + y^2 \leq 1/4, |z| \leq 1/2\}$,*

$$s_* = \frac{4L}{3(L-1)}, \quad p \notin L^{s_*}(K), \quad \int_K p_\delta^{s_*} dg \longrightarrow \infty.$$

For each $1 < s < s_$, $\sup_\delta \int_\Omega p_\delta^s < \infty$. In particular these are not entropy counterexamples. At $L = 3$, even qualitative local L^2 membership fails. The exponent describes this family, not a claimed optimal general integrability theorem.*

Proof. Formula and cancellation.

Put

$$\zeta = x + iy = re^{i\theta}, \quad u = r^2/4, \quad D = u - iz, \quad (16.61)$$

$$\eta = (L - 1)/2, \quad \kappa = \frac{\eta}{\eta + 1}, \quad \beta = \frac{\eta(\eta + 2)}{2\eta + 1}, \quad \alpha = 4\beta - 2\eta = \frac{6\eta}{2\eta + 1} < 3. \quad (16.62)$$

Away from the vertical axis define

$$q = u^\eta(u^2 + z^2)^{-\beta}, \quad w = \kappa q e^{-2i\theta} \frac{D}{\bar{D}}, \quad (16.63)$$

$$Q = \begin{pmatrix} q + \Re w & -\Im w \\ -\Im w & q - \Re w \end{pmatrix}. \quad (16.64)$$

Define the functions to be zero on the axis, a null set. Since $|w| = \kappa q$,

$$(1 - \kappa)qI \leq Q \leq (1 + \kappa)qI. \quad (16.65)$$

Thus $A = Q/q$ has eigenvalues $1 \pm \kappa$, with ratio L . Multiplying A by $(\lambda + \Lambda)/2$ realizes any prescribed pair $0 < \lambda < \Lambda$ of that ratio, without changing a zero stationary right side.

Let $Z = X + iY$. The tensor convention gives

$$X^2 Q_{11} + (XY + YX)Q_{12} + Y^2 Q_{22} = \Delta_H q + \Re(Z^2 w). \quad (16.66)$$

For radial functions of (u, z) , $\Delta_H q = q_u + u(q_{uu} + q_{zz})$. With $s = u^2 + z^2$, direct differentiation gives

$$\frac{\Delta_H q}{q} = \frac{\eta^2}{u} + 2\beta(2\beta - 2\eta - 1)\frac{u}{s}. \quad (16.67)$$

To check the other term, put $P = \partial_u + i\partial_z$. For any radial f ,

$$Z^2(\bar{\zeta}^2 f) = 8f + 16uPf + 4u^2 P^2 f. \quad (16.68)$$

Substitute $f = (\kappa/4)u^{\eta-1}D^{1-\beta}\bar{D}^{-1-\beta}$, for which $w = \bar{\zeta}^2 f$. This gives

$$\frac{\Re Z^2 w}{q} = \kappa \left[\frac{\eta(\eta + 1)(u^2 - z^2)}{us} + 4(\beta - 1)(\beta - \eta - 1)\frac{u}{s} \right]. \quad (16.69)$$

At $\kappa = \eta/(\eta + 1)$, their sum factors as

$$\frac{4(2\eta + 1)}{\eta + 1} \left(\beta - \frac{\eta(\eta + 2)}{2\eta + 1} \right) \left(\beta - \frac{\eta + 1}{2} \right) \frac{u}{s} = 0. \quad (16.70)$$

Thus stationarity holds off the axis. The fields are divergence free, so (16.66) has exactly the adjoint interpretation in (16.2).

No hidden axis or point source.

Both q, Q have homogeneous degree $-\alpha$ under $(x, y, z) \mapsto (ax, ay, a^2 z)$, whose volume dimension is four. They are locally integrable because $\alpha < 4$. Let $\mathfrak{D}$ denote the distribution in (16.66).

Away from the origin, $|Q| \leq Cr^{2\eta}$ near the axis. A horizontal cutoff supported in $r < 2a$, equal to one on $r < a$, gives for tests missing the origin

$$|\langle \mathfrak{D}, \varphi \rangle| \leq C a^{-2} \int_{r < 2a} r^{2\eta} dx dy dz \leq C a^{2\eta} \longrightarrow 0. \quad (16.71)$$

Hence $\mathfrak{D}$ is supported at the origin. If $\varphi(0) = 0$, use an intrinsic cutoff at radius a . On its support $|\varphi| \leq C a$ and the horizontal Hessian of the product is bounded by C/a . Homogeneity gives

$$|\langle \mathfrak{D}, \varphi \rangle| \leq C a^{-1} a^{4-\alpha} = C a^{3-\alpha} \longrightarrow 0. \quad (16.72)$$

Therefore $\mathfrak{D} = c \delta_0$ for some constant.

To compute c , take $\varphi = \psi(u^2 + z^2)$, with ψ smooth, compactly supported, and equal to one near zero. Differentiation gives

$$\langle \mathfrak{D}, \varphi \rangle = \int q \{ 6u\psi'(s) + 4(1 + \kappa)us\psi''(s) \}. \quad (16.73)$$

Use $dx dy = 4\pi du$, followed by $u = t \cos \vartheta$, $z = t \sin \vartheta$ on the right half-plane and $s = t^2$. With $C_\eta = \int_{-\pi/2}^{\pi/2} \cos^{\eta+1} \vartheta d\vartheta$ and $\gamma = (2 - \alpha)/4$, the expression becomes

$$2\pi C_\eta [6 - 4(1 + \kappa)(\gamma + 1)] \int_0^\infty s^\gamma \psi'(s) ds = 0, \quad (16.74)$$

because $\alpha = 6\kappa/(1 + \kappa)$. Integration by parts has no endpoint terms, since ψ' is supported away from both endpoints. As $\varphi(0) = 1$, this proves $c = 0$ and establishes exact distributional stationarity.

The second algebraic root $\beta = (\eta + 1)/2$ in (16.70) generally has a nonzero point source; it is not substituted for the selected branch. At $\eta = 1$ the roots coincide and the source vanishes.

Normalization, smoothing, and the endpoint.

Let $M = \int_\Omega q \in (0, \infty)$ and $p = q/M$. Then p has mass one and $\mathcal{L}_A^* p = 0$, with $\mu = 0$ and $\varepsilon = 0$.

For $s > 0$, polar coordinates in the (u, z) half-plane give near the origin

$$\int q^s dx dy dz = 4\pi \left(\int_{-\pi/2}^{\pi/2} \cos^{\eta s} \vartheta d\vartheta \right) \int_0^b t^{1-\alpha s/2} dt. \quad (16.75)$$

The angular factor is finite and positive. The radial integral is finite exactly for $s < 4/\alpha = s_*$ and diverges logarithmically at equality. Away from the origin the density is locally bounded, so this gives the full stated threshold.

Choose nonnegative compactly supported smooth group approximate identities ρ_δ , and set $q_\delta = \rho_\delta * q$, $Q_\delta = \rho_\delta * Q$, with convolution on the left. It preserves tensor positivity and commutes with the fixed left-invariant derivatives acting on the right factor. Thus Q_δ remains exactly stationary, $q_\delta > 0$, and the quotient $A_\delta = Q_\delta/q_\delta$ is smooth with the same bounds. Local L^1 approximate-identity convergence gives $M_\delta = \int_\Omega q_\delta \rightarrow M > 0$, so $p_\delta = q_\delta/M_\delta$ has mass one and converges to p in $L^1(\Omega)$.

For every fixed $1 < s < s_*$, Jensen and Haar invariance bound $\int_\Omega q_\delta^s$ by the finite integral of q^s on a fixed larger neighborhood. Hence p_δ has uniform L^s and entropy bounds. Pointwise convergence away from the axis and Fatou give

$$\int_K p_\delta^{s_*} \longrightarrow \infty. \quad (16.76)$$

At ratio $L = 3$, one has $s_* = 2$: even qualitative local L^2 membership can fail for an exact stationary density, while entropy remains finite. At every finite $L > 1$, the present family has $s_* > 4/3 > 1$ and therefore does not approach an entropy divergence within that fixed ratio. $\square$

16.4 All Euclidean convex tests can have bounded budgets while entropy diverges

Theorem 16.6 (Failure of Euclidean convex-budget coercivity). *For the fixed standard Heisenberg covariance $S = X \otimes X + Y \otimes Y$, there are smooth probability densities p_j , all supported in one fixed compact cube and positive in its interior, with*

$$\sup_{f: \mathbb{R}^3 \rightarrow \mathbb{R} \text{ convex, } \text{Lip}(f) \leq 1} \int p_j S : D_a^2 f \leq C, \quad \int p_j \log(1 + p_j) \rightarrow \infty.$$

Here $D_a^2 f$ denotes the almost-everywhere Alexandrov Hessian. These densities are geometric examples, and no stationary equation is asserted for them.

Proof. **Integrated convex chain-rule lemma.**

Let $Q \subset \mathbb{R}^n$ be a fixed cube and let v be convex on a neighborhood of $\overline{Q}$. In the piecewise quadratic setting used below, put $T = \nabla v$ and $H = D_a^2 v$. For every smooth globally convex 1-Lipschitz u ,

$$\boxed{\int_Q \text{tr}(H(y) D^2 u(T(y))) dy \leq \text{Per}(Q).} \quad (16.77)$$

To prove this, mollify v to obtain smooth convex v_ε . The ordinary chain rule gives

$$\text{div}_y [\nabla u(\nabla v_\varepsilon(y))] = \text{tr}(D^2 v_\varepsilon(y) D^2 u(\nabla v_\varepsilon(y))) \geq 0. \quad (16.78)$$

The divergence theorem and $|\nabla u| \leq 1$ bound its integral over Q by $\text{Per}(Q)$. At every interior point of a region where v equals one quadratic, $\nabla v_\varepsilon \rightarrow T$ and $D^2 v_\varepsilon \rightarrow H$. Such regions cover Q up to a null set in our construction. Fatou proves (16.77).

In particular, if

$$S(T(y)) \leq C_0 H(y) \quad \text{as quadratic forms for a.e. } y \in Q, \quad (16.79)$$

then the pushforward of normalized Lebesgue measure on Q obeys the full smooth convex-test budget

$$\frac{1}{|Q|} \int_Q \text{tr}(S(T(y)) D^2 u(T(y))) dy \leq C_0 \frac{\text{Per}(Q)}{|Q|}. \quad (16.80)$$

This proof also explains the role of the singular Hessian of v : it contributes nonnegative terms when v is regularized, so discarding it preserves the upper bound on the absolutely continuous integrand.

Piecewise quadratic construction.

Fix $Q = (-1, 1)^3$, partition it into a regular grid of subcubes of side $r = 2/N$, and denote the subcube centers by y_j . Assume $N \geq 4$, so $r \leq 1/2$. Set

$$\delta = r^2, \quad h = \frac{1}{2}, \quad n_j = \frac{(y_{j,2}/2, -y_{j,1}/2, 1)}{|(y_{j,2}/2, -y_{j,1}/2, 1)|}, \quad (16.81)$$

$$H_j = hI + (\delta - h)n_j \otimes n_j. \quad (16.82)$$

Thus H_j has eigenvalues h, h, δ , and its small eigendirection is the contact normal at y_j . Define

$$q_j(y) = \frac{|y|^2}{2} - \frac{1}{2}(y - y_j)^T (I - H_j)(y - y_j), \quad v_r(y) = \max_j q_j(y). \quad (16.83)$$

Each q_j is globally δ -strongly convex, so v_r is globally δ -strongly convex. Distinct q_j 's are distinct polynomials: if their Hessians coincide, their linear coefficients differ because $I - H_j$ is invertible and the centers differ. Consequently their tie sets have Lebesgue measure zero. The open active cells

$$C_j = \{y \in Q : q_j(y) > q_k(y) \text{ for all } k \neq j\} \quad (16.84)$$

cover Q up to a null set. On C_j ,

$$D_a^2 v_r = H_j, \quad T_r(y) := \nabla v_r(y) = y_j + H_j(y - y_j). \quad (16.85)$$

Active cells stay close to their centers.

The matrices $I - H_j$ have eigenvalues $1/2, 1/2, 1 - \delta$, hence

$$\frac{1}{2}I \leq I - H_j \leq I. \quad (16.86)$$

For any $y \in Q$, its nearest grid center y_k satisfies $|y - y_k| \leq \sqrt{3}r/2$. If j is active at y , maximizing (16.83) gives

$$\frac{1}{2}|y - y_j|^2 \leq (y - y_j)^T(I - H_j)(y - y_j) \leq (y - y_k)^T(I - H_k)(y - y_k) \leq \frac{3}{4}r^2. \quad (16.87)$$

Thus

$$|y - y_j| \leq \sqrt{3/2}r, \quad |T_r(y) - y_j| \leq \sqrt{3/8}r. \quad (16.88)$$

All the images $T_r(Q)$ therefore lie in a single fixed compact cube, for example $[-2, 2]^3$.

Uniform matrix domination by the active Hessian.

Write $a_1 = X$, $a_2 = Y$. By definition $n_j \cdot a_i(y_j) = 0$. Since the a_i 's are affine,

$$|n_j \cdot a_i(T_r(y))| \leq \frac{1}{2}|T_r(y) - y_j| \leq Cr. \quad (16.89)$$

The horizontal vectors are uniformly bounded on the fixed image cube. Since

$$H_j^{-1} = h^{-1}I + (\delta^{-1} - h^{-1})n_j \otimes n_j, \quad (16.90)$$

we get, uniformly in $r, j, y \in C_j$,

$$\text{tr}(H_j^{-1}S(T_r(y))) = \sum_{i=1}^2 [h^{-1}|a_i(T_r(y))|^2 + (\delta^{-1} - h^{-1})|n_j \cdot a_i(T_r(y))|^2] \leq C_0, \quad (16.91)$$

because $\delta = r^2$. The positive matrix $H_j^{-1/2}S(T_r(y))H_j^{-1/2}$ has operator norm at most its trace. Hence (16.91) implies

$$S(T_r(y)) \leq C_0 H_j. \quad (16.92)$$

Combining (16.92) with the lemma proves (16.80), uniformly in r .

Pushforward density and divergent entropy.

Strong convexity implies, at differentiability points,

$$(T_r(y) - T_r(z)) \cdot (y - z) \geq \delta|y - z|^2. \quad (16.93)$$

Therefore T_r is injective on the union of the active cells. Its affine restrictions in (16.85) have disjoint images and all have the same determinant

$$d_r = \det H_j = h^2 \delta = \frac{r^2}{4}. \quad (16.94)$$

Let

$$E_r = \bigcup_j T_r(C_j). \quad (16.95)$$

The pushforward of $\mathbf{1}_Q dy/|Q|$ has density

$$\boxed{p_r(x) = \frac{1}{|Q|d_r} \mathbf{1}_{E_r}(x), \quad |E_r| = |Q|d_r.} \quad (16.96)$$

It follows exactly that

$$\int p_r \log p_r = \log \frac{1}{|Q|d_r} \rightarrow \infty, \quad \int p_r \log(1 + p_r) = \log \left(1 + \frac{1}{|Q|d_r} \right) \rightarrow \infty. \quad (16.97)$$

For smooth convex 1-Lipschitz u , the pushforward formula and (16.80) give the required uniform budget. For an arbitrary globally convex 1-Lipschitz u , convolve u with a smooth compactly supported approximate identity. The resulting u_η remain convex and 1-Lipschitz. Their Hessians converge to the Alexandrov Hessian $D_a^2 u$ at almost every Lebesgue point of the Hessian's absolutely continuous part, with the singular part of the Hessian convolution tending to zero almost everywhere. Since p_r is absolutely continuous, Fatou gives the same budget for u .

Thus (16.96) proves the full stated counterexample.

Smooth strictly positive densities are also possible.

The preceding example is bounded for each fixed r , but is not smooth. This can be corrected without losing the uniform budget.

For the fixed affine Heisenberg fields,

$$S(x+z) \leq 2S(x) + \frac{|z|^2}{2} I. \quad (16.98)$$

Let $p_{r,\varepsilon} = p_r * \rho_\varepsilon$, where ρ_ε is a smooth nonnegative unit-mass kernel supported in B_ε , with $\varepsilon \leq r$. On a fixed cube K containing all these supports, (16.98) gives, for smooth convex 1-Lipschitz u ,

$$\begin{aligned} \int p_{r,\varepsilon} S : D^2 u &\leq 2B(p_r) + \frac{\varepsilon^2}{2} \|p_r\|_\infty \int_K \Delta u \\ &\leq 2B(p_r) + \frac{\varepsilon^2}{2} \|p_r\|_\infty \text{Per}(K). \end{aligned} \quad (16.99)$$

The first term uses the original budget on the translated tests $u(\cdot+z)$. Since $\|p_r\|_\infty \asymp r^{-2}$, the last expression is uniformly bounded for $\varepsilon \leq r$.

For each fixed r , compact support, boundedness, and approximate-identity convergence imply

$$\int p_{r,\varepsilon} \log p_{r,\varepsilon} \rightarrow \int p_r \log p_r \quad \text{as } \varepsilon \downarrow 0. \quad (16.100)$$

Choose $\varepsilon_r \leq r$ sufficiently small to retain divergent entropy. Finally let b be a fixed smooth probability density supported in K , positive at every point of $\text{int } K$, and set

$$\tilde{p}_r = \frac{1}{2} p_{r,\varepsilon_r} + \frac{1}{2} b. \quad (16.101)$$

It is smooth, positive in $\text{int } K$, supported in K , and has a uniformly bounded convex-test budget by linearity. Its entropy still diverges because

$$\int \tilde{p}_r \log \tilde{p}_r \geq \frac{1}{2} \int p_{r,\varepsilon_r} \log p_{r,\varepsilon_r} + \frac{1}{2} \int b \log b - \log 2. \quad (16.102)$$

The same arbitrary-convex-test extension by mollification and Fatou applies. $\square$

16.5 The stationary equation excludes thin microplates

Proposition 16.7 (Relative off-axis slab and microplate obstruction). *Let the ambient domain contain a fixed neighborhood of $[-1, 1]^3$, and let p satisfy the stationary weak equation in (16.2), with finite total mass M in place of the bound one. For $a = (a_x, a_y, a_z) \in [-1, 1]^3$, write*

$$u = x - a_x, \quad v = y - a_y, \quad \ell_a = z - a_z + \frac{1}{2}a_y(x - a_x) - \frac{1}{2}a_x(y - a_y),$$

$$B(a, r) = \{u^2 + v^2 < 4r^2, |\ell_a| < 2r^2\}, \quad F(a, r, \delta) = \{\kappa r \leq \sqrt{u^2 + v^2} \leq r, |\ell_a| \leq \delta r^2\}.$$

For every fixed $0 < \kappa < 1$, all sufficiently small $r > 0$, and every $0 < \delta \leq 1$,

$$\int_{F(a, r, \delta)} p \leq C\delta \int_{B(a, r)} p. \quad (16.103)$$

The constants are uniform over a , A , and $\varepsilon \in [0, 1]$. For the nonsmooth probabilities p_r constructed in the proof of Theorem 16.6, every stationary density dominating αp_r , with fixed $\alpha > 0$, must satisfy $M \geq c\alpha/r$. More generally, if η_σ is a nonnegative smooth unit-mass Euclidean mollifier supported in the radius- σ ball and $\sigma = o(r^2)$, then

$$p \geq \alpha(p_r * \eta_\sigma) \implies M \geq \frac{c\alpha}{r + \sigma/r^2}. \quad (16.104)$$

Thus arbitrary nonnegative added backgrounds cannot make these sufficiently thin microplates stationary while retaining a bounded total mass and one fixed ellipticity ratio.

Proof. Set $D = \mathcal{L}_A + \varepsilon\Delta$ and $\Gamma_D(f, g) = \nabla_H f^t A \nabla_H g + \varepsilon \nabla f \cdot \nabla g$. The affine function ℓ_a satisfies

$$X\ell_a = -v/2, \quad Y\ell_a = u/2, \quad X^2\ell_a = Y^2\ell_a = (XY + YX)\ell_a = \Delta\ell_a = 0.$$

Choose a nonnegative smooth cutoff χ , equal to one for horizontal radius at most r and $|\ell_a| \leq r^2$, supported in $B(a, r)$. Fixed cutoff profiles give

$$|\nabla_H \chi| \leq C/r, \quad |D\chi| \leq C(r^{-2} + \varepsilon r^{-4}), \quad |\Gamma_D(\chi, \ell_a)| \leq C(1 + \varepsilon r^{-2}).$$

Indeed horizontal derivatives of ℓ_a have size $O(r)$ on the support, while its Euclidean gradient is uniformly bounded and has z -component one. Let G be even, smooth, nonnegative, equal to one on $[-1, 1]$, supported in $(-2, 2)$, and at most one. Put

$$g(s) = G(s/(\delta r^2)), \quad h'(s) = \int_0^s g(t) dt, \quad h(s) = \int_0^s h'(t) dt.$$

Then $h \geq 0$, $h'' = g$, $|h'| \leq C\delta r^2$, and $|h| \leq C\delta r^4$ on $|s| \leq 2r^2$. Take $R_0 = C_0\delta r^4$ so that $\chi(R_0 - h(\ell_a)) \geq 0$. Its weak equation and the sink sign give

$$\int \chi g(\ell_a) \Gamma_D(\ell_a, \ell_a) p \leq C\delta(r^2 + \varepsilon) \int_{B(a, r)} p.$$

On $F(a, r, \delta)$, the left-hand integrand is at least $c_{\lambda, \kappa}(r^2 + \varepsilon)p$, since $\Gamma_D(\ell_a, \ell_a) \geq \lambda \kappa^2 r^2/4 + \varepsilon$. Cancellation proves (16.103) with no relation between r and ε .

For the microplate construction, let a_j be the grid centers and H_j its matrices. Fix $c = 1/10$. Each ball of radius cr around a_j lies in its active cell: the loss of its quadratic there is at most $c^2 r^2$, while every competing grid center is at distance at least $(1 - c)r$, with loss at least $(1 - c)^2 r^2/2$. The balls lie in $(-1, 1)^3$. Inside them take

$$D_j = a_j + \{\xi : cr/2 \leq |\xi_h| \leq 3cr/4, |\xi_z| \leq cr/10\}.$$

Their volume is βr^3 , where $\beta = \pi c^3/16 > 0$. At these centers the horizontal part of the unit contact normal satisfies $|n_{j,h}|^2 \leq 1/3$. Thus the horizontal block of H_j is at least $I_2/3$, and its off-diagonal horizontal-to-vertical column has norm at most $1/4$. Therefore

$$|(H_j \xi)_h| \geq \frac{1}{3}|\xi_h| - \frac{1}{4}|\xi_z| \geq 17cr/120, \quad |H_j \xi| < cr/2,$$

$$|\ell_{a_j}(a_j + H_j \xi)| = |\nabla \ell_{a_j}| r^2 |n_j \cdot \xi| \leq \sqrt{3/2} cr^3 < r^3.$$

Choose $\kappa = c/10$. The affine images $T_r(D_j)$ lie in $F_j = F(a_j, r, r)$. Since p_r is the pushforward of normalized Lebesgue measure on $(-1, 1)^3$, whose volume is eight, and the number of centers is $8/r^3$,

$$\sum_j \int_{F_j} p_r \geq \beta.$$

The boxes $B_j = B(a_j, r)$ have bounded overlap. At a point there are only a fixed number of horizontal grid pairs within radius $2r$; for each such pair, $|\ell_{a_j}| < 2r^2$ restricts the vertical grid coordinate, of spacing r , to an interval of length $4r^2$, containing at most three choices when $r \leq 1/2$. Summing (16.103) gives $\sum_j \int_{F_j} p \leq CrM$, and hence $M \geq \alpha\beta/(Cr)$ whenever $p \geq \alpha p_r$.

For $\sigma \leq r^3/100$, the displayed horizontal and central margins give $T_r(D_j) + B_\sigma \subset F_j$. Fubini preserves the mass lower bound for $p_r * \eta_\sigma$. If merely $\sigma = o(r^2)$, keep the same horizontal annulus and enlarge the central slab width to $C_1(r^3 + \sigma)$. Its parameter in (16.103) is $\delta = C_1(r + \sigma/r^2) \rightarrow 0$; the horizontal margins persist since $\sigma = o(r)$. The same overlap argument proves (16.104). The convex-budget construction permits choosing these arbitrarily fine mollification scales without losing its entropy divergence. $\square$

16.6 Relative scalar slab and ball bounds alone do not control entropy

Theorem 16.8 (Geometric counterexamples with exponent arbitrarily close to four). *The ambient group in this theorem remains $\mathbb{H}^1$. For every real $d < 4$, there exist a singular compactly supported probability measure σ , smooth compactly supported probability densities p_j , and constants $C_H, C_W, C_B < \infty$, such that $p_j dg \rightharpoonup \sigma$, all supports lie in one fixed compact set, and $\int p_j \log(1 + p_j) \rightarrow \infty$. The same measures $\tau = \sigma$ and $\tau = p_j dg$ satisfy all the following inequalities, with constants independent of j . Put $Q_R = \{|x| < R, |y| < R, |z| < R^2\}$, $E = q\mathcal{R}Q_R$, and $E^+ = q\mathcal{R}Q_{2R}$, where $q \in \mathbb{H}^1$, $0 < R \leq 1$, and $\mathcal{R}$ is any horizontal rotation. Write $(x', y', z') = q^{-1}g$. For every horizontal unit vector e , every $s \in \mathbb{R}$, and all $0 < a, \delta \leq 1$,*

$$\tau(E \cap \{|e \cdot (x', y') - s| < aR\}) \leq C_H a \tau(E^+), \quad (16.105)$$

$$\int_{E \cap \{|z' - s| < \delta R^2\}} (x'^2 + y'^2) d\tau \leq C_W \delta R^2 \tau(E^+), \quad (16.106)$$

$$\tau(E \cap \{|z' - s| < \delta R^2\} \cap \{x'^2 + y'^2 \geq a^2 R^2\}) \leq C_W \delta a^{-2} \tau(E^+), \quad (16.107)$$

$$\tau(E \cap \{|z' - s| < \delta R^2\}) \leq (C_H + C_W) \delta^{1/3} \tau(E^+). \quad (16.108)$$

Moreover, for every $q \in \mathbb{H}^1$ and $0 < r \leq R \leq 1$,

$$\tau(B_r(q)) \leq C_B(r/R)^d \tau(B_{2R}(q)). \quad (16.109)$$

No stationary PDE, coefficient tensor, Euclidean convex-test budget, or full horizontally convex-test budget is asserted for these examples. The constants need not remain bounded as $d \uparrow 4$.

Proof. Choose an integer $N \geq 2$, put $M = N^2 - 1$, $\alpha = \log M / (2 \log N) > 1/2$, and $D = 2 + \log(N^2 - 1) / \log N > d$. Such a choice exists because $D \rightarrow 4$. We prove the ball estimate with exponent D , which implies (16.109).

The finite-map probability measure.

Let $a_0 = N^{-2}$, and let the M vertical translations be

$$t_w = \frac{1 - a_0}{M - 1} w, \quad w = 0, \dots, M - 1. \quad (16.110)$$

Use the $N^2 M$ equally weighted maps

$$F_{u,v,w}(g) = (u/N, v/N, t_w) \delta_{1/N} g, \quad u, v \in \{0, \dots, N - 1\}. \quad (16.111)$$

An explicit construction avoids any existence assumption for an invariant measure. Take independent digits U_i, V_i , each uniform on $\{0, \dots, N - 1\}$, and independent digits W_i , uniform on $\{0, \dots, M - 1\}$. Define

$$X = \sum_{i \geq 1} U_i N^{-i}, \quad Y = \sum_{i \geq 1} V_i N^{-i}, \quad (16.112)$$

$$T = \sum_{i \geq 1} t_{W_i} N^{-2(i-1)}, \quad \mathcal{A} = \frac{1}{2} \sum_{i < k} (U_i V_k - V_i U_k) N^{-(i+k)}, \quad Z = T + \mathcal{A}. \quad (16.113)$$

All these series converge absolutely, uniformly over digit sequences. Let σ be the law of (X, Y, Z) , and let K_0 be the image of the compact digit-sequence space.

Splitting prefix and tail digits proves (16.111). The horizontal marginal is uniform on $[0, 1]^2$. The variable T lies in $[0, 1]$. Also

$$|\mathcal{A}| \leq \frac{(N - 1)^2}{2} \sum_{i < k} N^{-(i+k)} = \frac{1}{2(N + 1)} \leq \frac{1}{6}. \quad (16.114)$$

In particular the fixed, deliberately coarse bound

$$K_0 \subset [0, 1]^2 \times [-1/3, 4/3] \quad (16.115)$$

holds for every N . The support of σ equals K_0 : every finite prefix has positive probability, and its geometric diameter tends uniformly to zero.

A level- k symbolic cylinder has probability

$$(N^2 M)^{-k} = r^D, \quad r = N^{-k}, \quad (16.116)$$

and its conditional measure is the image of σ under

$$\xi \mapsto g_\omega \delta_r \xi. \quad (16.117)$$

Its horizontal marginal is uniform on a square of side r . The prefix center has horizontal coordinates in $[0, 1]^2$. Symbolic cylinders are disjoint events; geometric disjointness is not needed below.

The vertical noise and its interval bound.

The law ν of T is the equally weighted measure for the M interval maps

$$t \mapsto a_0 t + t_w. \quad (16.118)$$

Their intervals of length a_0 are separated by a positive gap

$$\frac{1 - a_0}{M - 1} - a_0 = \frac{a_0}{M - 1}. \quad (16.119)$$

At level k , basic intervals have length $b = a_0^k$, mass $M^{-k} = b^\alpha$, and pairwise disjoint interiors (in fact positive separating gaps).

For every interval I of length ℓ ,

$$\nu(I) \leq F_N \ell^\alpha, \quad F_N = N^2 + 2. \quad (16.120)$$

For $0 < \ell < 1$, choose $b \leq \ell < N^2 b$. Since distinct level- k basic intervals have disjoint interiors and length b , an interval of length less than $N^2 b$ meets at most $N^2 + 2$ of them. Each has mass $b^\alpha \leq \ell^\alpha$. For $\ell \geq 1$, total mass one proves the same bound. The exponent is greater than one half because $M > N$.

Exact N -ary jumps of the area term.

Fix $Y = \sum_i v_i N^{-i}$, away from the null set of ambiguous expansions. Rearrangement of the absolutely convergent area series gives

$$\mathcal{A}(X, Y) = -\frac{Y}{2}X + \beta_Y(X), \quad \beta_Y(X) = \sum_i U_i d_i(Y), \quad (16.121)$$

where

$$d_i(Y) = \frac{v_i}{2} N^{-2i} + \sum_{k>i} v_k N^{-(i+k)}. \quad (16.122)$$

At every N -ary point of generation i , namely j/N^i with j not divisible by N , the jump is

$$\begin{aligned} J_i(Y) &= d_i(Y) - (N-1) \sum_{k>i} d_k(Y) \\ &= \frac{v_i}{2} N^{-2i} + \frac{N+1}{2} \sum_{k>i} v_k N^{-2k}. \end{aligned} \quad (16.123)$$

There are $(N-1)N^{i-1}$ such points, and

$$0 \leq J_i(Y) \leq \frac{N}{2} N^{-2i}. \quad (16.124)$$

Comparing the first distinct digit of two expansions shows that β_Y is nondecreasing: a unit increase at digit i , even with the largest possible decrease in every later digit, changes its value by at least J_i .

Furthermore,

$$\sum_{i \geq 1} (N-1)N^{i-1} J_i(Y) = (N-1) \sum_i d_i(Y) = \beta_Y(1-) - \beta_Y(0+). \quad (16.125)$$

The identity follows by interchanging the absolutely convergent sums in the first expression of (16.123). Thus all its variation consists of these jumps. There is no continuous part in its variation measure. Choosing right-continuous representatives, finite sums of the jumps converge uniformly, with omitted variation at most

$$\sum_{i>k} (N-1)N^{i-1} \frac{N}{2} N^{-2i} = \frac{1}{2} N^{-k}. \quad (16.126)$$

The values at the countable ambiguous expansion points have no effect on any integral over X . Symmetrically,

$$\mathcal{A}(X, Y) = \frac{X}{2}Y - \beta_X(Y), \quad (16.127)$$

with the same jump bounds.

Projection estimate averaged against the vertical noise.

Put

$$S_N = \left(\frac{N}{2}\right)^\alpha \frac{N-1}{M-N}, \quad C_P = 1 + F_N S_N. \quad (16.128)$$

For either sign in $f(x) = cx \pm \beta(x) + b$, where β is a function just described, and for every nonnegative smooth compactly supported g ,

$$\int \nu(dt) \int_0^1 c^2 g(f(x) + t) dx \leq C_P |c| \int_{\mathbb{R}} g. \quad (16.129)$$

If $c = 0$, this is immediate. For a finite jump sum and $c \neq 0$, choose $H' = g$, $0 \leq H \leq G := \int g$. Integrating on the affine pieces gives

$$c \int_0^1 g(f(x) + t) dx = H(f(1-) + t) - H(f(0+) + t) - \sum_a [H(f(a+) + t) - H(f(a-) + t)]. \quad (16.130)$$

The endpoint term has magnitude at most G . After averaging over t , a jump of absolute size J contributes at most $F_N J^\alpha G$, by Fubini and (16.120). The entire jump contribution is finite, since

$$\sum_{i \geq 1} (N-1) N^{i-1} J_i^\alpha \leq \left(\frac{N}{2}\right)^\alpha \frac{N-1}{N} \sum_{i \geq 1} (N/M)^i = S_N. \quad (16.131)$$

Taking absolute values, then multiplying by $|c|$, proves (16.129). Uniform convergence of the finite jump sums passes this identity to the original β . This argument covers both jump signs.

For $q_h = (q_x, q_y)$, define

$$L_q(X, Y, Z) = Z - \frac{1}{2}(q_x Y - q_y X). \quad (16.132)$$

Conditioning on Y , (16.121) represents this variable as

$$T + \beta_Y(X) + cX - \frac{1}{2}q_x Y, \quad c = (q_y - Y)/2. \quad (16.133)$$

Using (16.129) and approximating the indicator of an interval of radius h , then using $(Y - q_y)^2 = 4c^2$, gives

$$\int (Y - q_y)^2 \mathbf{1}_{\{|L_q - s| < h\}} d\sigma \leq 4C_P h(1 + |q_y|). \quad (16.134)$$

Conditioning on X and using (16.127) gives the other term. Therefore, for every center, real s , and $h > 0$,

$$\int [(X - q_x)^2 + (Y - q_y)^2] \mathbf{1}_{\{|L_q - s| < h\}} d\sigma \leq 4C_P h(2 + |q_x| + |q_y|). \quad (16.135)$$

The central coordinate of q can be absorbed into the arbitrary level s .

Cylinder localization proves all relative slab inequalities.

Fix $E = q\mathcal{R}Q_R$, $0 < R \leq 1$, and choose a cylinder scale

$$\frac{R}{4N} < r = N^{-k} \leq \frac{R}{4}. \quad (16.136)$$

Any two points of K_0 have relative horizontal components of magnitude at most one and relative central component of magnitude at most

$$5/3 + 1/2 = 13/6 < 3. \quad (16.137)$$

Thus two points in a cylinder have relative horizontal length at most $\sqrt{2}r$ and central displacement at most $3r^2$. In a rotated coordinate frame, each horizontal component is at most $\sqrt{2}r$.

Every cylinder image meeting E is consequently contained in E^+ . Multiplication of a point with horizontal rotated components bounded by R , and central coordinate bounded by R^2 , by such a displacement gives horizontal components less than $R + \sqrt{2}r < 2R$, and central magnitude at most

$$R^2 + 3r^2 + Rr < 4R^2. \quad (16.138)$$

For a selected cylinder let $q_\omega = \delta_{1/r}(g_\omega^{-1}q)$. Since it meets the smaller rotated box and its normalized horizontal coordinates are in $[0, 1]^2$,

$$|(q_\omega)_x|, |(q_\omega)_y| \leq 1 + \sqrt{2}R/r < 1 + 4\sqrt{2}N =: L_N. \quad (16.139)$$

The exact identity

$$q^{-1}(g_\omega \delta_r \xi) = \delta_r(q_\omega^{-1} \xi) \quad (16.140)$$

shows that the horizontal weight scales by r^2 , while a central interval of radius δR^2 scales to one of radius $h = \delta R^2/r^2$. Applying (16.135) to the whole selected cylinder bounds its weighted contribution by its symbolic probability times

$$4C_P(2 + 2L_N)\delta R^2. \quad (16.141)$$

Summing selected disjoint symbolic events, whose images all lie in E^+ , proves (16.106) with

$$C_W = 4C_P(2 + 2L_N). \quad (16.142)$$

For (16.105), a uniform horizontal square of side r has projection density at most $\sqrt{2}/r$ in any unit direction. A strip of width $2aR$ has conditional probability at most

$$2\sqrt{2}aR/r \leq 8\sqrt{2}Na. \quad (16.143)$$

Summing the same cylinders proves (16.105) with $C_H = 8\sqrt{2}N$. All arguments allow every horizontal rotation. Inequality (16.107) follows by dividing (16.106) by a^2R^2 . For (16.108), split the central slab into horizontal radius less than aR and its complement, apply (16.105) to a containing centered horizontal strip and (16.107) to the complement, and set $a = \delta^{1/3}$.

Upper and lower ball masses.

For any center q , the horizontal projection of $B_R(q)$ is a disk of radius R . For each fixed horizontal point, its vertical fiber is contained in an interval of length $R^2/2$. Conditional on X, Y , the vertical law is a translate of ν . Thus (16.120) and the uniform horizontal marginal imply

$$\sigma(B_R(q)) \leq C_F R^D, \quad C_F = \pi F_N 2^{-\alpha}, \quad (16.144)$$

for all $R > 0$.

There is also a uniform lower bound at each support point. Indeed, for any two points of K_0 , (16.115) gives horizontal difference of length at most $\sqrt{2}$ and relative central coordinate of magnitude at most $13/6$. Their relative gauge therefore satisfies

$$\rho^4 \leq 4 + 16(13/6)^2 = \frac{712}{9} < 81. \quad (16.145)$$

Thus a cylinder of scale r has gauge diameter less than $3r$.

Fix $g \in K_0$ and $0 < R \leq 1$. Choose a coding of g , and choose its cylinder scale so that

$$\frac{R}{3N} \leq r = N^{-k} < \frac{R}{3}. \quad (16.146)$$

The entire cylinder lies inside $B_R(g)$, and its symbolic probability is r^D . Hence

$$\sigma(B_R(g)) \geq (3N)^{-D} R^D \quad (g \in K_0, 0 < R \leq 1). \quad (16.147)$$

Possible geometric overlaps only increase this mass; they do not affect the lower bound.

To obtain the relative estimate for an arbitrary center, suppose first that $0 < r \leq R/2$ and the numerator is nonzero. Choose $g \in K_0 \cap B_r(q)$. Then

$$B_R(g) \subset B_{2R}(q). \quad (16.148)$$

Here is an explicit check without invoking a metric triangle inequality. Write $q^{-1}g = (a, c)$ with horizontal vector a and central scalar c ; then $|a| \leq R/2$, $|c| \leq R^2/16$. For $g^{-1}t = (b, e)$ with $t \in B_R(g)$, one has $|b| \leq R$, $|e| \leq R^2/4$. Their product has horizontal length at most $3R/2$ and central magnitude at most $9R^2/16$. Its gauge fourth power is at most

$$(3R/2)^4 + 16(9R^2/16)^2 = \frac{81}{8}R^4 < 16R^4, \quad (16.149)$$

which proves (16.148). Combining (16.144)–(16.148) proves (16.109) for $r \leq R/2$, with constant $C_F(3N)^D$. For $r > R/2$, the inclusion of the smaller ball in the outer ball proves it with constant 2^D . Thus one may take

$$C_B = C_F(3N)^D + 2^D. \quad (16.150)$$

Singularity and smooth approximation.

There are $(N^2M)^j$ level- j cylinders. Cover each by the corresponding translate of the dilation of the bounding box in (16.115), of volume $(5/3)N^{-4j}$. Consequently

$$|K_0| \leq \frac{5}{3}(M/N^2)^j \rightarrow 0. \quad (16.151)$$

Thus σ is singular with respect to Lebesgue measure.

Choose a nonnegative smooth unit-mass density χ supported in Q_1 , and put

$$\chi_\varepsilon(g) = \varepsilon^{-4}\chi(\delta_{1/\varepsilon}g), \quad p_\varepsilon = \chi_\varepsilon * \sigma, \quad (16.152)$$

where the convolution is on the left. Equivalently, p_ε is the density of HG for independent H with density χ_ε and G with law σ .

These are smooth compactly supported probability densities. Smoothness follows from the kernel formula

$$p_\varepsilon(g) = \int \chi_\varepsilon(gk^{-1}) d\sigma(k). \quad (16.153)$$

They converge weakly to σ as $\varepsilon \downarrow 0$, by continuity of multiplication on the fixed compact sets.

Every displayed relative inequality is preserved with its identical constant. Indeed, under the fixed left translation $g = h\xi$, a center q becomes $h^{-1}q$, and

$$q^{-1}g = (h^{-1}q)^{-1}\xi. \quad (16.154)$$

Thus relative coordinates, rotations, slab levels, horizontal weights, and all gauge balls retain their exact form. Apply each inequality to the translate and average over h . This proves (16.105)–(16.108) and (16.109) for every $p_\varepsilon dg$.

Explicit entropy divergence.

Set $r_j = N^{-j}$, $\varepsilon_j = r_j^2$, and $p_j = p_{\varepsilon_j}$. All supports lie in one fixed compact box, for example

$$[-2, 3]^2 \times [-5, 5], \quad (16.155)$$

independently of j (and this coarse box works for every N).

For a prefix center g_ω , its horizontal coordinates lie in $[0, 1]^2$. If $h \in Q_{r_j^2}$, the conjugate $g_\omega^{-1}hg_\omega$ has horizontal components at most r_j^2 and central component at most

$$r_j^4 + 2r_j^2 \leq 3r_j^2. \quad (16.156)$$

Multiplying it by a point of $\delta_{r_j}K_0$, using (16.115), gives a point in

$$[-2r_j, 2r_j]^2 \times [-6r_j^2, 6r_j^2]. \quad (16.157)$$

Indeed its central magnitude is at most $3r_j^2 + (4/3)r_j^2 + r_j^3 < 6r_j^2$. This box has volume $192r_j^4$. Left translation preserves volume, so

$$|\text{supp } p_j| \leq 192(N^2M)^j N^{-4j} = 192(M/N^2)^j. \quad (16.158)$$

Jensen's inequality for a probability density on this support yields

$$\boxed{\int p_j \log(1 + p_j) \geq \int p_j \log p_j \geq j \log(N^2/M) - \log 192 \rightarrow \infty.} \quad (16.159)$$

All constants in the relative inequalities remain fixed while j tends to infinity. □

Corollary 16.9 (The explicit twelve-map example). *For $N = 2$, the construction uses exactly the twelve maps*

$$g \mapsto (u/2, v/2, 3w/8) \delta_{1/2}g, \quad u, v \in \{0, 1\}, \quad w \in \{0, 1, 2\}.$$

The resulting singular probability and its smooth approximations satisfy (16.105)–(16.108) with $C_H = 16\sqrt{2}$, $C_W = 560$, and admit smooth approximations whose entropies obey

$$\int p_j \log(1 + p_j) \geq j \log(4/3) - \log 192.$$

They also satisfy the relative ball inequality with exponent $D = 2 + \log 3 / \log 2$.

Proof. The three first-generation vertical intervals have length $1/4$, gap $1/8$, and equal mass $1/3$. Thus the vertical interval estimate in the proof improves to $\nu(I) \leq 4|I|^{\log 3 / \log 4}$: an interval of length $b \leq |I| < 4b$ meets at most four basic intervals of length b . The sum of the jump powers is at most one, so the projection constant becomes $C_P = 5$. Use cylinder scale $R/8 < r \leq R/4$. Each normalized horizontal center has both coordinates of absolute value less than $1 + 8\sqrt{2} < 13$. The weighted projection bound is then $4 \cdot 5(2 + 13 + 13)\delta R^2 = 560\delta R^2$, and the horizontal bound is $16\sqrt{2}a$. The support-volume estimate is $192(3/4)^j$, which gives the entropy bound. The general ball proof already applies to $N = 2$. □

Remark 16.10. For any fixed $\Lambda/\lambda > 1$, the PDE ball exponent $1 + 3\lambda/\Lambda$ is below four. Choosing D above this exponent gives one entropy-divergent geometric sequence satisfying both that exponent and every displayed scalar slab family. This establishes insufficiency with finite uniform constants, and does not identify those constants with optimal PDE constants. The convex-budget construction is a separate sequence; no conjunction of its convex budget with the IFS bounds is claimed.

16.7 A transmission condition excludes the twelve-map measure from the PDE

Proposition 16.11 (Tested normal-stress continuity). *In coordinates (x, y, w) , let $X = \partial_x$, $Y = \partial_y + x\partial_w$. Suppose locally $\rho = dx \rho_x(dy, dw)$ is a nonnegative measure whose conditional masses on each fixed compact (y, w) -set are uniformly bounded on compact x -intervals. Suppose $Q = A\rho$, where A is Borel symmetric and $\lambda I \leq A \leq \Lambda I$ ρ -almost everywhere. If $\mathcal{L}_A^* \rho = \mu$ for a locally finite signed measure μ , then for each $\psi \in C_c^\infty(\mathbb{R}^2)$, $F_\psi(x) = \int \psi dQ_{11,x}$ belongs to $W_{\text{loc}}^{1,\infty}$. If ρ_x has weak one-sided limits ρ_-, ρ_+ at an interior plane, then*

$$\lambda \rho_- \leq \Lambda \rho_+, \quad \lambda \rho_+ \leq \Lambda \rho_-. \quad (16.160)$$

The assertion remains valid for the standard-coordinate operator $\mathcal{L}_A + \varepsilon \Delta$, after the change $w = z + xy/2$, with $F_\psi = \int \psi d(Q_{11,x} + \varepsilon \rho_x)$ and constants $\lambda + \varepsilon, \Lambda + \varepsilon$ in (16.160). For admissible stationary measures with a mass bound and nonnegative sinks, the conditional-mass premise and uniform local bounds for each tested Lipschitz norm follow from the equation.

Proof. For zero viscosity set

$$\begin{aligned} F(x) &= \int \psi dQ_{11,x}, & G(x) &= 2 \int (\psi_y + x\psi_w) dQ_{12,x}, \\ H(x) &= \int \psi_w dQ_{12,x} + \int (\psi_{yy} + 2x\psi_{yw} + x^2\psi_{ww}) dQ_{22,x}. \end{aligned}$$

All three functions are locally bounded. Direct differentiation of $\eta(x)\psi(y, w)$ gives

$$\begin{aligned} X^2(\eta\psi) &= \eta''\psi, & (XY + YX)(\eta\psi) &= 2\eta'(\psi_y + x\psi_w) + \eta\psi_w, \\ Y^2(\eta\psi) &= \eta(\psi_{yy} + 2x\psi_{yw} + x^2\psi_{ww}). \end{aligned}$$

The distributional identity is therefore $F'' - G' + H = \mu_\psi$, where μ_ψ is the sink paired with ψ . On a compact interval choose a bounded-variation primitive B of $\mu_\psi - Hdx$. Then $(F' - G - B)' = 0$, hence $F' = G + B + c$ is bounded and F has a Lipschitz representative. For nonnegative ψ , ellipticity bounds this representative between $\lambda\rho_x(\psi)$ and $\Lambda\rho_x(\psi)$ almost everywhere. Taking one-sided limits along sequences outside the exceptional null sets proves (16.160). Radon regularity passes the smooth-test inequalities to measures; no trace of A is required.

The volume-preserving map $w = z + xy/2$ sends the standard horizontal fields to the displayed gauge. The original Laplacian becomes exactly

$$\mathcal{E} = \partial_{xx} + \partial_{yy} + y\partial_{xw} + x\partial_{yw} + \left(1 + \frac{x^2 + y^2}{4}\right) \partial_{ww}.$$

This follows by squaring $\partial_x + (y/2)\partial_w$, $\partial_y + (x/2)\partial_w$, and ∂_w ; no first-order term is generated. Add $\varepsilon \int \psi d\rho_x$ to F , $\varepsilon \int y\psi_w d\rho_x$ to G , and

$$\varepsilon \int \left[\psi_{yy} + x\psi_{yw} + \left(1 + \frac{x^2 + y^2}{4}\right) \psi_{ww} \right] d\rho_x$$

to H . The same distributional identity and proof apply, with the stated shifted ellipticity constants.

For general admissible stationary measures, the scalar occupation estimate of Lemma 12.1, applied to the original coordinate x , has nondegenerate horizontal gradient because $Xx = 1$. It bounds the local projection in x by a constant times Lebesgue measure. Disintegration on each compact product neighborhood supplies the bounded conditional masses. The local sink bound obtained by testing a nonnegative cutoff bounds $\|\mu_\psi\|$ uniformly. Thus F, G, H and B

are uniformly bounded. On nested intervals the additive constant c is bounded by integrating $F' = G + B + c$ and using the endpoint bound for F . In particular,

$$\|F'\|_{L^\infty(J')} \leq C_{J',J}(\|F\|_\infty + \|G\|_\infty + \|H\|_\infty + \|\mu_\psi\|), \quad J' \Subset J.$$

For a signed sink this regularity still holds, but the uniform constant additionally uses its local total variation. $\square$

Theorem 16.12 (The twelve-map IFS is not stationary). *The twelve-map measure of Corollary 16.9 cannot satisfy the stationary adjoint equation for any Borel symmetric A with fixed $0 < \lambda I \leq A \leq \Lambda I$, and any constant $\varepsilon \in [0, 1]$, even locally near the interface described below and even with a locally finite signed sink. No infinite sequence of its smooth probability approximations, weakly convergent to it on a fixed ambient domain containing its support, can satisfy the stationary hypotheses with common ellipticity and mass bounds and nonnegative sinks. The same conclusions hold after adding any fixed smooth nonnegative background with a fixed positive coefficient on the IFS measure.*

Proof. For $N = 2$ the vertical random variable in the IFS is $T = \sum_{i \geq 1} (3/2)W_i 4^{-i}$, with W_i independent uniform digits in $\{0, 1, 2\}$. Its law ν is supported in $[0, 1]$, singular, and satisfies $\nu([0, 1/32]) \geq 1/27$: the event that its first three digits vanish has that probability and puts T in $[0, 1/64]$. In coordinates $w = z + xy/2$, the IFS measure has conditional fibers

$$\begin{aligned} \rho_x(dy, dw) &= 1_{(0,1)}(y) dy \nu(dw - \beta_y(x)), \quad \beta_y(x) = \sum_i U_i d_i(y), \\ x &= \sum_i U_i 2^{-i}, \quad y = \sum_i V_i 2^{-i}, \quad d_i(y) = 2^{-i} \sum_{k > i} V_k 2^{-k} + \frac{1}{2} V_i 4^{-i}. \end{aligned}$$

All $d_i \geq 0$; dyadic ambiguity affects a null set only. For $1/2 < y < 9/16$,

$$d_1(y) = \frac{1}{8} + \frac{1}{2}(y - \frac{1}{2}) \geq \frac{1}{8}, \quad 0 \leq \sum_{i \geq 2} d_i(y) = \frac{1}{2}(y - \frac{1}{2}) - \frac{3}{2} \sum_{k > 1} V_k 4^{-k} \leq \frac{1}{32}.$$

The second identity follows by interchanging absolutely convergent sums. Thus $\beta_y(x) \in [0, 1/32]$ for $x < 1/2$, whereas $\beta_y(x) \geq 1/8$ for $x > 1/2$, almost everywhere. Choose nonzero $0 \leq \chi \in C_c^\infty((1/2, 9/16))$, and $0 \leq \zeta \in C_c^\infty((-1/64, 3/32))$ equal to one on $[0, 1/16]$. For $\psi(y, w) = \chi(y)\zeta(w)$,

$$\rho_x(\psi) \geq c_0 := \frac{1}{27} \int \chi > 0 \quad (x < 1/2), \quad \rho_x(\psi) = 0 \quad (x > 1/2).$$

These inequalities are only needed in a fixed small interval about $1/2$. Proposition 16.11 would make F_ψ continuous, while ellipticity forces $F_\psi \geq (\lambda + \varepsilon)c_0$ on the left and $F_\psi = 0$ on the right. This is impossible.

For smooth approximations $p_j dg \rightharpoonup \rho$, suppose an infinite admissible subsequence existed. Set $Q_j = A_j p_j dg$. The matrix measures have bounded total variation and satisfy $\lambda p_j I dg \leq Q_j \leq \Lambda p_j I dg$. The sinks have uniformly bounded mass on each compact interior set, since a nonnegative cutoff equal to one there gives $\mu_j(K) \leq \int p_j(\mathcal{L}_{A_j} + \varepsilon_j \Delta)\chi \leq C_\chi$. Weak compactness and a diagonal extraction give local limits Q, μ and $\varepsilon_j \rightarrow \varepsilon_*$, with $\mu \geq 0$ locally finite. The cone inequalities pass to the limit; Radon–Nikodym then gives $Q = A\rho$ with the same ellipticity bounds. The test derivatives are fixed smooth functions, so the weak equation passes to the limit and contradicts the interface obstruction.

Finally let $\tilde{\rho} = \alpha\rho + (1 - \alpha)b dx dy dw$, with $0 < \alpha \leq 1$ and fixed smooth $b \geq 0$. The one-sided IFS shifts at $x = 1/2$ are $\beta_-(y) = \sum_{i \geq 2} d_i(y)$, $\beta_+(y) = d_1(y)$; uniformly summable coefficients and dominated convergence justify these weak limits. The smooth background has

the same absolutely continuous trace on both sides. On $(1/2, 9/16) \times (-1/64, 3/32)$, the right IFS trace vanishes, while the left has positive mass concentrated on

$$E = \{(y, w) : 1/2 < y < 9/16, -1/64 < w < 3/32, w - \beta_-(y) \in \text{supp } \nu\}.$$

This is a Borel set of two-dimensional Lebesgue measure zero by Fubini. The right trace, including its background, gives E zero mass; the left trace gives it positive mass. This contradicts the two-sided measure domination in Proposition 16.11. The compactness argument just given also excludes smooth approximations with this fixed background. No assertion about an arbitrary varying background is made. $\square$

17 Further exact Heisenberg stresses and entropy identities

All results in this section concern the first Heisenberg group in its standard exponential coordinates. We fix

$$X = \partial_x - \frac{y}{2}\partial_z, \quad Y = \partial_y + \frac{x}{2}\partial_z, \quad T = \partial_z,$$

and write $dg = dx dy dz$, $\zeta = x + iy$, $r = |\zeta|$, $\rho = r^2/4$, $Z = X + iY$, $\bar{Z} = X - iY$, and $\Delta_H = X^2 + Y^2$. For a real symmetric matrix tensor Q , set

$$\mathcal{T}Q = X^2Q_{11} + (XY + YX)Q_{12} + Y^2Q_{22}.$$

The fields are divergence free, so $\mathcal{T}(Ap) = 0$ is exactly the stationary adjoint equation for the prescribed symmetrized operator, with zero Euclidean viscosity and zero sink. All the explicitly constructed stationary stresses below have this zero-viscosity setting. The pole-family obstructions and the angular-entropy identity have the wider viscosity ranges stated in their propositions.

17.1 A complex identity producing exact stationary stresses

Proposition 17.1 (CR stress identity). *Let F be a smooth complex function on an open subset of $\mathbb{H}^1$ such that $\bar{Z}F = 0$. Put*

$$q = \text{Re } F, \quad w = \frac{i}{8}\bar{\zeta}^2 T\bar{F}, \quad Q = \begin{pmatrix} q + \text{Re } w & -\text{Im } w \\ -\text{Im } w & q - \text{Re } w \end{pmatrix}. \quad (17.1)$$

Then $\Delta_H q + \text{Re}(Z^2 w) = 0$, equivalently $\mathcal{T}Q = 0$. The eigenvalues of Q are $q \pm |w|$. In particular, if $q > 0$ and $|w| \leq \kappa q$ with $0 \leq \kappa < 1$, then $A = Q/q$ has bounds $(1 - \kappa)I \leq A \leq (1 + \kappa)I$ and $\mathcal{L}_A^ q = 0$.*

Proof. The exact commutators are

$$[X, Y] = T, \quad [\bar{Z}, Z] = 2iT, \quad [Z, T] = [\bar{Z}, T] = 0, \quad \Delta_H = \frac{1}{2}(Z\bar{Z} + \bar{Z}Z).$$

Therefore $\Delta_H F = iTF$. Moreover $\bar{Z}\zeta = 2$ and $\bar{Z}(TF) = 0$, whence $\bar{Z}^2(\zeta^2 TF) = 8TF$. Since $\bar{w} = -(i/8)\zeta^2 TF$, it follows that $\bar{Z}^2 \bar{w} = -iTF$. Taking real parts proves the first identity. Expanding

$$\text{Re}(Z^2 w) = (X^2 - Y^2) \text{Re } w - (XY + YX) \text{Im } w$$

proves the tensor identity with the displayed minus sign in its off-diagonal entry. The eigenvalue and ellipticity statements follow by computing the characteristic polynomial of the trace-free part. $\square$

17.2 An explicit resolvent family and its exact singular limit

Theorem 17.2 (Resolvent stresses with an excluded strong L^2 endpoint). *Fix*

$$\Omega_0 = \{r < 1, |z| < 1\}, \quad K_0 = \{r \leq 1/2, |z| \leq 1/2\} \Subset \Omega_0.$$

For $0 < a \leq 1$, set $u_a = a + \rho$ and

$$q_a = \frac{u_a}{u_a^2 + z^2}, \quad w_a = \frac{\bar{\zeta}^2}{8(u_a + iz)^2}, \quad Q_a = \begin{pmatrix} q_a + \operatorname{Re} w_a & -\operatorname{Im} w_a \\ -\operatorname{Im} w_a & q_a - \operatorname{Re} w_a \end{pmatrix}. \quad (17.2)$$

Let $M_a = \int_{\Omega_0} q_a$, $p_a = q_a/M_a$, and $A_a = Q_a/q_a$. These are smooth probability densities and smooth symmetric coefficients satisfying

$$\frac{1}{2}I \leq A_a \leq \frac{3}{2}I, \quad \mathcal{L}_{A_a}^* p_a = 0.$$

As $a \downarrow 0$, the scalar densities and normalized tensor entries converge in $L^1(\Omega_0)$ to an exact stationary density and stress with the same bounds. The limiting density p_0 does not belong to $L^2(K_0)$. More precisely,

$$\int_{K_0} p_a^2 dg \geq \frac{2}{\pi^3} \log \left(1 + \frac{1}{16a} \right) \rightarrow \infty \quad (0 < a \leq 1/16). \quad (17.3)$$

Nevertheless, for every fixed $1 < s < 2$,

$$\sup_{0 \leq a \leq 1} \int_{\Omega_0} p_a^s dg < \infty, \quad \sup_{0 \leq a \leq 1} \int_{\Omega_0} p_a \log(1 + p_a) dg < \infty.$$

Thus this family disproves a uniform strong local L^2 conclusion at ellipticity ratio three, and its limit disproves qualitative local L^2 membership. It does not disprove entropy integrability.

Proof. For $D_a = a + \rho - iz$, direct differentiation gives $\bar{Z}D_a = 0$. Apply Proposition 17.1 to $F_a = D_a^{-1}$. Its formulas give exactly (17.2), and

$$\frac{|w_a|}{q_a} = \frac{r^2}{8u_a} = \frac{\rho}{2(a + \rho)} \leq \frac{1}{2}.$$

This proves the tensor equation and ellipticity. For each fixed horizontal point, $\int_{\mathbb{R}} u_a/(u_a^2 + z^2) dz = \pi$, hence $M_a \leq \pi^2$. On $1/2 < r < 1$, $|z| < 1$, one has $1/16 \leq u_a \leq 5/4$ and $q_a \geq (1/16)/((5/4)^2 + 1) = 1/41$. This region has volume $3\pi/2$, so

$$m_0 := 3\pi/82 \leq M_a \leq \pi^2. \quad (17.4)$$

Scalar normalization preserves the zero stationary equation.

Off the vertical axis define q_0, w_0 by (17.2) at $a = 0$, and set both to zero on that null set. On the tube $r < \eta$, integration in z bounds the q_a mass by $\pi^2\eta^2$, uniformly in a , and bounds the $|w_a|$ mass by half that amount. The same estimates hold at $a = 0$. On the complement, the rational formulas converge uniformly on the fixed bounded coordinate region. First letting $a \downarrow 0$ there and then $\eta \downarrow 0$ proves strong L^1 convergence of the scalar and every tensor entry. Thus $M_a \rightarrow M_0 \in [m_0, \pi^2]$. Define $A_0 = Q_0/q_0$ off the axis and $A_0 = I$ on it. This is Borel and obeys the same matrix bounds. The normalized tensors converge strongly in L^1 , so their zero weak equations pass to the limit against every compactly supported smooth test. In particular no hidden point or line source has been discarded.

If $a \leq 1/16$ and $0 \leq \rho \leq 1/16$, then $u_a \leq 1/8$, and $|z| \leq u_a$ lies in the central portion of K_0 . There $q_a \geq 1/(2u_a)$, so

$$\int_{K_0} q_a^2 \geq 4\pi \int_0^{1/16} \frac{d\rho}{2(a+\rho)} = 2\pi \log \left(1 + \frac{1}{16a} \right).$$

Here $dx dy = 4\pi d\rho$ after angular integration. Division by $M_a^2 \leq \pi^4$ proves (17.3); the same improper integral at $a = 0$ diverges. Finally, with $J_s = \int_{\mathbb{R}} (1+t^2)^{-s} dt$,

$$\int_{\mathbb{R}} q_a^s dz = J_s(a+\rho)^{1-s}, \quad \int_{\Omega_0} q_a^s \leq \frac{4\pi J_s}{2-s} (1/4)^{2-s} \quad (0 \leq a \leq 1).$$

The lower mass bound and $v \log(1+v) \leq C_s(v+v^s)$ prove the remaining conclusions. $\square$

17.3 Every positive-real one-variable holomorphic ansatz

Theorem 17.3 (Uniform powers for the positive-real CR class). *Let H be holomorphic on $\{\operatorname{Re} D > 0\}$, with $\operatorname{Re} H > 0$. For $0 < a \leq 1$, define on Ω_0*

$$q_{a,H} = \operatorname{Re} H(a + \rho - iz), \quad w_{a,H} = -\frac{\bar{\zeta}^2}{8} \overline{H'(a + \rho - iz)}, \quad (17.5)$$

and form $Q_{a,H}$ by the tensor convention in (17.1). Then $A_{a,H} = Q_{a,H}/q_{a,H}$ is smooth, symmetric, and satisfies $I/2 \leq A_{a,H} \leq 3I/2$. The normalized probability $p_{a,H} = q_{a,H}/\int_{\Omega_0} q_{a,H}$ satisfies $\mathcal{L}_{A_{a,H}}^* p_{a,H} = 0$, with zero sink and zero viscosity. For every fixed $1 < s < 2$,

$$\sup_{H, 0 < a \leq 1} \|p_{a,H}\|_{L^s(\Omega_0)} \leq C_s. \quad (17.6)$$

The constant depends only on s and the fixed cylinder, not on H . Uniform entropy follows. For each fixed H , the normalized scalar and tensor have exact L^1 stationary limits as $a \downarrow 0$, and the same estimates apply to these limits. The strong endpoint $s = 2$ cannot be added uniformly.

Proof. We give the positivity estimate and harmonic representation explicitly. Fix D_0 in the right half-plane and write $u_0 = \operatorname{Re} D_0$, $h_0 = \operatorname{Re} H(D_0)$. The maps

$$D \mapsto \frac{D - D_0}{D + \overline{D_0}}, \quad W \mapsto \frac{W - H(D_0)}{W + \overline{H(D_0)}}$$

map their half-planes to the disk and have derivatives $1/(2u_0)$ and $1/(2h_0)$ at the designated points. Conjugating H by them gives a disk self-map fixing zero. The Schwarz lemma therefore gives

$$|H'(D_0)| \leq \frac{\operatorname{Re} H(D_0)}{\operatorname{Re} D_0}. \quad (17.7)$$

Here the disk Schwarz lemma follows from the maximum principle applied to the quotient of the map by its variable on disks of radius tending to one. Since $\bar{Z}(a + \rho - iz) = 0$, Proposition 17.1 gives the displayed stress, and (17.7) yields $|w_{a,H}|/q_{a,H} \leq \rho/[2(a + \rho)] \leq 1/2$.

Put $h = \operatorname{Re} H$, $m(D) = (D - 1)/(D + 1)$, and $g(w) = h((1 + w)/(1 - w))$. The function g is positive harmonic in the disk and $g(0) = h(1) > 0$. For $0 < b < 1$, the circle measure $g(be^{it}) dt/(2\pi)$ has mass $h(1)$. Along a sequence $b \uparrow 1$, it has a weak limit σ of the same mass. The Poisson formula on the radius- b disk, at a fixed interior point w , uses kernel $P_{w/b}$. The kernels converge uniformly on the circle, so $g(w) = \int P_w(\xi) d\sigma(\xi)$. The Poisson formula here follows, for smooth boundary values on the smaller circle, by expanding its explicit harmonic kernel and using the maximum principle; no boundary trace of H on the unit circle is being assumed.

Separate the atom $b_\infty = \sigma(\{1\})$ and parameterize the rest by $\xi = m(it)$, $t \in \mathbb{R}$. Let τ be the image of the remaining finite measure. Direct computation of $P_w(\xi) = (1 - |w|^2)/|\xi - w|^2$ gives

$$h(u + iv) = b_\infty u + \int_{\mathbb{R}} \frac{(1 + t^2)u}{u^2 + (v - t)^2} d\tau(t), \quad b_\infty + \tau(\mathbb{R}) = h(1). \quad (17.8)$$

Indeed $1 - |m(D)|^2 = 4u/|D + 1|^2$, $|1 - m(D)|^2 = 4/|D + 1|^2$, and $|m(it) - m(D)|^2 = 4|D - it|^2/[(1 + t^2)|D + 1|^2]$. Thus the linear term, representing the boundary point at infinity, is included.

On $1/2 < r < 1$, $|z| < 1$, the variables $u = a + \rho$, $v = -z$ obey $1/16 \leq u \leq 5/4$, $|v| \leq 1$, and

$$u^2 + (v - t)^2 \leq 25/16 + 2 + 2t^2 \leq (57/16)(1 + t^2).$$

Every kernel in (17.8), and also its linear component u , is at least $1/57$ there. Since the region has volume $3\pi/2$,

$$M_{a,H} := \int_{\Omega_0} q_{a,H} \geq \frac{\pi}{38} h(1). \quad (17.9)$$

For fixed $1 < s < 2$, consider $k_{a,t} = (1 + t^2)(a + \rho)/[(a + \rho)^2 + (z + t)^2]$. If $|t| \geq 2$, then $|z + t| \geq |t|/2$ on the cylinder and $k_{a,t} \leq 5(a + \rho) \leq 25/4$. If $|t| \leq 2$, integration first on the whole central line gives

$$\int_{\Omega_0} k_{a,t}^s \leq \frac{5^s 4\pi J_s}{2 - s} (1/4)^{2-s}, \quad J_s = \int_{\mathbb{R}} (1 + v^2)^{-s} dv.$$

These bounds are uniform for $0 \leq a \leq 1$, interpreting the axis values at $a = 0$ arbitrarily. The linear component $a + \rho$ has the same type of uniform bound. Divide (17.8) by $h(1)$: it is a probability mixture of these functions. Scalar Jensen and Tonelli imply $\|q_{a,H}\|_{L^s(\Omega_0)} \leq B_s h(1)$, with B_s independent of a, H . Division by (17.9) proves (17.6), and a fixed subcritical exponent proves entropy.

For fixed H , define $q_{0,H}, w_{0,H}$ by their formulas off the axis. On the tube $r < \eta$, Hölder's inequality gives

$$\int_{\{r < \eta\} \cap \Omega_0} q_{a,H} \leq B_s h(1) (2\pi\eta^2)^{1-1/s}.$$

The same holds for the limit, and $|w_{a,H}| \leq q_{a,H}/2$ controls every tensor entry. Away from the tube the arguments of H, H' stay in a compact subset of the right half-plane, giving uniform convergence. This proves strong L^1 convergence of scalars and tensors; the positive lower normalization bound persists. Their weak zero equations therefore pass to the limit, including the entire axis. The limiting ratio is Borel and admissible after assigning I on the axis. Fatou preserves every subcritical power estimate. Finally $H(D) = 1/D$ gives Theorem 17.2 and excludes a uniform strong L^2 endpoint. $\square$

Corollary 17.4 (Exact oscillating stresses). *For $k > 0$, set*

$$q_k = 1 + e^{-kr^2/4} \cos(kz), \quad w_k = \frac{k}{8} \bar{\zeta}^2 e^{-kr^2/4} e^{-ikz}.$$

These smooth scalar functions and tensors satisfy $\mathcal{T}Q_k = 0$, with $0 \leq q_k \leq 2$ and $|w_k| \leq q_k/2$. Assigning an admissible coefficient value on $\{q_k = 0\}$ produces a Borel stationary coefficient matrix with bounds $I/2, 3I/2$. After normalization on Ω_0 , the densities are uniformly bounded:

$$\frac{q_k}{\int_{\Omega_0} q_k} \leq \frac{1}{\pi - 1} \quad (k > 0).$$

Proof. Take the entire function $H(D) = 1 + e^{-kD}$, whose real part is positive in the right half-plane, and let $a = 0$. The displayed formulas also satisfy the smooth CR identity directly on the axis. With $s = kr^2/4$,

$$\frac{|w_k|}{q_k} \leq \frac{s}{2(e^s - 1)} \leq \frac{1}{2}$$

where the denominator is nonzero; at a zero of q_k , $r = 0$ and $w_k = 0$. Direct integration gives

$$\int_{\Omega_0} q_k = 2\pi + \frac{8\pi(1 - e^{-k/4}) \sin k}{k^2} \geq 2\pi - 2.$$

For $k \leq \pi$ the added term is nonnegative. For $k \geq \pi$, use $1 - e^{-k/4} \leq k/4$ to bound its negative part by $2\pi/k \leq 2$. Since $q_k \leq 2$, the claimed bound follows. $\square$

17.4 What positivity of a CR real part does not guarantee

Proposition 17.5 (Failure of a particular elementary right inverse). *There is a family of smooth, strictly positive scalar functions $q_\eta \leq 2$, $0 < \eta < 1$, and smooth exact solutions of $\Delta_H q_\eta + \text{Re}(Z^2 w_\eta) = 0$, for which $|w_\eta|/q_\eta$ is unbounded as $\eta \uparrow 1$. Thus the elementary central-frequency right inverse below does not preserve positivity of the stationary tensor, even for a positive CR real part. This does not exclude a different positive tensor for the same density.*

Proof. For any CR function F , the central commutation relations give

$$\bar{Z}Z^2F = 4iTZF, \quad \bar{Z}^2Z^2F = -8T^2F.$$

At a nonzero central frequency the formal choice $\bar{w} = (i/8)Z^2T^{-1}F$ therefore satisfies $\bar{Z}^2\bar{w} = -iT\bar{w}$; here T^{-1} simply divides that frequency and has no zero-frequency ambiguity. Take

$$G = \sqrt{e}\bar{\zeta}e^{-r^2/2}e^{2iz}.$$

Using $\bar{Z} = 2\partial_{\bar{\zeta}} - (i/2)\bar{\zeta}T$ and $Z = 2\partial_{\zeta} + (i/2)\zeta T$, direct differentiation gives

$$\bar{Z}G = 0, \quad TG = 2iG, \quad ZG = 2\sqrt{e}(1 - r^2)e^{-r^2/2}e^{2iz}, \quad Z^2G = 4\sqrt{e}\zeta(r^2 - 2)e^{-r^2/2}e^{2iz}.$$

Moreover $|G| = \sqrt{e}re^{-r^2/2} \leq 1$. Set $q_\eta = 1 + \eta \text{Re} G$ and $\bar{w}_\eta = (\eta/16)Z^2G$. The frequency calculation proves the exact stress identity, and $q_\eta \geq 1 - \eta > 0$. At the fixed point $\zeta = 1, z = \pi/2$,

$$G = -1, \quad q_\eta = 1 - \eta, \quad Z^2G = 4, \quad \frac{|w_\eta|}{q_\eta} = \frac{\eta}{4(1 - \eta)} \rightarrow \infty.$$

For $\eta > 4/5$, the eigenvalue $q_\eta - |w_\eta|$ of the resulting tensor is negative there. Only this particular right inverse is ruled out. $\square$

Proposition 17.6 (The explicit squared-modulus pole family). *For a fixed $\beta > 0$ and $a > 0$, let*

$$q_{a,\beta} = [(a + r^2/4)^2 + z^2]^{-\beta}, \quad p_{a,\beta} = q_{a,\beta} / \int_{\Omega_0} q_{a,\beta}.$$

For each fixed $0 < \beta < 1$, the limit $q_{0,\beta}$ belongs to $L^s(\Omega_0)$ whenever $1 < s < 1/\beta$, and the normalized family has uniform entropy and these uniform power bounds as $a \downarrow 0$. The constants may depend on β ; no uniformity as $\beta \uparrow 1$ is asserted.

If $3/4 \leq \beta < 1$, the normalized limit cannot be stationary with any finite positive ellipticity bounds, a nonnegative finite sink, and any constant $\varepsilon \in [0, 1]$. For every fixed $\beta \geq 3/4$, no sequence $a_j \downarrow 0$ can make all $p_{a_j,\beta}$ stationary with one fixed pair λ, Λ , nonnegative finite sinks, and arbitrary constants $\varepsilon_j \in [0, 1]$. For $\beta \geq 1$, the normalized densities instead converge weakly to a point mass at the origin.

Proof. In (ρ, z) coordinates, polar integration in the horizontal variables gives $dg = 4\pi d\rho dz$. Near the origin the limiting power integral is a constant multiple of $\int_{\rho>0} (\rho^2 + z^2)^{-\beta s} d\rho dz$, which is finite exactly when $\beta s < 1$. Away from the origin the function is bounded. Also $q_{a,\beta} \leq q_{0,\beta}$ for $\beta < 1$, and the normalization masses have a positive lower bound for all sufficiently small a , by integration on any fixed annular region. Dominated convergence proves L^1 convergence and the stated powers and entropy, for each fixed β .

Every admissible stationary density has a bounded compactly localized marginal in x , uniformly under a unit mass bound: apply Lemma 12.1 to $f(x, y, z) = x$, since $Xx = 1$. For the proposed limit, work in a fixed inner rectangle about the origin. Integration first over $|z| < \rho$ gives a lower bound $c\rho^{1-2\beta}$. For $2|x| < |y| < c_0$, this is bounded below by a constant times $|y|^{2-4\beta}$. Hence the localized x -marginal is bounded below, as $x \rightarrow 0$, by

$$c \log(1/|x|) \quad (\beta = 3/4), \quad c|x|^{3-4\beta} \quad (\beta > 3/4).$$

These are unbounded on sets of positive measure approaching the origin, contradicting the scalar occupation estimate. For $3/4 \leq \beta < 1$, the strong L^1 convergence passes any common scalar occupation bound for an admissible sequence to the limit, giving the same contradiction. This argument is uniform in the allowed viscosities.

For fixed $\beta \geq 1$, monotone convergence shows $\int_{\Omega_0} q_{a,\beta} \rightarrow \infty$, while outside every fixed neighborhood of the origin the unnormalized integrals are bounded by the integrable restriction of $q_{0,\beta}$. Thus $p_{a,\beta} dg \rightarrow \delta_0$. Choose a fixed compact cutoff neighborhood of the origin and then an x -strip of sufficiently small width. Its mass tends to one, whereas the common stationary scalar occupation estimate would bound it by a constant times that width. This again excludes an admissible sequence with fixed coefficient bounds. $\square$

17.5 Sharpness of the inverse radial moment

Theorem 17.7 (The inverse-moment threshold is sharp). *Fix $0 < \lambda \leq \Lambda$, put $s = \lambda/\Lambda$, and on Ω_0 let*

$$p = \frac{s+1}{4\pi} r^{s-1}, \quad A = \Lambda e_r \otimes e_r + \lambda e_t \otimes e_t \quad (r > 0),$$

where $e_r = (x, y)/r$ and $e_t = (-y, x)/r$. Assign any admissible matrix value on the axis. This is an exact stationary probability density with zero viscosity and zero sink and the prescribed eigenvalues. For every compact cylinder containing an axis segment in its interior,

$$\int_K p \rho^{-\beta} dg = \infty \quad \text{if } \beta \geq \frac{s+1}{2} = \frac{\Lambda + \lambda}{2\Lambda}.$$

Thus the strict range in Theorem 16.2 is sharp for inverse moments, including when $\lambda = \Lambda$. This asserts neither sharpness of its density-power exponent nor failure of entropy.

Proof. Writing $c = (s+1)/(4\pi)$, normalization is $\int_{\Omega_0} p = 4\pi c \int_0^1 r^s dr = 1$. The tensor $Q = Ap$ is independent of z , and off the axis its planar divergence is

$$\operatorname{div}_{x,y} Q = \left[\Lambda p'(r) + (\Lambda - \lambda) \frac{p(r)}{r} \right] e_r = 0,$$

since $\Lambda(s-1) + \Lambda - \lambda = 0$. There is no distributional line source: integration by parts outside a radius- r tube against a fixed smooth vector test leaves an inner boundary term bounded by $Crp(r) = Cr^s \rightarrow 0$. Thus the first planar divergence vanishes distributionally across the axis. Since the tensor is independent of z , its horizontal row divergence is the planar one; a further horizontal divergence gives $\mathcal{T}Q = 0$. This uses precisely the symmetrized adjoint convention. Finally, near an interior axis segment, $\int_K p \rho^{-\beta}$ contains a positive constant times $\int_0^{r_0} r^{s-2\beta} dr$, which diverges exactly at and above the stated threshold. At equality the divergence is logarithmic. Entropy is finite because its radial integrand is bounded by $Cr^s(1 + |\log r|)$, and $s > 0$. $\square$

17.6 An exact conditional angular entropy identity

Proposition 17.8 (Marginal and angular entropy). *Assume the standard stationary density hypotheses (16.2), now allowing arbitrary constant $\varepsilon \in [0, 1]$, arbitrary Borel coefficients in the prescribed matrix bounds, and a nonnegative finite sink. Let $0 \leq \chi \leq 1$, $\chi \in C_c^\infty(\Omega)$, and extend $f = \chi p$ by zero. Put*

$$q(u, z) = 2 \int_0^{2\pi} f(2\sqrt{u} \cos t, 2\sqrt{u} \sin t, z) dt \quad (u > 0), \quad d\omega = \frac{dt}{2\pi}.$$

Set $q(u, z) = 0$ for $u \leq 0$. For almost every fiber with $q > 0$, define

$$F(u, z, t) = \frac{4\pi f(2\sqrt{u} \cos t, 2\sqrt{u} \sin t, z)}{q(u, z)};$$

put $F = 1$ on $q = 0$. Then $\int F d\omega = 1$, and the nonnegative extended quantity

$$\mathcal{E}_\chi(p) = \int q(u, z) \left[\int F(u, z, t) \log F(u, z, t) d\omega(t) \right] du dz$$

satisfies the exact identity

$$\int_{\mathbb{R}^3} f \log f dg = \int_{\mathbb{R}^2} q \log \frac{q}{4\pi} du dz + \mathcal{E}_\chi(p). \quad (17.10)$$

We use $0 \log 0 = 0$, and the identity is meaningful even before finiteness of the entropy of f is known. The marginal term has uniformly bounded absolute value. Consequently

$$\left| \int f \log(1 + f) dg - \mathcal{E}_\chi(p) \right| \leq C_\chi \quad (17.11)$$

whenever either term is finite; if one is infinite, both are infinite. Thus bounding these localized conditional angular entropies is equivalent to the stationary local entropy estimate. This identity by itself supplies no independent angular-entropy bound.

Proof. The Jacobian is $dx dy dz = 2 du dt dz = 4\pi du d\omega dz$. Hence $q = 4\pi \int f d\omega$, and the normalization of F follows. Fibers where $q = \infty$ form a null set and can be redefined arbitrarily; where $q = 0$, $f = 0$ for almost every angle. Substitution of $f = qF/(4\pi)$ gives (17.10) formally, and the following integrability argument justifies it with extended values.

Theorem 16.2, applied only to its projection assertion, gives $\int q^{1+\theta} \leq C_{\chi, \theta}$ for any fixed $0 < \theta < (\Lambda + \lambda)/(4\Lambda)$. Its use imposes no angular symmetry on p . The support of q lies in a fixed compact set, so its power bound and $q |\log q| \leq 1/e$ on $q \leq 1$ imply $\int q |\log(q/(4\pi))| < \infty$, uniformly. Meanwhile $F \log F \geq -1/e$, and its negative part against $q du dz d\omega$ has mass at most $\int q/e \leq 1/e$. Thus the separated marginal term is integrable and the other has integrable negative part; extended Fubini never subtracts two infinities. Jensen gives $\int F \log F d\omega \geq 0$, proving the stated nonnegativity.

For $v \geq 0$,

$$0 \leq v \log(1 + v) - v \log v = v \log(1 + 1/v) \leq 1,$$

with the value at zero interpreted by continuity. Integration over the fixed compact support of f , and the uniform absolute bound for the marginal entropy, prove (17.11). If $\chi = 1$ near K , a bound for $\mathcal{E}_\chi(p)$ bounds the entropy on K . Conversely, local entropy on the compact support of χ , monotonicity of $v \log(1 + v)$, and $0 \leq \chi \leq 1$ bound the entropy of f , hence $\mathcal{E}_\chi(p)$. $\square$

Corollary 17.9. *For the stationary class in Proposition 17.8, $\mathcal{E}_\chi(p) \leq C_\chi$ uniformly over the coefficients, the nonnegative sink, the density, and $\varepsilon \in [0, 1]$.*

Proof. Apply the stationary conclusion of Theorem 2.1 to $\text{supp } \chi$ in $\mathbb{H}^1$, and then use (17.11). The hypoelliptic entropy theorem, rather than the decomposition alone, supplies this bound. $\square$

18 Directional coefficient variance and stationary concentration

This section concerns a Carnot group of arbitrary step. The coefficient-variance statements do not assume any equation. The stationary concentration estimate uses the double-divergence equation with zero coordinate viscosity. The obstruction below is an obstruction to an unconditional comparison of regularizers; its input data are not asserted to solve a stationary equation.

18.1 Convolution orientation and nested smoothing

Let G have graded Lie algebra $V_1 \oplus \cdots \oplus V_s$, homogeneous dimension D , dilations δ_r , and Haar measure dx . Fix a basis $X_1, \dots, X_m$ of V_1 . For a Lie algebra vector v , our conventions are

$$X_v^L f(x) = \left. \frac{d}{dt} \right|_{t=0} f(x \exp(tv)), \quad X_v^R f(x) = \left. \frac{d}{dt} \right|_{t=0} f(\exp(tv)x).$$

Thus the $X_i = X_i^L$ are left-invariant and $X_i^* = -X_i$. We use

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) dy = \int_G f(xy^{-1})g(y) dy. \quad (18.1)$$

In particular, wherever the operations are defined,

$$X_v^L(\eta * f) = \eta * (X_v^L f), \quad X_v^R(\eta * f) = (X_v^R \eta) * f. \quad (18.2)$$

These two different identities will not be interchanged. Let $|x|$ denote the distance from the identity for a fixed left-invariant Carnot–Carathéodory distance. It is symmetric under inversion and subadditive under multiplication.

Let $p \in L^1(G)$ be nonnegative, $M = \int p$, and let $Q = pA$, where A is a measurable symmetric $m \times m$ matrix satisfying $\lambda I \leq A \leq \Lambda I$ $p dx$ -almost everywhere. The Frobenius norm is denoted by $|\cdot|_F$. If $M = 0$, all assertions below are interpreted as zero identities.

Lemma 18.1 (A positive nested approximate identity). *Put $r_j = 2^{-j}$ for $j \geq 0$. There exist smooth strictly positive probability densities K_j and η_{r_j} such that*

$$K_j = \eta_{r_j} * K_{j+1}, \quad K_j * f \longrightarrow f \quad \text{in } L^1(G),$$

and, for every fixed homogeneous right-invariant vector field Y^R of degree k ,

$$|Y^R \eta_{r_j}| \leq C_Y r_j^{-k} \eta_{r_j}. \quad (18.3)$$

The same construction permits bounds of this form for any fixed finite word of homogeneous derivatives, with its total degree in the exponent.

Proof. In exponential coordinates $x = (x_1, \dots, x_s)$ adapted to the grading, set

$$\rho(x) = \sum_{k=1}^s |x_k|^{2s!/k}, \quad \eta(x) = c_N (1 + \rho(x))^{-N}.$$

The exponents are even integers, so ρ is a smooth positive homogeneous polynomial of degree $2s!$, vanishing only at the identity. For a homogeneous field Y^R of degree k , homogeneity and compactness of $\{\rho = 1\}$ give

$$|Y^R \rho(x)| \leq C_Y (1 + \rho(x))^{1-k/(2s!)}.$$

Differentiation, and then iteration for higher derivatives, give $|Y^R \eta| \leq C_Y \eta$. Choose N large enough that η is integrable and has a finite first $|\cdot|$ -moment; normalize its integral to one. The dilates $\eta_r(x) = r^{-D} \eta(\delta_{1/r} x)$ satisfy (18.3).

For each $\ell \geq j$, let Z_ℓ have density η_{r_ℓ} , with these group-valued variables independent. Since the sum $\sum_{\ell \geq j} \mathbb{E}|Z_\ell|$ is finite, their ordered products are almost surely Cauchy. By left invariance, consecutive products are at distance $|Z_\ell|$. Completeness gives a limiting product. Its law is

$$K_j = \eta_{r_j} * \eta_{r_{j+1}} * \eta_{r_{j+2}} * \cdots$$

Convolution of its tail law with the first factor gives a strictly positive smooth density. The derivative bounds just established justify differentiation under this convolution, to every fixed order. The expected distance of the product from the identity is at most $C \sum_{\ell \geq j} r_\ell$. Consequently its law concentrates at the identity as $j \rightarrow \infty$. Continuity of translations in L^1 , followed by integration against these probability laws, proves the approximate-identity assertion. $\square$

Theorem 18.2 (Variance increments control right derivatives). *With the kernels of Lemma 18.1, define*

$$p_j = K_j * p, \quad Q_j = K_j * Q, \quad A_j = Q_j / p_j, \quad V_j = \int_G p_j |A_j|_F^2 dx.$$

If $M > 0$, $p_j > 0$ and $\lambda I \leq A_j \leq \Lambda I$. Moreover,

$$\Delta V_j := V_{j+1} - V_j = \int_G d_j(x) dx \geq 0, \quad (18.4)$$

$$d_j(x) = \int_G \eta_{r_j}(xy^{-1}) p_{j+1}(y) |A_{j+1}(y) - A_j(x)|_F^2 dy.$$

For a homogeneous right-invariant field Y^R of degree k ,

$$r_j^{2k} \int_G p_j |Y^R A_j|_F^2 dx \leq C_Y \Delta V_j, \quad \sum_{j \geq 0} r_j^{2k} \int_G p_j |Y^R A_j|_F^2 dx \leq C_Y m \Lambda^2 M. \quad (18.5)$$

Proof. The normalized kernel weights $\eta_{r_j}(xy^{-1}) p_{j+1}(y) / p_j(x)$ are a probability distribution in y , and $A_j(x)$ is their mean of $A_{j+1}(y)$. This proves cone preservation and the pointwise variance identity

$$d_j(x) = \int_G \eta_{r_j}(xy^{-1}) p_{j+1}(y) |A_{j+1}(y)|_F^2 dy - p_j(x) |A_j(x)|_F^2.$$

Integration in x proves (18.4). Differentiating the quotient in the right-invariant direction, with precisely the second identity in (18.2), gives

$$Y^R A_j(x) = \frac{1}{p_j(x)} \int_G (Y^R \eta_{r_j})(xy^{-1}) p_{j+1}(y) [A_{j+1}(y) - A_j(x)] dy.$$

The score bound and weighted Cauchy–Schwarz imply $p_j |Y^R A_j|_F^2 \leq C_Y r_j^{-2k} d_j$. Finally, $0 \leq V_j \leq m \Lambda^2 M$, so the nonnegative increments telescope. $\square$

Corollary 18.3 (Conversion on a ball of the smoothing scale). *For each fixed $R \geq 1$,*

$$r_j^2 \int_{B(e, Rr_j)} p_j \sum_{i=1}^m |X_i^L A_j|_F^2 dx \leq C_R \Delta V_j. \quad (18.6)$$

The sum of its left-hand side over $j \geq 0$ is at most $C_R m \Lambda^2 M$.

Proof. Choose homogeneous bases $Y_{k,\alpha}$ of all higher layers. The adjoint representation gives the exact frame conversion

$$X_i^L = X_i^R + \sum_{k=2}^s \sum_{\alpha} b_{i,k,\alpha}(x) Y_{k,\alpha}^R, \quad b_{i,k,\alpha}(\delta_r x) = r^{k-1} b_{i,k,\alpha}(x).$$

Indeed X_v^L at x equals $X_{\text{Ad}_x v}^R$ at x ; the Baker–Campbell–Hausdorff formula gives the stated polynomial degrees. On $B(e, Rr_j)$ the coefficient of a degree- k field is bounded by $C_R r_j^{k-1}$. Squaring the finite sum and applying (18.5) for each layer proves the assertion. $\square$

Remark 18.4. For any fixed $g \in G$, the preceding statements apply after pulling the data back by the left translation $x \mapsto gx$. This produces a smoothing depending on that anchor. The result does not identify the smoothings at different anchors. Nor does it supply an unconditional left/right comparison on a fixed spatial region. On a proper open set the kernels must be applied to localized, extended data, with all errors retained. Only for an equation defined globally, and convolutions which are defined, does (18.2) justify exact commutation with its left-invariant distributional derivatives.

18.2 Failure of a global left-direction replacement

Proposition 18.5 (A nonstationary regularizer obstruction). *Suppose G is nonabelian and $\lambda < \Lambda$. Let $P_t^R f = h_t * f$ be the heat semigroup generated by $L_R = \sum_i (X_i^R)^2$. There are smooth compactly supported unit-mass densities $p^{(r)}$ and smooth symmetric coefficient fields $A^{(r)}$ with $\lambda I \leq A^{(r)} \leq \Lambda I$, supported as stress data near one fixed point, such that, on a fixed compact neighborhood K ,*

$$\int_0^1 \int_K P_t^R p^{(r)} \left| X_v^L \left(\frac{P_t^R(p^{(r)} A^{(r)})}{P_t^R p^{(r)}} \right) \right|_F^2 dx dt \longrightarrow \infty$$

for some horizontal v . In contrast, the same expressions summed over the horizontal right-invariant derivatives and integrated on all of G remain bounded uniformly in r . No stationary positive-source equation is asserted for these data.

Proof. Noncommutativity and generation by V_1 provide $g \in G$ and $v \in V_1$ such that $\text{Ad}_g v$ has a nonzero higher-layer component. Write $W_\ell = (\text{Ad}_g v)_\ell$, and let $k \geq 2$ be its largest nonzero degree. Choose $b \in C_c^\infty(G)$ nonnegative of mass one. Choose a smooth bounded symmetric A_0 , taking values strictly between λI and ΛI , such that $W_k^L A_0 \neq 0$ on an open region where $b > 0$. For example, a small nonconstant smooth scalar perturbation of cI , in a fixed symmetric matrix direction, does this.

For $F_r(x) = \delta_{1/r}(xg^{-1})$, put

$$p^{(r)}(x) = r^{-D} b(F_r(x)), \quad A^{(r)}(x) = A_0(F_r(x)).$$

Right-translation invariance and homogeneity of L_R give, with $b_\tau = P_\tau^R b$ and $B_\tau = P_\tau^R(bA_0)/b_\tau$,

$$P_{r^2\tau}^R p^{(r)}(x) = r^{-D} b_\tau(F_r(x)), \quad \frac{P_{r^2\tau}^R(p^{(r)} A^{(r)})(x)}{P_{r^2\tau}^R p^{(r)}(x)} = B_\tau(F_r(x)). \quad (18.7)$$

The identity $x \exp(tv)g^{-1} = (xg^{-1}) \exp(t \text{Ad}_g v)$ then yields

$$X_v^L(B_\tau \circ F_r)(x) = \sum_{\ell=1}^k r^{-\ell} (W_\ell^L B_\tau)(F_r(x)). \quad (18.8)$$

Smooth convergence as $\tau \downarrow 0$ on a compact region where $b > 0$ provides a compact set U and $0 < a < b_*$ with

$$J = \int_a^{b_*} \int_U b_\tau |W_k^L B_\tau|_F^2 dz d\tau > 0.$$

Change variables $x = \delta_r(z)g$, $t = r^2\tau$ in the integral over $F_r^{-1}(U) \times (ar^2, b_*r^2)$. Equations (18.7)–(18.8) give

$$r^{2-2k} \int_a^{b_*} \int_U b_\tau \left| W_k^L B_\tau + \sum_{\ell < k} r^{k-\ell} W_\ell^L B_\tau \right|_F^2 dz d\tau = r^{2-2k} (J + o(1)) \rightarrow \infty.$$

All these sets lie in one fixed compact neighborhood of g for small r .

For completeness, if $p_t = P_t^R p$ and $Q_t = P_t^R(pA) = p_t A_t$, the diffusion chain rule gives the pointwise perspective identity

$$(\partial_t - L_R)(p_t |A_t|_F^2) = -2p_t \sum_i |X_i^R A_t|_F^2.$$

Integration, justified first on finite time intervals away from zero by the fixed heat-kernel bounds and then by approximation, gives

$$2 \int_0^1 \int_G p_t \sum_i |X_i^R A_t|_F^2 \leq \int_G p |A|_F^2 \leq m\Lambda^2 M.$$

This applies to the smooth data just constructed, with $M = 1$. $\square$

18.3 Poisson kernels and the anchored variance defect

Lemma 18.6 (A commuting choice of nested kernels). *For the fixed horizontal heat kernel h_t , set*

$$k_r(x) = \frac{r}{2\sqrt{\pi}} \int_0^\infty t^{-3/2} e^{-r^2/(4t)} h_t(x) dt. \quad (18.9)$$

*These are positive probability densities with $k_r * k_u = k_{r+u}$, $k_r(x) = r^{-D} k_1(\delta_{1/r}x)$, and, for a degree- k homogeneous right-invariant field Y^R ,*

$$k_r(x) \asymp \frac{r}{(r + |x|)^{D+1}}, \quad |Y^R k_r(x)| \leq \frac{C_Y r}{(r + |x|)^{D+1+k}} \leq C_Y r^{-k} k_r(x). \quad (18.10)$$

Theorems 18.2 and Corollary 18.3 hold with $K_j = k_{r_j}$ and the first convolution factor $\eta_j = k_{r_j - r_{j+1}} = k_{r_j/2}$.

Proof. Subordination of the fixed heat semigroup proves its semigroup and normalization assertions. We recall the fixed-kernel input used also in Theorem 14.6: two-sided Gaussian bounds for h_t and bounds

$$|Y^R h_t(x)| \leq C_Y t^{-(D+k)/2} \exp(-c_Y |x|^2/t).$$

Right derivatives follow from the corresponding left derivative bounds by inversion symmetry of the fixed heat kernel; a layer- k vector is a linear combination of commutators of k horizontal vectors. These are fixed smooth-operator estimates, not estimates for rough A ; see [2]. Substituting in (18.9) reduces the upper estimate to

$$C_r \int_0^\infty t^{-(D+k+3)/2} e^{-c(r^2 + |x|^2)/t} dt = C' r (r^2 + |x|^2)^{-(D+k+1)/2}.$$

The lower Gaussian bound gives the analogous lower bound for k_r . Since $(r^2 + |x|^2)^{1/2} \asymp r + |x|$, this proves (18.10). All derivatives used here are dominated by these integrable majorants. The nested identity and the score bound now give the earlier variance proof verbatim. In this choice no finite first moment of k_r is required. $\square$

Proposition 18.7 (The exact anchor-averaging identity). *Use the Poisson choice of Lemma 18.6. Define*

$$f_j(g, x) = \int_G K_j(y) p(gy^{-1}x) dy, \quad q_j(g, x) = \int_G K_j(y) Q(gy^{-1}x) dy, \quad a_j = q_j/f_j,$$

and $\mathcal{V}_j(x) = \int_G f_j(g, x) |a_j(g, x)|_F^2 dg$. For each fixed x , $0 \leq \mathcal{V}_j(x) \leq m\Lambda^2 M$ and $\mathcal{V}_{j+1}(x) \geq \mathcal{V}_j(x)$. However, the variance defect entering the derivative of the profile at anchor g is

$$d_j^g(x) = \int_G \eta_j(y) f_{j+1}(g, y^{-1}x) |a_{j+1}(g, y^{-1}x) - a_j(g, x)|_F^2 dy,$$

and its exact anchor average is

$$\int_G d_j^g(x) dg = (\eta_j * \mathcal{V}_{j+1})(x) - \mathcal{V}_j(x). \quad (18.11)$$

It is not, in general, $\mathcal{V}_{j+1}(x) - \mathcal{V}_j(x)$.

Proof. For fixed x , put $p_x(g) = p(gx)$. Then $f_j(\cdot, x) = p_x * K_j$. The kernels commute, so $K_j = K_{j+1} * \eta_j$ as well as $K_j = \eta_j * K_{j+1}$. Consequently, at fixed x , the pair (f_j, q_j) is obtained from (f_{j+1}, q_{j+1}) by the same positive right convolution in g . Convexity of the perspective $|q|_F^2/f$ proves the fixed- x monotonicity and its mass bound.

At fixed g , instead, $f_j(g, \cdot) = K_j * (p \circ L_g)$ and its recursion uses the variables $y^{-1}x$. The variance identity in these variables is

$$d_j^g(x) = \int_G \eta_j(y) f_{j+1}(g, y^{-1}x) |a_{j+1}(g, y^{-1}x)|_F^2 dy - f_j(g, x) |a_j(g, x)|_F^2.$$

Integrating in g proves (18.11). The additional term relative to a pointwise telescoping difference is exactly $(\eta_j * \mathcal{V}_{j+1})(x) - \mathcal{V}_{j+1}(x)$. $\square$

18.4 A stationary concentration cost

Hereafter G is nonabelian, so $D > 2$. Write

$$\mathcal{T}Q = \sum_{i,j} X_i X_j Q_{ij}, \quad \mathcal{D}U = \sum_i X_i^* U_i = - \sum_i X_i U_i.$$

For symmetric Q this $\mathcal{T}$ is the adjoint of contraction with the symmetrized horizontal Hessian. Let globally defined data satisfy

$$q \geq 0, \quad q \in L^1(G), \quad Q = qA, \quad \lambda I \leq A \leq \Lambda I \quad q \, dx\text{-a.e.}, \quad \mathcal{T}Q + \mathcal{D}U = \kappa - \beta, \quad (18.12)$$

where κ, β are finite nonnegative measures and

$$E_U = \int_G \frac{|U|^2}{q} dx < \infty, \quad B = \beta(G).$$

Here $U = 0$ on $\{q = 0\}$, and the integrand there is zero. Finite mass and Cauchy–Schwarz imply $U \in L^1$. The statements remain valid for a nonnegative locally finite κ whenever its displayed convolutions are defined and the convolved distributional identity holds. No estimate charges the mass of κ .

For the Poisson kernels define

$$p_j = K_j * q, \quad Q_j = K_j * Q = p_j A_j, \quad U_j = K_j * U, \quad \kappa_j = K_j * \kappa, \quad \beta_j = K_j * \beta.$$

Left-invariant differentiation commutes with these left convolutions, so the smoothed data satisfy the same equation. Define V_j and ΔV_j using q, Q , as in Theorem 18.2.

Lemma 18.8 (Horizontal Sobolev inequality used below). *For $v \in C_c^\infty(G)$,*

$$\|v\|_{2D/(D-2)}^2 \leq C_G \sum_i \|X_i v\|_2^2. \quad (18.13)$$

Proof. Here is a derivation from the fixed heat-kernel bounds already used above. For the nonnegative self-adjoint operator $S = -\sum_i X_i^2$, its inverse square root has convolution kernel

$$I_1(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} h_t(x) dt \leq C|x|^{1-D} \quad (x \neq e).$$

The integral and bound follow by the same Gaussian substitution as in Lemma 18.6. If $g = S^{1/2}v$, the spectral theorem gives $v = S^{-1/2}g$ and $\|g\|_2^2 = \sum_i \|X_i v\|_2^2$. These identities can first be read with spectral cutoffs and then passed to the limit.

Let $\mathcal{M}g$ be the centered Hardy–Littlewood maximal function for Carnot balls. Their volumes are $c_G R^D$. Splitting the potential integral at radius R , summing dyadic annuli inside the ball, and using Cauchy–Schwarz outside it, gives

$$|g * I_1(x)| \leq C(R\mathcal{M}g(x) + R^{1-D/2}\|g\|_2).$$

Here $S^{-1/2}g = g * I_1$, consistently with the left-invariant operator S ; the ball in the integral is described by $|y^{-1}x| < R$. Choose $R = (\|g\|_2/\mathcal{M}g(x))^{2/D}$, with the zero cases obtained by a limit. Thus

$$|S^{-1/2}g(x)| \leq C(\mathcal{M}g(x))^{1-2/D}\|g\|_2^{2/D}.$$

The maximal operator is bounded on L^2 : the disjoint-ball covering argument on this doubling metric space gives its weak $(1,1)$ estimate, and applying that estimate to $g\mathbf{1}_{\{|g|>a/2\}}$ in each superlevel set, then integrating $2a da$, gives the strong $(2,2)$ bound. Raising the preceding potential estimate to $2D/(D-2)$ makes the exponent of $\mathcal{M}g$ equal to two. Integration therefore proves $\|S^{-1/2}g\|_{2D/(D-2)} \leq C\|g\|_2$ and the assertion. $\square$

Theorem 18.9 (Stationary concentration with a linear variance cost). *Under (18.12), there are fixed constants $R_0 \geq 2$ and C_{R_0} , depending only on the group, the horizontal frame, and λ, Λ , such that for every $j \geq 0$,*

$$\int_{B(e, r_j)} p_j dx \leq \frac{1}{2} \int_{B(e, 2R_0 r_j)} p_j dx + C_{R_0} \Delta V_j + C_{R_0} r_j^2 (E_U + B). \quad (18.14)$$

More generally, for each fixed $L \geq 1$ there are $R = R(L) \geq 2$ and $C_{R,L}$ such that

$$\int_{B(e, Lr_j)} p_j dx \leq \frac{1}{2} \int_{B(e, 2RLr_j)} p_j dx + C_{R,L} \Delta V_j + C_{R,L} r_j^2 (E_U + B). \quad (18.15)$$

The error terms are summable over j for each fixed anchor.

Proof. If $\int q = 0$, the conclusion is immediate. Otherwise put temporarily $r = r_j$, $p = p_j$, $A = A_j$, and fix $R \geq 2$. Choose a smooth cutoff $0 \leq \chi \leq 1$ equal to one on $B(e, Rr)$, supported in $B(e, 2Rr)$, with

$$|\nabla_H \chi| \leq C/(Rr), \quad |X_i X_j (\chi^2)| \leq C/(Rr)^2.$$

Such cutoffs are the dilates of a single fixed smooth cutoff. Set

$$M_R = \int_{B(e, 2Rr)} p, \quad b = M_R/|B(e, 2Rr)| > 0, \quad f = \log(1 + p/b).$$

The Poisson bounds imply, for $x, y \in B(e, 2Rr)$,

$$p(x) \leq C(1 + R)^{D+1} p(y).$$

Indeed $p(x) = \int k_r(xz^{-1})q(z) dz$, and $|yz^{-1}| \leq |yx^{-1}| + |xz^{-1}| \leq 4Rr + |xz^{-1}|$. Apply (18.10) to the two kernels and integrate. Averaging in y gives the essential logarithmic bound

$$0 \leq f \leq C(1 + \log R) \quad \text{on } \text{supp } \chi. \quad (18.16)$$

The smooth nonnegative test $\phi = \chi^2 f$ is compactly supported. The favorable measure in (18.12) gives

$$0 \leq \langle \kappa_j, \phi \rangle = \langle \mathcal{T}(pA) + \mathcal{D}U_j, \phi \rangle + \langle \beta_j, \phi \rangle.$$

Expand the derivatives of $\chi^2 f$ and integrate one derivative in the term containing $\chi^2 X_i X_j f$. Since $X_j f = X_j p / (p + b)$ and A is symmetric, the exact result is

$$\begin{aligned} 0 \leq & - \int \chi^2 \frac{(\nabla_H p)^T A \nabla_H p}{p + b} - \int \chi^2 \frac{p}{p + b} \sum_{i,j} (X_i A_{ij}) X_j p \\ & + 2 \int \chi \frac{p}{p + b} (A \nabla_H \chi) \cdot \nabla_H p + \int p \sum_{i,j} A_{ij} X_i X_j (\chi^2) f \\ & + \int \chi^2 U_j \cdot \frac{\nabla_H p}{p + b} + 2 \int \chi f U_j \cdot \nabla_H \chi + \int \chi^2 f d\beta_j. \end{aligned} \quad (18.17)$$

All unsigned integrals in this proof are over G . In particular, the sign of the flux follows from $\mathcal{D} = \sum_i X_i^*$, rather than from the opposite divergence convention.

Let $F = \int \chi^2 |\nabla_H p|^2 / (p + b)$. Ellipticity bounds the first term by $-\lambda F$. For any fixed sufficiently small $\delta > 0$, the absolute values of the coefficient-gradient, first cutoff-gradient, and principal flux terms are bounded respectively by

$$\begin{aligned} \delta F + C_\delta \int_{B(e, 2Rr)} p |\nabla_H A|_F^2, \\ \delta F + C_\delta (Rr)^{-2} M_R, \\ \delta F + C_\delta \int_G |U_j|^2 / p. \end{aligned}$$

Here we used $p^2 / (p + b) \leq p$ and $(p + b)^{-1} \leq p^{-1}$. For the remaining flux term use

$$2\chi f |U_j| |\nabla_H \chi| \leq C |U_j|^2 / p + C p f^2 |\nabla_H \chi|^2.$$

The Hessian-cutoff term is bounded by $C(Rr)^{-2}(1 + \log R)M_R$, and the measure term by $C(1 + \log R)\beta_j(G)$. Absorbing the three small multiples of F therefore proves

$$F \leq C \int_{B(e, 2Rr)} p |\nabla_H A|_F^2 + C \int_G |U_j|^2 / p + \frac{C(1 + \log R)^2}{(Rr)^2} M_R + C(1 + \log R)\beta_j(G). \quad (18.18)$$

The constants denoted by C here do not depend on $R \geq 2$.

Apply Lemma 18.8 to $v = \chi \sqrt{p + b}$. Direct differentiation, followed by $b|B(e, 2Rr)| = M_R$, gives

$$\|v\|_{2D/(D-2)}^2 \leq C \int |\nabla_H v|^2 \leq CF + C(Rr)^{-2} M_R.$$

Since $\chi = 1$ on $B(e, r)$, Hölder's inequality supplies the factor $|B(e, r)|^{2/D} = C_G r^2$. Substituting (18.18) yields

$$\begin{aligned} \int_{B(e, r)} p \leq & Cr^2 \int_{B(e, 2Rr)} p |\nabla_H A|_F^2 + Cr^2 \int_G |U_j|^2 / p \\ & + \frac{C(1 + \log R)^2}{R^2} M_R + Cr^2 (1 + \log R) \beta_j(G). \end{aligned} \quad (18.19)$$

Choose R_0 so large that the coefficient of M_{R_0} is at most $1/2$. Corollary 18.3 on $B(e, 2R_0r_j)$ bounds the first term by $C_{R_0}\Delta V_j$. The positive convolution and weighted Cauchy–Schwarz give

$$|U_j|^2/p_j \leq K_j * (|U|^2/q), \quad \int_G |U_j|^2/p_j \leq E_U, \quad \beta_j(G) = B.$$

This proves (18.14).

For (18.15), use the inner ball $B(e, Lr)$ and a cutoff equal to one on $B(e, RLr)$, supported in $B(e, 2RLr)$. The logarithmic bound is now $C(1 + \log(RL))$ and Hölder gives $C(Lr)^2$. The coefficient of the outer mass becomes $C(1 + \log(RL))^2/R^2$, which is at most $1/2$ for large enough R at each fixed L . Apply (18.6) with radius $2RLr_j$ and absorb the fixed powers of L into $C_{R,L}$. Finally,

$$\sum_{j \geq 0} \Delta V_j \leq m\Lambda^2 \int_G q, \quad \sum_{j \geq 0} r_j^2(E_U + B) = \frac{4}{3}(E_U + B).$$

These are precisely the asserted summable costs for a fixed anchor. $\square$

Remark 18.10 (Content and limits of the concentration statement). One can choose L so large that $\int_{B(e,L)} k_1 > 3/4$. For the Poisson smoothing of a unit point mass at the identity, $\int_{B(e,Lr_j)} p_j > 3/4$, whereas one half of the mass of any outer ball is at most $1/2$. Smooth sufficiently concentrated approximations have the same strict inequality. Thus the pure contraction in (18.15), with its error terms deleted, is not an automatic property of arbitrary Poisson-smoothed densities. These inputs are not asserted to obey (18.12).

The theorem compares two balls for the same smoothed density and the same anchor. It does not compare the density at two successive smoothing scales. The exact averaging term in (18.11) prevents one from declaring the coefficient costs globally telescoping merely by averaging over new anchors in every spatial cell. No global packing or entropy-increment conclusion is included in this section.

19 Quantitative concentration from positivity on a Carnot group

This section concerns the stationary problem with zero coordinate viscosity on a fixed Carnot group. Its unconditional conclusion is a small-set estimate with an explicit coefficient-shape oscillation error. A power estimate is proved under an additional uniform small-oscillation hypothesis. A separate stopped-martingale theorem states a sufficient entropy criterion; its premise is not asserted to follow from the stationary equation.

19.1 A logarithmic improvement supplied by positivity

Lemma 19.1 (Weak-to-strong estimate with a positive sum). *Let V be a measurable set of finite measure, let $q > 1$, and let f, h be measurable functions such that $f + h \geq 0$ almost everywhere on V . Suppose*

$$|V \cap \{|f| > t\}| \leq a/t, \quad |V \cap \{|h| > t\}| \leq b^q/t^q \quad (t > 0).$$

If $0 < a \leq 1$ and $|V| + b \leq C_0$, then

$$\int_V f^- \leq a|V| + \frac{q}{q-1}a|\log a| + \frac{b^q}{q-1}a. \quad (19.1)$$

Suppose in addition that f, h are integrable on the support of a bounded nonnegative function ψ supported in V , that $\psi \geq 1$ on W , and that $|\int \psi f| \leq C_1 a$. Then

$$\int_W |f| \leq Ca(1 + |\log a|), \quad (19.2)$$

where C depends only on $q, C_0, C_1, \|\psi\|_\infty$. Consequently, for every measurable $E \subset W$,

$$\int_E (f + h) \leq \frac{q}{q-1} b|E|^{1-1/q} + Ca(1 + |\log a|). \quad (19.3)$$

When $a = 0$, the error term in the last two estimates is zero.

Proof. Positivity gives $f^- \leq |h|$. Thus, for $t > 0$,

$$|\{f^- > t\} \cap V| \leq \min\{|V|, a/t, b^q/t^q\}.$$

Split the layer-cake integral at the thresholds a and $a^{-1/(q-1)}$. On the three resulting intervals, use respectively the three displayed bounds. The resulting integrals are the three terms in (19.1). Since

$$\int \psi f^+ = \int \psi f + \int \psi f^-,$$

we obtain (19.2). The weak- L^q bound and layer cake imply

$$\int_E |h| \leq \int_0^\infty \min\{|E|, b^q t^{-q}\} dt = \frac{q}{q-1} b|E|^{1-1/q}.$$

Adding (19.2) proves (19.3). If $a = 0$, then $f = 0$ almost everywhere. The local integrability hypothesis on the pairing is essential: weak L^1 alone does not justify integrating f against ψ . $\square$

19.2 Uniform constant-matrix kernel bounds

Let G be a Carnot group, let $X_1, \dots, X_m$ be fixed left-invariant degree-one generators, and let $D > 2$ denote the homogeneous dimension. Fix a left-invariant homogeneous control distance d . Balls have measure $|B(g, r)| = v_G r^D$. For a constant symmetric matrix $B_0 > 0$, set

$$P_{B_0} = - \sum_{i,j} (B_0)_{ij} X_i X_j, \quad R_{B_0} = (I + P_{B_0})^{-1}, \quad \mathcal{T}Q = \sum_{i,j} X_i X_j Q_{ij}.$$

The last formula agrees with $\sum X_j^* X_i^* Q_{ij}$ on symmetric tensors.

Lemma 19.2 (Uniformity on a compact ellipticity family). *Let $\mathcal{K}$ be a compact family of positive definite symmetric $m \times m$ matrices with fixed positive lower and upper eigenvalue bounds. The fixed-matrix resolvent bounds of Lemma 8.1 hold uniformly for $B_0 \in \mathcal{K}$. In particular,*

$$\|R_{B_0} \mathcal{T}E\|_{L^{1,\infty}} \leq C \|E\|_{L^1} \quad (E \in L^1(G; \text{Sym}_m)),$$

with the operator interpreted as its weak-(1,1) extension. No such function-valued assertion is made here for arbitrary tensor measures, since a possible identity term can retain a singular part. If F_i, G_0 are finite signed measures supported in a fixed bounded set, then on any fixed bounded target set

$$\left\| \sum_i R_{B_0} X_i F_i + R_{B_0} G_0 \right\|_{L^{q,\infty}} \leq C \left(\sum_i \|F_i\|_{\mathcal{M}} + \|G_0\|_{\mathcal{M}} \right), \quad q = \frac{D}{D-1}. \quad (19.4)$$

For every fixed smooth compactly supported ψ , the adjoint image $(R_{B_0} \mathcal{T})^* \psi$ is uniformly bounded on a fixed compact set.

Proof. We give the uniformity argument, including the point that a horizontal change of basis need not induce an automorphism of G . Let s be the step of G and let $\tilde{G}$ be the free step- s Carnot group on m generators $\tilde{X}_i$. There is a fixed surjective graded homomorphism $\pi : \tilde{G} \rightarrow G$ with $d\pi(\tilde{X}_i) = X_i$. This remains true for redundant generators. Its kernel H is a closed homogeneous normal subgroup. Write $\tilde{D}$ for the homogeneous dimension upstairs and $D_H = \tilde{D} - D$ for that of H .

For $S = B_0^{1/2}$, freeness extends the linear map S on the first layer to a graded automorphism Φ_S of $\tilde{G}$. The heat kernel of $-\sum (B_0)_{ij} \tilde{X}_i \tilde{X}_j$ is therefore

$$\tilde{h}_{B_0}(t, z) = J_S^{-1} \tilde{h}_I(t, \Phi_S^{-1} z),$$

where J_S is the Haar Jacobian. Compactness of $\mathcal{K}$ makes S, S^{-1} , their induced maps on all homogeneous layers, and $J_S^{\pm 1}$ uniformly bounded. Homogeneous norms are uniformly comparable under these maps. A prescribed finite sequence of left- and right-horizontal derivatives transforms into a finite linear combination of derivatives of the same type, with uniformly bounded coefficients. The fixed smooth heat-kernel derivative estimates [2, Theorem 2.4] thus give

$$|\mathcal{D}^k \tilde{h}_{B_0}(t, z)| \leq C_k t^{-(\tilde{D}+k)/2} \exp\{-c_k \tilde{d}(e, z)^2/t\} \quad (19.5)$$

for every prescribed order k , uniformly in B_0 .

Push this probability convolution semigroup forward under π . The generator downstairs is $-P_{B_0}$, and its density is the integral of $\tilde{h}_{B_0}$ over the fibers. Left and right derivatives push forward to the corresponding derivatives downstairs, because the fields are π -related. Integration by parts on fibers is justified by (19.5), first with compact cutoffs.

Here is the required uniform fiber estimate. If a coset zH meets a radius- R ball upstairs, choose zh_0 in that intersection. Every other zh in the intersection satisfies $\tilde{d}(e, h_0^{-1}h) \leq 2R$. Haar invariance on H and homogeneous scaling yield

$$|\{h \in H : \tilde{d}(e, zh) \leq R\}|_H \leq CR^{D_H}.$$

Extracting half of the Gaussian exponent using the quotient distance and integrating the remaining half by this volume bound gives

$$\int_H e^{-c\tilde{d}(e, zh)^2/t} dh \leq Ct^{D_H/2} e^{-c'd(e, \pi(z))^2/t}.$$

The quotient distance and the fixed distance on G are equivalent. Consequently all prescribed source and target derivatives of the two-point heat kernel downstairs obey the uniform bounds

$$|\mathcal{D}^k h_{B_0}(t, x)| \leq C_k t^{-(D+k)/2} e^{-c_k d(e, x)^2/t}. \quad (19.6)$$

The maps Φ_S have been used only upstairs; no assertion that they descend to G is required.

Integrating (19.6) against $e^{-t} dt$ gives the resolvent bounds $Cd(x, y)^{2-D}$ for its kernel and $Cd(x, y)^{1-D}$ for its first source derivatives. Second source derivatives have the uniform cancellation, annular size, and smoothness constants used in the Calderón-Zygmund proof in Lemma 8.1. That proof, including its integrable resolvent remainder and its possible bounded identity term, therefore gives the asserted uniform weak-(1, 1) bound. Smooth-test subtraction in the same principal-value representation gives the uniform adjoint-test bound.

For completeness, the weak potential estimate is elementary. For a finite positive measure τ , split its order-one potential at distance ρ . The far part is at most $C\rho^{1-D}\|\tau\|$, and Fubini and the ball-volume formula bound the integral of the near part by $C\rho\|\tau\|$. At level $t > 0$, choose $\rho = (C\|\tau\|/t)^{1/(D-1)}$ so that the far part is at most $t/2$. Chebyshev's inequality gives a distribution-function bound $C(\|\tau\|/t)^{D/(D-1)}$. On bounded source and target sets the order-two kernel is bounded by a fixed multiple of the order-one kernel; applying this argument to the total variations proves (19.4). $\square$

19.3 An unconditional small-set estimate with an oscillation error

Theorem 19.3 (Stationary normalized-shape estimate). *Let $U \subset G$ be open. Suppose $p \geq 0$ is locally integrable and A is a Borel symmetric matrix such that $\lambda I \leq A \leq \Lambda I$ almost everywhere for the measure $p dx$, where $0 < \lambda \leq \Lambda < \infty$. Suppose*

$$\mathcal{T}(Ap) = \mu \geq 0 \quad \text{in } \mathcal{D}'(U), \quad (19.7)$$

where μ is locally finite. Define

$$d\sigma = (\operatorname{tr} A)p dx, \quad B = A / \operatorname{tr} A \quad \text{on } \{p > 0\}.$$

Choose any admissible value of B on $\{p = 0\}$. Let

$$\mathcal{K} = \left\{ C = C^\top : \operatorname{tr} C = 1, \frac{\lambda}{m\Lambda} I \leq C \leq \frac{\Lambda}{m\lambda} I \right\}.$$

For $B(g, 4r) \Subset U$, set $M = \sigma(B(g, 4r))$. If $M > 0$, put

$$e_{g,r} = \inf_{B_0 \in \mathcal{K}} \frac{1}{M} \int_{B(g, 3r)} |B - B_0|_F d\sigma.$$

Then for every Borel $E \subset B(g, r)$,

$$\sigma(E) \leq CM \left[\left(\frac{|E|}{r^D} \right)^{1/D} + \omega(e_{g,r}) \right], \quad \omega(a) = a(1 + |\log a|), \quad \omega(0) = 0. \quad (19.8)$$

The constant depends only on the group, the fixed generators and distance, and λ, Λ . When $M = 0$, the assertion is interpreted as $\sigma(E) = 0$. Since

$$m\lambda p dx \leq \sigma \leq m\Lambda p dx, \quad (19.9)$$

the corresponding estimate also holds for $p dx$ with its outer mass, after changing the constant and retaining the same shape error.

Proof. The trace normalization gives $B \in \mathcal{K}$ and (19.9). The map $F(x) = \delta_{1/r}(g^{-1}x)$ transforms the equation into the same equation for

$$\sigma' = M^{-1}F_*\sigma, \quad Q' = M^{-1}F_*(B\sigma), \quad \mu' = r^2 M^{-1}F_*\mu.$$

Moreover $|F(E)| = r^{-D}|E|$. It suffices to take $g = e$, $r = 1$, and $\sigma(B_4) = 1$, where $B_a = B(e, a)$.

Choose $0 \leq \theta \in C_c^\infty(B_4)$ equal to one near $\overline{B_3}$. Positivity of the source gives

$$\mu(B_3) \leq \int \theta d\mu = \int \sum_{i,j} B_{ij} X_i X_j \theta d\sigma \leq C. \quad (19.10)$$

Only the entries of B and the fixed cutoff derivatives are bounded; no derivative of B has been taken.

Choose $0 \leq \chi \in C_c^\infty(B_3)$ equal to one near $\overline{B_2}$, fix $B_0 \in \mathcal{K}$, and put $E_0 = (B_0 - B)\sigma$. Extend compactly supported products below by zero. For $P = P_{B_0}$, direct distributional multiplication gives

$$(I + P)(\chi\sigma) = -\mathcal{T}(\chi E_0) + \sum_i X_i F_i + G_0, \quad (19.11)$$

where the measures can be written explicitly as

$$F_i = -2 \sum_j B_{ij} (X_j \chi) \sigma, \quad G_0 = \chi \sigma - \chi \mu + \sum_{i,j} B_{ij} (X_i X_j \chi) \sigma.$$

Indeed, symmetry gives $\mathcal{T}(\chi B\sigma) = \chi\mu + 2\sum_i X_i \sum_j B_{ij}(X_j\chi)\sigma - \sum_{i,j} B_{ij}(X_i X_j\chi)\sigma$, and $B_0\chi\sigma = \chi B\sigma + \chi E_0$. Equation (19.10) implies

$$\sum_i \|F_i\|_{\mathcal{M}} + \|G_0\|_{\mathcal{M}} \leq C. \quad (19.12)$$

Smooth the original equation (19.7) before applying the cutoff, using a common nonnegative left group convolution approximate identity $\eta_\delta*$. Restrict the original measures to a neighborhood of all cutoff supports compactly contained in B_4 before convolution; for sufficiently small δ the equation on those supports is unchanged, since left-invariant derivatives commute with this operation. Write $\sigma_\delta = \eta_\delta * \sigma$, $Q_\delta = \eta_\delta * (B\sigma)$, and $\mu_\delta = \eta_\delta * \mu$. Positivity and the matrix cone are preserved, and $B_\delta = Q_\delta/\sigma_\delta$ is admissible where $\sigma_\delta > 0$, with any admissible value elsewhere. Set $E_{0,\delta} = B_0\sigma_\delta - Q_\delta = \eta_\delta * E_0$ on the cutoff supports. Now derive (19.11) anew for these smoothed quantities. In particular $F_{i,\delta}, G_{0,\delta}$ are rebuilt from $\chi, Q_\delta, \sigma_\delta, \mu_\delta$ using the displayed formulas; they are not obtained by convolving the old localized sources. The same mass bounds hold, with harmless fixed constants. Since $\text{supp } \chi \subseteq B_3$, positivity and Fubini also give, for all sufficiently small δ ,

$$\|\chi(\eta_\delta * E_0)\|_1 \leq \int_{B_3} |B - B_0|_F d\sigma =: \alpha. \quad (19.13)$$

Nested supports guarantee this estimate without any assertion that noncommutative convolution preserves a particular ball exactly.

Apply $R = R_{B_0}$ to (19.11) for the smoothed data, and write

$$u_\delta = \chi\sigma_\delta = f_\delta + h_\delta, \quad f_\delta = -RT(\chi E_{0,\delta}), \quad h_\delta = \sum_i RX_i F_{i,\delta} + RG_{0,\delta}.$$

These are genuine locally integrable functions and $u_\delta \geq 0$. Lemma 19.2 gives

$$\|f_\delta\|_{L^{1,\infty}} \leq C\alpha, \quad \|h_\delta\|_{L^{q,\infty}(B_3)} \leq C, \quad q = D/(D-1).$$

Choose a nonnegative $\psi \in C_c^\infty(B_3)$ with $\psi \geq 1$ on B_2 . The uniform adjoint-test bound gives

$$|\langle f_\delta, \psi \rangle| \leq C\alpha.$$

For small positive α , Lemma 19.1, with $a = C\alpha$, implies for every Borel $E \subset B_1$

$$\int_E \sigma_\delta \leq C|E|^{1/D} + C\omega(\alpha). \quad (19.14)$$

The case $\alpha = 0$ follows from the zero-error part of that lemma. The parameter α is uniformly bounded by the diameter of $\mathcal{K}$. When α is bounded away from zero, (19.14) follows instead from the uniform mass bound, after enlarging C . Thus it holds for every admissible B_0 .

The assumed local L^1 density of σ implies local L^1 convergence of σ_δ to σ . Therefore (19.14) passes to every fixed Borel set E . The infimum defining the oscillation is attained: its objective is continuous on the compact set $\mathcal{K}$. Choose a minimizer and undo the normalization to obtain (19.8). $\square$

19.4 A power estimate under a fixed small-oscillation threshold

Corollary 19.4 (Additional uniform small shape oscillation). *There exist $e_* > 0$, $s > 0$, and $C < \infty$, depending only on the data in Theorem 19.3, with the following property. Fix $B(g, 4r) \subseteq U$ and assume, in addition to that theorem's hypotheses,*

$$e_{h,\rho} \leq e_* \quad \text{whenever } B(h, 4\rho) \subseteq B(g, 4r) \text{ and } \sigma(B(h, 4\rho)) > 0.$$

If $w = d\sigma/dx$, then

$$\int_{B(g,r)} w^{1+s} dx \leq Cr^{-Ds} \sigma(B(g,4r))^{1+s}. \quad (19.15)$$

The same conclusion holds for p with its outer p mass, after changing C . No rate of decay of $e_{h,\rho}$ as $\rho \downarrow 0$ is required.

Proof. We include the local iteration because the extra hypothesis does not follow uniformly from mere measurable differentiation. Normalize $g = e$, $r = 1$, and $\sigma(B_4) = 1$; zero outer mass is trivial. Extend $w\mathbf{1}_{B_4}$ by zero and denote its centered Hardy–Littlewood maximal function by F . The doubling-ball covering argument gives

$$|\{F > t\}| \leq C/t \quad (t > 0), \quad F \geq w \quad \text{a.e. on } B_4. \quad (19.16)$$

For the weak estimate, choose disjoint balls from the family witnessing averages above t , so that their fixed dilates cover the level set; doubling and the total mass bound prove the assertion. The second assertion is differentiation of L^1 functions.

We will use a Whitney ball cover $\{B_i = B(x_i, \rho_i)\}$ of the open set $O_t = \{F > t\}$. It can be chosen so that $4B_i \subset O_t$, the balls $4B_i$ have overlap at most a fixed number N_4 , and each B_i has a point $z_i \notin O_t$ with $d(x_i, z_i) \leq C_W \rho_i$. One construction takes a maximal disjoint subfamily of balls $B(x, d(x, O_t^c)/1000)$ and uses fixed fivefold enlargements to cover O_t . When enlarged balls intersect, their distances to O_t^c , and hence their radii, are comparable. The disjoint smaller balls and the formula $|B(x, a)| = v_G a^D$ then bound the multiplicity of the fourfold enlargements. These observations prove all the properties just listed, with fixed constants. As O_t has finite measure, its complement is nonempty and the construction applies.

Choose $a_* > 0$ and then $e_* > 0$ so small that Theorem 19.3 implies

$$\sigma(E) \leq \frac{1}{4N_4} \sigma(4B) \quad \text{if } E \subset B, \quad |E| \leq a_* |B|, \quad 4B \Subset B_4. \quad (19.17)$$

Explicitly, it suffices that $C[(v_G a_*)^{1/D} + \omega(e_*)] \leq (4N_4)^{-1}$.

If $x \in B_i$, then every ball centered at x with radius at least ρ_i is contained in a ball centered at z_i with a fixed multiple of that radius. Since $F(z_i) \leq t$, its w average is at most Ct . Consequently, for a fixed sufficiently large A , the condition $F(x) > At$ can only be witnessed by balls of radius less than ρ_i ; these lie in $2B_i$. The weak maximal estimate applied to $w\mathbf{1}_{2B_i}$, and then the bound on the average over a ball centered at z_i containing $2B_i$, show

$$|B_i \cap \{F > At\}| \leq \frac{C}{At} \int_{2B_i} w \leq \frac{C}{A} |B_i|. \quad (19.18)$$

Increase the fixed A until $C/A \leq a_*$.

For $1 \leq a < b \leq 2$, select the Whitney balls that meet $B_a \cap \{F > At\}$. Each such ball lies in O_t , so (19.16) gives $\rho_i \leq Ct^{-1/D}$. Hence $4B_i \Subset B_b$ whenever

$$t \geq t_0 := C_*(b-a)^{-D},$$

with a sufficiently large fixed C_* . Apply (19.17) to $E = B_i \cap B_a \cap \{F > At\}$ using (19.18), and sum over these balls. Their fourfold enlargements lie in $B_b \cap O_t$ and have overlap at most N_4 . We obtain the local good-lambda inequality

$$\sigma(B_a \cap \{F > At\}) \leq \frac{1}{4} \sigma(B_b \cap \{F > t\}) \quad (t \geq C_*(b-a)^{-D}). \quad (19.19)$$

To avoid assuming the conclusion is already finite, let

$$\Psi_N(a) = \int_{B_a} w \min(F, N)^s dx, \quad N < \infty.$$

Layer cake, (19.19), and the mass bound imply

$$\Psi_N(a) \leq C(b-a)^{-Ds} + \vartheta \Psi_N(b), \quad \vartheta = A^s/4. \quad (19.20)$$

Indeed, integrate the distribution function up to At_0 using mass at most one, and above At_0 use $t = Au$ and (19.19); the resulting upper integral is bounded by $A^s \Psi_N(b)/4$. Choose $s > 0$ so small that $\vartheta 2^{Ds} < 1$. Iterate (19.20) along $a_j = 2 - 2^{-j}$. The resulting geometric series is bounded independently of N , whereas its remainder is at most $\vartheta^k N^s$ and tends to zero as $k \rightarrow \infty$. Thus $\Psi_N(1) \leq C$ uniformly in N . Monotone convergence and $F \geq w$ give $\int_{B_1} w^{1+s} \leq C$. Translation, dilation, and mass normalization give (19.15). Finally use (19.9) for p . $\square$

Remark 19.5. For a general Borel matrix B , differentiation gives small oscillation at almost every point at sufficiently fine scales, but no uniform threshold scale or bound on the number of preceding scales. Thus neither Theorem 19.3 nor Corollary 19.4 proves an entropy estimate for unrestricted matrices on all Carnot groups. Scalar variation along one fixed positive matrix ray has zero shape oscillation, but arbitrary matrix variation does not.

19.5 A conditional stopped-martingale entropy theorem

Theorem 19.6 (Corona entropy criterion). *Let Q_* be a finite-volume root cube, equipped with nested finite dyadic partitions whose generated sigma algebra agrees with the measurable sets modulo Haar-null sets. Let $d\sigma = w dx$ on Q_* , where $w \geq 0$, $w \in L^1$, and $M = \sigma(Q_*) > 0$. Let B be a bounded matrix-valued measurable function with $|B|_F \leq L$. For positive-mass cubes put*

$$B_Q = \frac{1}{\sigma(Q)} \int_Q B d\sigma, \quad d_Q = \frac{\sigma(Q)}{|Q|}.$$

Fix $\delta > 0$. For each finite terminal generation N , form a corona tree as follows. From a root Q stop along every branch at the first strict descendant R for which $|B_R - B_Q|_F \geq \delta$, or at generation N , whichever comes first. The resulting partition is denoted $\text{Stop}_N(Q)$. Threshold hits strictly before generation N become new roots, to which the same rule is applied. Cubes of zero σ mass are omitted.

Assume that one constant C_0 works for every N and every resulting corona root Q in the estimate

$$\sum_{R \in \text{Stop}_N(Q)} \sigma(R) \log(d_R/d_Q) \leq C_0 \sigma(Q). \quad (19.21)$$

Then

$$\sum_{R \in \mathcal{D}_N(Q_*)} \sigma(R) \log \frac{d_R}{M/|Q_*|} \leq C_0 \left(1 + \frac{4L^2}{\delta^2}\right) M \quad (19.22)$$

for every N , and

$$\int_{Q_*} w \log \frac{w}{M/|Q_*|} dx \leq C_0 \left(1 + \frac{4L^2}{\delta^2}\right) M. \quad (19.23)$$

Here $0 \log 0$ is interpreted as zero. This theorem is conditional on (19.21); it does not assert that a differential constraint implies that premise.

Proof. Use the probability measure $d\mathbb{P} = d\sigma/M$ and let B_n be the conditional expectation of B at ordinary dyadic generation n . The successive matrix-threshold stopping times τ_k are bounded by N ; once N is reached, set all subsequent $\tau_k = N$. The process B_{τ_k} is a square-integrable martingale for the stopped sigma algebras, by conditional expectation and optional sampling for bounded stopping times. Orthogonality of its increments gives

$$\sum_k \mathbb{E}|B_{\tau_{k+1}} - B_{\tau_k}|_F^2 = \mathbb{E}|B_N - B_0|_F^2 \leq \mathbb{E}|B - B_0|_F^2.$$

Multiplying by M and writing the finite sum cube by cube yields

$$\sum_{Q \text{ corona root}} \sum_{R \in \text{Stop}_N(Q)} \sigma(R) |B_R - B_Q|_F^2 \leq \int_{Q_*} |B - B_{Q_*}|_F^2 d\sigma \leq 4L^2 M. \quad (19.24)$$

Every root other than Q_* is a threshold hit strictly before generation N . Its mass is charged once to its unique parent-corona increment, which costs at least $\delta^2 \sigma(R)$. Hence

$$\sum_{Q \text{ corona root}} \sigma(Q) \leq (1 + 4L^2/\delta^2) M. \quad (19.25)$$

Threshold hits at the terminal generation need not become roots and require no additional charge.

For a terminal cube S , the logarithms of the density ratios along its successive stopping ancestors telescope:

$$\log(d_S/d_{Q_*}) = \sum_{\text{successive stopping edges } Q \rightarrow R \text{ on the branch to } S} \log(d_R/d_Q).$$

Multiply by $\sigma(S)$ and sum over terminal cubes. The mass of the terminal descendants of a stopped cube R is exactly $\sigma(R)$. Rearranging these finite sums proves that the left side of (19.22) is the sum, over all roots, of the left sides of (19.21). Equations (19.21) and (19.25) prove (19.22).

The terminal density $w_N = \sum_{R \in \mathcal{D}_N} d_R \mathbf{1}_R$ is the conditional expectation of w with respect to Haar measure. The generating-filtration assumption implies $w_N \rightarrow w$ almost everywhere and in L^1 . The function $z \mapsto z \log(z/(M/|Q_*|))$ is bounded below by $-M/(e|Q_*|)$. Adding this constant and applying Fatou proves (19.23), without an initial entropy assumption. $\square$

Remark 19.7 (The unproved PDE premise). The energy identity (19.24) controls actual successive stopped increments. It does not control the sum, over all ordinary scales, of the full future oscillation within each cube. Likewise, summability of numbers ΔV_j does not imply summability of $\omega(\sqrt{\Delta V_j})$; take $\Delta V_j = j^{-2}$. Ordinary dyadic averaging does not preserve $\mathcal{T}(B\sigma) = \mu$. Replacing the stress by its averages inside stopped cubes creates second-order interface distributions. No estimate for these distributions that proves (19.21) is included here. Accordingly Theorem 19.6 supplies a precise sufficient criterion, not an unconditional entropy theorem for the full matrix cone.

20 Further exact feasibility and convexity statements

This section records three independently useful conclusions not contained in the entropy theorems: a fixed-density feasibility criterion, an exact dual representation of a horizontal-convex test budget, and finite-interface bounds for a specified Heisenberg digit construction. Their conclusions are distinct from entropy coercivity, and their restrictions are retained.

20.1 A fixed-density Pucci feasibility criterion

Theorem 20.1 (Exact local positive-stress feasibility). *Let $U \subset \mathbb{R}^n$ be open, let $X_1, \dots, X_m$ be smooth real vector fields on U , and fix $0 < \lambda \leq \Lambda$ and $\varepsilon \geq 0$. Let $p \geq 0$ belong to $L^1(U)$. Define*

$$H_X \varphi = \left(\frac{X_i X_j + X_j X_i}{2} \varphi \right)_{ij}, \quad \mathcal{P}^+(H) = \max_{\lambda I \leq B \leq \Lambda I} \text{tr}(BH).$$

The following are equivalent:

1. There exists a Borel symmetric field A , with $\lambda I \leq A \leq \Lambda I$ for $p dx$ -almost every point, such that $\mathcal{L}_A^*(p dx) + \varepsilon \Delta(p dx)$ is a nonnegative distribution on U .
2. Every nonnegative $\varphi \in C_c^\infty(U)$ satisfies

$$\int_U p [\mathcal{P}^+(H_X \varphi) + \varepsilon \Delta \varphi] dx \geq 0.$$

The distribution in the first assertion is a locally finite nonnegative Radon measure. It need not have finite total mass on U . Its restriction to every $V \Subset U$ is finite. No bracket assumption is needed.

Proof. Let $\eta = p dx$ and let $\mathcal{C}$ be the set of $A \in L^\infty(\eta; \text{Sym}_m)$ with $\lambda I \leq A \leq \Lambda I$. This set is nonempty, bounded and convex. It is weak-star closed: test each quadratic matrix inequality against nonnegative $L^1(\eta)$ functions and a countable dense family of vectors, then use continuity in the vector. Banach–Alaoglu makes $\mathcal{C}$ weak-star compact. For a nonnegative smooth compact test put

$$F_\varphi(A) = \int_U p [\text{tr}(AH_X \varphi) + \varepsilon \Delta \varphi] dx.$$

This is affine and weak-star continuous. Suppose a finite family $\varphi_1, \dots, \varphi_k$ has no common $A \in \mathcal{C}$ with $F_{\varphi_j}(A) \geq 0$. Its compact convex image in $\mathbb{R}^k$ is disjoint from the closed positive orthant. Strict separation supplies $\theta_j \geq 0$, not all zero, for which $\sup_{A \in \mathcal{C}} \sum_j \theta_j F_{\varphi_j}(A) < 0$. The signs follow because a separating functional bounded below on the positive orthant must have nonnegative coefficients, and its infimum there is zero. With $\varphi = \sum_j \theta_j \varphi_j$, this says $\sup_{A \in \mathcal{C}} F_\varphi(A) < 0$.

On the other hand pointwise maximization gives exactly

$$\sup_{A \in \mathcal{C}} F_\varphi(A) = \int_U p [\mathcal{P}^+(H_X \varphi) + \varepsilon \Delta \varphi] dx.$$

For attainment use $A(x) = \lambda I + (\Lambda - \lambda) \mathbf{1}_{(0, \infty)}(H_X \varphi(x))$, where the last factor is the positive spectral projection. Finite-dimensional spectral calculus makes this a Borel field in the matrix interval and it maximizes the trace at every point. Thus the second assertion gives the finite intersection property for the closed sets $\{A : F_\varphi(A) \geq 0\}$. Compactness yields one common coefficient class for all tests. It has a Borel representative; reset that representative to λI on its null bound-violation set.

The resulting distribution is positive. For tests ψ supported in a compact $K \Subset U$, choose $\chi \geq 0$ equal to one near K . Positivity applied to $\|\psi\|_\infty \chi \pm \psi$ bounds its value by $\|\psi\|_\infty F_\chi(A)$. Hence it has order zero locally, and the local Riesz representation theorem gives a positive Radon measure μ . In particular

$$\mu(K) \leq F_\chi(A) \leq \sup_{\substack{x \in \text{supp } \chi \\ \lambda I \leq B \leq \Lambda I}} |\text{tr}(BH_X \chi(x)) + \varepsilon \Delta \chi(x)| \int_U p.$$

This proves the local finiteness statement and finite restriction to $V \Subset U$. Conversely, any field from the first assertion satisfies $0 \leq F_\varphi(A) \leq \int p [\mathcal{P}^+(H_X \varphi) + \varepsilon \Delta \varphi]$, proving the second assertion. $\square$

Example 20.2 (The global sink-mass qualification is necessary). On $U = (0, 1)$ take $X = \partial_x$, $\lambda = \Lambda = 1$, $\varepsilon = 0$, and $p(x) = (1 - a)(1 - x)^{-a}$ with $0 < a < 1$. Then $\int_U p = 1$ and $p'' = (1 - a)a(a + 1)(1 - x)^{-a-2} > 0$. All tests in Theorem 20.1 are nonnegative by compactly supported integration by parts, but $\int_U p'' = \infty$. The coefficient is forced to be $A = 1$, so the only distributional sink is $p'' dx$. Thus the criterion cannot be strengthened to a globally finite-sink characterization without an additional hypothesis. This example concerns sink finiteness, not failure of an entropy bound.

Unlike Proposition 15.1, which varies both density and stress in a compact relaxed class, Theorem 20.1 fixes p and characterizes whether one common coefficient field can satisfy the entire stationary test family. Testing only a selected convex or scalar-chain subfamily does not establish its premise.

20.2 Exact dual representation of a horizontal-convex budget

Use $X = \partial_x - (y/2)\partial_z$, $Y = \partial_y + (x/2)\partial_z$ on $\mathbb{H}^1$. Let $U \subset \mathbb{R}^3$ be bounded and open and $K \Subset U$. Let $\mathcal{V} = C^\infty(\bar{U})$ mean restrictions of functions smooth on a neighborhood of $\bar{U}$. Derivatives on the closure are independent of the representative, because they agree on U and extend continuously. Put

$$Hu = \begin{pmatrix} X^2u & (XY + YX)u/2 \\ (XY + YX)u/2 & Y^2u \end{pmatrix}, \quad N(u) = \max\{\|u\|_{\infty, \bar{U}}, \|Xu\|_{\infty, \bar{U}}, \|Yu\|_{\infty, \bar{U}}\}.$$

For a finite nonnegative Borel measure ρ supported in K , set $\ell_\rho(u) = \int_K \text{tr}(Hu) d\rho$.

Theorem 20.3 (Horizontal-convex budget duality on $\mathbb{H}^1$). *For $B \geq 0$, the inequality*

$$\ell_\rho(u) \leq BN(u) \quad \text{for every } u \in \mathcal{V} \text{ with } Hu \geq 0 \text{ on } \bar{U}$$

is equivalent to the existence of finite signed Borel measures η_0, η_1, η_2 and a finite symmetric positive-semidefinite matrix measure M on $\bar{U}$ such that

$$\sum_{j=0}^2 \|\eta_j\|_{\text{TV}} \leq B, \quad \ell_\rho(u) + \int_{\bar{U}} Hu : dM = \int u d\eta_0 + \int Xu d\eta_1 + \int Yu d\eta_2 \quad (20.1)$$

for every $u \in \mathcal{V}$. In addition, with $u_0 = (x^2 + y^2)/2$, the measures may be chosen so that

$$\text{tr } M(\bar{U}) \leq BN(u_0) - 2\rho(K).$$

Positivity of M means positivity of the measure $e^T M e$ for every constant $e \in \mathbb{R}^2$. The identity includes boundary measures.

Proof. Let $\mathcal{C} = \{c \in \mathcal{V} : Hc \geq 0\}$ and assume the budget. Define

$$P(v) = \inf_{c \in \mathcal{C}} \{BN(v + c) - \ell_\rho(c)\}.$$

The budget and triangle inequality give $-BN(v) \leq P(v) \leq BN(v)$. Addition and positive scaling in the cone give subadditivity and positive homogeneity; $P(0) = 0$. Algebraic real Hahn–Banach supplies a linear functional F with $F(v) \leq P(v) \leq BN(v)$. Applying this to $-v$ gives $|F(v)| \leq BN(v)$. Taking c itself in the infimum defining $P(-c)$ gives $F(c) \geq \ell_\rho(c)$, so $G = F - \ell_\rho$ is nonnegative on $\mathcal{C}$.

If $Hu = 0$, both signs of u are in $\mathcal{C}$, hence $G(u) = 0$. Thus $g(Hu) = G(u)$ defines a positive linear functional on the Hessian range $\mathcal{R} \subset C(\bar{U}; \text{Sym}_2)$. The range contains the order unit $I_2 = Hu_0$. For $A \in \mathcal{R}$, $-\|A\|_\infty I_2 \leq A \leq \|A\|_\infty I_2$, with the uniform operator norm, whence $\|g\| = g(I_2)$. Extend g with its norm preserved to the full continuous matrix space. The extension $\tilde{g}$ is positive: for $A \geq 0$, $a = \|A\|_\infty > 0$, $\|I_2 - A/a\|_\infty \leq 1$, and therefore $\tilde{g}(A/a) = g(I_2) - \tilde{g}(I_2 - A/a) \geq 0$. The zero case is immediate. Coordinatewise Riesz representation gives $\tilde{g}(A) = \int A : dM$ for a symmetric matrix measure. Tests $A = \phi e e^T$, $\phi \geq 0$, prove positivity of M . Its trace mass is

$$g(I_2) = F(u_0) - \ell_\rho(u_0) \leq BN(u_0) - 2\rho(K).$$

The map $J : u \mapsto (u, Xu, Yu)$ is an isometric embedding of $(\mathcal{V}, N)$ into $C(\overline{U})^3$ with its maximum norm. Extend $F \circ J^{-1}$ from its image by norm-preserving Hahn–Banach. Coordinatewise Riesz representation supplies η_j with $F(u) = \int u d\eta_0 + \int Xu d\eta_1 + \int Yu d\eta_2$. The dual norm of the maximum product norm is the sum of the coordinate total variations, giving (20.1). Since $G(u) = \int Hu : dM$, the full asserted identity follows. Conversely, if $Hu \geq 0$, positivity gives $\int Hu : dM \geq 0$, and the three signed measures bound the remaining terms by $BN(u)$. $\square$

Corollary 20.4 (What the stationary equation supplies). *Let A be Borel symmetric with $\lambda I_2 \leq A \leq \Lambda I_2$ for pdx -almost every point. For $\varepsilon = 0$, a nonnegative stationary density on U with finite mass and $\mathcal{L}_A^* p = \mu \geq 0$ implies the budget of Theorem 20.3 for $\rho = \mathbf{1}_K p dx$, with $B \leq C \int_{\text{supp } \chi} p$, where χ is a fixed nonnegative compact cutoff equal to one near K . The resulting dual stress is bounded below by $I_2 \rho$ and has controlled total mass; no pointwise upper domination by $CI_2 \rho$ is asserted.*

Proof. For $Hu \geq 0$ set $R = \|u\|_{\infty, \overline{U}}$ and test with $\chi(R - u) \geq 0$. The exact product formula is

$$\mathcal{L}_A[\chi(R - u)] = (R - u)\mathcal{L}_A\chi - \chi A : Hu - 2 \sum_{i,j} a_{ij}(X_i \chi)(X_j u), \quad (X_1, X_2) = (X, Y).$$

The source sign and $A : Hu \geq \lambda \text{tr}(Hu)$ give $\lambda \ell_\rho(u) \leq CN(u) \int_{\text{supp } \chi} p$. No derivative of A appears. Apply the preceding theorem and note $R_* = I_2 \rho + M \geq I_2 \rho$; its trace mass is at most $BN(u_0)$. The theorem provides only that trace bound, signed lower-order sources and possible boundary measures. It supplies no upper domination by ρ . Horizontal convexity does not control the sign of the coordinate Euclidean Laplacian, so the argument here retains $\varepsilon = 0$. $\square$

For completeness, throughout the following scaling observation all densities are supported in K . The least admissible budget $\mathcal{B}(\rho)$ is positively homogeneous. If one budget ball $\{p \geq 0 : \int p \leq 1, \mathcal{B}(p dx) \leq B_0\}$, $B_0 > 0$, had a uniform L^q bound for some fixed $q > 1$, then every density of finite mass and finite budget would belong qualitatively to the same L^q space. Indeed multiply a nonzero such density by $\theta = \min\{1, (\int p)^{-1}, B_0/\mathcal{B}(p dx)\} > 0$, apply the bound, and divide by θ . A zero budget forces zero mass by the test u_0 . This scaling observation is not an asserted power embedding, nor does the duality theorem prove entropy from the budget alone.

20.3 Finite-interface bounds in a fixed Heisenberg digit geometry

We give the digit data explicitly, so the following bounds require no unstated properties of an iterated-function system. For $y \in [0, 1]$ fix binary digits $v_k \in \{0, 1\}$ with $y = \sum_k v_k 2^{-k}$; choose any consistent convention at dyadic points. For the binary digits $x = \sum_k u_k 2^{-k}$ define

$$d_k(y) = \frac{v_k}{2} 4^{-k} + \sum_{\ell > k} v_\ell 2^{-k-\ell}, \quad \beta_y(x) = \sum_k u_k d_k(y).$$

Let ν be the probability law of $T = \frac{3}{2} \sum_{k \geq 1} W_k 4^{-k}$, with independent uniform digits $W_k \in \{0, 1, 2\}$. This is the same binary-horizontal, ternary-vertical digit law used in the twelve-map Heisenberg construction; only the explicit formulas here are needed. Its support is in $[0, 1]$. Put $\alpha = \log 3 / \log 4 > 1/2$.

Lemma 20.5 (Jump and interval bounds). *The function β_y is nondecreasing. At each odd dyadic point of generation i its jump is*

$$J_i(y) = \frac{v_i}{2} 4^{-i} + \frac{3}{2} \sum_{k > i} v_k 4^{-k}, \quad 0 \leq J_i(y) \leq 4^{-i}.$$

Inside an open dyadic cell of length $H = 2^{-\ell}$ its total variation is at most $H^2/2$. Moreover $\nu(I) \leq 4|I|^\alpha$ for every interval I .

Proof. At a generation- i binary carry the digit u_i changes from zero to one and all later digits change from ones to zeros, so the jump is $d_i - \sum_{k>i} d_k$. Summing the geometric coefficients gives the formula displayed, which is nonnegative. This proves monotonicity at all dyadic carries; uniform convergence of the digit expansions proves monotonicity everywhere, with one-sided conventions at jumps. Since $d_k \leq \frac{3}{2}4^{-k}$, the total possible change within a cell with the first ℓ digits fixed is at most $\sum_{k>\ell} d_k \leq \frac{1}{2}4^{-\ell} = H^2/2$. The jump sums exhaust the total variation: the first n generations have total jump mass tending to $\sum_k d_k$, with residual bounded by $2^n \sum_{k>n} d_k \rightarrow 0$. For the Cantor estimate, its level- n cylinders have length 4^{-n} , probability 3^{-n} , and distinct left endpoints separated by at least $\frac{3}{2}4^{-n}$. If $\ell = 4^{-n} \leq |I| < 4\ell$, at most four such cylinders meet I . Hence $\nu(I) \leq 4 \cdot 3^{-n} = 4\ell^\alpha \leq 4|I|^\alpha$. The case $|I| \geq 1$ is immediate. $\square$

Proposition 20.6 (Same-generation packing for every convex test). *Let u be a smooth globally Euclidean-convex, one-Lipschitz function on $\mathbb{R}^3$. For fixed $y \in [0, 1]$ and $t \in \mathbb{R}$ set*

$$P(x) = (x, y, t - yx/2 + \beta_y(x)), \quad v = (1, 0, -y/2), \quad g(x) = v \cdot \nabla u(P(x)) = Xu(P(x)).$$

If $\mathcal{I}_i$ is the set of generation- i dyadic interfaces in $(0, 1)$, then

$$\sum_{a \in \mathcal{I}_i} [g(a-) - g(a+)]_+ \leq (\sqrt{5} + \sqrt{2})2^{i/2}.$$

The same bound holds after averaging in $y \in [0, 1]$ and t against ν . The bound grows with i and is not a uniform convex-budget estimate.

Proof. One has $|g| \leq \sqrt{5}/2$ and $|u_z| \leq 1$. Convex-gradient monotonicity at $P(a-), P(b+)$, $b > a$, gives

$$g(a-) - g(b+) \leq \frac{2[\beta_y(b+) - \beta_y(a-)]}{b - a}.$$

Indeed their displacement is $(b - a)v + [\beta_y(b+) - \beta_y(a-)]e_z$, and the difference of their vertical gradients has absolute value at most two. Put $h = 2^{-i}$. In an open level- ℓ cell of length $H = 2^{-\ell}$, $\ell < i$, select exactly the generation- i points with negative jumps of g , say $a_1 < \dots < a_m$. If $m \leq 1$, the bound $\sqrt{5} + H^2/h$ is immediate. Otherwise

$$\sum_{j=1}^m [g(a_j-) - g(a_j+)] = g(a_m-) - g(a_1+) + \sum_{j=1}^{m-1} [g(a_j-) - g(a_{j+1}+)].$$

The first term is at most $\sqrt{5}$. Selected consecutive interfaces are separated by at least $2h$. Their intervening variation intervals count each contribution at most twice, including shared endpoint jumps. The preceding inequality and Lemma 20.5 therefore bound the remaining sum by H^2/h . All generation- i points lie strictly inside level- ℓ cells. Summing those cells yields $\sqrt{5}2^\ell + 2^{i-\ell}$. Choose $\ell = \lfloor i/2 \rfloor$ to obtain the displayed bound. All selected terms are their positive negative-jump magnitudes, so the positive-part operation has been accounted for. Averaging preserves the estimate. $\square$

Proposition 20.7 (Uniform bound for a restricted common-trace gluing class). *Let $0 < a_1 < \dots < a_m < 1$ be finitely many distinct dyadic interfaces of any generations. Suppose the same smooth globally convex one-Lipschitz function u satisfies the additional condition*

$$u(a_j, y, z) = \psi(y, z) + c_j \quad \text{for all } (y, z) \in [-2, 2]^2,$$

where ψ is independent of j . Put $b_{j,-}(y) = -ya_j/2 + \beta_y(a_j-)$ and $b_{j,+}(y) = b_{j,-}(y) + J_j(y)$, with J_j the jump at the selected interface. Then

$$\sum_{j=1}^m \int_0^1 \int [Xu(a_j, y, t + b_{j,-}(y)) - Xu(a_j, y, t + b_{j,+}(y))]_+ d\nu(t) dy \leq 10.$$

This conclusion is restricted to the displayed common-trace class.

Proof. For $m \geq 2$ define $s_j = (c_{j+1} - c_j)/(a_{j+1} - a_j)$. Convexity along lines with fixed (y, z) makes these secant slopes nondecreasing and in $[-1, 1]$. At each interior plane, $s_{j-1} \leq u_x(a_j, y, z) \leq s_j$, so its normal-derivative oscillation on the rectangle is at most $s_j - s_{j-1}$. These oscillations sum to at most two. Each endpoint plane contributes at most two. Thus a uniform bound of six holds also when $m = 1$ or $m = 2$.

The common trace gives $u_z = \psi_z$ on every selected plane. For fixed y , $\psi_z(y, \cdot)$ is nondecreasing and lies in $[-1, 1]$. The horizontal before-minus-after difference equals

$$u_x(a_j, y, t + b_{j,-}) - u_x(a_j, y, t + b_{j,+}) + \frac{y}{2}[\psi_z(y, t + b_{j,+}) - \psi_z(y, t + b_{j,-})].$$

The bracket is nonnegative. All sampled z values lie in $[-2, 2]$: $0 \leq \beta_y \leq 1/2$, $0 \leq t \leq 1$, and $0 \leq ya_j/2 \leq 1/2$. Fubini with the positive measure $\partial_z \psi_z$ and the Cantor interval bound therefore make the t -average of the second term at most

$$\frac{y}{2} 4J_j(y)^\alpha \int_{-2}^2 d(\psi_z(y, \cdot)) \leq 4J_j(y)^\alpha.$$

Generation i has 2^{i-1} interfaces, each of jump at most 4^{-i} . Even summing over all possible selected generations gives $\sum_{i \geq 1} 2^{i-1} 4^{-i\alpha} = \sum_{i \geq 1} 2^{i-1} 3^{-i} = 1$. Hence the total averaged central contribution is at most four. Its nonnegativity permits taking positive parts at the cost of the normal oscillation bound, and the total is at most $6 + 4 = 10$. $\square$

Neither finite-interface proposition determines whether the unrestricted digit measure satisfies a uniform Euclidean convex-test budget. The first bound still grows with generation; the second imposes an essential common tangential trace. They do not furnish a stationary PDE or an entropy counterexample.

21 Scope of the conclusions

The unrestricted parabolic conclusion is established in the step-two Carnot setting of Theorem 2.1, including measure input and uniform coordinate viscosity. Every extension retains the additional assumptions in its statement. Qualitative absolute continuity alone does not imply a uniform entropy bound. The general commutator, exponential-duality, and stopped-martingale criteria retain premises not proved here for arbitrary coefficients. The corresponding unrestricted entropy problem for arbitrary smooth Hörmander families or higher-step Carnot groups remains outside the conclusions of this manuscript.

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